



100

ULTRA-CUT™

PLASMA CUTTING SYSTEM

W/AUTOMATED GAS CONTROL



Art# A-08579

Operating Manual

Rev. AG
Operating Features:

Date: October 28, 2010

Manual # 0-5132

100
AMP

DC

3
PHASE

208-
230
V

400
V

400
V (CE)

460
V

600
V



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This Operating Manual has been designed to instruct you on the correct use and operation of your Thermal Dynamics product. Your satisfaction with this product and its safe operation is our ultimate concern. Therefore please take the time to read the entire manual, especially the Safety Precautions. They will help you to avoid potential hazards that may exist when working with this product.

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Above all, we are committed to developing technologically advanced products to achieve a safer working environment within the welding industry.



WARNINGS

Read and understand this entire Manual and your employer's safety practices before installing, operating, or servicing the equipment.

While the information contained in this Manual represents the Manufacturer's best judgment, the Manufacturer assumes no liability for its use.

Plasma Cutting Power Supply, Ultra-Cut® 100
Automated Gas Control, DFC-3000

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Record the following information for Warranty purposes:

Where Purchased: _____

Purchase Date: _____

Power Supply Serial #: _____

Torch Serial #: _____

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SECTION 1: GENERAL INFORMATION

1.01 Notes, Cautions and Warnings

Throughout this manual, notes, cautions, and warnings are used to highlight important information. These highlights are categorized as follows:

NOTE

An operation, procedure, or background information which requires additional emphasis or is helpful in efficient operation of the system.



CAUTION

A procedure which, if not properly followed, may cause damage to the equipment.



WARNING

A procedure which, if not properly followed, may cause injury to the operator or others in the operating area.

1.02 Important Safety Precautions



WARNINGS

OPERATION AND MAINTENANCE OF PLASMA ARC EQUIPMENT CAN BE DANGEROUS AND HAZARDOUS TO YOUR HEALTH.

Plasma arc cutting produces intense electric and magnetic emissions that may interfere with the proper function of cardiac pacemakers, hearing aids, or other electronic health equipment. Persons who work near plasma arc cutting applications should consult their medical health professional and the manufacturer of the health equipment to determine whether a hazard exists.

To prevent possible injury, read, understand and follow all warnings, safety precautions and instructions before using the equipment. Call 1-603-298-5711 or your local distributor if you have any questions.



GASES AND FUMES

Gases and fumes produced during the plasma cutting process can be dangerous and hazardous to your health.

- Keep all fumes and gases from the breathing area. Keep your head out of the welding fume plume.

- Use an air-supplied respirator if ventilation is not adequate to remove all fumes and gases.
- The kinds of fumes and gases from the plasma arc depend on the kind of metal being used, coatings on the metal, and the different processes. You must be very careful when cutting or welding any metals which may contain one or more of the following:

Antimony	Chromium	Mercury
Arsenic	Cobalt	Nickel
Barium	Copper	Selenium
Beryllium	Lead	Silver
Cadmium	Manganese	Vanadium

- Always read the Material Safety Data Sheets (MSDS) that should be supplied with the material you are using. These MSDSs will give you the information regarding the kind and amount of fumes and gases that may be dangerous to your health.
- For information on how to test for fumes and gases in your workplace, refer to item 1 in Subsection 1.03, Publications in this manual.
- Use special equipment, such as water or down draft cutting tables, to capture fumes and gases.
- Do not use the plasma torch in an area where combustible or explosive gases or materials are located.
- Phosgene, a toxic gas, is generated from the vapors of chlorinated solvents and cleansers. Remove all sources of these vapors.
- This product, when used for welding or cutting, produces fumes or gases which contain chemicals known to the State of California to cause birth defects and, in some cases, cancer. (California Health & Safety Code Sec. 25249.5 et seq.)



ELECTRIC SHOCK

Electric Shock can injure or kill. The plasma arc process uses and produces high voltage electrical energy. This electric energy can cause severe or fatal shock to the operator or others in the workplace.

- Never touch any parts that are electrically "live" or "hot."
- Wear dry gloves and clothing. Insulate yourself from the work piece or other parts of the welding circuit.
- Repair or replace all worn or damaged parts.
- Extra care must be taken when the workplace is moist or damp.
- Install and maintain equipment according to NEC code, refer to item 9 in Subsection 1.03, Publications.
- Disconnect power source before performing any service or repairs.
- Read and follow all the instructions in the Operating Manual.



FIRE AND EXPLOSION

Fire and explosion can be caused by hot slag, sparks, or the plasma arc.

- Be sure there is no combustible or flammable material in the workplace. Any material that cannot be removed must be protected.
- Ventilate all flammable or explosive vapors from the workplace.
- Do not cut or weld on containers that may have held combustibles.

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- Provide a fire watch when working in an area where fire hazards may exist.
- Hydrogen gas may be formed and trapped under aluminum workpieces when they are cut underwater or while using a water table. **DO NOT** cut aluminum alloys underwater or on a water table unless the hydrogen gas can be eliminated or dissipated. Trapped hydrogen gas that is ignited will cause an explosion.



NOISE

Noise can cause permanent hearing loss. Plasma arc processes can cause noise levels to exceed safe limits. You must protect your ears from loud noise to prevent permanent loss of hearing.

- To protect your hearing from loud noise, wear protective ear plugs and/or ear muffs. Protect others in the workplace.
- Noise levels should be measured to be sure the decibels (sound) do not exceed safe levels.
- For information on how to test for noise, see item 1 in Subsection 1.03, Publications, in this manual.



PLASMA ARC RAYS

Plasma Arc Rays can injure your eyes and burn your skin. The plasma arc process produces very bright ultra violet and infra red light. These arc rays will damage your eyes and burn your skin if you are not properly protected.

- To protect your eyes, always wear a welding helmet or shield. Also always wear safety glasses with side shields, goggles or other protective eye wear.
- Wear welding gloves and suitable clothing to protect your skin from the arc rays and sparks.
- Keep helmet and safety glasses in good condition. Replace lenses when cracked, chipped or dirty.
- Protect others in the work area from the arc rays. Use protective booths, screens or shields.
- Use the shade of lens as suggested in the following per ANSI/ASC Z49.1:

Arc Current	Minimum Protective Shade No.	Suggested Shade No.
Less Than 300*	8	9
300 - 400*	9	12
400 - 800*	10	14

** These values apply where the actual arc is clearly seen. Experience has shown that lighter filters may be used when the arc is hidden by the workpiece.*



LEAD WARNING

This product contains chemicals, including lead, or otherwise produces chemicals known to the State of California to cause cancer, birth defects and other reproductive harm. **Wash hands after handling.** (California Health & Safety Code § 25249.5 et seq.)

1.03 Publications

Refer to the following standards or their latest revisions for more information:

1. OSHA, SAFETY AND HEALTH STANDARDS, 29CFR 1910, obtainable from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402
2. ANSI Standard Z49.1, SAFETY IN WELDING AND CUTTING, obtainable from the American Welding Society, 550 N.W. LeJeune Rd, Miami, FL 33126
3. NIOSH, SAFETY AND HEALTH IN ARC WELDING AND GAS WELDING AND CUTTING, obtainable from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402
4. ANSI Standard Z87.1, SAFE PRACTICES FOR OCCUPATION AND EDUCATIONAL EYE AND FACE PROTECTION, obtainable from American National Standards Institute, 1430 Broadway, New York, NY 10018
5. ANSI Standard Z41.1, STANDARD FOR MEN'S SAFETY-TOE FOOTWEAR, obtainable from the American National Standards Institute, 1430 Broadway, New York, NY 10018
6. ANSI Standard Z49.2, FIRE PREVENTION IN THE USE OF CUTTING AND WELDING PROCESSES, obtainable from American National Standards Institute, 1430 Broadway, New York, NY 10018
7. AWS Standard A6.0, WELDING AND CUTTING CONTAINERS WHICH HAVE HELD COMBUSTIBLES, obtainable from American Welding Society, 550 N.W. LeJeune Rd, Miami, FL 33126
8. NFPA Standard 51, OXYGEN-FUEL GAS SYSTEMS FOR WELDING, CUTTING AND ALLIED PROCESSES, obtainable from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269
9. NFPA Standard 70, NATIONAL ELECTRICAL CODE, obtainable from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269
10. NFPA Standard 51B, CUTTING AND WELDING PROCESSES, obtainable from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269
11. CGA Pamphlet P-1, SAFE HANDLING OF COMPRESSED GASES IN CYLINDERS, obtainable from the Compressed Gas Association, 1235 Jefferson Davis Highway, Suite 501, Arlington, VA 22202
12. CSA Standard W117.2, CODE FOR SAFETY IN WELDING AND CUTTING, obtainable from the Canadian Standards Association, Standards Sales, 178 Rexdale Boulevard, Rexdale, Ontario, Canada M9W 1R3
13. NWSA booklet, WELDING SAFETY BIBLIOGRAPHY obtainable from the National Welding Supply Association, 1900 Arch Street, Philadelphia, PA 19103
14. American Welding Society Standard AWSF4.1, RECOMMENDED SAFE PRACTICES FOR THE PREPARATION FOR WELDING AND CUTTING OF CONTAINERS AND PIPING

THAT HAVE HELD HAZARDOUS SUBSTANCES, obtainable from the American Welding Society, 550 N.W. LeJeune Rd, Miami, FL 33126

15. ANSI Standard Z88.2, PRACTICE FOR RESPIRATORY PROTECTION, obtainable from American National Standards Institute, 1430 Broadway, New York, NY 10018

1.04 Note, Attention et Avertissement

Dans ce manuel, les mots “note,” “attention,” et “avertissement” sont utilisés pour mettre en relief des informations à caractère important. Ces mises en relief sont classifiées comme suit :

NOTE

Toute opération, procédure ou renseignement général sur lequel il importe d'insister davantage ou qui contribue à l'efficacité de fonctionnement du système.



ATTENTION

Toute procédure pouvant résulter l'endommagement du matériel en cas de non-respect de la procédure en question.



AVERTISSEMENT

Toute procédure pouvant provoquer des blessures de l'opérateur ou des autres personnes se trouvant dans la zone de travail en cas de non-respect de la procédure en question.

1.05 Precautions De Securite Importantes



AVERTISSEMENTS

L'OPÉRATION ET LA MAINTENANCE DU MATÉRIEL DE SOUDAGE À L'ARC AU JET DE PLASMA PEUVENT PRÉSENTER DES RISQUES ET DES DANGERS DE SANTÉ.

Coupant à l'arc au jet de plasma produit de l'énergie électrique haute tension et des émissions magnétique qui peuvent interférer la fonction propre d'un "pacemaker" cardiaque, les appareils auditif, ou autre matériel de santé électronique. Ceux qui travail près d'une application à l'arc au jet de plasma devrait consulter leur membre professionnel de médication et le manufacturier de matériel de santé pour déterminer s'il existe des risques de santé.

Il faut communiquer aux opérateurs et au personnel TOUS les dangers possibles. Afin d'éviter les blessures possibles, lisez, comprenez et suivez tous les avertissements, toutes les précautions de sécurité et toutes les consignes avant d'utiliser le matériel. Composez le + 603-298-5711 ou votre distributeur local si vous avez des questions.



FUMÉE et GAZ

La fumée et les gaz produits par le procédé de jet de plasma peuvent présenter des risques et des dangers de santé.

- Eloignez toute fumée et gaz de votre zone de respiration. Gardez votre tête hors de la plume de fumée provenant du chalumeau.
- Utilisez un appareil respiratoire à alimentation en air si l'aération fournie ne permet pas d'éliminer la fumée et les gaz.
- Les sortes de gaz et de fumée provenant de l'arc de plasma dépendent du genre de métal utilisé, des revêtements se trouvant sur le métal et des différents procédés. Vous devez prendre soin lorsque vous coupez ou soudez tout métal pouvant contenir un ou plusieurs des éléments suivants:

antimoine	cadmium	mercure
argent	chrome	nickel
arsenic	cobalt	plomb
baryum	cuivre	sélénium
béryllium	manganèse	vanadium

- Lisez toujours les fiches de données sur la sécurité des matières (sigle américain "MSDS"); celles-ci devraient être fournies avec le matériel que vous utilisez. Les MSDS contiennent des renseignements quant à la quantité et la nature de la fumée et des gaz pouvant poser des dangers de santé.
- Pour des informations sur la manière de tester la fumée et les gaz de votre lieu de travail, consultez l'article 1 et les documents cités à la page 5.
- Utilisez un équipement spécial tel que des tables de coupe à débit d'eau ou à courant descendant pour capter la fumée et les gaz.
- N'utilisez pas le chalumeau au jet de plasma dans une zone où se trouvent des matières ou des gaz combustibles ou explosifs.
- Le phosgène, un gaz toxique, est généré par la fumée provenant des solvants et des produits de nettoyage chlorés. Éliminez toute source de telle fumée.
- Ce produit, dans le procédé de soudage et de coupe, produit de la fumée ou des gaz pouvant contenir des éléments reconnu dans L'état de la Californie, qui peuvent causer des défauts de naissance et le cancer. (La sécurité de santé en Californie et la code sécurité Sec. 25249.5 et seq.)



CHOC ELECTRIQUE

Les chocs électriques peuvent blesser ou même tuer. Le procédé au jet de plasma requiert et produit de l'énergie électrique haute tension. Cette énergie électrique peut produire des chocs graves, voire mortels, pour l'opérateur et les autres personnes sur le lieu de travail.

- Ne touchez jamais une pièce "sous tension" ou "vive"; portez des gants et des vêtements secs. Isolez-vous de la pièce de travail ou des autres parties du circuit de soudage.
- Réparez ou remplacez toute pièce usée ou endommagée.
- Prenez des soins particuliers lorsque la zone de travail est humide ou moite.

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- Montez et maintenez le matériel conformément au Code électrique national des Etats-Unis. (Voir la page 5, article 9.)
- Débranchez l'alimentation électrique avant tout travail d'entretien ou de réparation.
- Lisez et respectez toutes les consignes du Manuel de consignes.



INCENDIE ET EXPLOSION

Les incendies et les explosions peuvent résulter des scories chaudes, des étincelles ou de l'arc de plasma. Le procédé à l'arc de plasma produit du métal, des étincelles, des scories chaudes pouvant mettre le feu aux matières combustibles ou provoquer l'explosion de fumées inflammables.

- Soyez certain qu'aucune matière combustible ou inflammable ne se trouve sur le lieu de travail. Protégez toute telle matière qu'il est impossible de retirer de la zone de travail.
- Procurez une bonne aération de toutes les fumées inflammables ou explosives.
- Ne coupez pas et ne soudez pas les conteneurs ayant pu renfermer des matières combustibles.
- Prévoyez une veille d'incendie lors de tout travail dans une zone présentant des dangers d'incendie.
- Le gas hydrogène peut se former ou s'accumuler sous les pièces de travail en aluminium lorsqu'elles sont coupées sous l'eau ou sur une table d'eau. NE PAS couper les alliages en aluminium sous l'eau ou sur une table d'eau à moins que le gas hydrogène peut s'échapper ou se dissiper. Le gas hydrogène accumulé explosera si enflammé.



RAYONS D'ARC DE PLASMA

Les rayons provenant de l'arc de plasma peuvent blesser vos yeux et brûler votre peau. Le procédé à l'arc de plasma produit une lumière infra-rouge et des rayons ultra-violets très forts. Ces rayons d'arc nuiront à vos yeux et brûleront votre peau si vous ne vous protégez pas correctement.

- Pour protéger vos yeux, portez toujours un casque ou un écran de soudeur. Portez toujours des lunettes de sécurité munies de parois latérales ou des lunettes de protection ou une autre sorte de protection oculaire.
- Portez des gants de soudeur et un vêtement protecteur approprié pour protéger votre peau contre les étincelles et les rayons de l'arc.
- Maintenez votre casque et vos lunettes de protection en bon état. Remplacez toute lentille sale ou comportant fissure ou rognure.
- Protégez les autres personnes se trouvant sur la zone de travail contre les rayons de l'arc en fournissant des cabines ou des écrans de protection.

- Utilisez la nuance de lentille qui est suggérée dans le recommandation qui suivent ANSI/ASC Z49.1:

Courant Arc	Nuance Minimum Protective Numéro	Nuance Suggestée Numéro
Moins de 300*	8	9
300 - 400*	9	12
400 - 800*	10	14

* Ces valeurs s'appliquent ou l'arc actuel est observé clairement. L'expérience a démontré que les filtres moins foncés peuvent être utilisés quand l'arc est caché par moicéau de travail.



BRUIT

Le bruit peut provoquer une perte permanente de l'ouïe. Les procédés de soudage à l'arc de plasma peuvent provoquer des niveaux sonores supérieurs aux limites normalement acceptables. Vous devez vous protéger les oreilles contre les bruits forts afin d'éviter une perte permanente de l'ouïe.

- Pour protéger votre ouïe contre les bruits forts, portez des tampons protecteurs et/ou des protections auriculaires. Protégez également les autres personnes se trouvant sur le lieu de travail.
- Il faut mesurer les niveaux sonores afin d'assurer que les décibels (le bruit) ne dépassent pas les niveaux sûrs.
- Pour des renseignements sur la manière de tester le bruit, consultez l'article 1, page 5.



PLOMB AVERTISSEMENT

Ce produit contient des produits chimiques, comme le plomb, ou engendre des produits chimiques, reconnus par l'état de Californie comme pouvant être à l'origine de cancer, de malformations fœtales ou d'autres problèmes de reproduction. **Il faut se laver les mains après toute manipulation.** (Code de Californie de la sécurité et santé, paragraphe 25249.5 et suivants)

1.06 Documents De Reference

Consultez les normes suivantes ou les révisions les plus récentes ayant été faites à celles-ci pour de plus amples renseignements :

1. OSHA, NORMES DE SÉCURITÉ DU TRAVAIL ET DE PROTECTION DE LA SANTÉ, 29CFR 1910, disponible auprès du Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402
2. Norme ANSI Z49.1, LA SÉCURITÉ DES OPÉRATIONS DE COUPE ET DE SOUDAGE, disponible auprès de la Société Américaine de Soudage (American Welding Society), 550 N.W. LeJeune Rd., Miami, FL 33126
3. NIOSH, LA SÉCURITÉ ET LA SANTÉ LORS DES OPÉRATIONS DE COUPE ET DE SOUDAGE À L'ARC ET AU GAZ, disponible auprès du Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402
4. Norme ANSI Z87.1, PRATIQUES SURES POUR LA PROTECTION DES YEUX ET DU VISAGE AU TRAVAIL ET DANS LES ECOLES, disponible de l'Institut Américain des Normes Nationales (American National Standards Institute), 1430 Broadway, New York, NY 10018
5. Norme ANSI Z41.1, NORMES POUR LES CHAUSSURES PROTECTRICES, disponible auprès de l'American National Standards Institute, 1430 Broadway, New York, NY 10018
6. Norme ANSI Z49.2, PRÉVENTION DES INCENDIES LORS DE L'EMPLOI DE PROCÉDÉS DE COUPE ET DE SOUDAGE, disponible auprès de l'American National Standards Institute, 1430 Broadway, New York, NY 10018
7. Norme A6.0 de l'Association Américaine du Soudage (AWS), LE SOUDAGE ET LA COUPE DE CONTENEURS AYANT RENFERMÉ DES PRODUITS COMBUSTIBLES, disponible auprès de la American Welding Society, 550 N.W. LeJeune Rd., Miami, FL 33126
8. Norme 51 de l'Association Américaine pour la Protection contre les Incendies (NFPA), LES SYSTEMES À GAZ AVEC ALIMENTATION EN OXYGÈNE POUR LE SOUDAGE, LA COUPE ET LES PROCÉDÉS ASSOCIÉS, disponible auprès de la National Fire Protection Association, Batterymarch Park, Quincy, MA 02269
9. Norme 70 de la NFPA, CODE ELECTRIQUE NATIONAL, disponible auprès de la National Fire Protection Association, Batterymarch Park, Quincy, MA 02269
10. Norme 51B de la NFPA, LES PROCÉDÉS DE COUPE ET DE SOUDAGE, disponible auprès de la National Fire Protection Association, Batterymarch Park, Quincy, MA 02269
11. Brochure GCA P-1, LA MANIPULATION SANS RISQUE DES GAZ COMPRIMÉS EN CYLINDRES, disponible auprès de l'Association des Gaz Comprimés (Compressed Gas Association), 1235 Jefferson Davis Highway, Suite 501, Arlington, VA 22202
12. Norme CSA W117.2, CODE DE SÉCURITÉ POUR LE SOUDAGE ET LA COUPE, disponible auprès de l'Association des Normes Canadiennes, Standards Sales, 178 Rexdale Boulevard, Rexdale, Ontario, Canada, M9W 1R3
13. Livret NWSA, BIBLIOGRAPHIE SUR LA SÉCURITÉ DU SOUDAGE, disponible auprès de l'Association Nationale de Fournitures de Soudage (National Welding Supply Association), 1900 Arch Street, Philadelphia, PA 19103
14. Norme AWSF4.1 de l'Association Américaine de Soudage, RECOMMANDATIONS DE PRATIQUES SURES POUR LA PRÉPARATION À LA COUPE ET AU SOUDAGE DE CONTENEURS ET TUYAUX AYANT RENFERMÉ DES PRODUITS DANGEREUX , disponible auprès de la American Welding Society, 550 N.W. LeJeune Rd., Miami, FL 33126
15. Norme ANSI Z88.2, PRATIQUES DE PROTECTION RESPIRATOIRE, disponible auprès de l'American National Standards Institute, 1430 Broadway, New York, NY 10018

1.07 Declaration of Conformity

Manufacturer: Thermal Dynamics Corporation
Address: 82 Benning Street
West Lebanon, New Hampshire 03784
USA

The equipment described in this manual conforms to all applicable aspects and regulations of the 'Low Voltage Directive' (2006/95 EC) and to the National legislation for the enforcement of this Directive.

The equipment described in this manual conforms to all applicable aspects and regulations of the "EMC Directive" (European Council Directive 89/336/EEC) and to the National legislation for the enforcement of this Directive.

Serial numbers are unique with each individual piece of equipment and details description, parts used to manufacture a unit and date of manufacture.

National Standard and Technical Specifications

The product is designed and manufactured to a number of standards and technical requirements. Among them are:

- * CSA (Canadian Standards Association) standard C22.2 number 60 for Arc welding equipment.
- * UL (Underwriters Laboratory) rating 94VO flammability testing for all printed-circuit boards used.
- * CENELEC EN50199 EMC Product Standard for Arc Welding Equipment.
- * ISO/IEC 60974-1 (BS 638-PT10) (EN 60 974-1) (EN50192) (EN50078) applicable to plasma cutting equipment and associated accessories.
- * For environments with increased hazard of electrical shock, Power Supplies bearing the 'S' mark conform to EN50192 when used in conjunction with hand torches with exposed cutting tips, if equipped with properly installed standoff guides.
- * Extensive product design verification is conducted at the manufacturing facility as part of the routine design and manufacturing process. This is to ensure the product is safe, when used according to instructions in this manual and related industry standards, and performs as specified. Rigorous testing is incorporated into the manufacturing process to ensure the manufactured product meets or exceeds all design specifications.

Thermal Dynamics has been manufacturing products for more than 30 years, and will continue to achieve excellence in our area of manufacture.

Manufacturers responsible representative: Steve Ward
Operations Director
Thermadyne Europe
Europa Building
Chorley N Industrial Park
Chorley, Lancashire,
England PR6 7BX



1.08 Statement of Warranty

LIMITED WARRANTY: Thermal Dynamics® Corporation (hereinafter "Thermal") warrants that its products will be free of defects in workmanship or material. Should any failure to conform to this warranty appear within the time period applicable to the Thermal products as stated below, Thermal shall, upon notification thereof and substantiation that the product has been stored, installed, operated, and maintained in accordance with Thermal's specifications, instructions, recommendations and recognized standard industry practice, and not subject to misuse, repair, neglect, alteration, or accident, correct such defects by suitable repair or replacement, at Thermal's sole option, of any components or parts of the product determined by Thermal to be defective.

THIS WARRANTY IS EXCLUSIVE AND IS IN LIEU OF ANY WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

LIMITATION OF LIABILITY: Thermal shall not under any circumstances be liable for special or consequential damages, such as, but not limited to, damage or loss of purchased or replacement goods, or claims of customers of distributor (hereinafter "Purchaser") for service interruption. The remedies of the Purchaser set forth herein are exclusive and the liability of Thermal with respect to any contract, or anything done in connection therewith such as the performance or breach thereof, or from the manufacture, sale, delivery, resale, or use of any goods covered by or furnished by Thermal whether arising out of contract, negligence, strict tort, or under any warranty, or otherwise, shall not, except as expressly provided herein, exceed the price of the goods upon which such liability is based.

THIS WARRANTY BECOMES INVALID IF REPLACEMENT PARTS OR ACCESSORIES ARE USED WHICH MAY IMPAIR THE SAFETY OR PERFORMANCE OF ANY THERMAL PRODUCT.

THIS WARRANTY IS INVALID IF THE PRODUCT IS SOLD BY NON-AUTHORIZED PERSONS.

The limited warranty periods for this product shall be: A maximum of three (3) years from date of sale to an authorized distributor and a maximum of two (2) years from date of sale by such distributor to the Purchaser, and with further limitations on such two (2) year period (see chart below).

	Parts	<u>Labor</u>
AutoCut® and UltraCut® Power Supplies and Components	2 Years	1 Year
<u>Torch And Leads</u>		
XT™-300 / XT™-301 Torch (Excluding Consumable Parts)	1 Year	1 Year
<u>Repair/Replacement Parts</u>	90 Days	90 Days

Warranty repairs or replacement claims under this limited warranty must be submitted by an authorized Thermal Dynamics® repair facility within thirty (30) days of the repair. No transportation costs of any kind will be paid under this warranty. Transportation charges to send products to an authorized warranty repair facility shall be the responsibility of the customer. All returned goods shall be at the customer's risk and expense. This warranty supersedes all previous Thermal warranties.

Effective August 19, 2005

SECTION 2: SPECIFICATIONS

2.01 General Description Of The System

A typical Ultra-Cut® 100 system configuration will include:

- One Power Supply
- Remote Arc Starter
- Gas Control Module
- Torch Valve Assembly
- Precision Plasma Cutting Torch
- Set Of Connecting Leads
- Torch Spare Parts Kit

The components are connected at installation.

2.02 Plasma Power Supply

The power supply provides the necessary current for cutting operations. The power supply also monitors system performance, and cools and circulates the liquid coolant for the torch and leads.

2.03 Remote Arc Starter

This unit produces a temporary HF pulse to start the pilot arc. The pilot arc creates a path for the main arc to transfer to the work. When the main arc is established, the pilot arc shuts off.

2.04 Gas Control Module

This module allows remote setting of gas selection, pressures, and flows together with setting of cutting current.

2.05 Precision Plasma Cutting Torch

The torch delivers the controlled current to the work through the main arc, causing the metal to be cut.

2.06 Specifications & Electrical Requirements

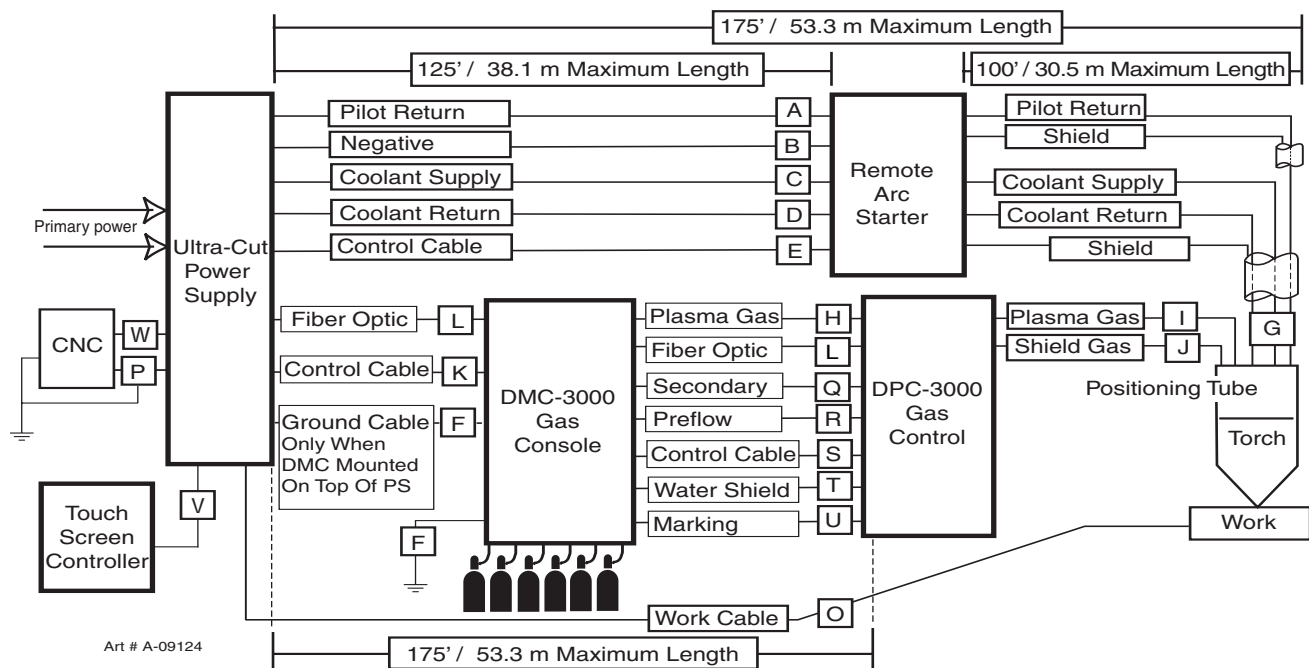
Ultra-Cut 100 Specifications & Design Features	
Input Power & Current	208/230V - 460V, 1/3 ph, 50 / 60 Hz
Max OCV (U ₀)	380 vdc
Minimum Output Current	10 Amps
Max Output Current	100 Amps
Output Voltage	60 - 230 vdc
Duty Cycle Rating	100% @ 100A, 180V
Ambient Temperature for Duty Cycle Rating	104F° (40°C)
Operating range	14°F to 122°F (-10°C to + 50°C)
Power Factor	0.70 @ 100 A DC Output
Cooling	Forced Air (Class F)

Ultra-Cut 100 Power Supply					
Input		Power Input	Current	Suggested Sizes (See Note)	
Voltage (Volts)	Freq. (Hz)	3-Ph (kVA)	3-Ph (Amps)	Fuse (Amps) 3-Ph	Wire (AWG) 3-Ph
208	50/60	23	64	80	#4 ¹
230	50/60	24	60	75	#4 ¹
400	50	28	40	50	#6 ¹
400 (CE)	50	28	40	50	#6 ¹
460	50/60	26	33	40	#8 ¹
600	50/60	25	24	30	#8

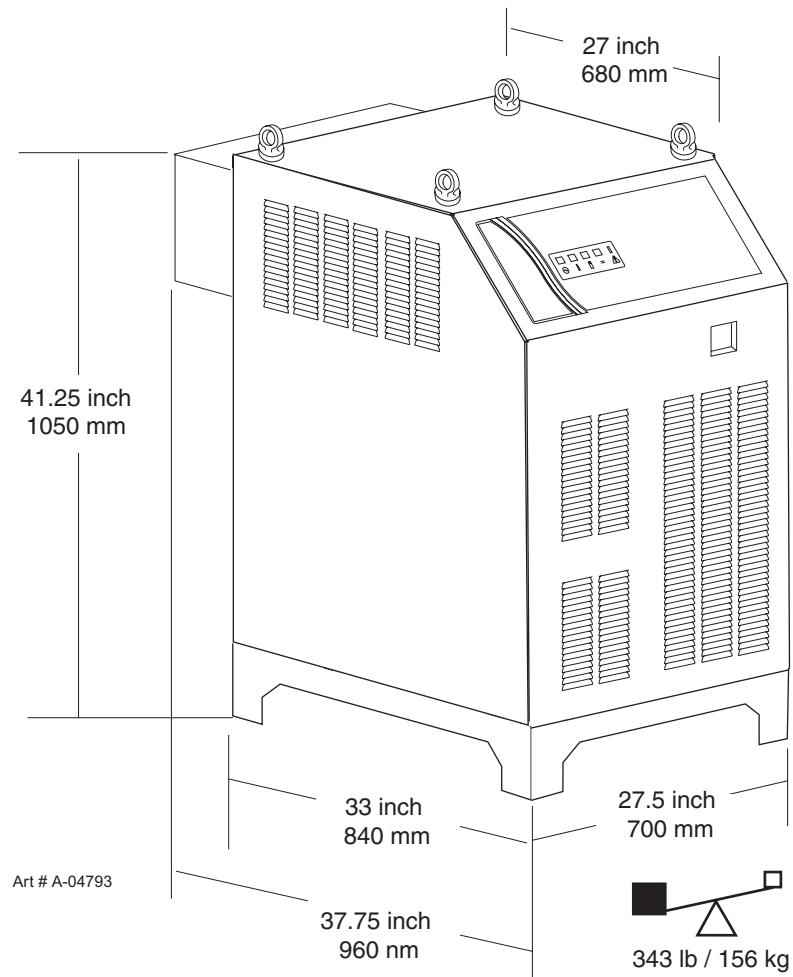
Line Voltages with Suggested Circuit Protection and Wire Sizes
Based on National Electric Code and Canadian Electrical Code

¹ Extra Hard Usage Type SO, SOW, SOO, SOOW, ST, STW, STO, STOW, STOO, STOOW

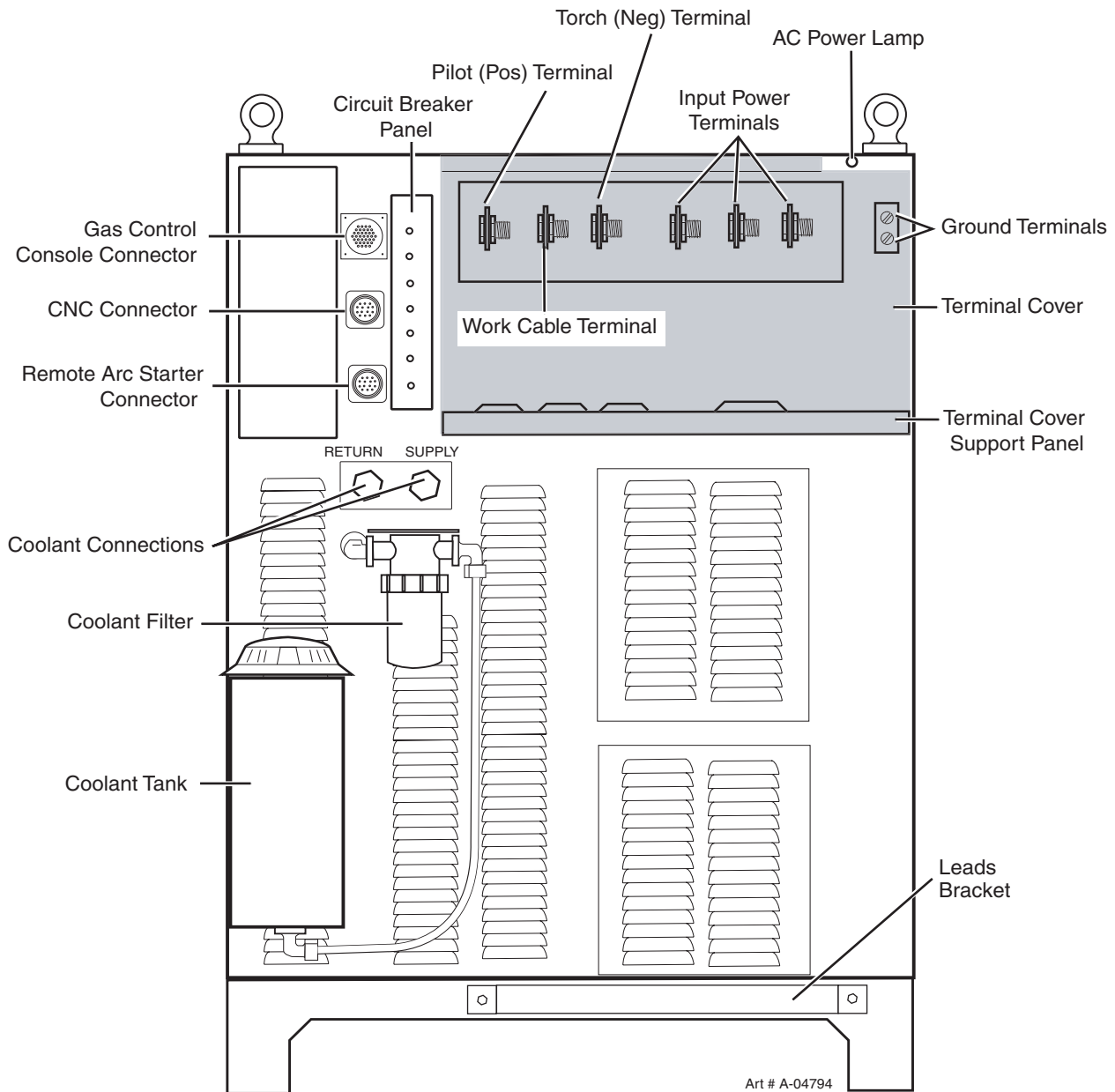
2.07 System Component Layout



2.08 Power Supply Dimensions



2.09 Power Supply Rear Panel Features



2.10 Gas Requirements

The customer will provide all gases and pressure regulators. Gases must be of high quality. Pressure regulators shall be double stage. The chart includes gases which may not be used in all applications.

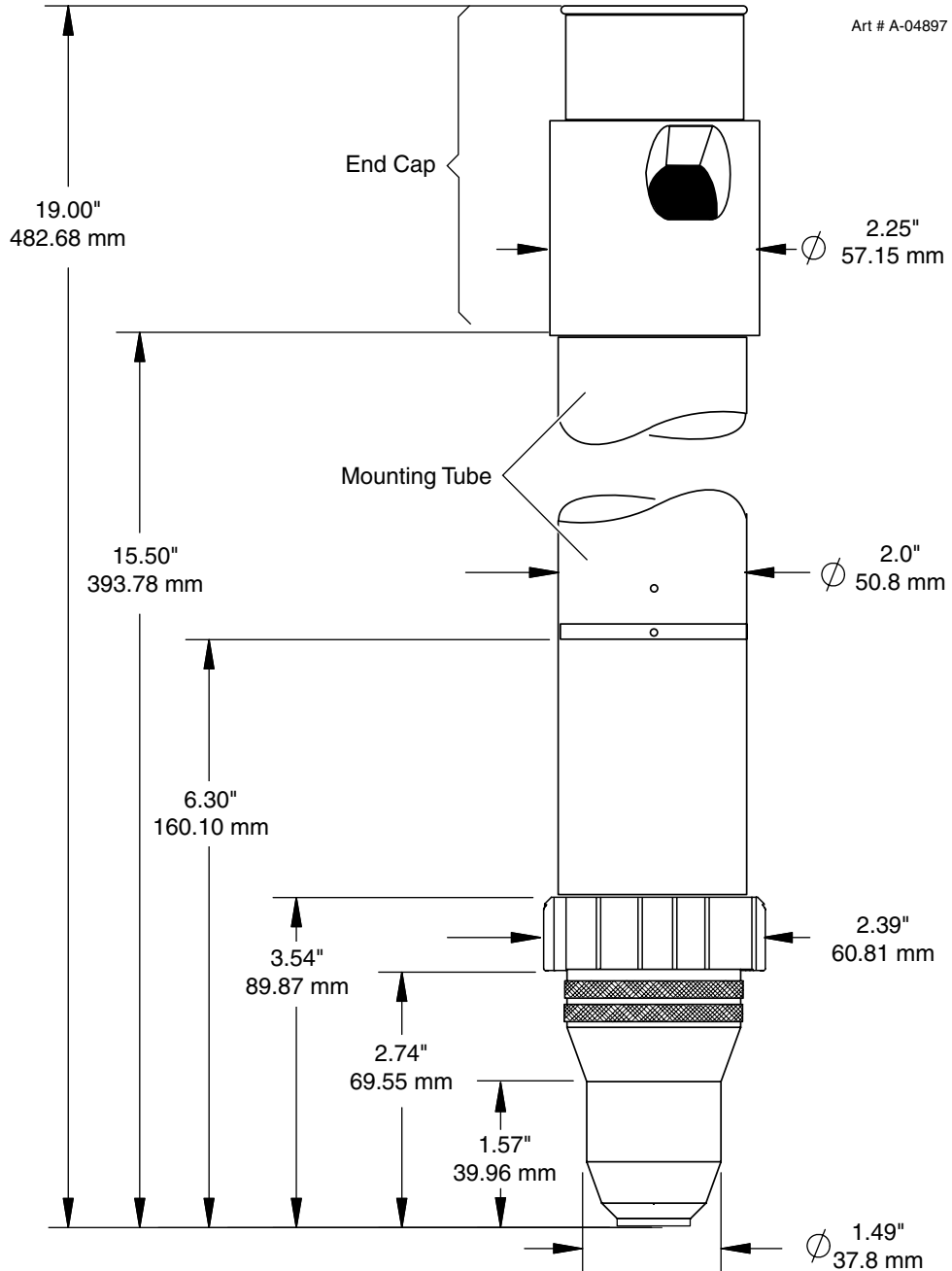
Ultra-Cut 100 Power Supply: Gas Pressures, Flows, and Quality Requirements			
Gas	Quality	Minimum Pressure	Flow
O ₂ (Oxygen)	99.5% Purity (Liquid recommended)	120 psi 8.3 bar / 827 kPa	70 scfh (1982 l/h)
N ₂ (Nitrogen)	99.5% Purity (Liquid recommended) <1000 ppm O ₂ , <32 ppm H ₂ O)	120 psi 8.3 bar / 827 kPa	200 scfh (5700 l/h)
Compressed or Bottled Air	Clean, Dry, Free of Oil (see Note 1)	120 psi 8.3 bar / 827 kPa	244 scfh (6910 l/h)
H35 (Argon-Hydrogen) H35 = 35% Hydrogen, 65% Argon	99.995% Purity (gas recommended)	120 psi 8.3 bar / 827 kPa	90 scfh (2550 l/h)
H ₂ O (Water)	See Note 2	50 psi (3.5 bar)	10 gph (38 lph)
Note 1: The air source must be adequately filtered to remove all oil or grease. Oil or grease contamination from compressed or bottled air can cause fires in conjunction with oxygen. For filtering, a coalescing filter able to filter to 0.01 microns should be placed as close as possible to the gas inlets on the Gas Control Module.			
Note 2: The tap water source does not need to be deionized, but in water systems with extremely high mineral content a water softener is recommended. Tap water with high levels of particulate matter must be filtered.			
Note 3: Water Pressure Regulator No. 8-6118 is recommended to ensure proper water pressure.			

2.11 Gas Applications

Ultra-Cut-100 Gas Applications									
MATERIAL	MILD STEEL			STAINLESS STEEL			ALUMINUM		
OPERATION	GAS TYPE			GAS TYPE			GAS TYPE		
	PREFLOW	PLASMA	SHIELD	PREFLOW	PLASMA	SHIELD	PREFLOW	PLASMA	SHIELD
30A CUT	Air	O ₂	O ₂	Air	Air	Air	Air	Air	Air
				N ₂	N ₂	H ₂ O	N ₂	N ₂	H ₂ O
50 A CUT	Air	O ₂	Air	Air	Air	Air	Air	Air	Air
				N ₂	N ₂	H ₂ O	N ₂	N ₂	H ₂ O
70 A CUT	Air	O ₂	Air	Air	Air	Air	Air	Air	Air
				N ₂	N ₂	H ₂ O	N ₂	N ₂	H ₂ O
85A CUT	Air	Air	Air						
100 A CUT	Air	O ₂	Air	N ₂	H35	N ₂	N ₂	H35	N ₂
				N ₂	N ₂	H ₂ O	N ₂	N ₂	H ₂ O

2.12 XT™-300 Torch Specifications

A. Torch Dimensions



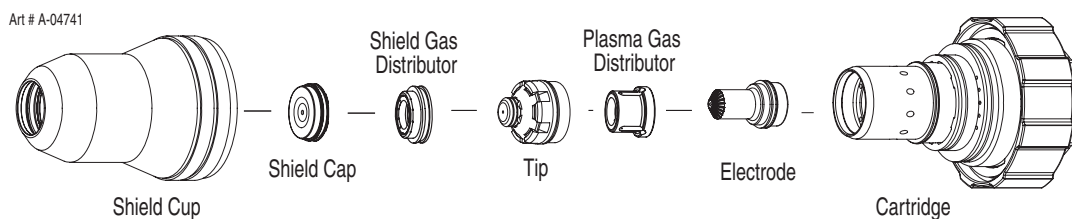
B. Torch Specifications

XT TM -300 Torch Ratings for use with Ultra-Cut 100 Power Supply	
Ambient Temperature	104° F 40° C
Duty Cycle	100% @ 100 Amps
Maximum Current	100 Amps
Voltage (V _{peak})	500V
Arc Striking Voltage	10kV
Current	Up to 100 Amps, DC, Straight Polarity
XT TM -300 Torch Gas Specifications	
Plasma Gases:	Compressed Air, Oxygen, Nitrogen, H35
Shield Gases:	Compressed Air, Oxygen, Nitrogen, Water
Operating Pressure	125 psi ± 10 psi 8.6 bar ± 0.7 bar
Maximum Input Pressure	135 psi / 9.3 bar
Gas flow	10 - 300 scfh

C. Leads Lengths

Gas Lead Assemblies	
Length	
Feet	Meters
10	3.05
15	4.6
25	7.6
35	10.6
50	15.2
75	22.8
100	30.4
125	38.2
150	45.7
175	53.3

D. Torch Parts (Generic Parts Shown)



E. Parts - In - Place (PIP)

The torch is designed for use with a power supply which senses coolant return flow to confirm that torch parts are in place. If coolant return flow to the power supply is absent or insufficient the power supply will not provide power to the torch. Coolant leakage from the torch also indicates that torch parts are absent or installed improperly.

F. Type of Cooling

Combination of gas stream through torch and liquid cooling.

SECTION 3: INSTALLATION

3.01 Installation Requirements

Electric Supply

The electrical supply network, the gas and water supply system must meet local safety standards. This conformity shall be checked by qualified personnel.

Ultra-Cut 100 Power Supply					
Input		Power Input	Current	Suggested Sizes (See Note)	
Voltage (Volts)	Freq. (Hz)	3-Ph (kVA)	3-Ph (Amps)	Fuse (Amps) 3-Ph	Wire (AWG) 3-Ph
208	50/60	23	64	80	#4 ¹
230	50/60	24	60	75	#4 ¹
400	50	28	40	50	#6 ¹
400 (CE)	50	28	40	50	#6 ¹
460	50/60	26	33	40	#8 ¹
600	50/60	25	24	30	#8
Line Voltages with Suggested Circuit Protection and Wire Sizes Based on National Electric Code and Canadian Electrical Code					
¹ Extra Hard Usage Type SO, SOW, SOO, SOOW, ST, STW, STO, STOW, STOO, STOOW					



CAUTION

Fuse and wire sizes are for reference only. The installation must conform to national and local codes for the type and method of wire being used.

Gas Supply

The customer must supply all gases and pressure regulators. Gases must be of high quality. Pressure regulators must be double-stage and installed as close as possible to the gas console. Contaminated gas can cause one or more of the following problems:

- Reduced cutting speed
- Poor cut quality
- Poor cutting precision
- Reduced consumables life.
- Oil or grease contamination from compressed or bottled air can cause fires in conjunction with oxygen.

Cooling System Requirements

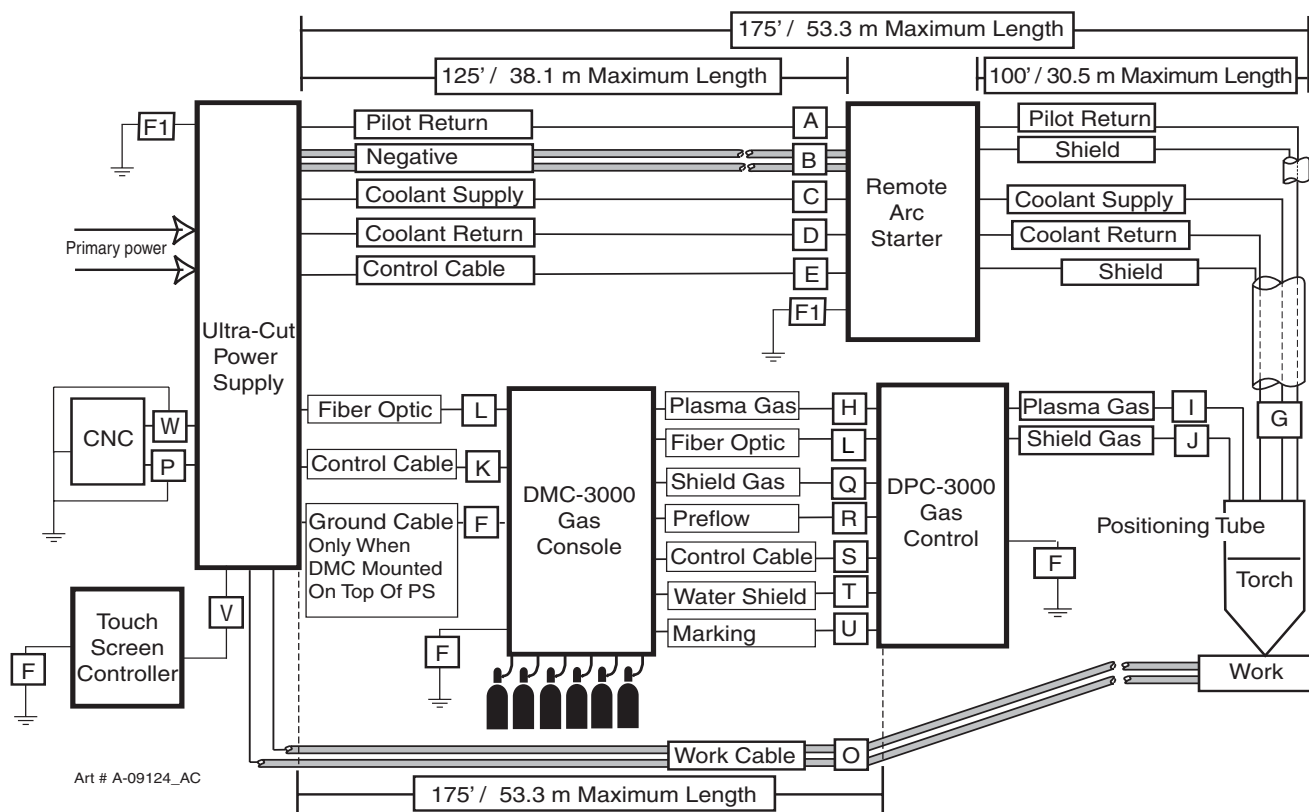
Coolant must be added to the system on installation. The amount required varies with torch leads length.

Thermal Dynamics recommends the use of its coolants 7-3580 and 7-3581 (for low temperatures).

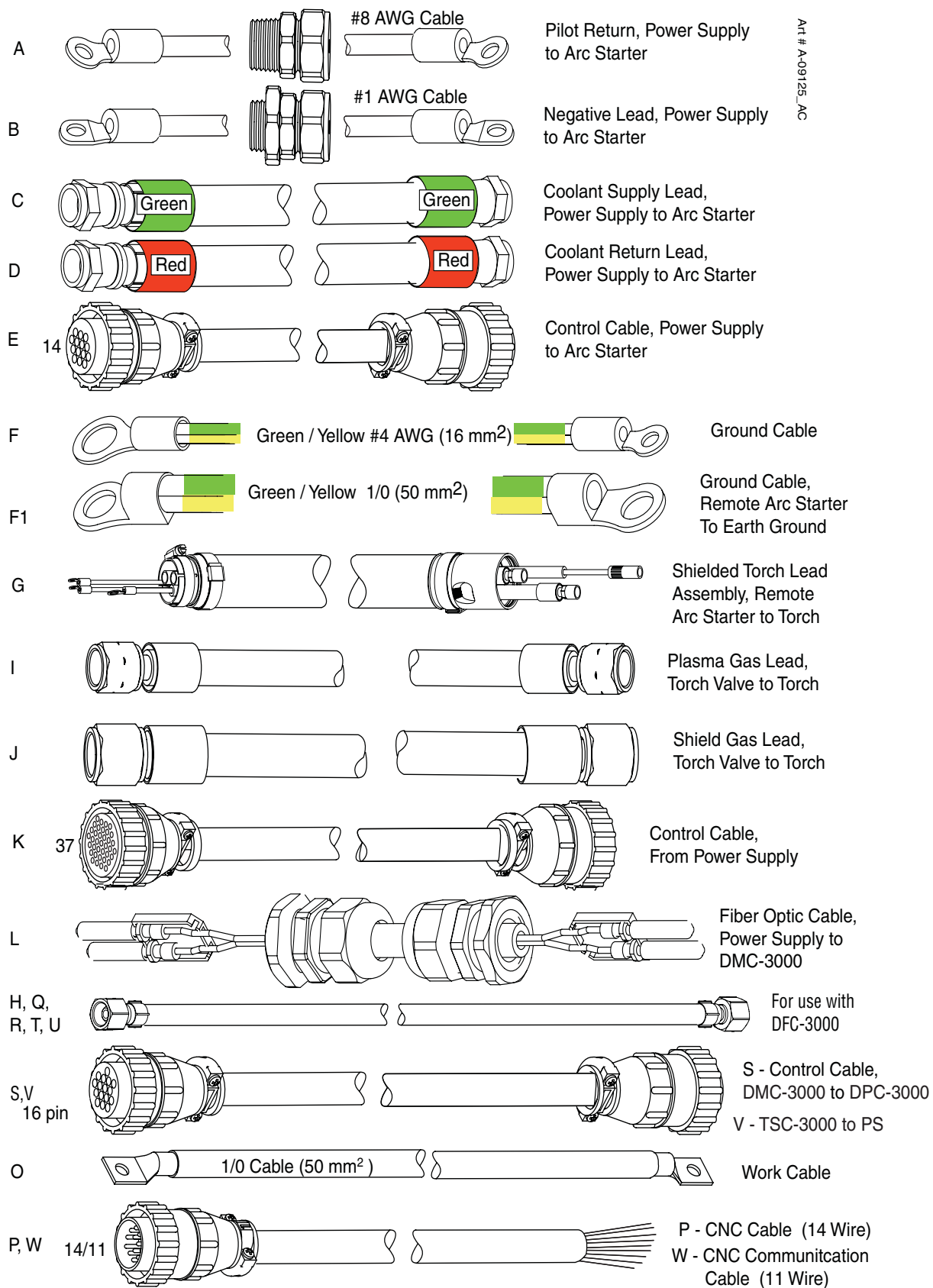
Coolant Capabilities		
Cat. Number and Mixture	Mixture	Protects To
7-3580 'Extra-Cool' TM	25 / 75	10° F / -12° C
7-3581 'Ultra-Cool' TM	50 / 50	-27° F / -33° C
7-3582 'Extreme Cool' TM	Concentrate*	-76° F / -60° C
* For mixing with D-I Cool TM 7-3583		

3.02 System Layout

Refer to section 3.05 for ground connections and ground cables.



3.03 Cables & Leads Identification



3.04 Lift the Power Supply



WARNINGS

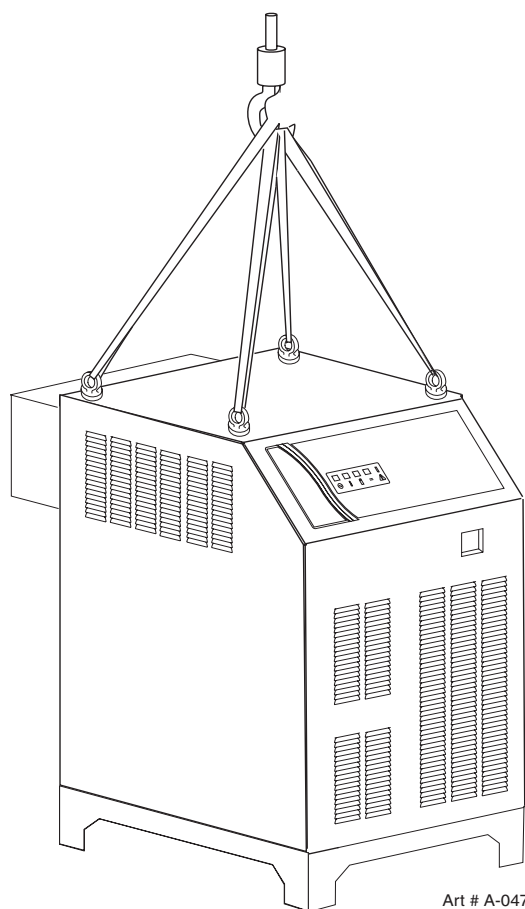
Do not touch live electrical parts.

Disconnect input power conductors from de-energized supply line before moving unit.

FALLING EQUIPMENT can cause serious personal injury and equipment damage.

Use all four lifting eyes when using lifting straps to lift the power supply.

Use a forklift, crane, or hoist to lift the unit off the shipping pallet as shown. Keep the power supply steady and vertical. Do not lift it any further than necessary to clear the shipping pallet. Ensure all panels and screws are secure prior to lifting.



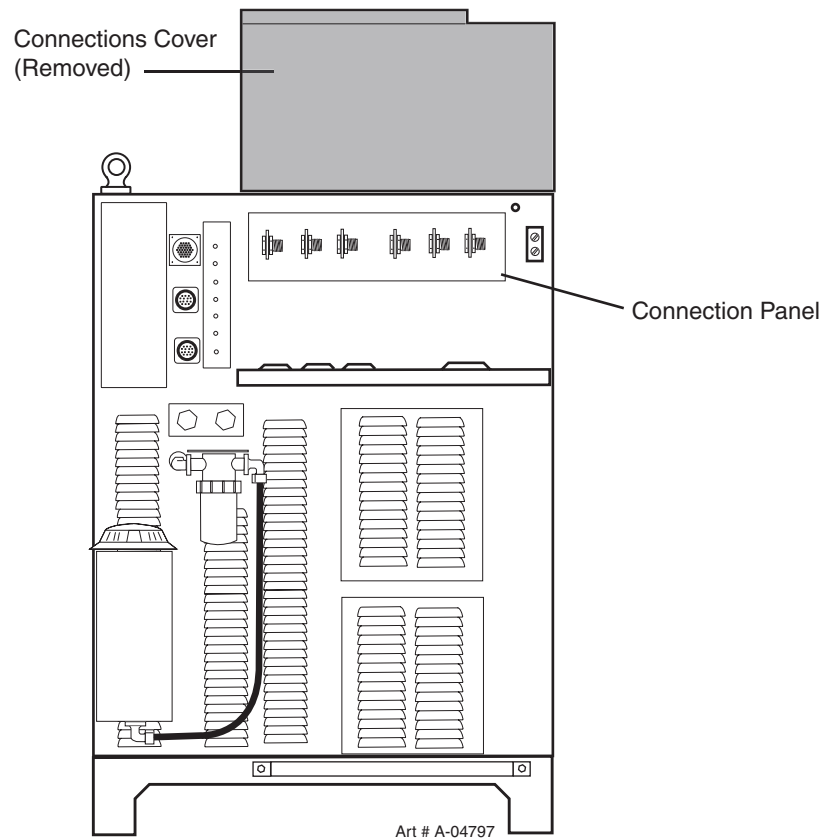
Art # A-04796

Set the power supply on a solid, level surface. The installer may fasten the power supply to the floor or a supporting fixture with hardware passing through the horizontal parts of the power supply feet.

3.05 Remove the Connections Cover

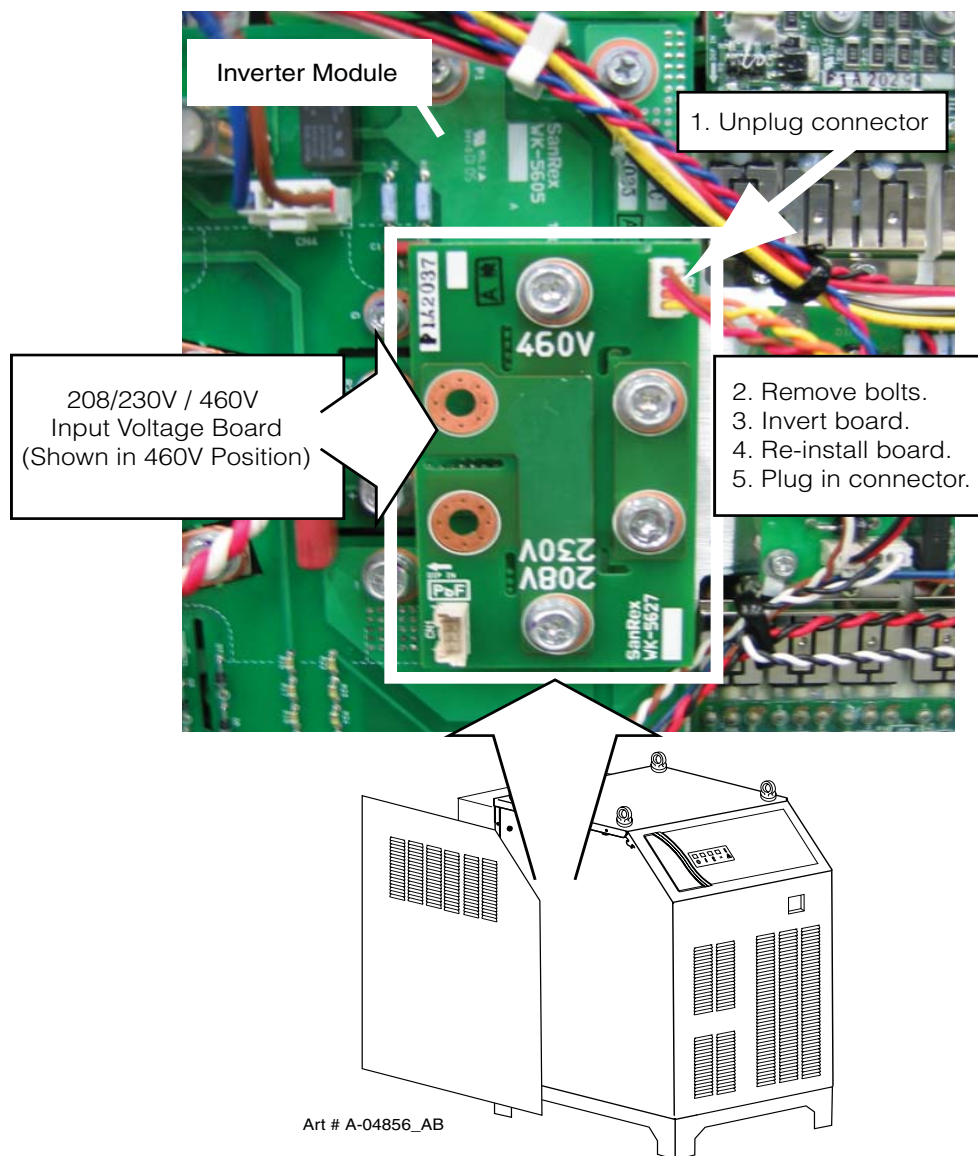
The primary power cable must be supplied by the end user and connected to the power supply.

Remove the connections cover on the rear of the Power Supply. Use caution when removing the panel; there is a ground wire connected to the inside of the panel. Do not disconnect this wire.



Check / Adjust Input Voltage Configuration for 208/230 and 460V Systems

1. The 208/230 and 460 V power supply includes a voltage configuration board which must be positioned to match the primary input voltage. Remove the power supply left side panel and locate the voltage configuration board. The input voltage configuration is shown at the top of the board.
2. If necessary, disconnect the jumper at the top right corner of the board, remove the board and re-install it with the correct input voltage shown at the top of the board.
3. Re-connect the jumper. Re-install the power supply side panel.



Connect Input Power and System Ground Cables

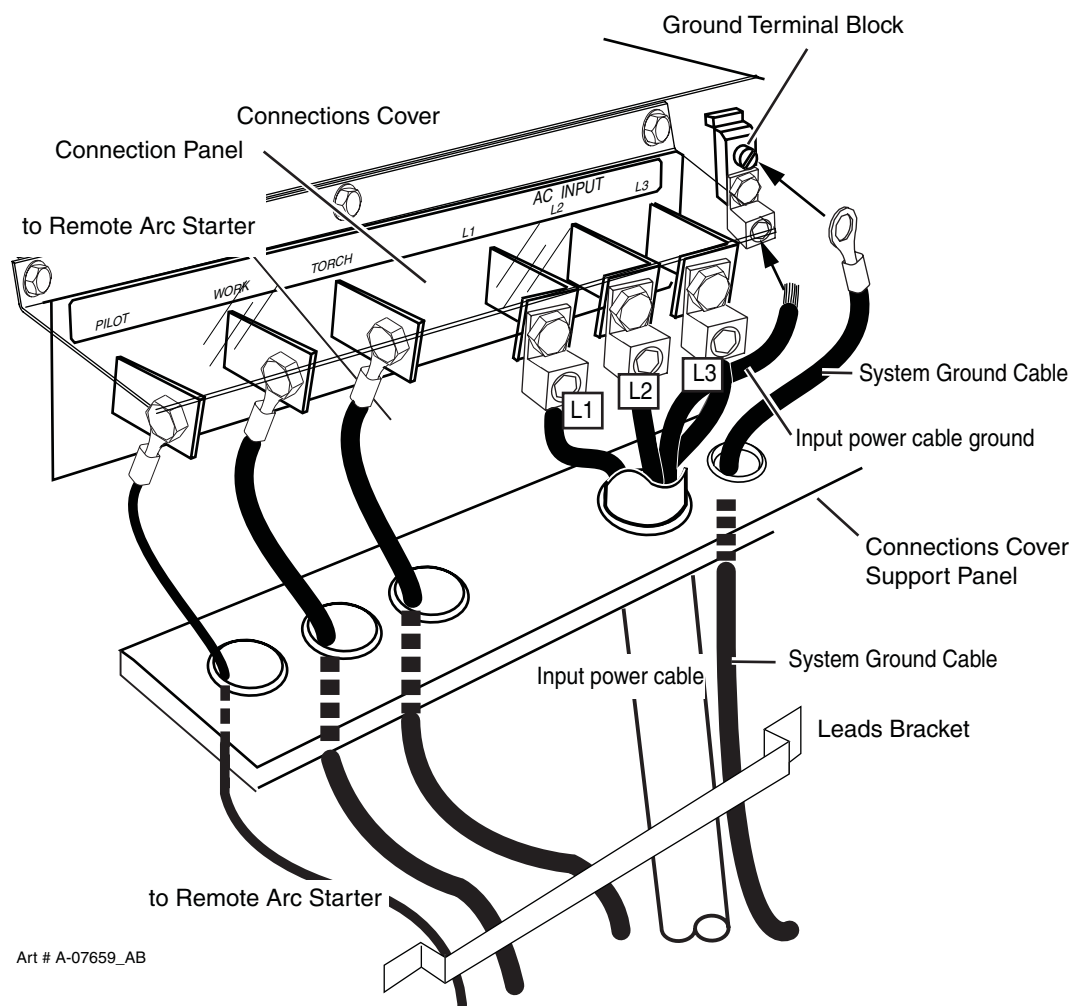
1. Carefully cut back the outer sheath on the primary input power cable to expose the individual wires. Cut back the insulation on the individual wires. Route the cable upward through leads bracket at the bottom of the rear panel, then through the connections cover support panel on the rear panel of the power supply.
2. Install stripped end of 3 phase wires into the terminal block L1, L2 and L3.
3. Connect the individual cables as shown. Connect the power cable ground cord to the ground terminal block.



CAUTION

The clear connections cover must remain in place.

4. If required pass a system ground cable (F1) through the last opening in the connections cover support panel next to the input power cable. Connect the cable to the ground terminal block on the power supply rear panel. Refer to the Ground Connections Section for full details and procedures on proper system grounding.



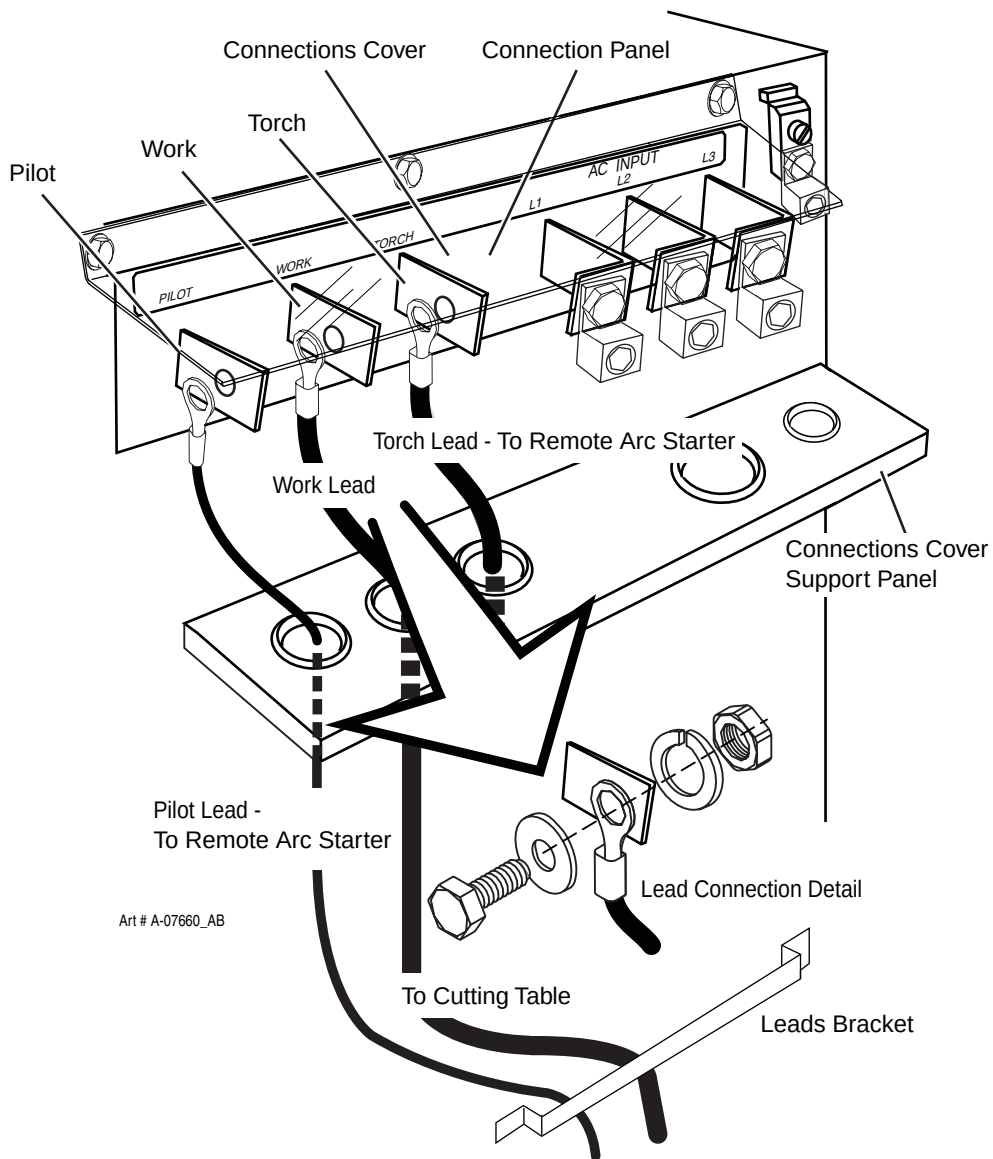
3.06 Connect Work Cable and Pilot and Negative Leads

1. Pass the ends of the work cable and pilot and negative leads upward through the leads bracket at the bottom edge of the rear panel, then through the openings in the connections cover support panel.
2. Refer to the illustration. Connect the leads as shown. Tighten securely. Do not overtighten.



CAUTION

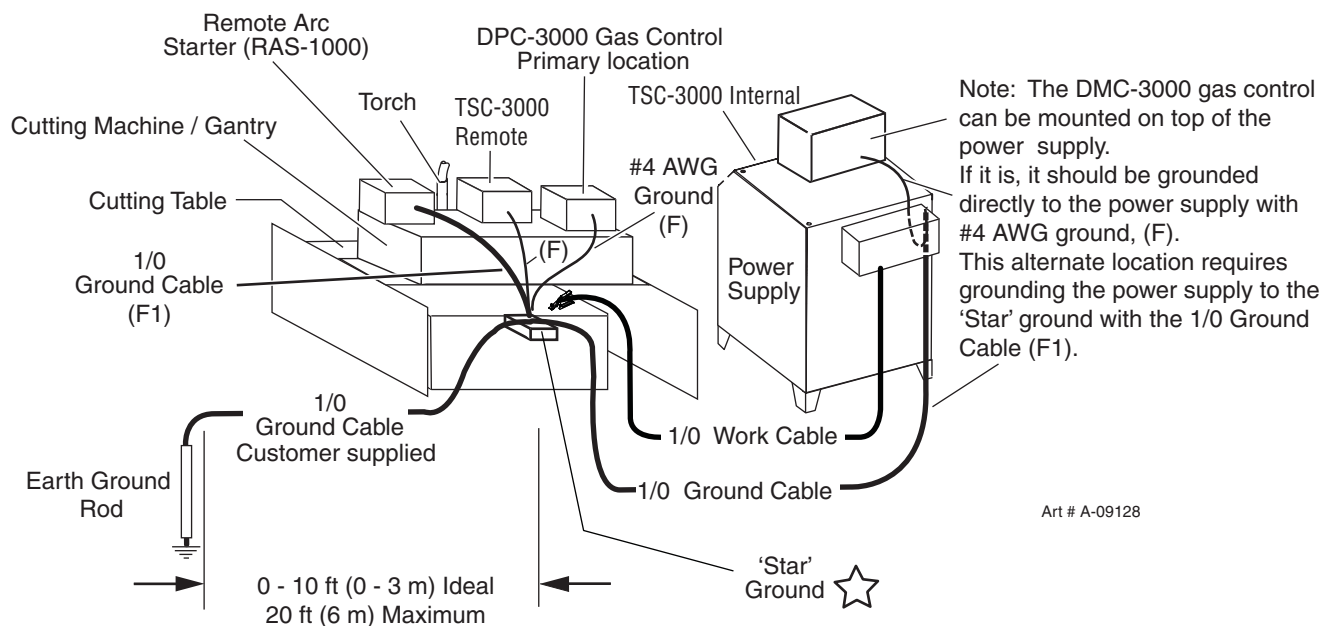
The clear connections cover must remain in place.



3. Re-install the connections cover on the power supply. Snug the hardware securely by hand. Do not overtighten.

3.07 Ground Connections

Star Ground on Cutting Table



A. Electromagnetic Interference (EMI)

Pilot arc starting generates a certain amount of electromagnetic interference (EMI), commonly called RF noise. This RF noise may interfere with other electronic equipment such as CNC controllers, remote controls, height controllers, etc. To minimize RF interference, follow these grounding procedures when installing mechanized systems:

B. Grounding

1. The preferred grounding arrangement is a single point or "Star" ground. The single point, usually on the cutting table, is connected with 1/0 AWG (European 50 mm²) or larger wire to a good earth ground (measuring less than 3 ohms; an ideal ground measures 1 ohm or less. Refer to paragraph 'C', Creating An Earth Ground. The ground rod must be placed as close as possible to the cutting table, ideally less than 10 ft (3.0 m), but no more than 20 ft (6.1 m) from the cutting table.

NOTE

All ground wires should be as short as possible. Long wires will have increased resistance to RF frequencies. Smaller diameter wire has increased resistance to RF frequencies, so using a larger diameter wire is better.

2. Grounding for components mounted on the cutting table (CNC controllers, height controllers, plasma remote controls, etc.) should follow the manufacturer's recommendations for wire size, type, and connection point locations.

For Thermal Dynamics components (except Remote Arc Starter and Gas Control Module) it is recommended to use a minimum of 10 AWG (European 6 mm²) wire or flat copper braid with cross section equal to or greater than 10 AWG connected to the cutting table frame. The Remote Arc Starter uses 1/0 earth ground wire and the Gas Control Module should use minimum # 4 AWG wire. The connection point must be to clean bare metal; rust and paint make poor connections. For all components, wires larger than the recommended minimum can be used and may improve noise protection.

3. The cutting machine frame is then connected to the “Star” point using 1/0 AWG (European 50 mm²) or larger wire.
4. The plasma power supply work cable (see NOTE) is connected to the cutting table at the single point “Star” ground.

NOTE

Do Not connect the work cable directly to the ground rod. Do not coil up excess ground or power cables. Cut to proper length and reterminate as needed.

5. Make sure work cable and ground cables are properly connected. The work cable must have a solid connection to the cutting table. The work and ground connections must be free from rust, dirt, grease, oil and paint. If necessary grind or sand down to bare metal. Use lock washers to keep the connections tight. Using electrical joint compound to prevent corrosion is also recommended.
6. The plasma power supply chassis is connected to the power distribution system ground as required by electrical codes. If the plasma supply is close to the cutting table (see NOTE) a second ground rod is not usually needed, in fact it could be detrimental as it can set up ground loop currents that cause interference.

When the plasma power supply is far away from the ground rod and interference is experienced, it may help to install a second earth ground rod next to the plasma power supply. The plasma power supply chassis would then be connected to this ground rod.

NOTE

It is recommended that the Plasma Power Supply be within 20 - 30 ft (6.1 – 9.1 m) of the cutting table, if possible.

7. The plasma control cable should be shielded with the shield connected only at the cutting machine end. Connecting the shield at both ends will allow ground loop currents which may cause more interference than with no shield at all.

Creating An Earth Ground

1. To create a solid, low resistance, earth ground, drive a 1/2 in (12 mm) or greater diameter copper clad ground rod at least 6 - 8 ft (1.8 - 2.4 m) into the earth so that the rod contacts moist soil over most of its length. Depending on location, a greater depth may be required to obtain a low resistance ground (see NOTE). Ground rods, typically 10 ft (3.0 m) long, may be welded end to end for greater lengths. Locate the rod as close as possible to the work table. Install a ground wire, 1/0 AWG (European 50 mm²) or greater, between the ground rod and the star ground point on the cutting table.

NOTE

Ideally, a properly installed ground rod will have a resistance of three ohms or less.

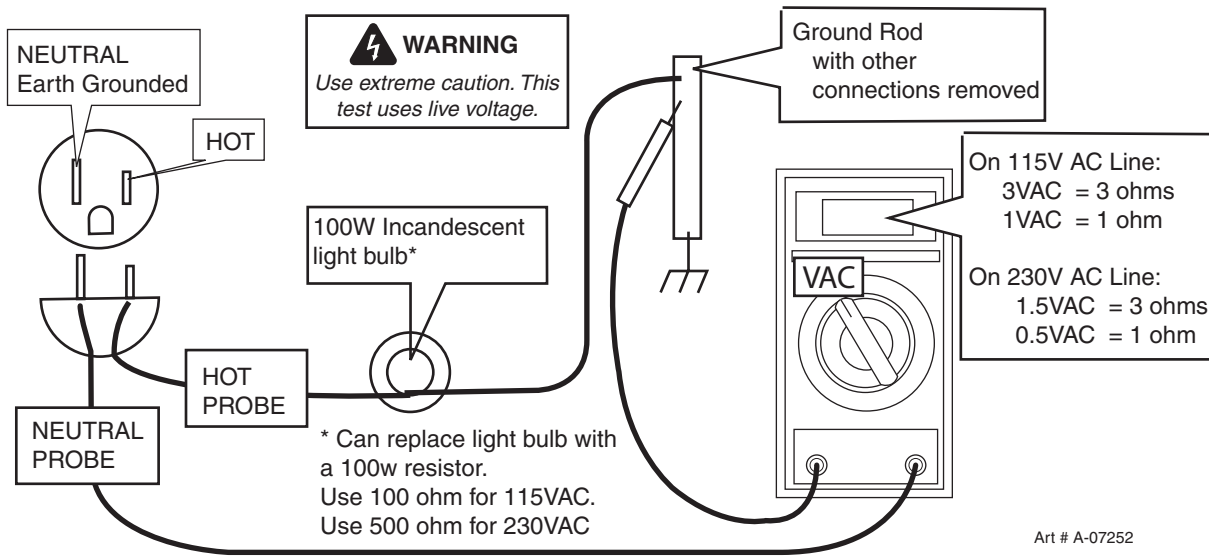
To test for a proper earth ground, refer to the following diagram. Ideally, the reading on the multimeter should be as shown.



CAUTION

No other connections should be made at the ground rod being tested.

This test assumes the 115 or 230 VAC source neutral is connected to the utility earth ground.



Ground Testing

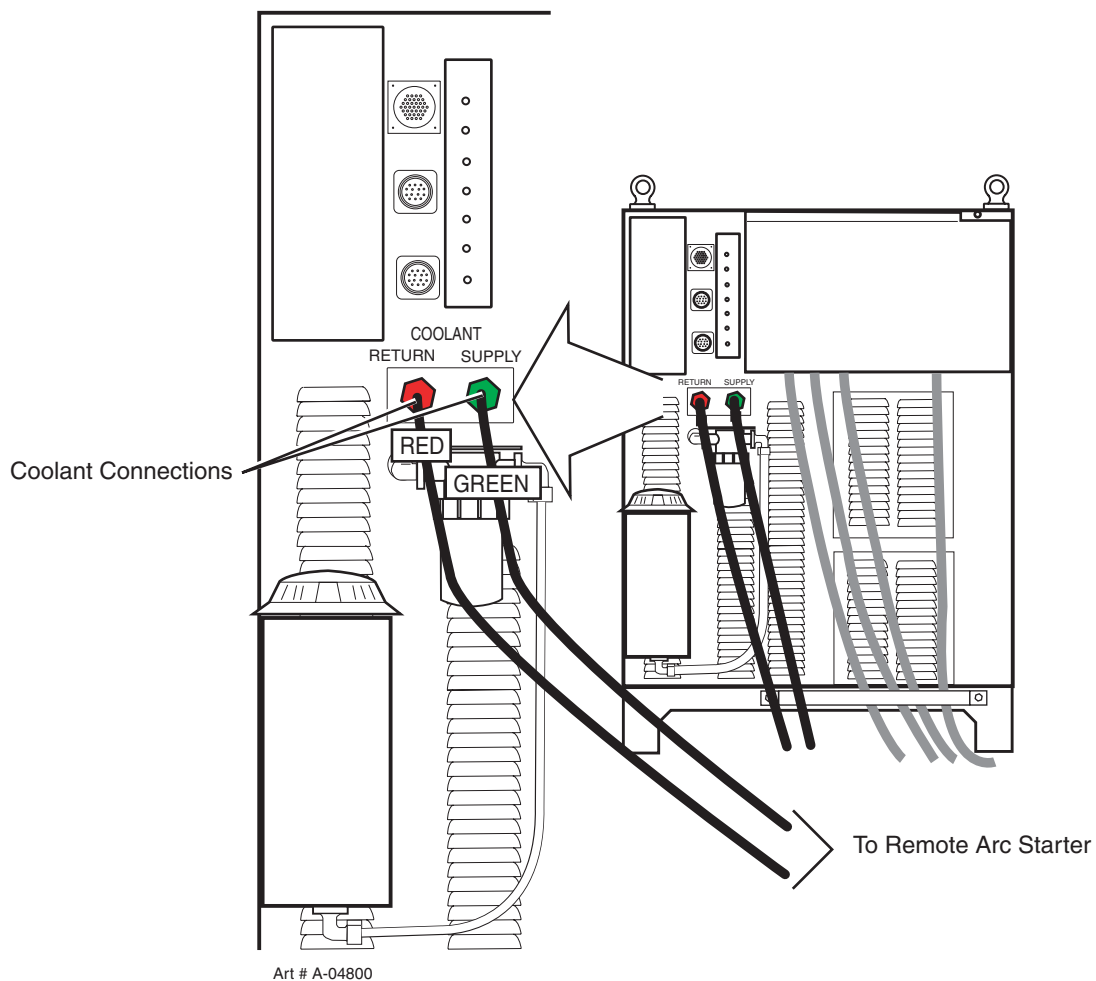
- Increasing the ground rod length beyond 20 - 30 ft (6.1 – 9.1 m) does not generally increase the effectiveness of the ground rod. A larger diameter rod which has more surface area may help. Sometimes keeping the soil around the ground rod moist by continuously running a small amount of water into it will work. Adding salt to the soil by soaking it in salt water may also reduce its resistance. You may also try a chemical ground rod devise. When these methods are used, periodic checking of the ground resistance is required to make sure the ground is still good.

D. Routing Of Torch Leads

- To minimize RF interference, position torch leads as far as possible from any CNC components, drive motors, control cables, or primary power lines. If cables have to pass over torch leads, do so at an angle. Do not run the plasma control or other control cables in parallel with the torch leads in power tracts.
- Keep torch leads clean. Dirt and metal particles bleed off energy, which causes difficult starting and increased chance of RF interference.

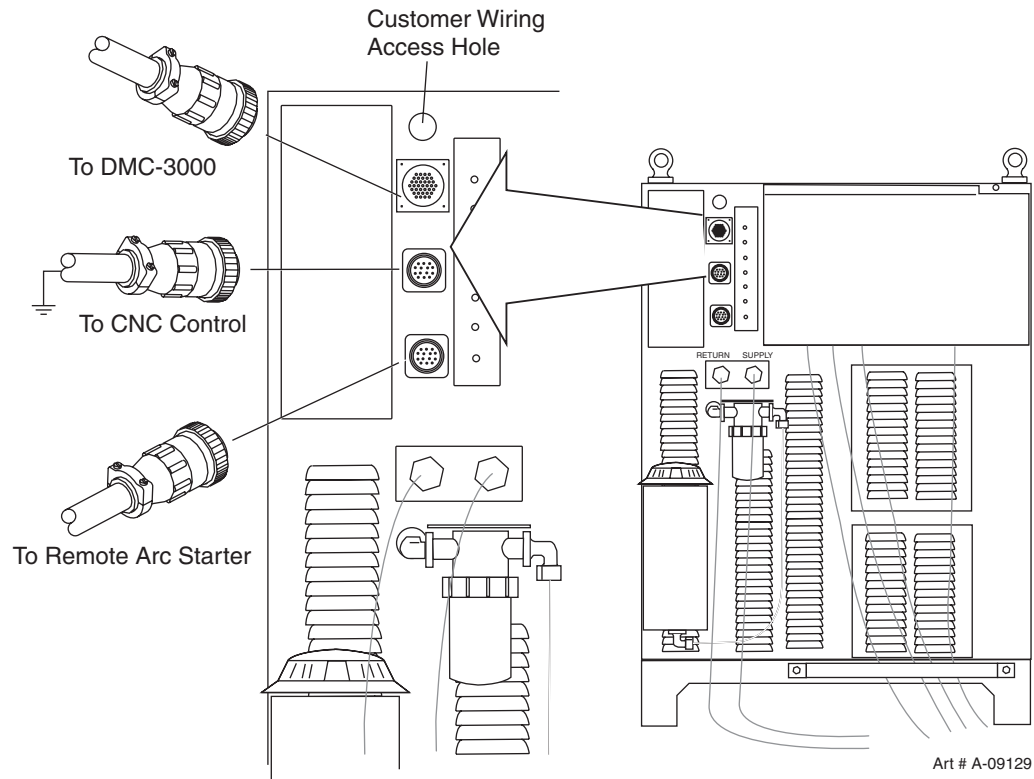
3.08 Connect Coolant Leads

1. Connect the color-coded coolant hoses to the coolant connections on the power supply rear panel. The supply line (out) is flagged green, the return line (in) is flagged red.



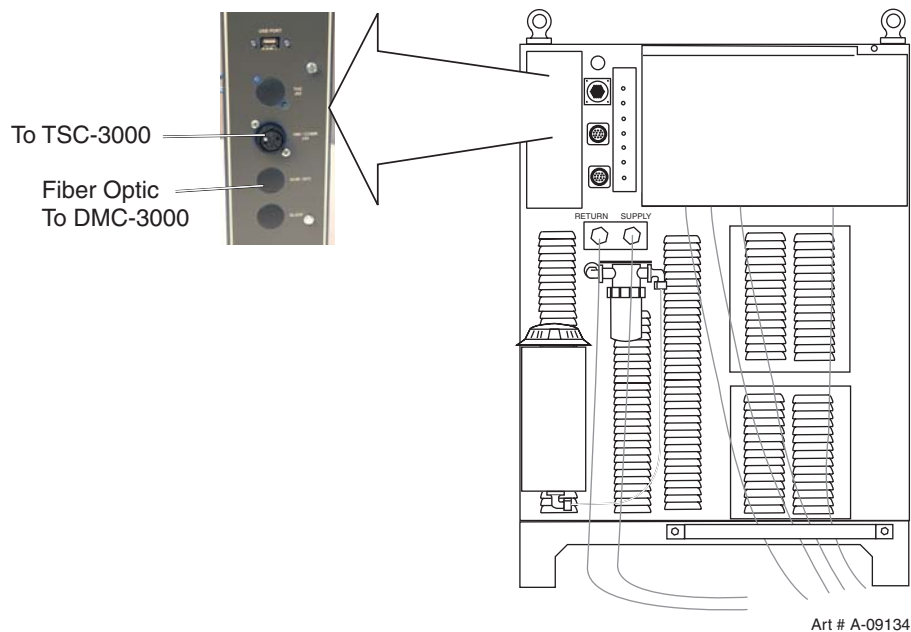
3.09 Connect Control Cables for CNC, Remote Arc Starter, and DMC-3000

1. Connect one end of each cable to the power supply.
2. Connect the other end of the CNC cable to the CNC device.
3. The CNC cable shield must be attached to ground.



3.10 Connect TSC-3000 Cable and DMC-3000 Fiber Optic Cable to CCM

1. Remove the plastic plug in the CCM at the rear of the Power Supply labelled GCM/DCM.

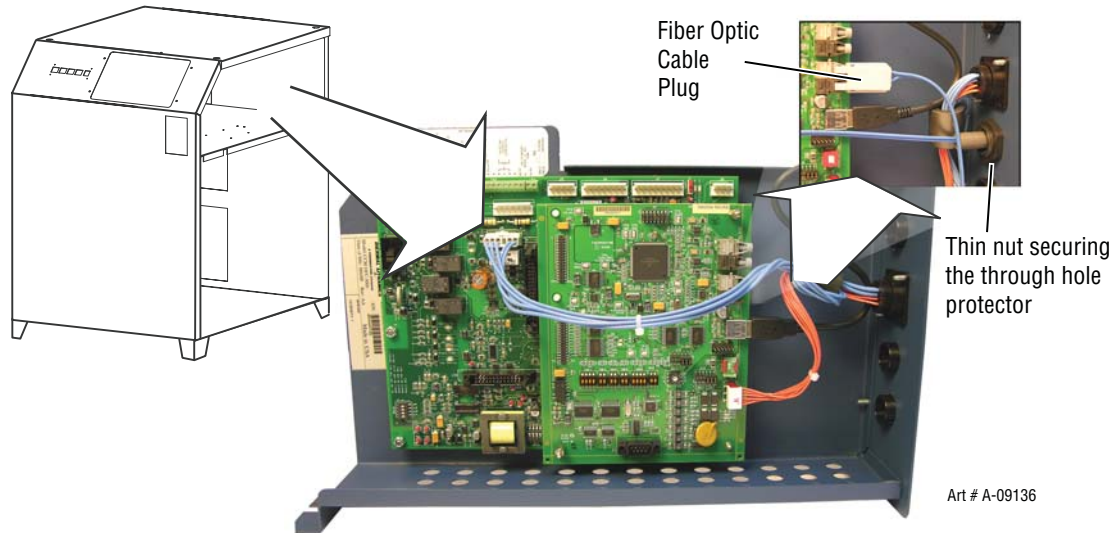


CAUTION

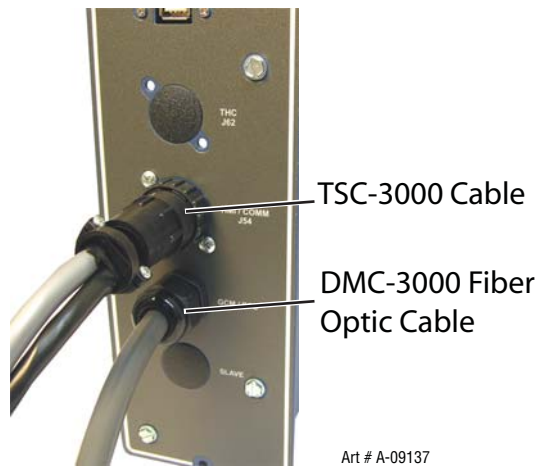
Avoid kinking, twisting, or bunching the fiber optic cable. The cable can be damaged by being forced into tight-radius turns.

2. Remove outer thin nut from through hole protector at one end of Fiber Optic Cable (L) that connects between the CCM portion of the Power Supply and the DMC-3000.

3. Feed the Fiber Optic Cable plug and wires through the hole where the plastic plug was removed labelled GCM/DCM and slide the thin nut back over the Fiber Optic Cable.
4. Secure the thin nut on the through hole protector so both nut faces are tight against the sheet metal inside and out.
5. Plug the Fiber Optic Cable into the PCB as shown below. Make sure the locking tabs are engaged.



6. Notice the alignment pin and then connect the TSC-3000 cable (V) to the rear of the CCM as shown below.



3.11 Set Switches on the Command - Control Module

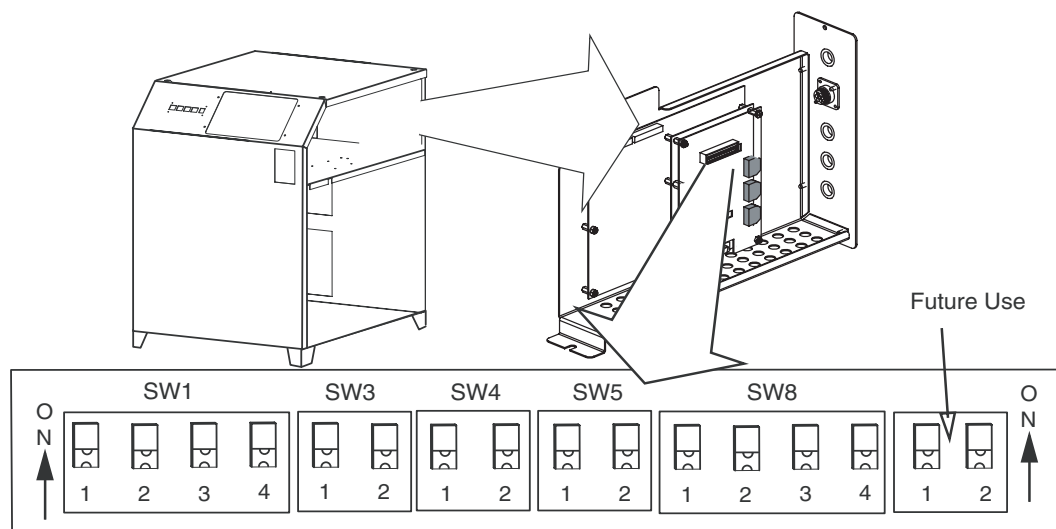
Compare the command - control module to the illustration. Follow instructions in this section for a module without an external connection cover.

Remove the power supply right side. Set switches on the CCM (Command-Control Module) per the illustrations. Switch settings and connection details are provided in the Appendix.



CAUTION

Printed circuit boards in the Command - Control Module are static - sensitive. Discharge any built-up static charges in your body or surroundings before touching the printed circuit boards.



SW-1-1: Auto Pilot Restart. 1 = ON = Auto Pilot Function enabled.

1 = OFF = Auto Pilot Function disabled (Factory default setting).

SW-1-2: Pilot Delay 2 = OFF, 3 = OFF, 4 = OFF: 0 Seconds (Factory default setting).

SW-1-3: Pilot Delay 2 = ON, 3 = OFF, 4 = OFF: 0.1 Seconds

SW-1-4: Pilot Delay 2 = OFF, 3 = ON, 4 = OFF: 0.2 Seconds

2 = ON, 3 = ON, 4 = OFF: 0.4 Seconds

2 = OFF, 3 = OFF, 4 = ON: 0.8 Seconds

2 = ON, 3 = OFF, 4 = ON: 1.0 Seconds

2 = OFF, 3 = ON, 4 = ON: 1.5 Seconds

2 = ON, 3 = ON, 4 = ON: 2.0 Seconds

Active only when
SW-1-1 is set to ON.

SW-3: Gas Preflow Time 1 = Off, 2 = OFF: 2 seconds

1 = ON, 2 = OFF: 4 seconds

1 = OFF, 2 = ON: 6 seconds

1 = ON, 2 = ON: 8 seconds

SW-4: Postflow Time 1 = OFF, 2 = OFF: 10 Seconds (Factory default setting).

1 = ON, 2 = OFF: 20 Seconds

1 = OFF, 2 = ON: 5 Seconds

1 = ON, 2 = ON: 0 Seconds

SW-5-1: Tip Saver Reserved for Factory use.

SW-5-2: Off Plate Reserved for Factory use.

SW 8-1: Pilot Time OFF = Short (85 ms.) (Factory default setting).
ON = Long (3 s.)

SW 8-2: Remote Current OFF = Disabled (Factory default setting).
ON = (Remote Analog Current Control)

*SW 8-3: Auto Transfer Retry OFF = Enabled Up to 3 tries (Factory default setting).
ON = Disabled

SW 8-4: Remote Marking OFF = Disabled (Factory default setting).
ON = Remote Marking SW Enabled at TB3-1&2

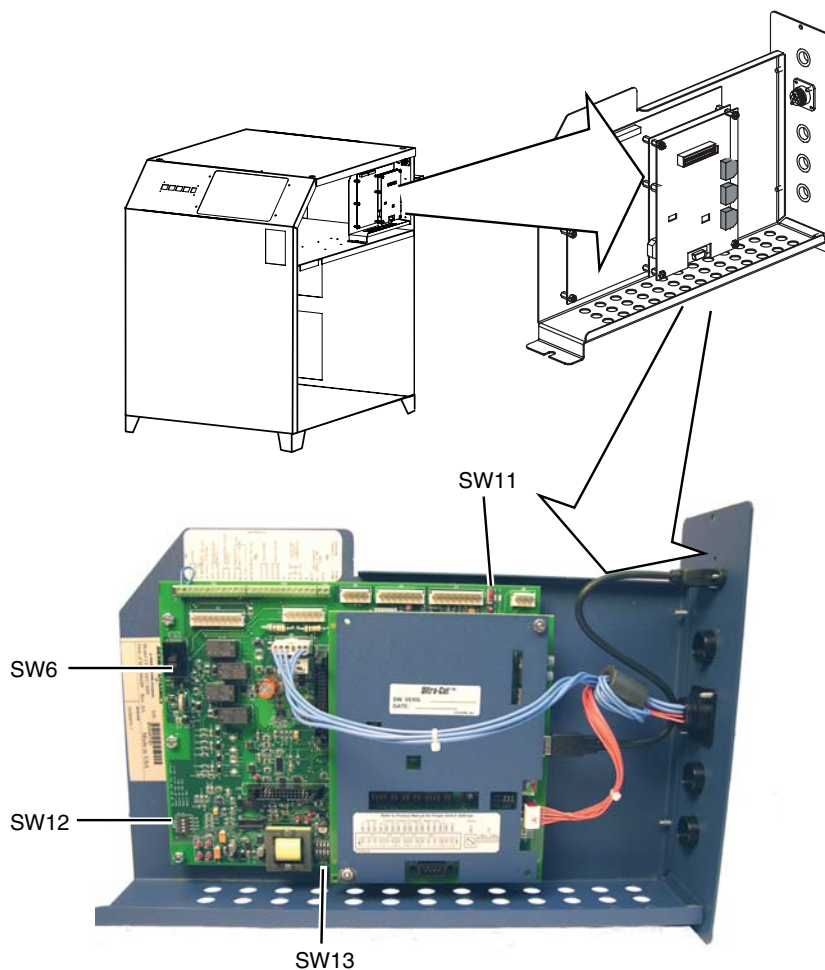
* SW 8-3 is reserved for
Factory use prior to
Firmware V3.5

Art # A-09166



CAUTION

Printed circuit boards in the Command - Control Module are static - sensitive. Discharge any built-up static charges in your body or surroundings before touching the printed circuit boards.



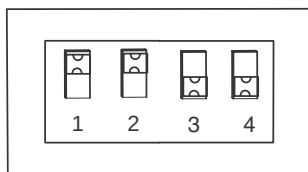
SW-6: OK-to-Move: Contact closure, 120VAC @ 1A (Factory default setting) or DC Volts (16-18vdc @ up to 100 ma.)

SW-11: Analog Current Control. B = from Gas Control (Factory default setting) or A = Remote Analog Source. Position A requires that SW-8-2 be ON.

SW-12-1/2/3/4: Divided Arc signal All = OFF = 50:1 (Factory default setting)

1 = ON = 16.6:1	} Only 1 on at a time.
2 = ON = 30:1	
3 = ON = 40:1	
4 = ON = 25:1	

SW13: Auto Gas Switch positions

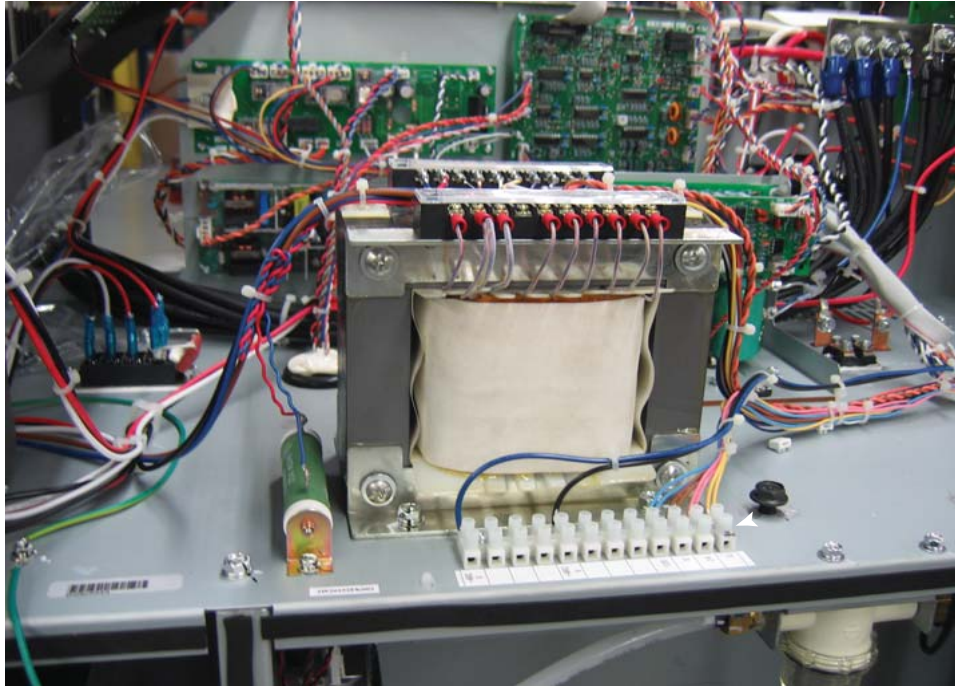


SW13 (Note positions 3-4 are not yet used)

Art # A-09167

3.12 Height Control Connections

The terminal strip has Arc Volts (-) connection to power supply negative output TORCH terminal, Arc Volts (+) connection to power supply positive output WORK terminal. These are for a height control that requires connection to the full non-divided arc voltage. Also available on the terminal strip are 120VAC (120,0) and 24 VAC (24, 0). Note that the two 0's are not common. The allowable current draw is 100ma @ 120VAC and 1Amp @ 24 VAC.



Art # A-08404

NOTE

There is also a hole added in the rear panel above the J55-GCM receptacle for customer wiring. This, rather than the one in the CCM will be the preferred place for customer added wiring (and strain relief) for connections to height controls, etc..

3.13 DMC-3000 Gas Manifold Control Installation

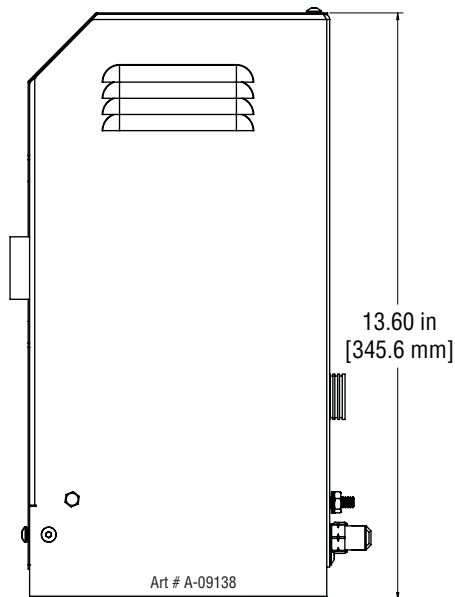
The DMC-3000 Gas Control must be installed in a suitable location where it is easily accessible to the system operator. The unit must be mounted to a flat horizontal surface. If the Module is mounted to any support subject to vibration or motion, the installer must fasten the module to the support securely.

The Module should be located as far away as possible from the Arc Starter due to electromagnetic interference. It is acceptable to locate the control cable in the same track as the cables from the Arc Starter.

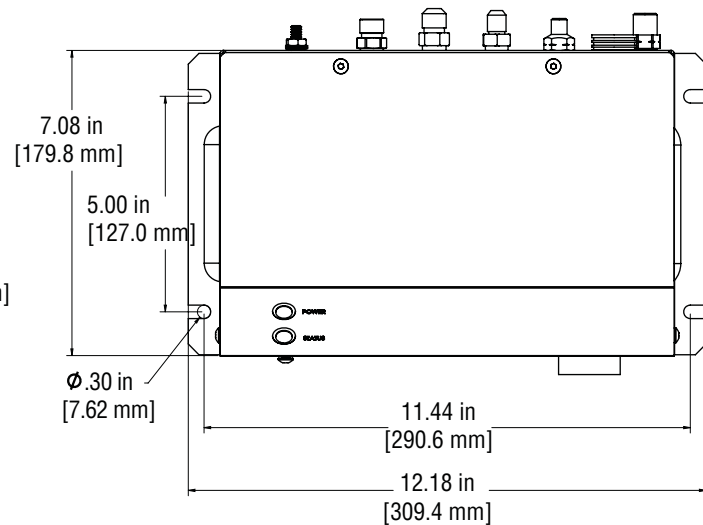
The Module includes feet which lift the bottom panel off the mounting surface. There are ventilation holes on the bottom panel; the space between the bottom panel and the mounting surface must remain open for ventilating air to enter the module. Louvers on the back panel of the module must also remain unblocked, for the free passage of ventilating air.

Mounting Dimensions

DMC-3000 Profile



DMC-3000 Top



Preparation

1. Remove the screws securing the Cover Panel to the Module.



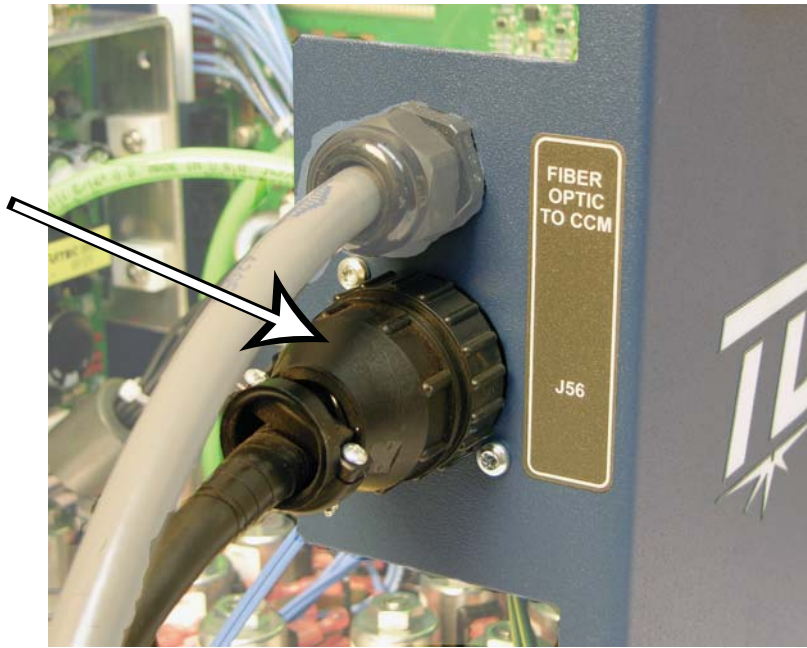


NOTE

NOTE

All gas fittings are 1/4" and the air fitting is 3/8".

4. Connect the ground cable to rear of DMC-3000 shown in previous illustration.
5. Connect Control cable from the Power Supply to J56 on the front of the DMC-3000 as shown below. The Fiber Optic cable will be covered a little later.



Art # A-09142

3.14 DPC-3000 Gas Pressure Control Installation

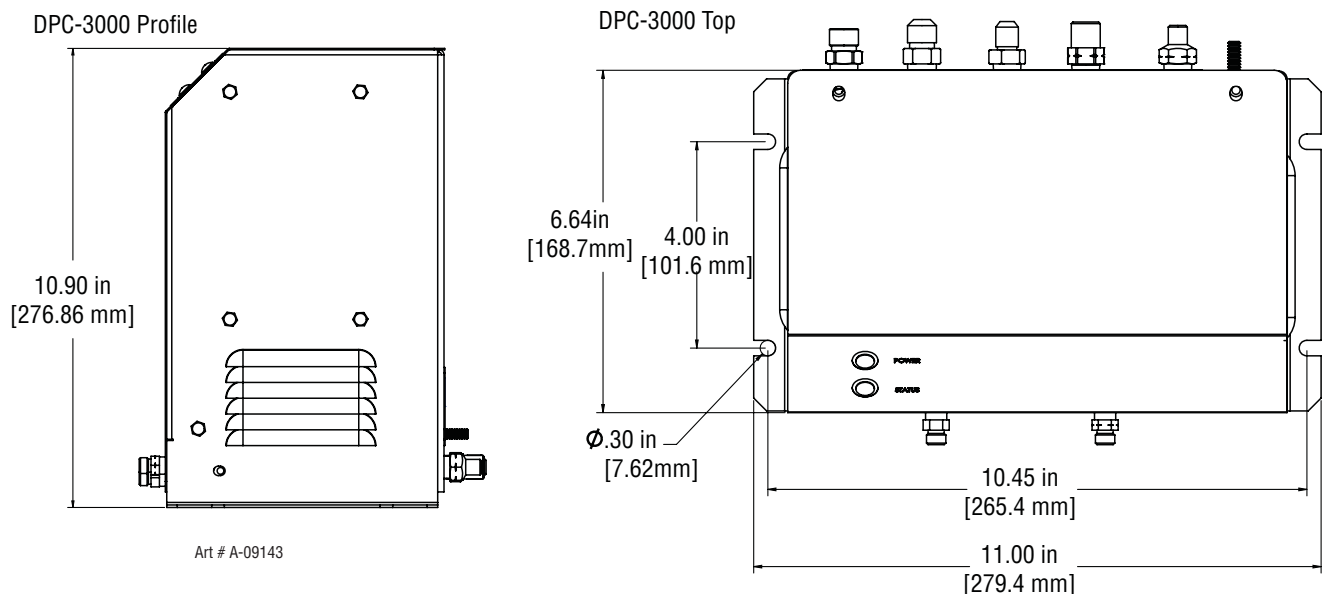
The DPC-3000 Gas Control must be installed in a suitable location close to the Torch such as the gantry. The unit must be mounted to a flat horizontal surface. If the Module is mounted to any support subject to vibration or motion, the installer must fasten the module to the support securely.

The Module should be located as far away as possible from the Arc Starter due to electromagnetic interference. It is acceptable to locate the control cable in the same track as the cables from the Arc Starter.

The Module includes feet which lift the bottom panel off the mounting surface. There are ventilation holes on the bottom panel; the space between the bottom panel and the mounting surface must remain open for ventilating air to enter the module. Louvers on the back panel of the module must also remain unblocked, for the free passage of ventilating air.

The Module also comes with non metallic isolation grommets for mounting. These must be used in all four mounting slots to raise the Module so there is no metal to metal contact between the Module and the mounting surface.

Mounting Dimensions

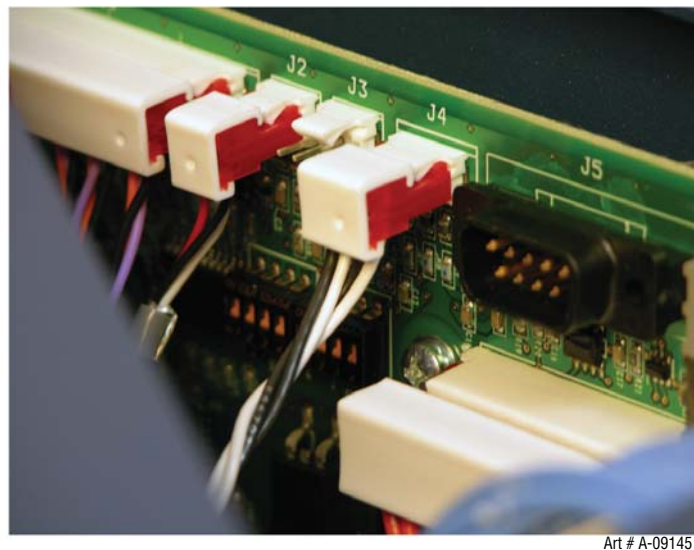


Preparation

1. Remove the screws securing the Cover Panel to the Module.



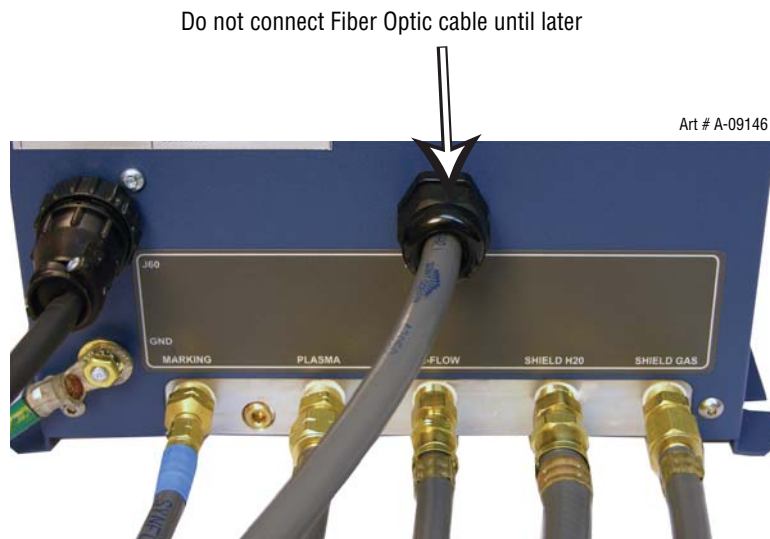
2. Carefully remove the cover from the module noting the attached wire harness that connects to J4. Remove the wire harness and then set the cover to the side.



Cover Removal

ULTRA-CUT 100

3. Attach all hoses and cables to the back of the DPC-3000 except the Fiber Optic cable which will be covered a little later. Avoid turning any fittings already mounted in the unit by placing a wrench on it before tightening the hose fitting to it.



4. Connect hoses from the front of the DMC-3000 where shown below to the torch.

NOTE

One fitting with the notches is left hand thread and the other is standard right hand thread.



3.15 Fiber Optic Cable Installation from CCM to DMC-3000.

1. Remove outer thin nut from through hole protector at the end of Fiber Optic Cable (L) that is connectsed to the CCM.



CAUTION

Avoid kinking, twisting, or bunching the fiber optic cable. The cable can be damaged by being forced into tight-radius turns.

2. Ensure that the cable is exposed beyond the through hole protector approximately 1" (as shown below) and secure by hand tightening the wide nut with the rounded end.
3. Feed the Fiber Optic Cable plug and wires through the hole where shown then slide the thin nut back over the Fiber Optic Cable.



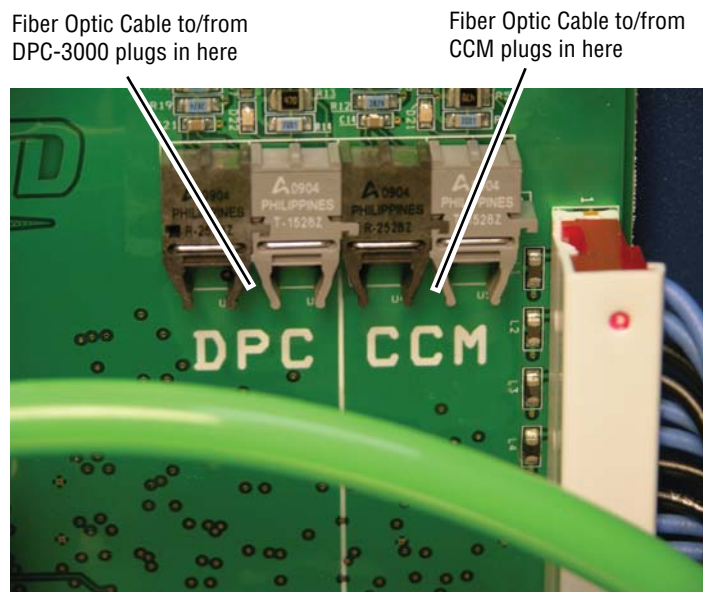
Art # A-09148

4. Secure the thin nut on the through hole protector so both nut faces are tight against the sheet metal inside and out.



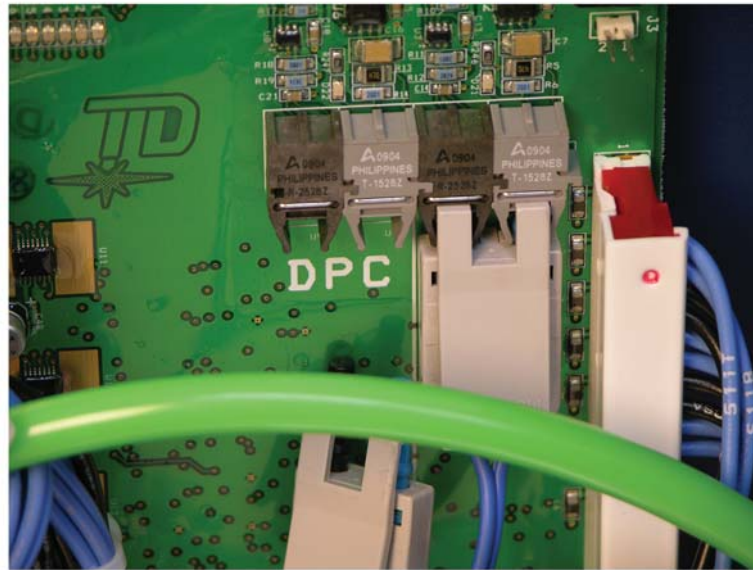
Art # A-09149

5. Plug the Fiber Optic Cable into the PCB as shown below. Make sure the locking tabs are engaged.



Art # A-09150

Before



Art # A-09151

After

3.16 Fiber Optic Cable Installation From DMC-3000 to DPC-3000

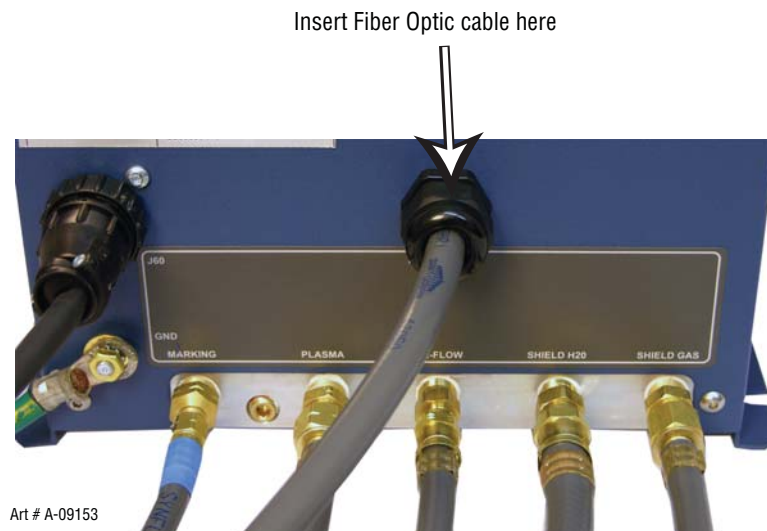
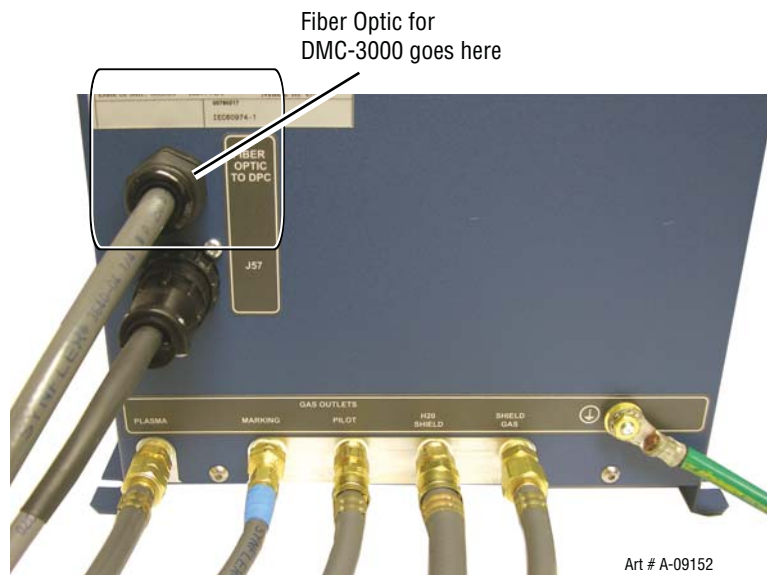
1. Remove outer thin nuts from through hole protector at both ends of Fiber Optic Cable (L) that connects between the DCM-3000 and DPC-3000.



CAUTION

Avoid kinking, twisting, or bunching the fiber optic cable. The cable can be damaged by being forced into tight-radius turns.

2. Ensure that the cable is exposed beyond the through hole protector approximately 1" and secure by hand tightening the wide nut with the rounded end.
3. Feed the Fiber Optic Cable plug and wires through the back of each module (DMC-DPC) where indicated and slide the thin nut back over the Fiber Optic Cable.



4. Secure each of the thin nuts on the through hole protectors so both nut faces are tight against the sheet metal inside and out.
5. Plug the Fiber Optic Cable into the PCB as shown below for the DMC-3000. Make sure the locking tabs are engaged.

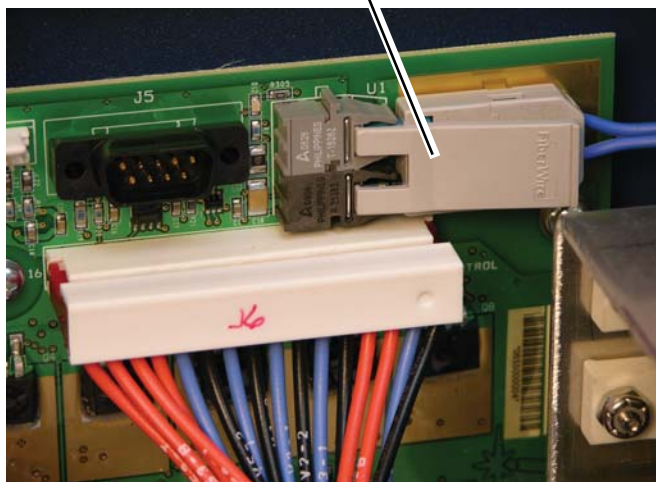
Plug Fiber Optic cable to/from DPC-3000 here



Art # A-09154

6. Plug the Fiber Optic Cable into the PCB as shown below for the DPC-3000. Make sure the locking tabs are engaged.

Fiber Optic to/from DMC-3000 plugs in here

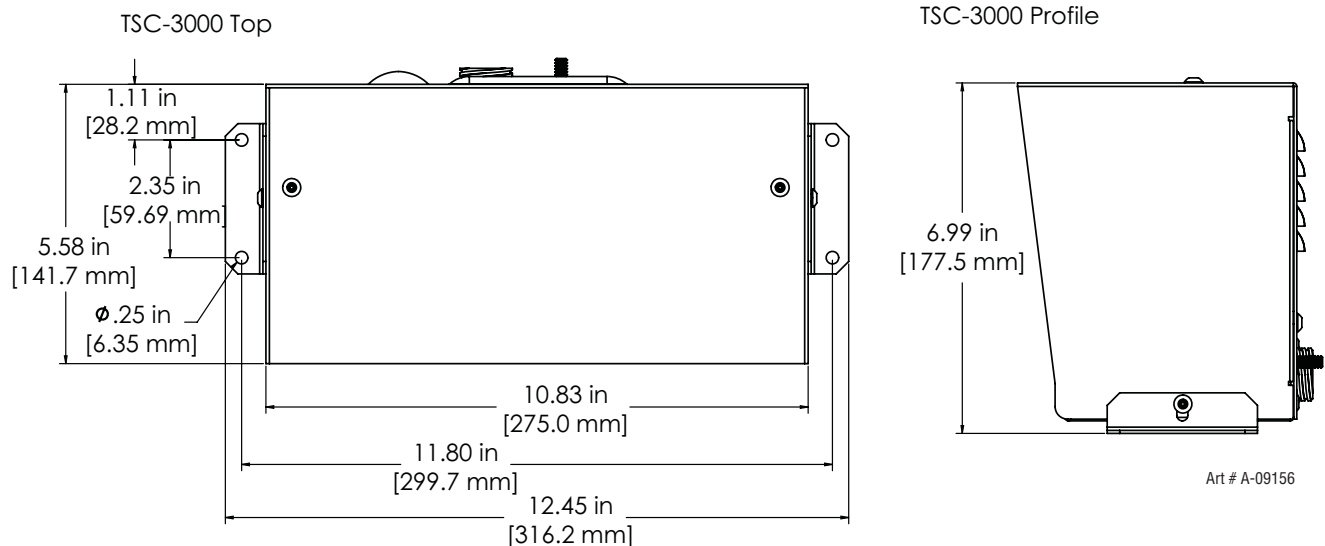


Art # A-09155

7. Reinstall the Cover Panels making sure the wireharnesses are attached. Reverse steps 1 and 2 in sub section 3.13 and 3.14

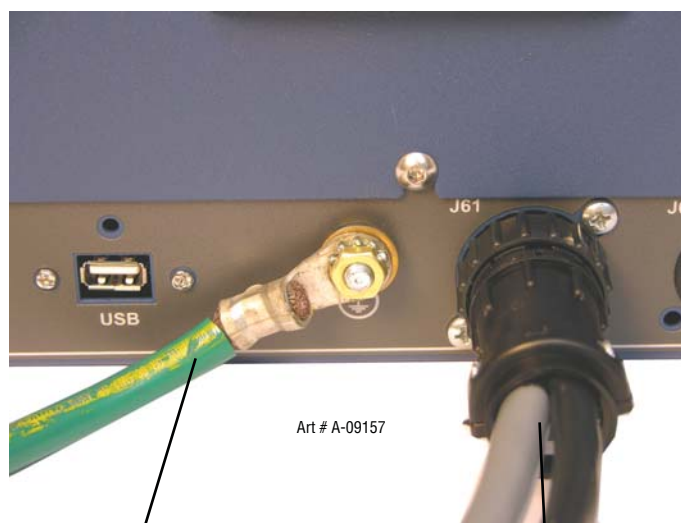
3.17 Install TSC-3000 Touch Screen Control

Mounting Dimensions



Preparation

1. Select a clean, dry location with good ventilation and adequate working space around all components. Review the safety precautions in the front of this manual to be sure that the location meets all safety requirements. The TSC-3000 mounting brackets allow for mounting on or under a horizontal surface as well as left or right of a vertical surface. Choose which ever works best for protection against moving equipment and flying metal/cutting debris etc..
2. Once the unit has been secured to a flat surface, attach Communication cable ("V" which is already attached to the CCM done in sub section 3.10) and ground cable ("F") to the rear of the unit.



Art # A-09157

Ground Cable to/from
"Star" ground on table

Communication Cable to/from
CCM rear of power supply

3.18 Install Remote Arc Starter

Site Location

Select a clean, dry location with good ventilation and adequate working space around all components.

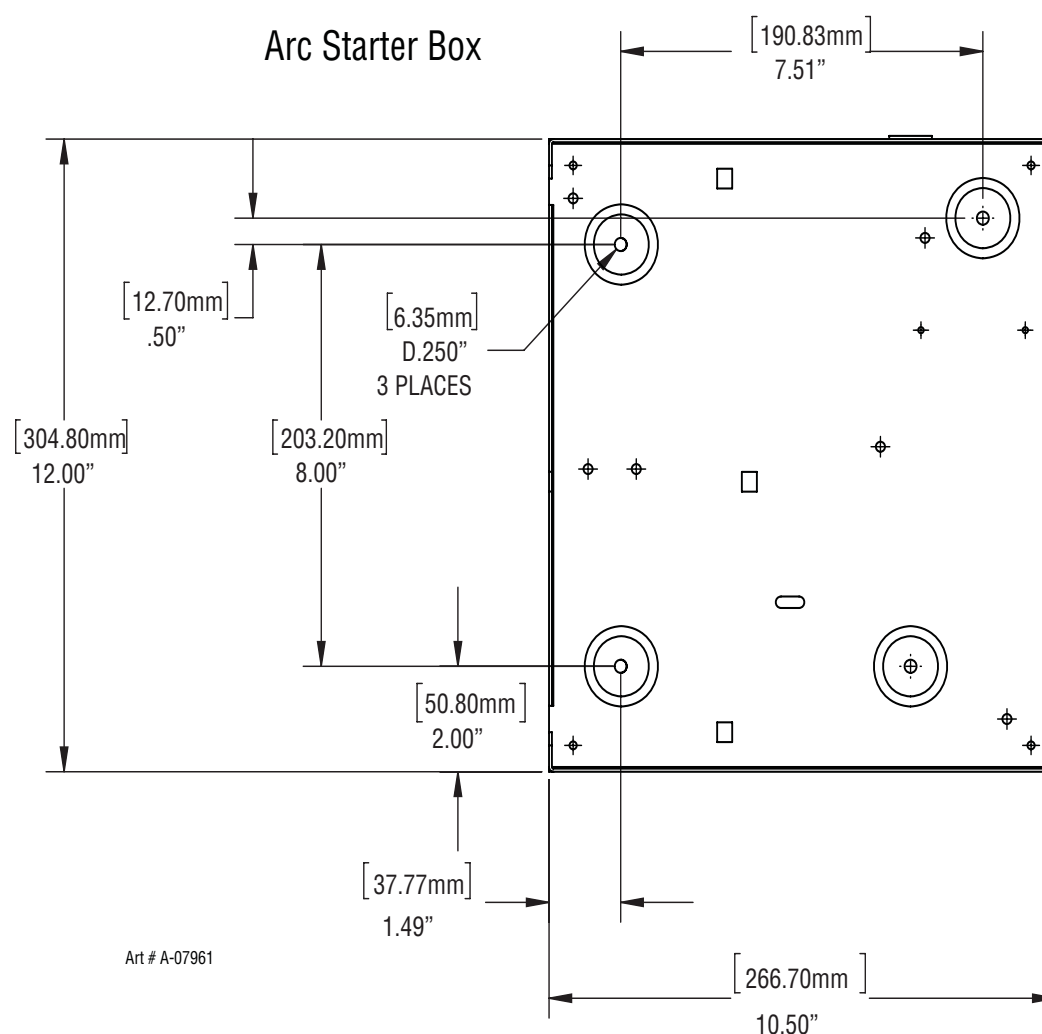
Review the safety precautions in the front of this manual to be sure that the location meets all safety requirements.

Interconnecting cables and hoses attach to the Arc Starter. There must be adequate space around the Arc Starter for these connections without crimping.

Mounting Dimensions

NOTE

Height not shown is 7.375" (187mm)



Installation

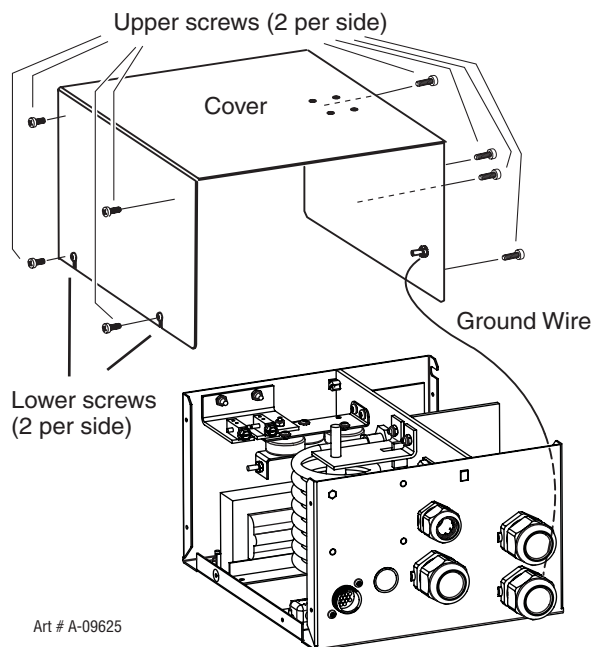
The Remote Arc Starter must be installed in a suitable location near the torch head. If the Arc Starter is mounted to a gantry or to any other support subject to motion or vibration, fasten the Arc Starter to the support securely.

1. Loosen, but do not remove, the lower screws securing the cover to the Arc Starter. Remove the upper screws securing the Cover Panel to the Arc Starter.

NOTE

A ground wire connects the cover to the Arc Starter base. This wire must remain in place.

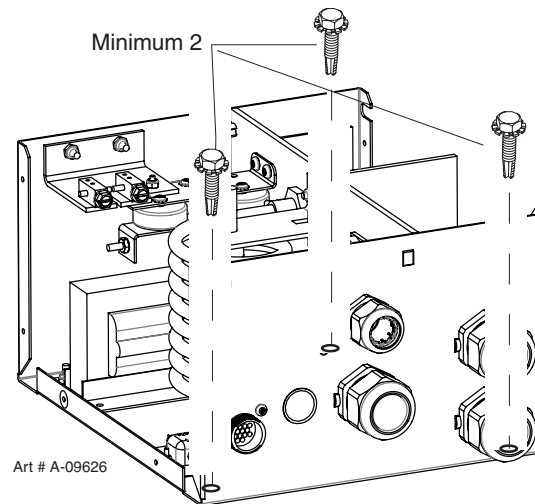
2. Remove the Cover Panel from the Arc Starter.



Cover Removal

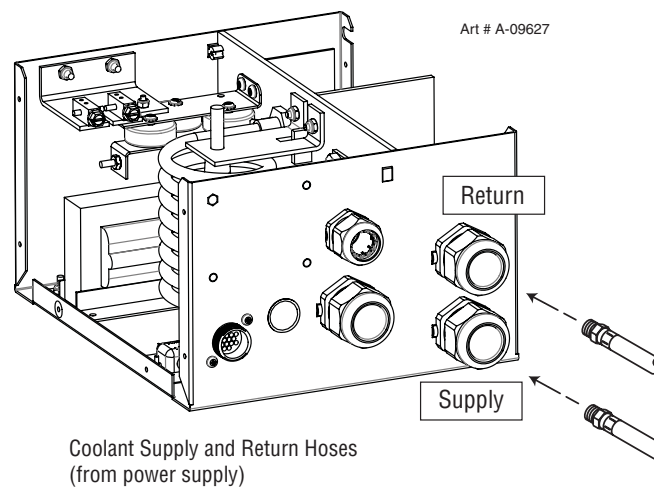
3. Position the Arc Starter on a flat, horizontal mounting surface.

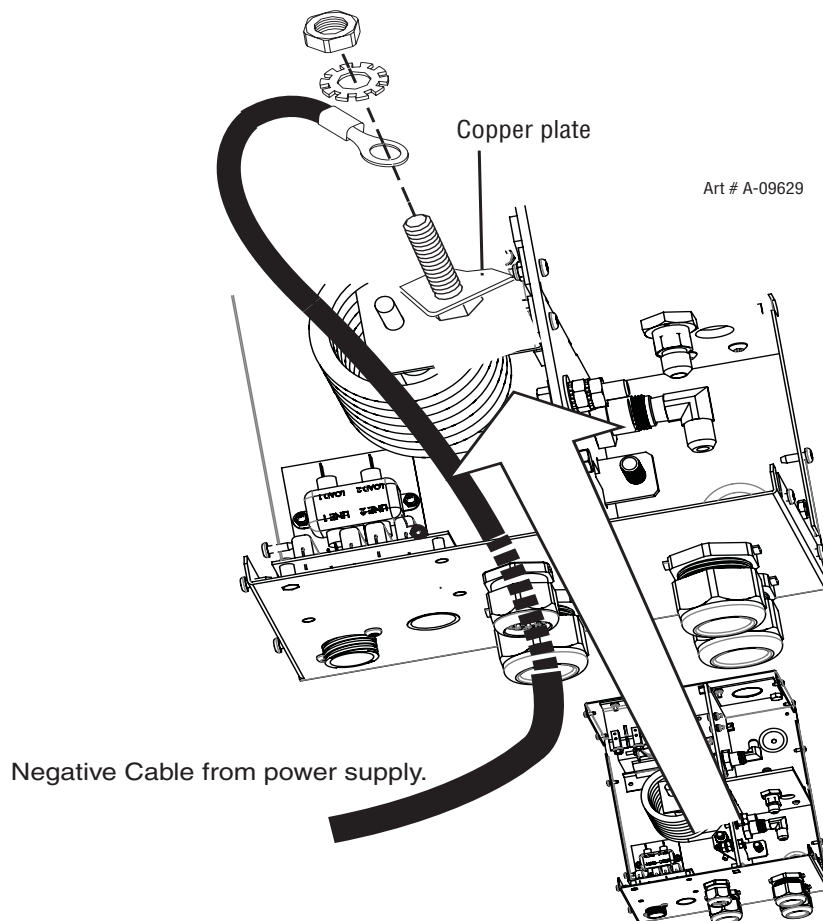
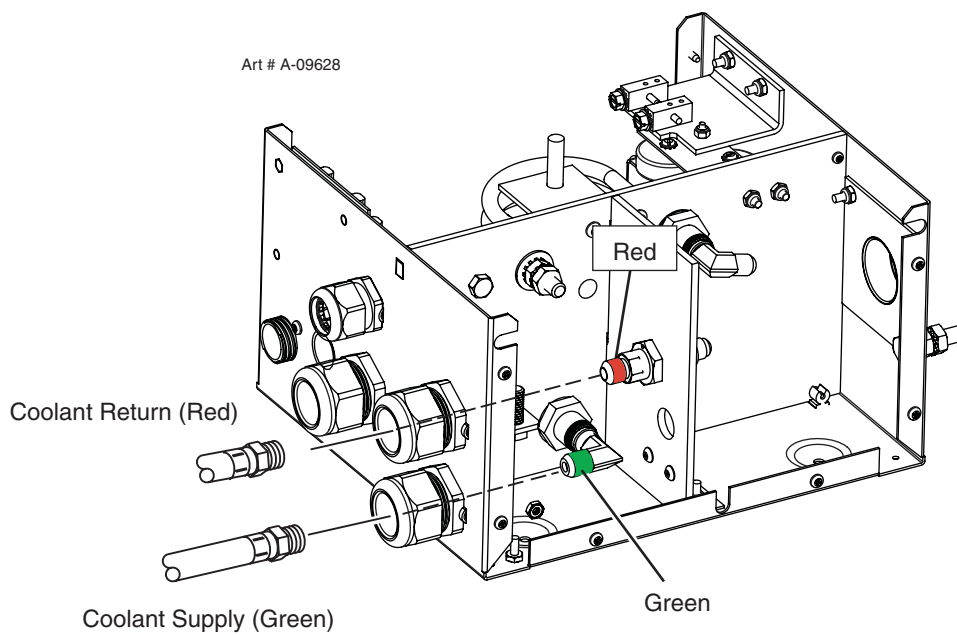
4. Use pre-drilled holes in at least two of the feet on the bottom of the Arc Starter to secure the Arc Starter to the mounting surface.



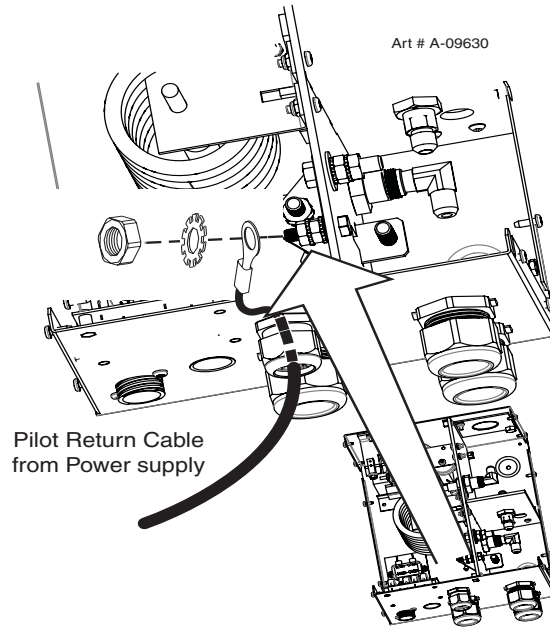
Input Connections

1. Refer to the illustrations. Make the following input connections to the Arc Starter.
 - Coolant Supply and Return Hoses (from power supply). Hoses and connectors are color-coded; Red for Return, Green for Supply.

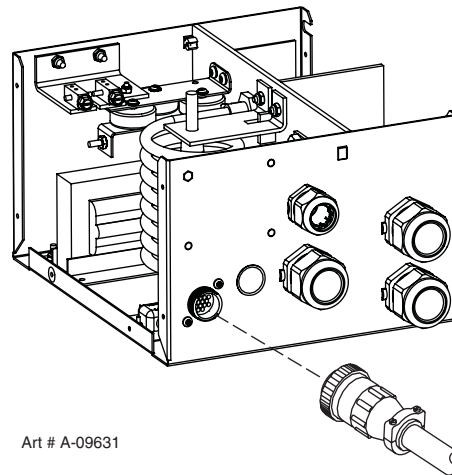




Negative Cable (from Power Supply rear panel)



PILOT Return Cable (from Power Supply rear panel)



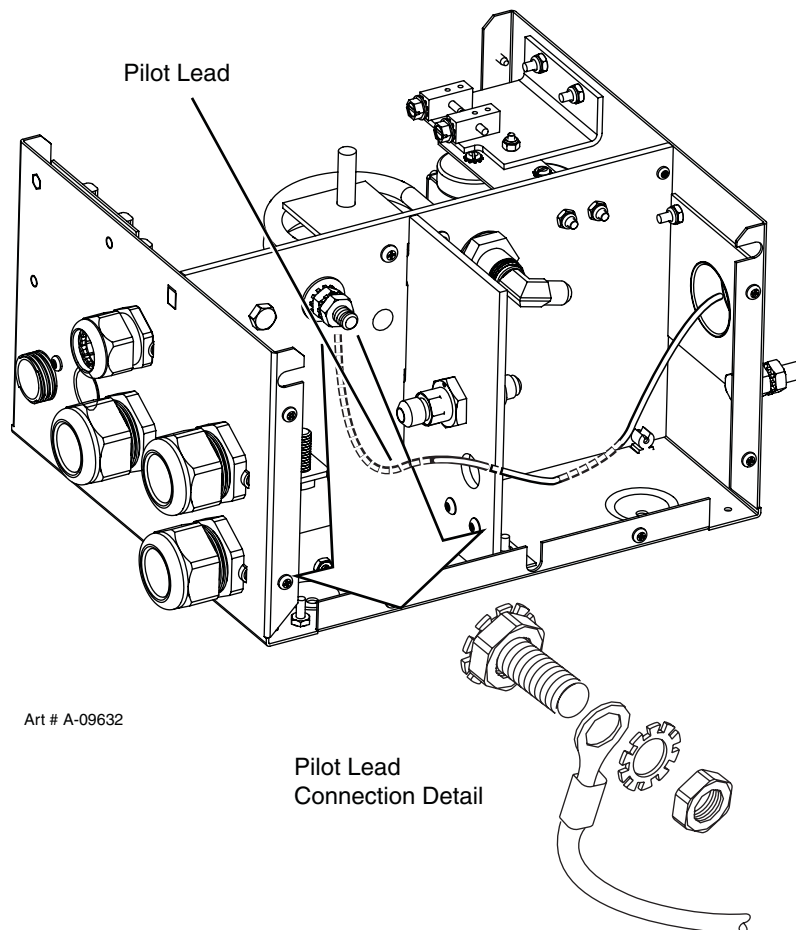
Control Cable from Power Supply rear panel

Output Connections

1. Refer to the illustrations. Make the following output connections to the Arc Starter.

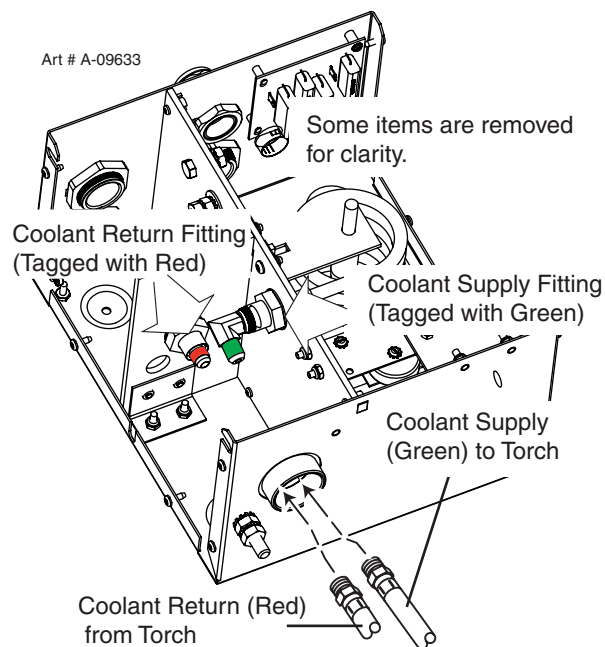
NOTE

For newer leads with only one black wire, it must be connected to the pilot return. There are no inner shield wires.



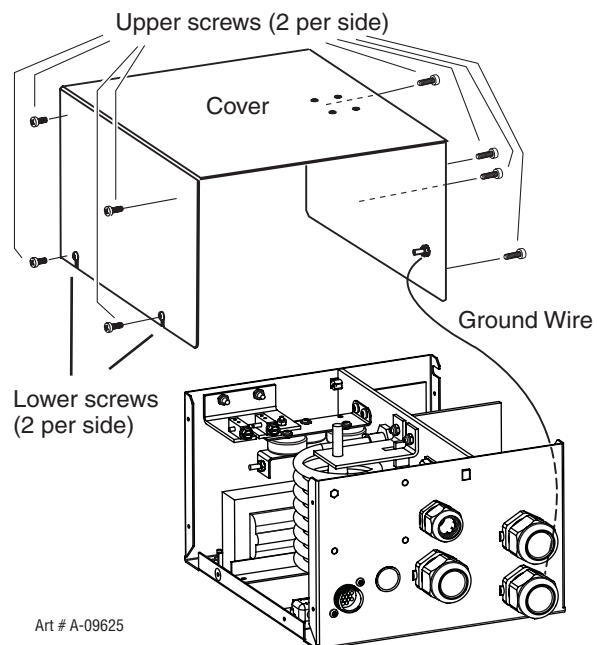
Art # A-09632

Pilot Return Cable (from Torch Leads)

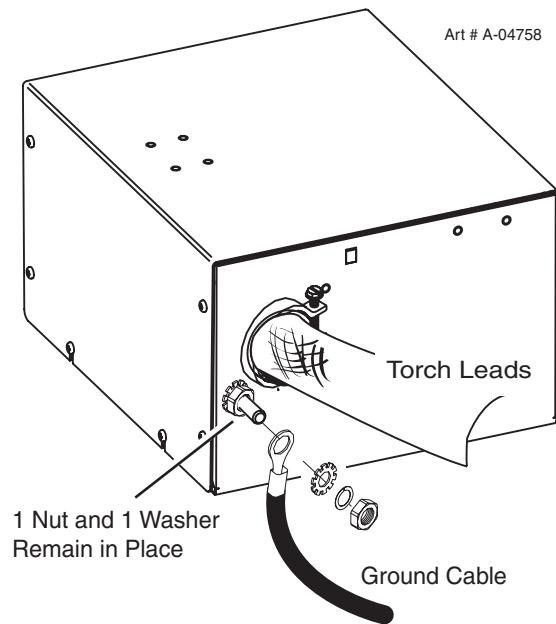


Coolant Supply and Return Hoses (from Torch)

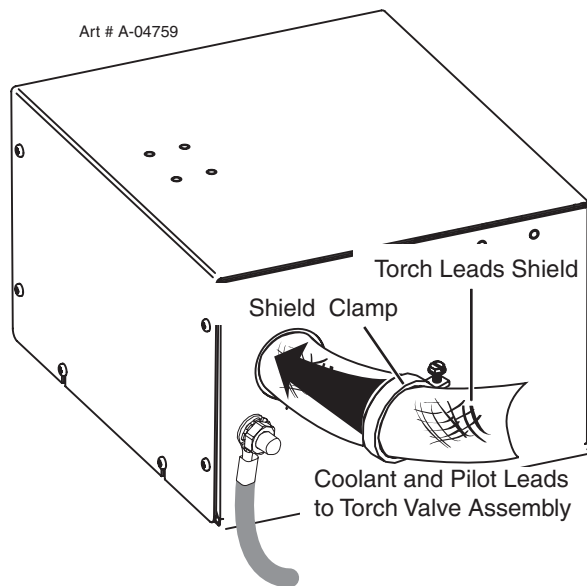
Reinstall the Arc Starter Cover. Ensure that the ground wire is not crimped between the cover and the base.



The Arc Starter must be grounded; the grounding terminal is marked $\perp$. Refer to the previous section for grounding details.

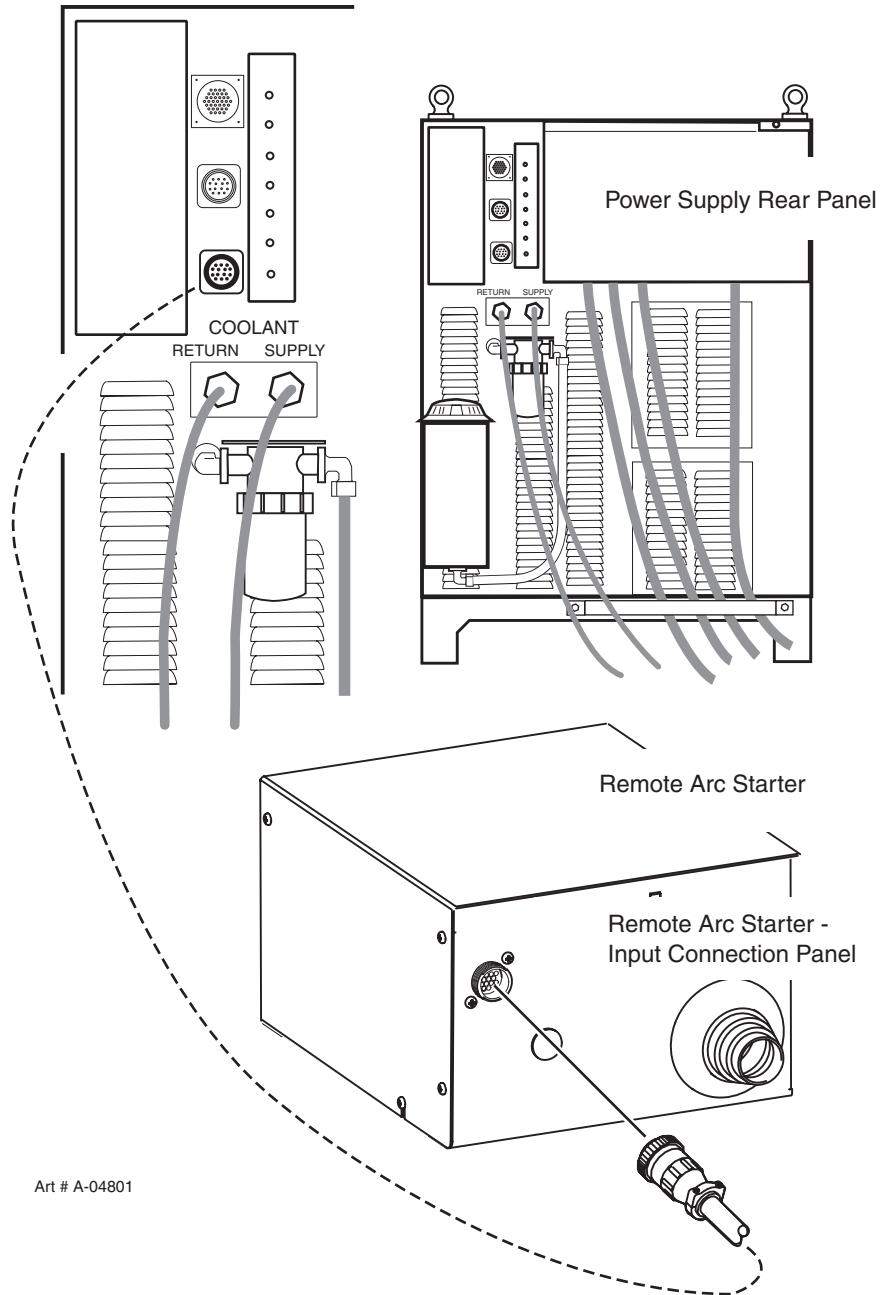


- Use a clamp to secure the Torch Lead Shield braid to the port on the Remote Arc Starter as shown.



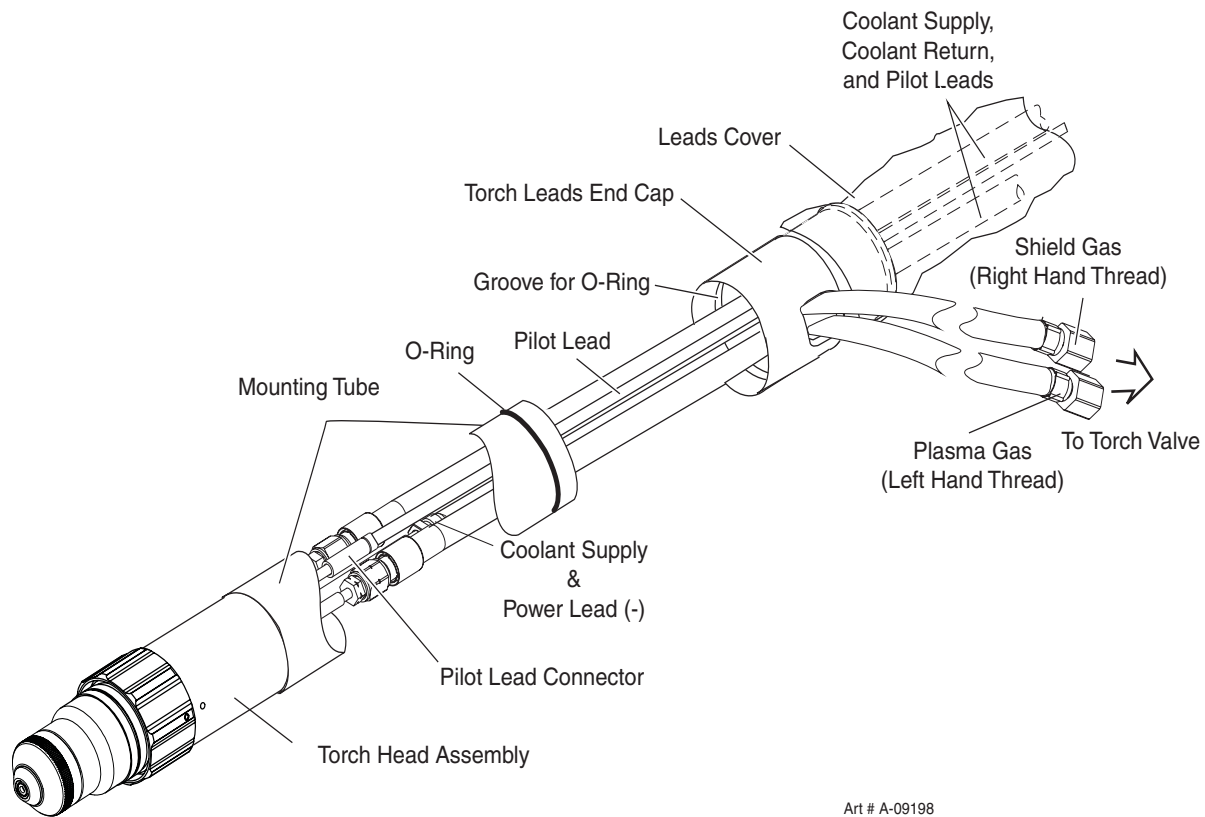
Connect Control Cable

1. Connect the Remote Arc Starter cable to the Remote Arc Starter receptacle.



3.19 Connecting Torch

Connect the Torch as follows:



1. Lay out the torch leads on a clean, dry working surface.
2. Hold the Torch Leads End Cap stationary. Pull approximately 18" (0.5 m) of leads through the End Cap.
3. Remove and discard the protective end caps from the Mounting Tube.
4. Install the O-ring in the groove at the upper end of the Mounting Tube.
5. Install the Mounting Tube as follows:
 - a. Position the Mounting Tube at the end of the leads assemblies as shown.
 - b. Slide the Mounting Tube upward onto the leads assemblies.
 - c. Press the upper end of the Mounting Tube into the lower end of the Torch Leads End Cap. Ensure that the O-Ring on the Tube engages the mating groove inside the Torch Leads End Cap.
 - d. Ensure that the Mounting Tube is free to rotate within the Torch Leads End Cap.

6. Connect the gas and coolant leads to the Torch Head.
 - a. Coolant supply and return connections to the Torch Head are of different lengths.
 - b. Plasma and secondary gas connections to the Torch Head are threaded differently; the plasma gas connection is left-hand thread, the shield gas connection is right-hand thread.
 - c. Hold the Torch Head leads connectors stationary; turn the leads fittings with a wrench to secure the leads to the Torch Head. Do not overtighten.



CAUTION

The gas and coolant leads include compression fittings. Do not use sealant on these connections. Use two wrenches for each connection: one to hold the non hose fitting in place while using the other to tighten the hose fitting.

Slowly apply pressure to the gas lines. Check for leaks at all connections before continuing. If there are no leaks, shut off the gas supplies and continue with installation.

7. Connect the pilot lead to the Torch Head. Press the two ends of the connector firmly together. Fold the excess wire in a serpentine shape and tuck inside cover. Thread the plastic lead cover/connector onto the mating Torch Head connector.
8. Press the Torch Head Assembly upward to connect to the Mounting Tube. Pull the leads back as needed to ensure a proper fit through the Mounting Tube and Torch Leads End Cap. Hold the Torch Head Assembly stationary; rotate the Mounting Tube to thread it onto the Torch Head.



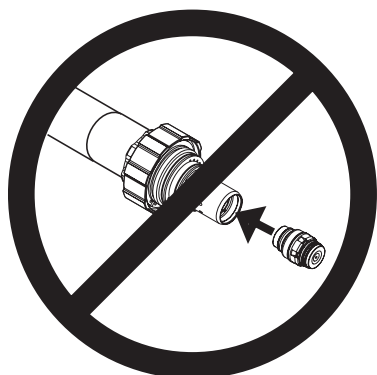
CAUTION

Ensure that the leads do not twist within the mounting tube. Leads must lie as shown in the installation sketch.

9. The lower end of the Mounting Tube includes four threaded holes.
10. Install the appropriate consumable parts as shown on the following pages. The torch manual includes diagrams showing the correct parts to install, depending on the metal to be cut and the gases in use.

3.20 Install Consumable Torch Parts

Install the consumable parts as follows to ensure proper operation. These steps will help ensure that parts are seated correctly.

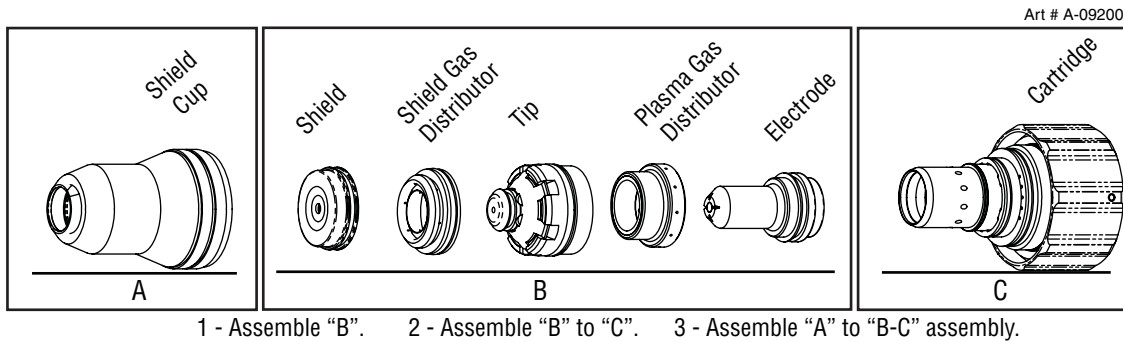


WARNINGS

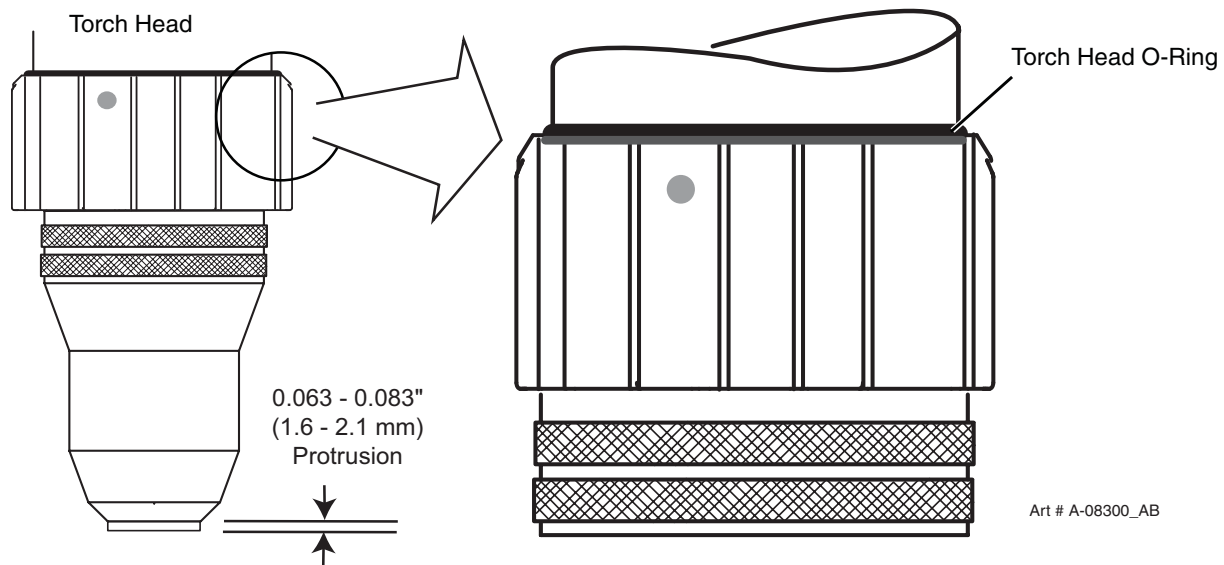
Do not install consumables into the Cartridge while the Cartridge is attached to the Torch Head. Keep foreign materials out of the consumables and Cartridge. Handle all parts carefully to avoid damage, which may affect torch performance.

Art # A-03887

1. Check the appropriate cut chart for the right combination of parts for the cutting application.
2. Stack the consumable parts together.



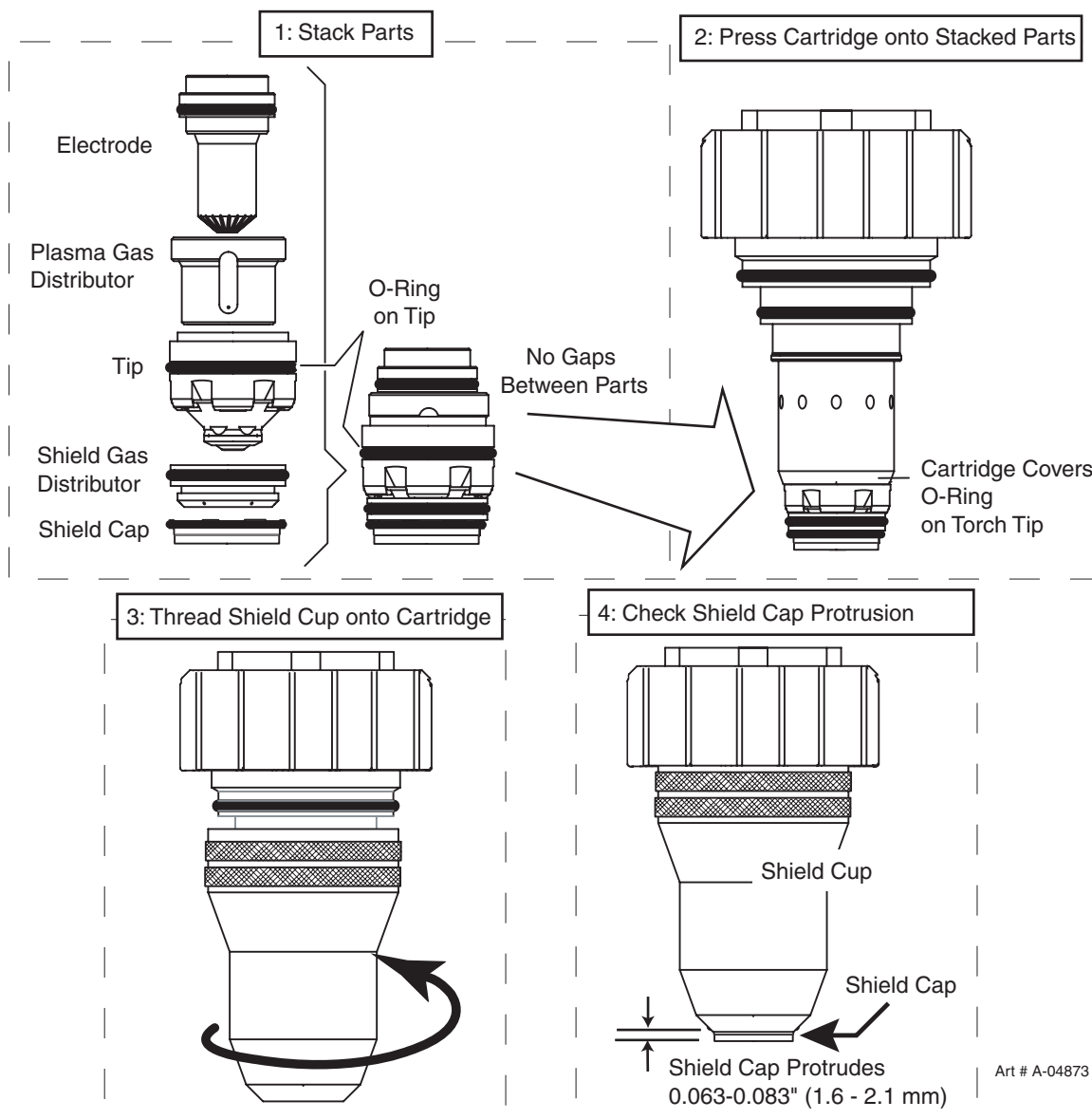
3. Insert the stack of consumable parts into the cartridge. Ensure that the large O-ring on the torch tip fits completely into the cartridge. If any part of the O-ring protrudes from the cartridge, the parts are not seated properly.
4. Use the cartridge tool to hold the cartridge assembly, while turning the shield cup or shield retainer for onto the cartridge assembly. When this group is fully assembled, the shield should protrude from the front of the shield cup or shield retainer. Without this protrusion the shield cup is not properly tightened onto the cartridge assembly.
5. Take the cartridge tool off the cartridge. Fit the cartridge assembly onto the torch head. The Speed Lok ring should click into place and the cartridge assembly should touch the large O-ring on the torch body.



Installing Assembled Cartridge Onto Torch Head

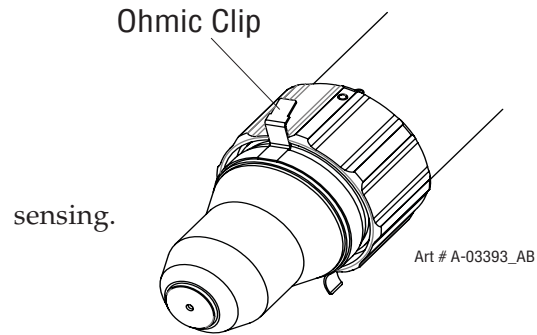
6. Slide the ohmic clip over the shield cup if using ohmic torch height sensing.

55 - 100 Amp Parts Assembly



Installing Assembled Cartridge Onto Torch Head

7. Connect the wire lead from the height finder to the ohmic clip if using ohmic torch height



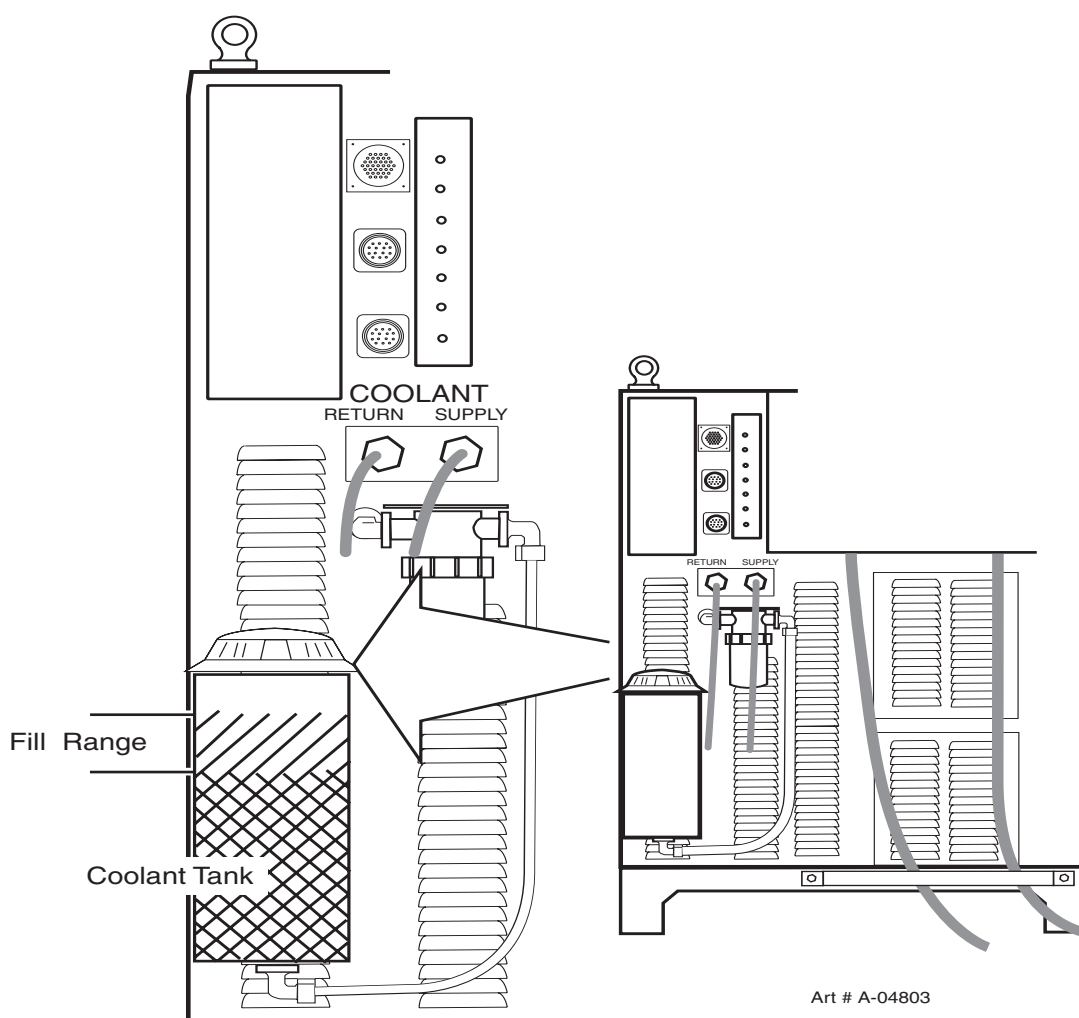
NOTE

Ohmic height sensing is not recommended with water shield. Water on the plate interferes electrically with the ohmic sensing circuit.

3.21 Complete the Installation

1. Remove the cap from the coolant tank. Fill the coolant tank to the level shown, with Thermal Dynamics coolant. The coolant level is visible through the translucent coolant tank. The amount of coolant required varies with torch leads length.

Coolant Capabilities		
Cat. Number and Mixture	Mixture	Protects To
7-3580 'Extra-Cool™	25 / 75	10° F / -12° C
7-3581 'Ultra-Cool™	50 / 50	-27° F / -33° C
7-3582 'Extreme Cool™	Concentrate*	-76° F / -60° C
* For mixing with D-I Cool™ 7-3583		



2. After the complete system has been installed, check that the coolant has been pumped through the system as follows (see NOTE):

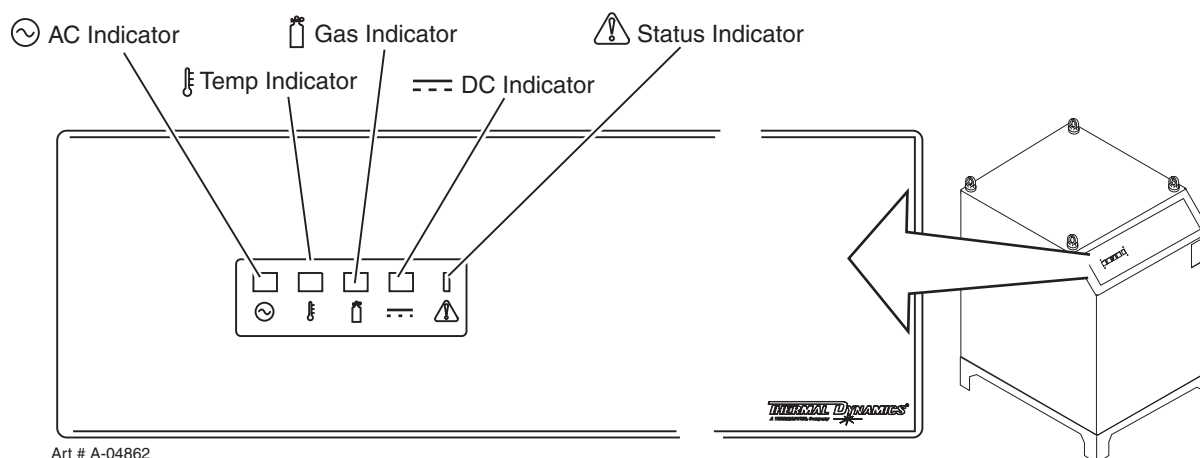
NOTE

Depending on the length of the torch leads, the system may require more coolant after turning the system ON for the first time.

- a. Place the ON/OFF Switch to ON.
 - b. After about 30 seconds the system may shut down if the leads are not full of coolant.
 - c. Place the ON/OFF switch to OFF.
 - d. After 10 seconds place the ON/OFF switch to ON again.
 - e. Repeat steps 'b' through 'd' until the system no longer shuts down. Depending on the length of the torch leads this sequence may need to be done three to five times and coolant may need to be added.
 - f. After the system stays operational allow the pump to operate for ten minutes to properly purge any air from the coolant lines before using the system.
3. Re-fill the reservoir and re-install the filler cap.
 4. Purge coolant from the torch before firing the torch. Ensure there are no leaks before use. If leaks are evident, consult the coolant leak trouble-shooting guide in the maintenance section of this manual.

SECTION 4: OPERATION

4.01 Power Supply Control Panel



AC Power Indicator

Indicates AC power is being supplied to the inverters when the ON/OFF switch is in ON position. When switch is first set to ON, the indicator stays off until the inrush cycle is complete and the correct voltage is confirmed.

TEMP Indicator: Normally OFF. Indicator turns ON when the internal temperature sensors detect temperatures above normal limits. Let the unit cool before continuing operation.

GAS Indicator: Normally ON when gas is flowing. The indicator blinks when the gas control is not ready, such as during the purge cycle or when the gas control is not in the RUN mode or when there is a communication error. Indicator is off when there is no call for gas flow.

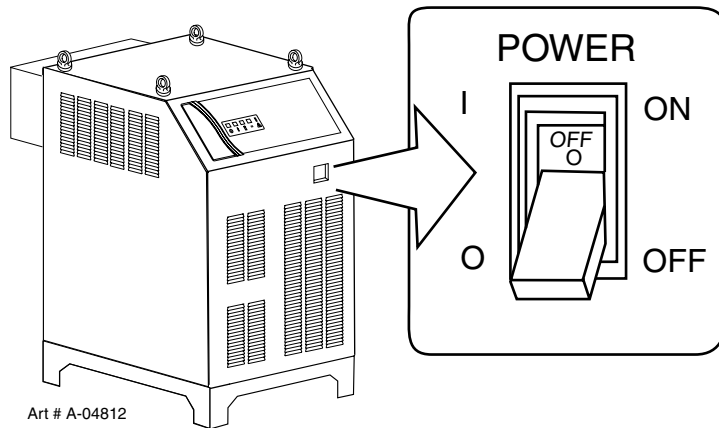
DC Indicator: Indicates the power supply is generating output DC voltage.



Status Indicator: Shows system status. The number of flashes indicates the status. Refer to the Status Code Section for details. On power supply start-up, the indicator flashes to show the revision level of the operating software installed in the systems CCM.

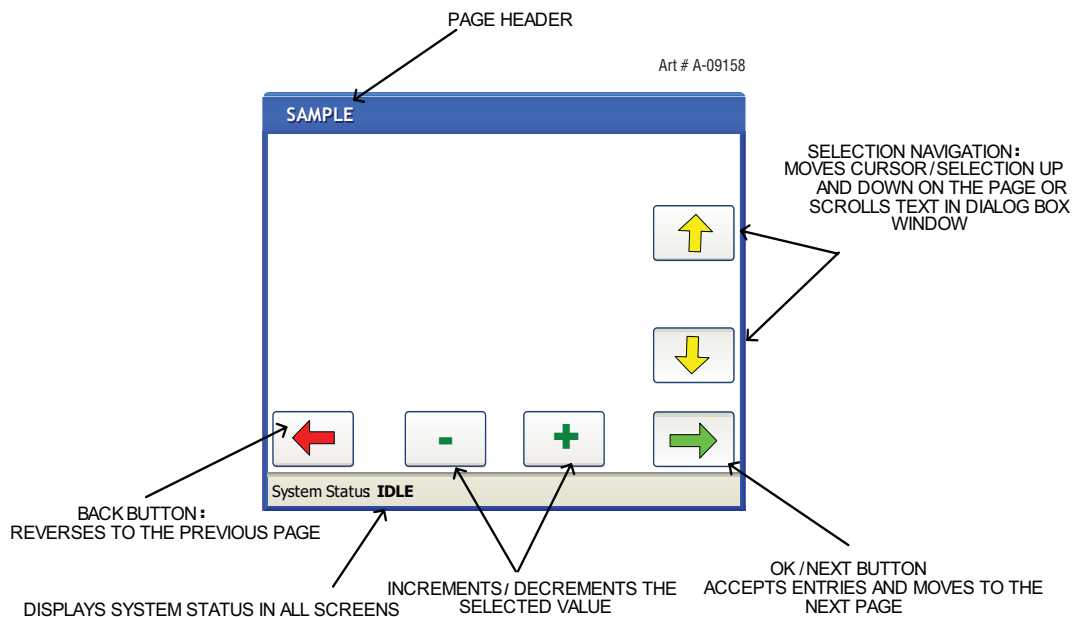
4.02 Start-Up Sequence

1. Connect system to primary input power.
2. Turn ON/OFF switch to ON (up) position. AC Indicator  turns ON after about 5 seconds.



3. Check the AC indicator. If the inverter modules are correctly configured for the input voltage and the switch is ON, then voltage has been applied to the inverter modules (input contactors are ON). If the indicator does not light after a few seconds disconnect power and check that each inverter is correctly configured for the input voltage and all 3 phases are present. Check Status Indicator for any codes that can help. Once it is determined that input is OK, it is necessary to select and load a process.

4.03 TSC-3000 Navigation Functions



4.04 TSC-3000 Initial Setup ONLY

Initial steps required before First operation only. Note 400 Amp system requires a Height Controller with Elevation Height (EH) feature.

When power is turned on TSC 3000 goes to the Home screen. For new installation it is required to perform initial SETUP.

Press SETUP button (1).
Enter Initial default PASSWORD 00000 (2)

Select language and measurement units.

Enter SYSTEM UTILITY (3).

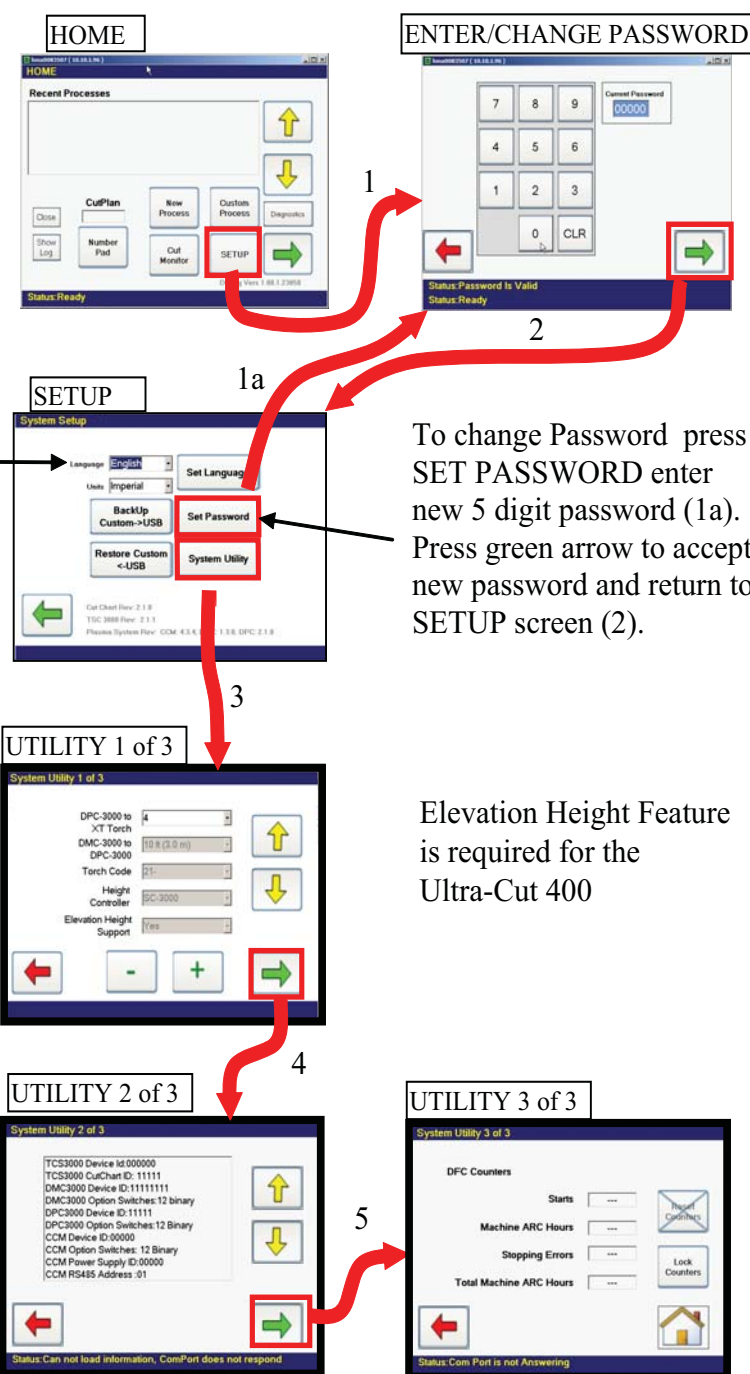
Enter XT lead (DPC to Torch); only 4' current allowed.

Enter DFC 3000 lead length (DMC to DPC);

Utility Screen 2 is for information only. Displays Device and Cutchart IDs, setting of options switches etc.

Utility screen 3 (5) allows viewing consumable statistics, Starts, Hours, Errors.

Return to Home Screen.

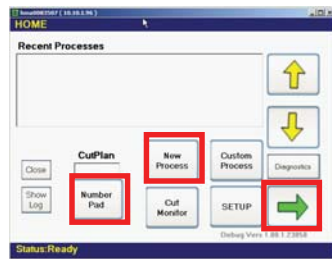


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4.05 TSC-3000 Selecting New Process

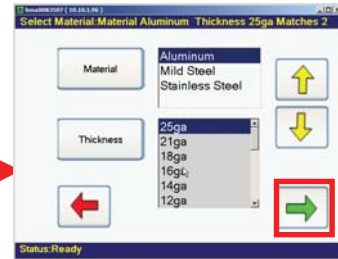
Select new standard process by material; thickness; amperage (1) or enter process # using pop-up number pad (2).

HOME

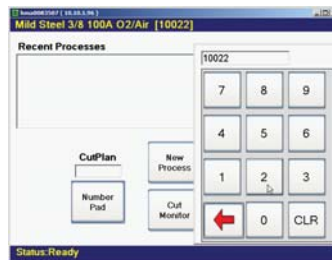


Material –
Scroll to select,
Thickness –
Scroll to select,
Accept & advance (1).

SELECT MATERIAL, THICKNESS

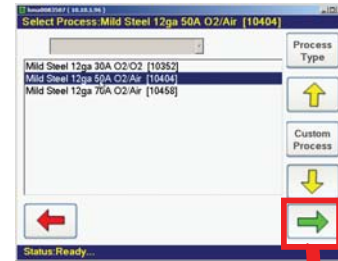


2
SELECT PROCESS NUMBER

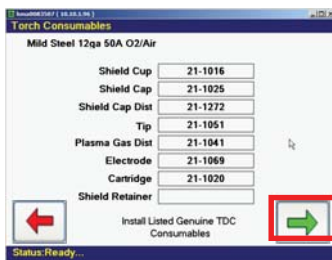


Select process by # (2) or by name (3).
If by name, first select process type; best cut, fastest, MAX Pierce, etc.
Then press
To accept & advance to Torch Consumables. (4).

3
SELECT PROCESS



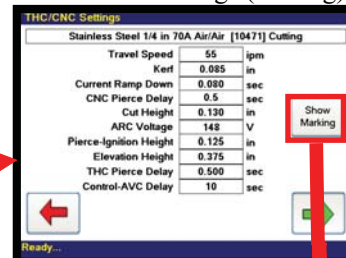
TORCH CONSUMABLES



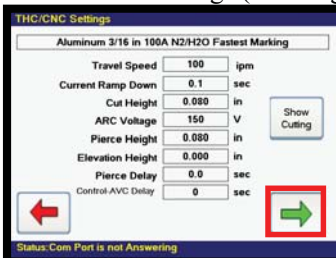
Confirm the listed torch consumables are installed in the torch.

Go to THC / CNC settings for cutting (5).

THC / CNC Settings (Cutting)



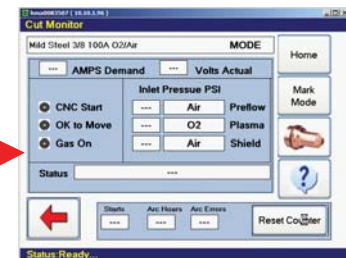
THC / CNC Settings (Marking)



Toggle between Cut & Marking settings (6).

Proceed to Process Monitor (7).

PROCESS MONITOR

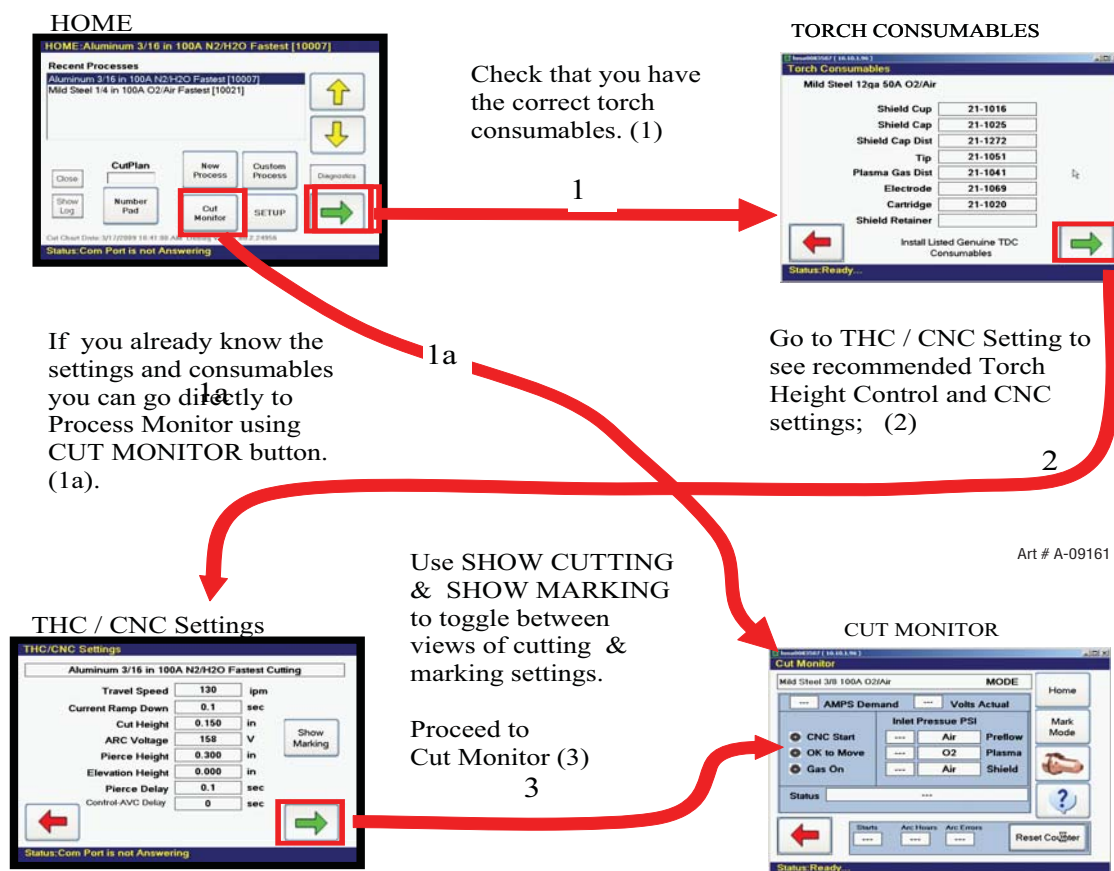


Now you are ready to start cutting. Displayed Amps Demand is the output current setting, not the actual current.

As soon as Start is applied Process Monitor shows output voltage, gas pressures, status of Start, OK to Move and Gas On signals. Here you can toggle between Plasma Marking with the selected consumables and cutting.

4.06 TSC-3000 Selecting Recently Used Process

Home screen shows some of the most recently used processes with the last one used highlighted. Select the highlighted process by pressing the Green arrow or scroll with up/down arrows to select another recent process then press the green arrow.



Now you are ready to start cutting. Displayed Amps Demand is the output current setting, not the actual current.

As soon as Start is applied Process Monitor shows output voltage, gas pressures, status of Start, OK to Move and Gas On signals. Here you can toggle between Plasma Marking with the selected consumables and cutting.

4.07 TSC-3000 Creating a Custom Process

Considerable effort has been made to define the best parameters for each cutting and marking process. However for various reasons, variations in material composition, variations in cutting table performance, user preferences, etc. you may desire to modify one or more processes. Or perhaps you want to keep the same cutting current and gas pressures but save different THC / CNC parameters such as cutting speed, arc voltage, pierce height, etc. That will still be treated as a custom process.

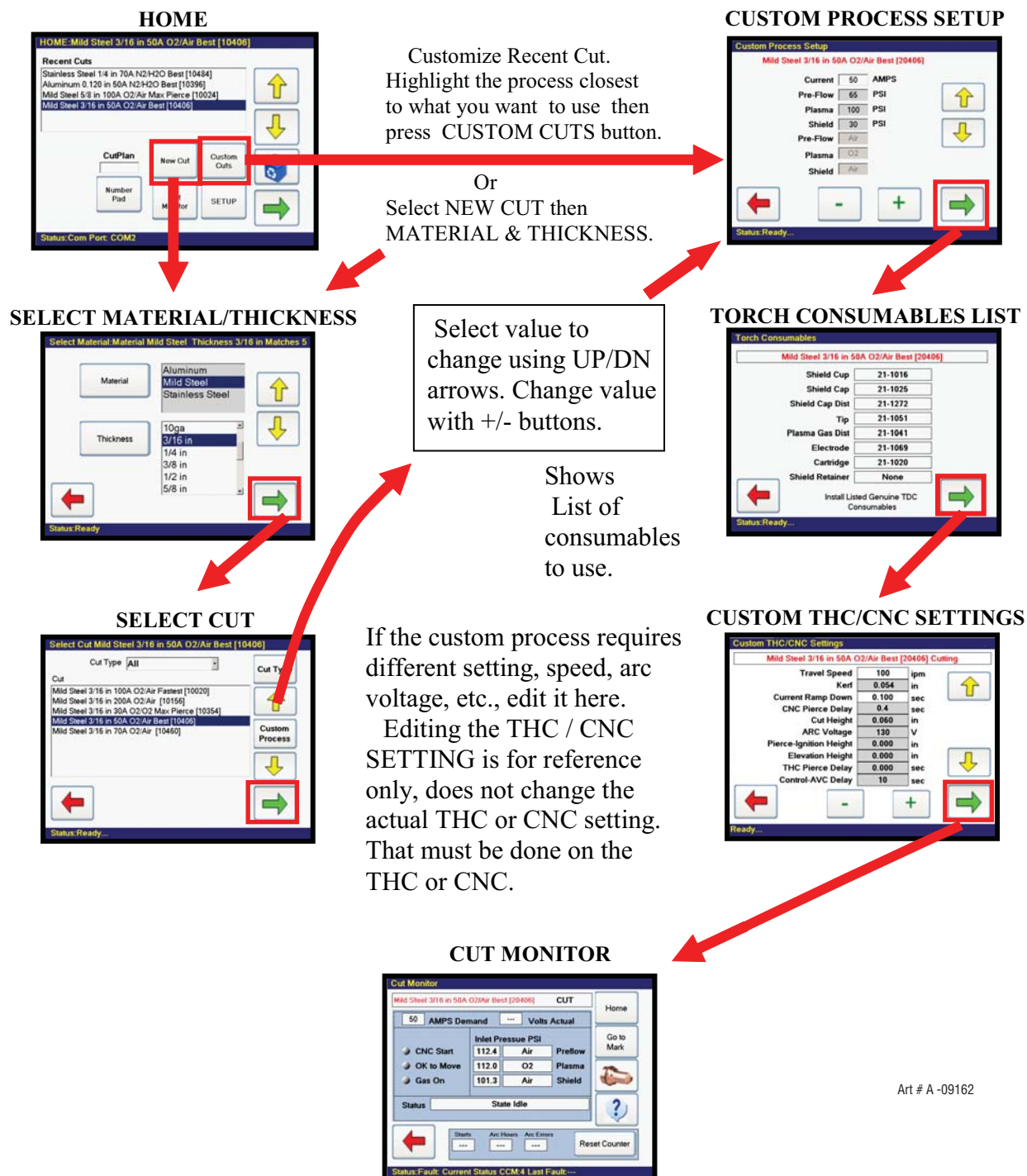
Creating a **Custom Process** starts with an existing standard process. From there you may adjust the cutting current and the gas pressures. It's up to you to determine and enter the correct values. There no limits in these values, if you change too far it may cause faults.

You may also change the values displayed on THC/CNC Settings screen. THC / CNC settings on the TSC-3000 at this time are for information only. They do not connect to the THC or CNC so do not automatically modify setting in those devices. Still, when you determine the correct values for your modified (custom) process you may want to record them here.

The custom process will have the same name as the standard base process but it will be automatically assigned a new number and the whole name and number will be in red text wherever it appears.

You cannot change the gas type. If you want a different gas types find a standard process that uses those types and edit it.

First step is to select a standard process. Usually you want a process that is for the same metal type and thickness that you want to cut. From the HOME screen you may select a recent process (sect 4.06, Selecting Recently Used Process) or a new process (sect 4.05, Selecting New Process).



Art # A -09162

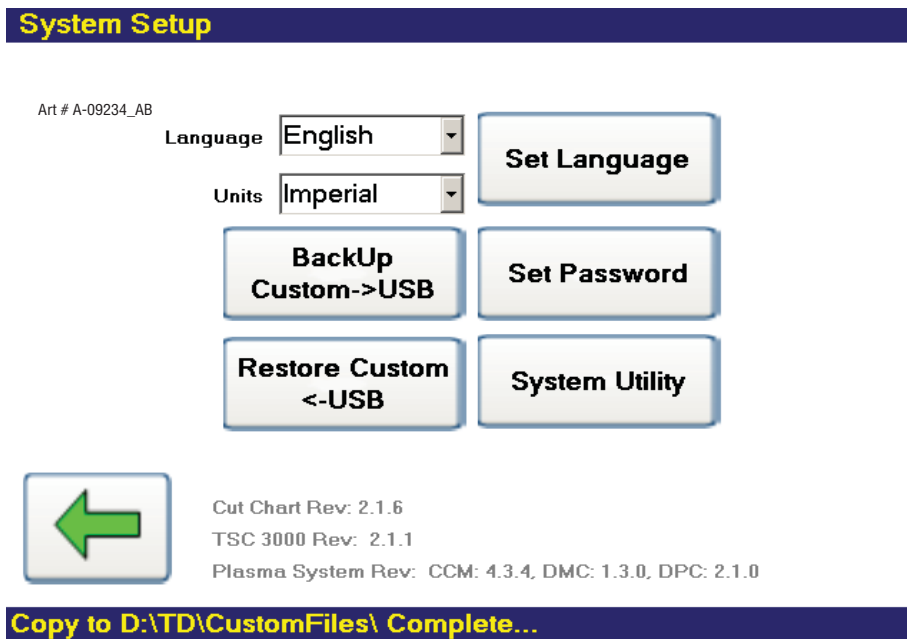
Going to the Cut Monitor loads the new custom process with its name/number in red. Following purging of the new process you are ready to cut. Your custom process will appear in the recent cuts menu in red.

4.08 Back up and Restoration of Custom Processes.

Users can create their own custom cut processes. These can be lost by accidentally deleting them on the TDC 3000 or they can be over written when updating programs. They can also be lost if the DSC 3000 should be replaced. We recommend you back up your custom processes every time a new one is created. To back up you will need a USB Flash drive.

Backing Up Custom Process:

1. For the remotely mounted DSC 3000 insert the flash drive in the USB connector on the back. If using the TSC 3000 mounted in the plasma supply front panel the USB connector is in the front under the clear plastic protective screen.
2. Go to the HOME screen. Select SETUP. Enter the 5 digit password (00000) unless you've changed it.
3. Press button "Back Up Custom>USB"
4. Very quickly, depending on how many custom processes there are, you should see along the bottom of the screen the message "Copy to D:\TD\CustomFiles\Complete" at which point you can remove the flash drive.



Restoring Custom Processes:

During file backup (see previous process) a folder and sub folders called “TD” were placed on your flash drive. You should copy this folder with its subfolders to some secure place. Making more than one copy is advised. If the need arises to restore from this backup, place the entire TD folder in the root directory (not in another folder) on a USB flash drive and plug it into the USB port on the TDC 3000.

1. For the remotely mounted DSC 3000 insert the flash drive in the USB connector on the back. If using the TSC 3000 mounted in the plasma supply front panel the USB connector is in the front under the clear plastic protective screen.
2. Go to the HOME screen. Select SETUP. Enter the 5 digit password (00000) unless you’ve changed it.
3. Press the “Restore Custom <USB.

System Setup

Art # A-09235_AB

Language	English	Set Language
Units	Imperial	
BackUp Custom->USB		Set Password
Restore Custom <-USB		System Utility



Cut Chart Rev: 2.1.6

TSC 3000 Rev: 2.1.1

Plasma System Rev: CCM: 4.3.4, DMC: 1.3.0, DPC: 2.1.0

Copy to .\Files\Custom\ Complete...

4. Shortly, almost instantly if only a few files, you should see at the bottom “Status: Copy to .\Files\Custom/ Complete” then you can remove the USB flash drive.

Renaming Custom Files

You cannot rename the files while on the TSC 3000 but once you backup a file on the flash drive you can use a computer to rename the file on the flash drive and perform the restore to put the renamed file back on the TSC 3000.

1. Use Windows Explorer to find the folder **TD** in the root directory of the flash drive. Then find and open the folder **CustomFiles**. Here you will find all the custom processes saved.

Art # A-09236

Name	Size	Type	Date Modified
Aluminum 5mm 47A Air--Air Max Pierce [20392].cus	9 KB	CUS File	6/1/2009 11:40 AM
Mild Steel 5mm 47A O2--Air Best [20411].cus	9 KB	CUS File	6/1/2009 11:41 AM
Mild Steel 12mm 96A O2--Air Best [20031].cus	9 KB	CUS File	6/1/2009 11:40 AM

2. Use Windows Rename to change the file name to whatever you want. **Do not change the file extension (.cus) or the file number in the brackets!** Suggest not making the name too long as the TSC 3000 screen is limited in the number characters on one line. Here I've renamed the top file.

Art # A-09237

Name	Size	Type	Date Modified
John Doe's Aluminum 5mm 47A Air--Air [20392].cus	9 KB	CUS File	6/1/2009 11:40 AM
Mild Steel 5mm 47A O2--Air Best [20411].cus	9 KB	CUS File	6/1/2009 11:41 AM
Mild Steel 12mm 96A O2--Air Best [20031].cus	9 KB	CUS File	6/1/2009 11:40 AM

3. Now put the flash drive with the renamed file back into the TSC 3000 and perform the file restore.
4. Go back to the Home screen. You will find your new file name (John Doe's) and the old name with the same file number [20392]. They are the same process, either one will work. You can delete the file with the old name if you don't want both of them showing.

4.09 Sequence of Operation

Ultracut with DFC 3000 using TSC 3000

The purpose of this section is to explain the steps an operator should follow when using the DFC 3000 with the TSC 3000 touch screen control panel for a specific cutting process. Also included are steps to follow for changing torch consumables.

NOTE

For units not using the TSC 3000 where control is embedded into the cutting table controller most of this sequence still applies except for the parts that are specific to the TSC 3000. For those sections refer to the cutting table controller documentation.

This assumes the system setup; language and units, lead lengths, etc., have already been performed by the installer and the system is confirmed as operational.

1. Before Applying power:

- a) Ensure that the required gasses are connected to the DMC 3000 inlet and gasses are turned on and set for required inlet pressure.
- b) Ensure you have a torch cartridge with consumables installed on the torch.

If you don't know what consumables you need to use and need to select the cutting process in order to read the consumables list on the TSC 3000, you can start up with Plasma Disabled. In this case the operation sequence will skip to step 3.

- c) Set Plasma Enable SW on the TSC 3000 to Enabled.

2. Turn on 3 phase power at the main disconnect (Plasma Enable on TSC 3000 on). Turn on the power supply ON/Off switch on the front panel.

- a) The AC indicator on the front panel will come on and the inverter fans will start after an initial 4-5 second self check.
- b) After another 5 seconds the DC indicator will blink twice indicating this is an Ultracut.
- c) Immediately following this the Temp, Gas, DC & Status indicators will blink the CCM firmware version. Example, version 4.1 will blink 4 times followed by a short space then blink once more.
- d) At the same time the DMC & DPC, both Green & Red LEDs, also blink their firmware version in the same fashion.
- e) Following the firmware version indication, while CCM establishes communication with the gas control (Step e.), the coolant pump and fans come on to "prime" the system. The "gas" indicator on the front panel blinks until correct coolant flow is detected. Normally this takes only a few seconds but can last up to 4 minutes if there is a problem. Should there be a problem with priming the coolant system the Status indicator will flash code 4-4 and the pump will stop until you rectify the problem and recycle the power.

NOTE

If communication is not established in step e. the Gas indicator will not blink at this time, also pump shuts off after 15 sec and shows code 4-2.

- f) Once communication with the DFC components is established the Green power indicator on both DMC & DPC will be on and their Red indicator will be off.
- g) Skip to step 4.

3. Turn on 3 phase power at the main disconnect (Plasma Enable on TSC 3000 Disabled).
 - a) AC indicator on front panel does not come on, pump and fans do not start.
 - b) Approximately 10 sec after power up the front panel DC indicator will blink twice indicating this is an Ultracut.
 - c) Immediately following this the Temp, Gas, DC & Status indicators will blink the CCM firmware version. Example, version 4.1 will blink 4 times followed by a short space then blink once more.
 - d) At the same time the DMC & DPC, both Green & Red LEDs, also blink their firmware version in the same fashion.
 - e) After blinking the CCM firmware version on the front panel indicators the CCM establishes communication with the DMC & DPC. Once communication is established the Green LEDs are on steady and the Red LEDs should be off.
4. At power on while all the items in step 2 (or 3) are going on the TSC 3000 has been starting up Windows XP™ which takes about 2 ¼ minutes. Another 15 sec the application starts and displays the HOME screen.
5. Now you must select the cutting (and plasma marking) process. Refer to sections 4.05, 4.06 and 4.07 for step by step instructions in selecting New, Recently Used or Creating Custom Processes.
6. During process selection you will come to the “Torch Consumables” screen. At this point if you have not installed correct consumables and “Plasma” is not already disabled, shut off (disable) the Plasma Enable SW. Install the consumables, Set the Plasma Enable SW to “Enable”.
 - a) Coolant Pump and fan start up to prime the torch cooling system. The “Gas” indicator on the front panel blinks until correct coolant flow is detected. Normally only a few seconds. See step 2.d. for full details.
7. The next screen following Torch Consumables is a table of recommended CNC and THC (Torch Height Control) settings. You may toggle between cutting and marking settings using the “Show Marking / Show Cutting” button. Set your cutting table controller and height control to the recommended setting.
 - a) If you prefer other settings for cutting you can create a “Custom” process where you edit the cutting CNC/THC setting if you want. Currently, modifying the marking CNC/THC settings is not allowed. A custom process does not have to change the cutting current or gas pressures unless you want to. You can use customs to modify the recommended CNC/THC settings as well.
8. Continue the process selection until you reach the Cut Monitor screen.
 - a) Purging of the gasses used for the selected process begins. Depending on your lead length and consumable type this can take some time.

Purge times for different lead lengths have been set to allow time to fully pressurize the leads and to remove any coolant that gets into the consumables during a parts change. Make sure that during setup the correct lead length has been selected, the default time is for max lead length and may be longer than you require.
9. Once purge is complete you are ready to start cutting.
 - a) If marking is desired press button “Go to Marking”. The screen will show recommended CNC/THC setting. Green arrow takes you back to the Cut Monitor set up for Marking.
 - b) To return to cutting on the Cut Monitor press button “Got to Cutting”. Same as with the marking this takes you to the CNC/THC screen except this displays the cutting settings.

10. Changing torch consumables:

Following a consumables change or inspection the system will initiate a purge of the gas. This serves two purposes:

- a) Remove the coolant that gets into the consumables when they have been removed.
- b) If the gas type has changed "Purge" removes the old gas from the lines and replaces it with the new gas. If the gas type has changed from a fuel gas like H35 to an oxidizing gas like oxygen or air the purge inserts a buffer of inert gas, nitrogen, so the H35 and oxygen don't mix.

You can change consumables by removing power or using the "Plasma Disable" SW.

If you remove power the system does not remember what gasses were last used so it will do a complete purge including nitrogen buffer even if you do not change gas types.

With "Plasma Disable" the system remembers what was used last and does only the amount of purge necessary possibly saving time.

11. Use "Plasma Disable".

- a) You may use the Plasma Disable when removing the torch cartridge to change or inspect consumables. Plasma disable stops coolant pump, shuts off all gas solenoids, removes power from the inverter (power supply) and pilot circuits, inhibits the arc starter ignition (HF) circuit. It does not remove power from the TSC 3000 or the systems logic and communications circuits so you do not need to go through as long a power up process after replacing the consumables.
- b) If changing to a process with a different gas type you may want to select it before enabling the plasma otherwise it will start purging the old process and when you do select the new one it will purge again taking more time.
- c) If you are not changing gas type you may want to turn on the Enable first to start the purge while you enter the process.

12. Shutting off power.

- a) If you chose to shut off power for inspecting or changing consumables, or any other purpose, the system does not remember what you were doing last so you will have to go through the complete sequence from the beginning starting with step 1.

4.10 Gas Selection

A. Plasma Gases

1. Air Plasma

- Most often used on ferrous or carbon base materials for good quality at faster cutting speeds.
- Air plasma is normally used with air shield.
- Only clean, dry air is recommended for use as plasma gas. Any oil or moisture in the air supply will substantially reduce torch parts life.
- Provides satisfactory results on nonferrous materials and reduced weldability on ferrous materials.

2. Argon/Hydrogen (H35) Plasma

- Recommended for use on 3/4 in (19 mm) and thicker stainless steel. Recommended for 1/2 inch (12 mm) and thicker nonferrous materials. Ar/H₂ is not normally used for thinner nonferrous materials because less expensive gases can achieve similar cut quality.
- Poor cut quality on ferrous materials.
- Provides faster cutting speeds and high cut quality on thicker materials to offset the higher cost.
- A 65% argon / 35% hydrogen mixture should be used.

3. Oxygen (O₂) Plasma

- Oxygen is recommended for cutting ferrous materials.
- Provides faster cutting speeds.
- Provides very smooth finishes and minimizes nitride build-up on cut surface (nitride build-up can cause difficulties in producing high quality welds if not removed).

4. Nitrogen (N₂) Plasma

- Provides better cut quality on nonferrous materials such as stainless steel and aluminum.
- Can be used in place of air plasma with air shield or carbon dioxide (CO₂).
- Provides much better parts life than air.
- A good clean welding grade nitrogen should be used.

5. H17 Plasma (17.5% Hydrogen / 32.5% Argon / 50% Nitrogen)

- Recommended for 1/2 inch (12 mm) and thicker nonferrous materials. H17 is not normally used for thinner nonferrous materials because less expensive gases can achieve similar cut quality.
- Alternative to H35 Plasma. Yields slightly higher cut speeds and similar cut quality.
- When using H17, the gas should be supplied to the gas supply port marked "H35".
- Poor cut quality on ferrous materials.

B. Shield Gases**1. Compressed Air Shield**

- Air shield is normally used when operating with air plasma and oxygen plasma.
- Improves cut quality on some ferrous materials.
- Inexpensive - reduces operating costs.

2. Nitrogen (N2) Shield

- Nitrogen shield is used with Ar/H2 (H35) plasma.
- Provides smooth finishes on nonferrous materials.
- May reduce smoke when used with Ar/H2 plasma.

3. Water Shield

- Water shield should be used only in mechanized applications - never in hand cutting!
- Normally used with nitrogen.
- Provides very smooth cut surface.
- Reduces smoke and heat input to the workpiece.
- Effective up to 3/4 inch (19.1 mm) maximum material thickness.
- Tap water provides low operating expense.

4.11 Power Supply Status Codes

On start-up and during operation, the power supply control circuitry performs various tests. If the circuitry detects a condition requiring operator attention, the status indicator on the control panel flashes a 2-part code to indicate a code group and then a particular condition within the group. After 4 seconds, the sequence repeats.

Example: Indicator flashes 4 times; the condition is in group 4. After a 1.2 second delay, the indicator blinks 3 times; the condition code is 4-3, indicating that the coolant is overheating. After a 4-second delay, the indicator repeats the sequence until the condition is corrected.

Some conditions can be active indefinitely, while others are momentary. The power supply latches momentary conditions; some momentary conditions can shut down the system. The indicator may show multiple conditions in sequence; it is important to recognize all possible conditions that may be displayed.

Note

These codes are for systems with Firmware versions 2.4 or later. If your system's Firmware is a prior version, contact your distributor for updates.

CCM Status Code		
Code	Message	Remedy / Comments
Group 1 -- Plasma Process		
1-1	Plasma Disabled or Missing AC Input Phase	Plasma Enable Off ; External Disable / E-Stop Activated or CCM TB1-1 & 2 jumper missing; Control cable to DMC not connected; Missing AC Phase.
1-2	Pilot Ignition Failure	Pilot did not start within 15 seconds. Ensure process loaded agrees with torch parts installed; Torch consumable parts worn; Defective Arc Starter. Defective chopper (100-300A unit); Defective #2 inverter (400A unit).
1-3	Lost Pilot	Pilot went out while CNC Start active. Ensure process loaded agrees with torch parts installed; If custom process cut current too low or preflow pressure too high. Torch consumable parts worn
1-4	Transfer Lost	Arc was transferred to work for more than 50 ms and then went out while CNC Start still active. Standoff too high; Ensure process loaded agrees with torch parts installed; If custom process current set too low or wrong plasma pressure.
1-5	Off the Plate	Function not currently enabled
1-6	Pilot Timeout, no Transfer	Must transfer from Pilot to Cutting Arc in 85 ms. (SW8-1 OFF) or 3 sec. (SW8-1 ON). Standoff too high or void in work under torch; Wrong cut process selected resulting in: Pilot current too low for consumables; Preflow pressure too low.
1-7	Reserved	No information available; Contact customer service
1-8	Possible Shorted Torch	Tip voltage too close to electrode voltage. Plasma supply pressure too low; Leak in Plasma hose to torch; Ensure process loaded agrees with torch parts installed; If custom process, plasma pressure may be too low or cut current too high. Torch consumable parts worn; Shorted torch body
1-9	Part Process not configured	CNC Start applied without a process loaded. Select process and send to plasma system
1-10	Device Locked	Process loading, wait until finished

CCM Status Code		
Code	Message	Remedy / Comments
Group 2 -- Plasma Power Supply		
2-1	Missing AC Phase	Blown fuse, Broken or loose connection on power cable
2-2	Wrong input voltage	Inverter(s) not configured correctly for input voltage; Poor power quality (brownouts, dropouts); Input power capacity / wiring too small causing voltage drop; broken or loose power cable connections.
2-3	Inverter Over Temperature	Ambient above 40 deg C. (104 F); Blocked airflow.Failed fan; Defective inverter.
2-4	Power Supply not Ready	Defective inverter; PCB4; PCB7; defective input contactor.
2-5	DC Output Low	Less than 60 VDC; Defective inverter, shorted output; CCM voltage sense (J6) wire open or disconnected. Negative lead short to gnd. Shorted chopper (100-300A only).
2-6	Inverter input fault	Capacitor bank voltage imbalance (400-460-600V only) or inverter primary switching circuit overcurrent. If occasional may be line voltage surge; remove power to reset. If persists defective inverter; LED on inverter indicate which one.
2-7	Unexpected current	Current above 20A in work or pilot leads before pilot ignition; Possible shorted torch; Defective current sensor.
2-8	Unexpected current in pilot circuit	Current above 5A in pilot circuit; wrong or mismatched consumables; Pilot lead shorted to negative in torch tube; Possible shorted torch
2-9	Unexpected current in work lead	Current above 5A in work lead; Short to chassis in RAS; Negative lead short to ground.
2-10	Output Current Too High	Current greater than 20% above process setting. Measure current of each inverter, only one high likely defective inverter. All high likely defective PCB5, CCM.
2-11	Output Current Too Low	Current more than 20% below process setting. Measure current of each inverter, only one low likely defective inverter. All low likely defective PCB5, CCM.
Group 3 -- Gas Controller Status & Protocol		
3-1	Gas Control Communication fault.	Cannot establish Communication with gas control. DMC to DPC Control Cable not connected or Fiber-optic cable not properly connected, dirty or defective; DPC power supply or Control PCB defective. DMC Control PCB defective; CCM defective, replace.
3-2	Gas Control reply fault.	Connection was established but Gas Control did not reply to the CCM in the allowed time (15 ms.). Dirt on fiber ends or in connectors, blow out with clean dry air. Fiber not locked into connector; sharp bends in fiber; fiber defective. If it's determined the fiber is OK then DPC or DMC Control or Power Supply PCB likely defective, replace board.
3-3	Gas Supply Pressure out of range.	DPC has detected a pressure error, Check DPC status indicator for specific problem.
3-4	Gas Control Purging	The time to purge depends on Lead Length, previous fluids used, and current process fluids selected. Wait until finished.
3-5	Gas Control Protocol Error	Verify Firmware revision for compatibility with DMC and DPC
3-6	Reserved	No information available; Contact customer service
3-7	Gas Control sequencing error	Verify Firmware revision for compatibility with DMC and DPC
3-8	Gas Control Type Mismatch	CCM must be Ultracut type with USB for use with Auto Gas Control (DFC-3000)
3-9	Gas Control Command fault	Verify Firmware revision for compatibility with DMC and DPC; Electromagnetic interference with Arc Starter; inspect grounding; bonding; and isolation
3-10	DPC-fault	Check DPC status indicator for specific problem
3-11	DPC valve control fault	Check DPC status indicator for specific problem
3-12	DMC fault	Check DMC status indicator for specific problem

CCM Status Code		
Code	Message	Remedy / Comments
Group 4 -- Torch Coolant System		
4-1	Coolant Level low	Check coolant level, add as needed.
4-2	Low coolant flow	Fault response time: Process Never started since last power cycle: Below 0.7 GPH for 15 Seconds. Else below 0.5 GPH no delay or after 3 seconds if below 0.7 GPH Possible cause: Suction leak introducing air into coolant, suspect rear panel filter seal; clogged filter; defective pump.
4-3	Coolant overheated	Coolant return temperature exceeded 70 degrees Celcius (158F). Coolant fan failed; radiator fins clogged with dirt; operating with side panel loose or removed. Ambient temperature > 40 deg C.
4-4	Coolant System not ready.	Proper coolant flow of 0.7 GPH was not obtained during up to 4 minutes of Priming. New installation can require additional Priming cycle(s) to fill hoses with coolant; cycle power to restart Priming; Suction leak introducing air into coolant; suspect rear panel filter seal; Clogged coolant filter; Damaged torch coolant tube; Defective pump assembly.
4-5	Low Coolant Level Warning	Low coolant level during cut, does not stop cut. Add coolant as required.
Group 5 -- Gas Control Communications Port		
5-1	CANBUS Acknowledge fault.	Message sent over CANBUS (Fiber-optic) was not acknowledged as received by DMC. CCM to DMC Fiber-optic cable not properly connected, dirty or defective; DMC power supply or Control PCB defective; CCM defective.
5-2	CANBUS Off	Excessive communication errors resulted in communication with DMC to turn off. Dirt on fiber ends or in connectors, blow out with clean dry air; fiber not locked into connector; sharp bends in fiber; fiber defective;
5-3	CANBUS Errors Warning.	Communication with DMC has required a significant number of message retries due to communication errors. Dirt on fiber ends or in connectors, blow out with clean dry air; fiber not locked into connector; sharp bends in fiber; fiber defective;
5-4	Reserved	No information available; Contact customer service
Group 6 -- CCM		
6-1	CCM Analog Voltage Error	Replace CCM
6-2	CCM ADC or DAC error	Replace CCM
6-3	Coolant Flow too High	Flow greater than 2.7 gal/min Torch coolant tube broken or missing; CCM fault, replace CCM
6-4	CCM Data Memory error	Replace CCM
6-5	CCM Program memory fault	Replace CCM
6-6	+5V Logic supply low	Replace CCM
6-7	CCM processor over temp	Reduce ambient temperature; Defective CCM; replace
6-8	5V supply for RS 485/422 communication low.	Replace CCM

4.12 DMC-3000 Status Codes

Status Code indicators:

1. At power up, RED & GREEN indicators flash program firmware revision number in a 2 digit code as explained below. Following the revision code
2. Green indicator
 - On steady = OK, communication established.
 - Blinking = no communication established with CCM
3. Red indicator
 - Off = Status OK
 - Rapid blinking = downloading new program
 - Blinking 2 digit code = Status (table below)

Status Indicators on the DMC & DPC modules flash a 2-part code. Both green and red flash the firmware version at power up. Only the red flashes for a fault.

First part of the code indicates a code group, the second part a particular condition within that group. DMC and DPM may flash codes at the same time and they may be different codes. For example, a power supply fault, such as 1-3, in the DMC could prevent communicating with the DPM so the DPC would flash one of its group 2 codes. You have to consider the codes in all the modules, CCM, DMC & DPC before determining which hardware has the fault.

The 2 digit code is separated by a 1.2 second space between digits and a 4 second space before the sequence repeats or flashes another code group. Codes displayed may be currently active or may represent a fault that shut the process down but is no longer active. If the fault that shut down the process is no longer active the first blink of the first digit is extra long.

Example: Indicator flashes 1 long followed by 3 shorter blinks the condition is in group 4. After 1.2 seconds delay, the indicator blinks 3 times; the condition code is 4-3 (the first long blink counts as part of the first digit), indicating the DMC has detected a DPC time-out error and it is not currently active. After a 4 second delay, the indicator repeats the sequence until the condition is corrected.

DMC Status Code Key		
Code	Message	Cause/Remedy
Group 1 - DMC		
1-1	Plasma Disabled	Enable Plasma; Defective DMC Power Supply (E-Stop relay)
1-2	24 VDC fault	Check /replace DMC power supply fuse (F2); Replace DMC Power Supply; Replace DMC control PCB
1-3	5 VDC fault	Check /replace DMC power supply; Replace DMC control PCB
1-4	No gas process selected	Select and send process from TSC 3000 or CNC;
1-5	DMC is locked	Process loading, wait until finished
1-6	Gas not Purged	DPC fault prevented purge from occurring, see DPC fault status
Group 2 - DPC Communication Port *		
2-1	DPC Acknowledge error	Check DMC to DPC fiber optic cable and connections; Dirty or defective fiber-optic; DPC power supply, fuse or Control PCB. DMC Control PCB
2-2	DPC Bus off error	Check DMC to DPC fiber optic cable and connections; Dirty or defective fiber-optic; DPC power supply, fuse or Control PCB. DMC Control PCB.
2-3	Reserved	No information available; Contact customer service
Group 3 - CCM Communication Port *		
3-1	CCM Acknowledge error	Check DMC to CCM fiber optic cable and connections; Dirty or defective fiber-optic; Replace DMC Control PCB; Replace CCM.
3-2	CCM Bus off error	Check DMC to CCM fiber optic cable and connections; Dirty or defective fiber-optic; Replace DMC Control PCB; Replace CCM.
3-3	Reserved	No information available; Contact customer service
Group 4 - Gas Controller Status and Protocol Errors		
4-1	CCM Connection loss	Loss of communication with either DPC or DMC; check for dirty or defective fiber optic to either DPC or CCM; See DPC and CCM status codes
4-2	CCM Timeout	Loss of communication with either DPC or DMC, check for dirty or defective fiber-optic to either DPC or CCM. See DPC and CCM status codes.
4-3	DPC Timeout	Loss of communication with DPC, check for dirty or defective fiber-optic to DPC. See DPC status code.
4-4	DPC error	Loss of communication with DPC, check for dirty or defective fiber-optic to DPC. See DPC status code.
4-5	Reserved	No information available; Contact customer service
4-6	Reserved	No information available; Contact customer service
Group 5 - Shorted Solenoid Faults		
5-1	Sol 1 – H35 Plasma	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-2	Sol 2 – Oxygen Plasma	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-3	Sol 3 – Air Plasma	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-4	Sol 4 – Nitrogen Plasma	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-5	Sol 5 – Aux Plasma	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-6	Sol 6 – Oxygen Shield	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-7	Sol 7 – Air Shield	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-8	Sol 8 – Nitrogen Shield	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-9	Sol 9 – H ² O Shield	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-10	Sol 10 – Oxygen Preflow	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-11	Sol 11 – Air Preflow	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-12	Sol 12 – Nitrogen Preflow	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-13	Sol 13 – Argon Marking	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-14	Sol 14 – Air Marking	Replace solenoid; Shorted wire harness; Defective DMC Control PCB
5-15	Sol 15 – Nitrogen Marking	Replace solenoid; Shorted wire harness; Defective DMC Control PCB

* Communication faults shown on the DMC can be caused by the device on the other end of the communication line. Before assuming the DMC is at fault see if either the CCM or the DPC are showing any codes that might indicate they are at fault instead.

4.13 DPC-3000 Status Codes

DPC Status Code Key		
Code	Message	Cause/Remedy
Group 1 - DPC		
1-1	Plasma Disabled	Enable Plasma; DMC Power Supply input fuse F1 blown (no lights on DMC); Defective DMC Power Supply (E-Stop relay). Defective DPC Power Supply (E-Stop relay). Defective DPC control PCB
1-2	24 VDC fault	Check /replace DPC power supply output fuse (F2); Defective DPC power supply; Replace DPC control PCB
1-3	+12 VDC fault	Check /replace DPC power supply; Replace DPC control PCB
1-4	-12 VDC fault	Check /replace DPC power supply; Replace DPC control PCB
1-5	5 VDC fault	Check /replace DPC power supply; Replace DPC control PCB
1-6	Analog Reference fault	Replace DPC Control PCB
1-7	ADC voltage fault	Replace DPC Control PCB
1-8	Reserved	
1-9	Valve not configured	Resend process from TSC 3000 / CNC
1-10	DPC is locked	Process loading, wait until finished
Group 2 - Control Communication		
2-1	Acknowledge error	Fiber optic cable; DPC Control PCB
2-2	Timeout	Communication loss: Check DMC codes; if DMC 2-1 check DPC to DMC Fiber optic cable, DMC 4-2 check DMC to CCM fiber-optic cable.; DPC Control PCB; DMC Control PCB; CCM.
2-3	Protocol	Program firmware fault, consult factory
2-4	Bus off	Fiber optic cable; DPC Control PCB
2-5	Physical	Fiber optic cable; DPC Control PCB
Group 3 - Inlet Pressure Faults		
3-1	Plasma Gas High (>145 PSI)	Reduce pressure from gas supply; defective pressure sensor (PS4).
3-2	Pilot Gas High (>145 PSI)	Reduce pressure from gas supply; defective pressure sensor-(PS3).
3-3	Shield Gas High (>145 PSI)	Reduce pressure from gas supply; defective pressure sensor (PS1).
3-4	Shield H2O High (>145 PSI)	Reduce pressure from gas supply; defective pressure sensor (PS2).
Group 4 - Proportional Valves & Solenoids Electrical Faults		
4-1	Plasma High (V5) shorted	Check for shorted wire harness; Replace valve
4-2	Plasma Low (V4) shorted	Check for shorted wire harness; Replace valve
4-3	Pilot (V3) shorted	Check for shorted wire harness; Replace valve
4-4	Shield Gas (V1) shorted	Check for shorted wire harness; Replace valve
4-5	Shield H2O (V2) shorted	Check for shorted wire harness; Replace valve
4-6	Vent (SOL2) shorted	Check for shorted wire harness; Replace valve
4-7	Cut (SOL3) shorted	Check for shorted wire harness; Replace valve
4-8	Mark (SOL1) shorted	Check for shorted wire harness; Replace valve
4-9	Vent (SOL2) open	Check for open connection; Replace valve.
4-10	Cut (SOL3) open	Check for open connection; Replace valve.
4-11	Mark (SOL1) open	Check for open connection; Replace valve.
Group 5 - Plasma High Proportional Valve (V5)		
5-1	Valve open	Check for open connection; Replace valve.
5-2	DPC inlet Supply pressure low	Raise plasma inlet pressure; check for DMC valve not open; restriction in gas supply / hoses.
5-3	Outlet pressure too low	Valve full open. Leak in plasma hose to torch; defective control valve (V5); defective pressure sensor (PS6); DPC Control PCB
5-4	Outlet pressure low	Tracking error Leak in plasma hose to torch; defective control valve (V5); defective pressure sensor (PS6); DPC Control PCB
5-5	Outlet pressure too high.	Valve at minimum setting. Restriction in plasma hose, torch, DPC manifold; defective control valve (V5); defective pressure sensor (PS6); DPC Control PCB
5-6	Outlet pressure high.	Tracking error. Restriction in plasma hose, torch, DPC manifold; defective control valve (V5); defective pressure sensor (PS6); DPC Control PCB

DPC Status Code Key		
Code	Message	Cause/Remedy
Group 6 - Plasma Low Proportional Valve (V4)		
6-1	Valve open	Check for open connection; Replace valve.
6-2	DPC inlet Supply pressure low	Raise plasma inlet pressure; check for DMC valve not open; restriction in gas supply/hoses.
6-3	Outlet pressure too low	Valve full open but pressure still too low. Leak in plasma hose to torch; Defective control valve (V4); Defective pressure sensor (PS6); DPC Control PCB
6-4	Outlet pressure low	Tracking error. Leak in plasma hose to torch; Defective control valve (V4); Defective pressure sensor (PS6); DPC Control PCB
6-5	Outlet pressure Too high	Valve at minimum setting but pressure still too high. Restriction in plasma hose to torch; torch or DPC manifold; Defective control valve (V4); Defective pressure sensor (PS6); DPC Control PCB
6-6	Outlet pressure high	Tracking error. Restriction in plasma hose; torch or DPC manifold; Defective control valve (V4); Defective pressure sensor (PS6); DPC Control PCB
Group 7 - Pilot Proportional Valve (V3)		
7-1	Valve open	Check for open connection; Replace valve.
7-2	DPC inlet Supply pressure low	Raise plasma inlet pressure; check for DMC valve not open; restriction in gas supply/hoses.
7-3	Outlet pressure too low	Valve full open but pressure still too low. Leak in plasma hose to torch; defective control valve; defective pressure sensor (PS6); DPC Control PCB
7-4	Outlet pressure low	Tracking error. Leak in plasma hose to torch; defective control valve; defective pressure sensor (PS6); DPC Control PCB
7-5	Outlet pressure Too high	Valve at minimum setting but pressure still too high. Restriction in plasma hose, torch, DPC manifold; defective control valve; defective pressure sensor (PS6); DPC Control PCB
7-6	Outlet pressure high	Tracking error. Restriction in plasma hose, torch, DPC manifold; defective control valve; defective pressure sensor (PS6); DPC Control PCB
Group 8 - Shield Gas Proportional Valve (V1)		
8-1	Valve open	Check for open connection; Replace valve.
8-2	DPC inlet Supply pressure low	Raise plasma inlet pressure; check for DMC valve not open; restriction in gas supply/hoses.
8-3	Outlet pressure too low	Valve full open but pressure still too low. Leak in plasma hose to torch; defective control valve; defective pressure sensor (PS5); DPC Control PCB
8-4	Outlet pressure low	Tracking error. Leak in plasma hose to torch; defective control valve; defective pressure sensor (PS5); DPC Control PCB
8-5	Outlet pressure Too high	Valve at minimum setting but pressure still too high. Restriction in plasma hose, torch, DPC manifold; defective control valve; defective pressure sensor (PS5); DPC Control PCB
8-6	Outlet pressure high	Tracking error. Restriction in plasma hose, torch, DPC manifold; defective control valve; defective pressure sensor (PS5); DPC Control PCB
Group 9 - Shield H2O Proportional Valve (V2)		
9-1	Valve open	Check for open connection; Replace valve.
9-2	DPC inlet Supply pressure low	Raise plasma inlet pressure; check for DMC valve not open; restriction in gas supply/hoses.
9-3	Outlet flow too low	Valve full on but flow still too low. Restriction in shield hose, torch, DPC manifold; defective control valve; defective flow sensor (FS-1)); DPC Control PCB
9-4	Outlet flow low	Tracking error. Restriction in shield hose, torch, DPC manifold; defective control valve; defective flow sensor (FS-1)); DPC Control PCB
9-5	Outlet flow Too high	Valve set to minimum but flow still too high. Leak in shield hose to torch; defective control valve; DPC Control PCB
9-6	Outlet flow high	Tracking error. Leak in shield hose to torch; defective control valve; DPC Control PCB

SECTION 5: MAINTENANCE

5.01 General Maintenance

Perform the following checks periodically to ensure proper system performance.

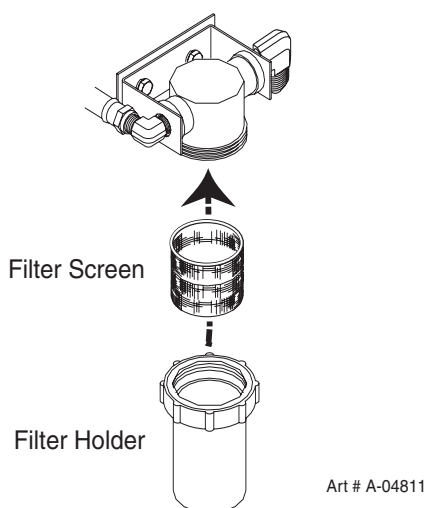
Power Supply Maintenance Schedule	
Daily	
Check coolant level; add coolant as needed.	
Check gas hose connections and pressures.	
Monthly	
Check cooling fan and radiator; clean as needed.	
Check gas hoses for cracks, leaks, or abrasion. Replace as needed.	
Check all electrical connections for cracks or abrasion. Replace as needed.	
Six Months	
Clean or replace external coolant filter.	
Clean coolant tank.	
Vacuum out any dust buildup inside power supply.	
Check internal coolant filter	

5.02 External Coolant Filter Cleaning Procedure

Periodic cleaning of the coolant filter ensures maximum coolant flow efficiency. Poor coolant flow causes inefficient torch parts cooling with consequent faster consumable wear.

Clean the coolant filter as follows:

1. Disconnect the system from main input power.



2. Unscrew and remove the filter bowl by hand. Be sure to keep the O-ring.
3. Remove and clean the filter. Re-install the bowl, tightening it **by hand**. Be sure the O-ring is in place.
4. Turn the system on and check for leaks.

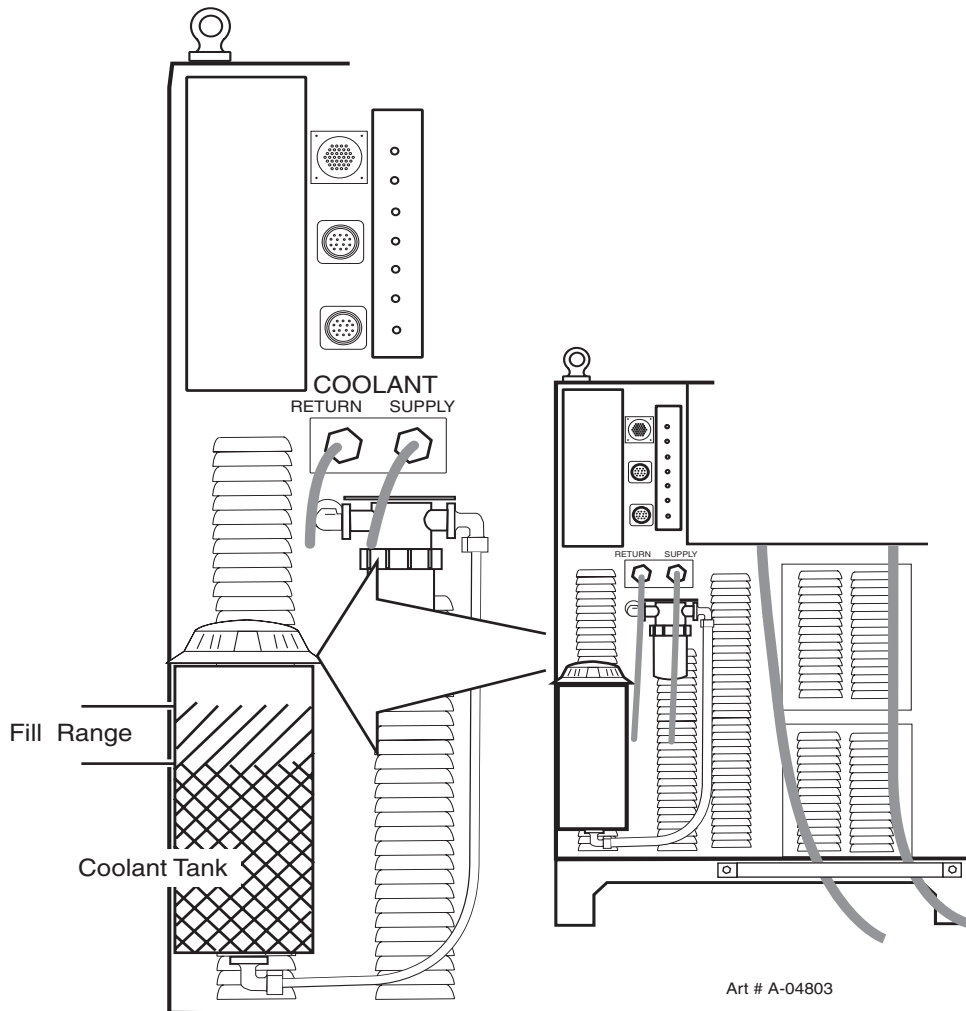
5.03 Internal Coolant Filter Cleaning Procedure

The in-line filter screen should be cleaned periodically. To gain access to the In-Line Filter Assembly remove the right side panel (viewed from the front of unit) of the Power Supply. Remove the filter screen by unscrewing the filter holder from the In-Line Filter Assembly. Clean the filter screen by rinsing with hot soapy water. Remove soap residue by rinsing with clean hot water. Be sure that all the soap has been removed and the screen is dry of water before re-installing in the In-Line Filter Assembly. Make sure the sealing O-ring is in place.

5.04 Coolant Replacement Procedure

Replace coolant as follows:

1. Disconnect the system from main input power. At the bottom of the coolant tank, disconnect the coolant line fitting and collect coolant in a disposable container.
2. Connect the system to main input power and turn it on to allow the pump to empty the tank. The pump will run approximately for 20 seconds before the flow switch causes an interlock. This procedure may be repeated a few times until the tank is completely empty.
3. Connect the coolant line fitting and fill the tank with fresh coolant until the right level is reached.

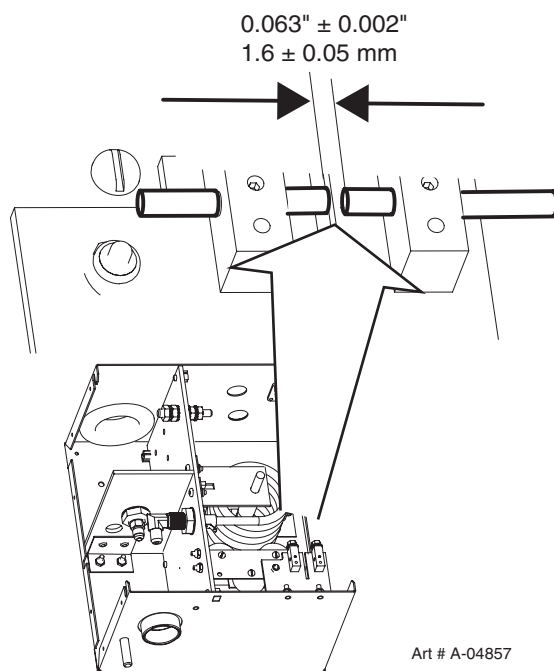


4. Turn system on, let it run for a few minutes and check coolant level, refill if necessary.

5.05 Remote Arc Starter: Service Chart

Arc Starter Service Chart			
Symptom	Cause	Check	Remedy
No Pilot ignition: Spark in Arc Starter but no ignition	Coolant has become conductive	Use conductivity meter	Flush system, replace coolant.
	Pilot return wire not connected	Visual inspection	Connect Wire.
	Spark gap set too close	Check with feeler gauge	Set to 0.063" ±0.002"
	High Frequency cap (C4) possibly open	Use capacitance meter	Reconnect or replace.
	Broken or missing ferrites	Visual inspection	Replace.
	Short across in inductor (L1)	Visual inspection	Remove short; increase coil gaps.
	Spark gap bus caps (C1, C2, C3) broken or defective	Capacitance meter	Replace.
	Negative supply not connected correctly	Visual inspection	Reconnect.
No Pilot ignition: No Spark in Arc Starter	Spark gap set too large	Check with feeler gauge	Set to 0.063" ±0.002"
	Faulty transformer	Resistance measurement	Replace.
	No 120V supply	Check input voltage at EMI filter	Make connections; replace harness.
	No/ loose connection to spark gap	Visual inspection	Reconnect.
	Faulty EMI filter	Voltage/ Resistor measurement	Replace.
No cooling or insufficient cooling: Leaks coolant	Loose fitting(s)	Visual inspection	Tighten fittings.
	Failure to braze joints (L1)	Visual inspection	Replace HF Coil.
	Damaged or punctured coolant line(s).	Visual inspection	Replace coolant line(s).
No cooling or insufficient cooling: No Coolant Flow	Supply & return hose reversed	Visual Inspection of color-coded connections	Match coolant connection colors to arc starter fitting colors.
	Blockage in coil or supply/return hoses	Loosen fitting slightly and check for coolant flow	Flush system.
Erratic System Behavior (EMI Interference)	Shield Drain / Torch lead inner shield connector missing or loose.	Visual inspection of lead attachment to Arc Starter	Reconnect / tighten lead connectors.
	Missing or loose ground connection	Visual inspection of ground wire to Arc Starter	Make or tighten connections to good ground.
	Cap C5 not connected, open or loose	Visual inspection / capacitor measurement	Replace PCB.

5.06 Remote Arc Starter: Spark Gap Adjustment



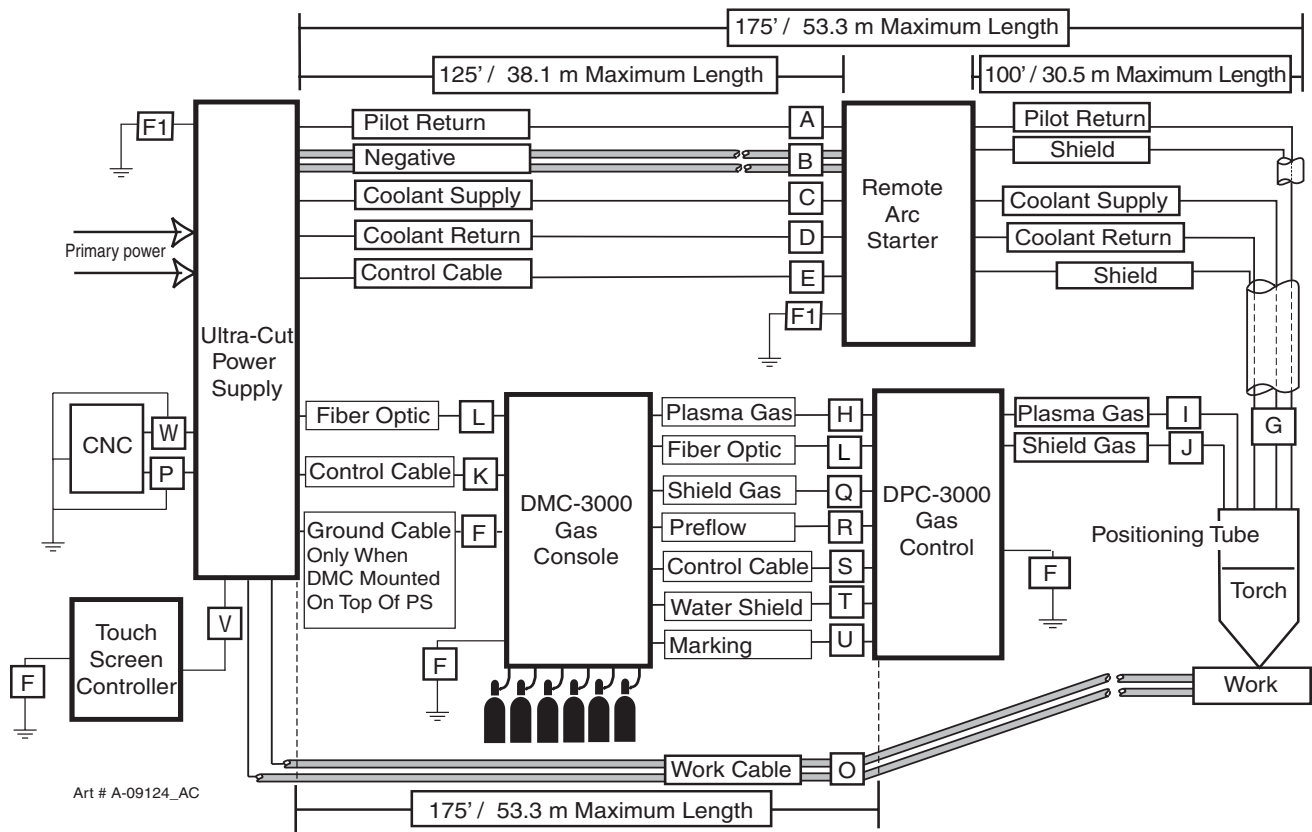
SECTION 6: REPLACEMENT ASSEMBLIES & PARTS

6.01 Main Component / System Replacement

Complete Unit / Component	Catalog Number
Ultra-Cut® 100 Power Supply, 208/230V	3-9116-1
Ultra-Cut® 100 Power Supply, 460V	3-9116-2
Ultra-Cut® 100 Power Supply, 400V	3-9116-3
Ultra-Cut® 100 Power Supply, 400V (CE)	3-9116-4
Ultra-Cut® 100 Power Supply, 600V	3-9116-5
Remote Arc Starter (RAS-1000)	3-9130
OPTIONAL EQUIPMENT:	
Wheel Kit	9-9379

6.02 System Layout

Refer to section 3.05 for ground connections and ground cables.



6.03 Recommended Gas Supply Hose

Item #	Qty	Description	Catalog #
	1	1/4" Gray Synflex Hose. No fittings included. Catalog number per foot 9-3743	
	1	3/8" Gray Synflex Hose. No fittings included. Catalog number per foot 9-3616	
	1	1/2" Gray Synflex Hose. No fittings included. Catalog number per foot 9-2551	

6.04 Leads and Cables

A		#8 AWG Cable	Pilot Return, Power Supply to Arc Starter	Art # A-09125, AC
B		#1 AWG Cable	Negative Lead, Power Supply to Arc Starter	
C		Green	Coolant Supply Lead, Power Supply to Arc Starter	
D		Red	Coolant Return Lead, Power Supply to Arc Starter	
E		14	Control Cable, Power Supply to Arc Starter	
F		Green / Yellow #4 AWG (16 mm ²)	Ground Cable	
F1		Green / Yellow 1/0 (50 mm ²)	Ground Cable, Remote Arc Starter To Earth Ground	
G			Shielded Torch Lead Assembly, Remote Arc Starter to Torch	
I			Plasma Gas Lead, Torch Valve to Torch	
J			Shield Gas Lead, Torch Valve to Torch	
K		37	Control Cable, From Power Supply	
L			Fiber Optic Cable, Power Supply to DMC-3000	
H, Q, R, T, U			For use with DFC-3000	
S, V		16 pin	S - Control Cable, DMC-3000 to DPC-3000 V - TSC-3000 to PS	
O		1/0 Cable (50 mm ²)	Work Cable	
P, W		14/11	P - CNC Cable (14 Wire) W - CNC Communication Cable (11 Wire)	

Key	Description	Catalog #
A,B,C,D,E	Supply Lead Set, 4' / 1.3 m (See Note 1)	4-3027
A,B,C,D,E	Supply Lead Set, 10' / 3.05 m (See Note 1)	4-3028
A,B,C,D,E	Supply Lead Set, 15' / 4.5 m (See Note 1)	4-3029
A,B,C,D,E	Supply Lead Set, 25' / 7.6 m (See Note 1)	4-3030
A,B,C,D,E	Supply Lead Set, 35' / 10.6 m (See Note 1)	4-3050
A,B,C,D,E	Supply Lead Set, 50' / 15.2 m (See Note 1)	4-3031
A,B,C,D,E	Supply Lead Set, 75' / 22.9 m (See Note 1)	4-3032
A,B,C,D,E	Supply Lead Set, 100' / 30.5 m (See Note 1)	4-3033
A,B,C,D,E	Supply Lead Set, 125' / 38.1 m (See Note 1)	4-3034
A	Pilot Return Cable (only), 4' / 1.2 m	9-4890
A	Pilot Return Cable (only), 10' / 3.05 m	9-4891
A	Pilot Return Cable (only), 15' / 4.5 m	9-4790
A	Pilot Return Cable (only), 25' / 7.6 m	9-4791
A	Pilot Return Cable (only), 35' / 10.6 m	9-9426
A	Pilot Return Cable (only), 50' / 15.2 m	9-4792
A	Pilot Return Cable (only), 75' / 22.8 m	9-4793
A	Pilot Return Cable (only), 100' / 30.5 m	9-4794
A	Pilot Return Cable (only), 125' / 38.1m	9-4796
B	Negative Cable (only), 3' / 1 m	9-4892
B	Negative Cable (only), 10' / 3.05 m	9-4897
B	Negative Cable (only), 15' / 4.5 m	9-4798
B	Negative Cable (only), 25' / 7.6 m	9-4799
B	Negative Cable (only), 35' / 10.6 m	9-9427
B	Negative Cable (only), 50' / 15.2 m	9-4800
B	Negative Cable (only), 75' / 22.8 m	9-4801
B	Negative Cable (only), 100' / 30.5 m	9-4802
B	Negative Cable (only), 125' / 38.1 m	9-4804
C	Hose, Coolant Supply, (only), 3' / 1 m	9-4886
C	Hose, Coolant Supply, (only), 10' / 3.05 m	9-4887
C	Hose, Coolant Supply, (only), 15' / 4.6 m	9-4780
C	Hose, Coolant Supply, (only), 20' / 6 m	9-4781
C	Hose, Coolant Supply, (only), 25' / 7.6 m	9-4782
C	Hose, Coolant Supply, (only), 30' / 9.1 m	9-4783
C	Hose, Coolant Supply, (only), 35' / 10.6 m	9-4784
C	Hose, Coolant Supply, (only), 40' / 12 m	9-4785

NOTE 1:

Supply lead sets include Pilot Return Cable, Negative Cable, Coolant Supply & Return Hoses, and Control Cable.

Key	Description	Catalog #
C	Hose, Coolant Supply, (only), 50' / 15.2 m	9-4786
C	Hose, Coolant Supply, (only), 75' / 23 m	9-4787
C	Hose, Coolant Supply, (only), 100' / 30.5 m	9-4788
D	Hose, Coolant Return, (only), 3' / 1 m	9-4888
D	Hose, Coolant Return, (only), 10' / 3.05 m	9-4889
D	Hose, Coolant Return, (only), 15' / 4.6 m	9-4762
D	Hose, Coolant Return, (only), 20' / 6 m	9-4763
D	Hose, Coolant Return, (only), 25' / 7.6 m	9-4764
D	Hose, Coolant Return, (only), 30' / 9.1 m	9-4765
D	Hose, Coolant Return, (only), 35' / 10.6 m	9-4766
D	Hose, Coolant Return, (only), 40' / 12 m	9-4767
D	Hose, Coolant Return, (only), 50' / 15.2 m	9-4768
D	Hose, Coolant Return, (only), 75' / 23 m	9-4769
D	Hose, Coolant Return, (only), 100' / 30.5 m	9-4770
E	Control Cable (only), Power Supply to Arc Starter 3' / 1 m	9-4941
E	Control Cable (only), Power Supply to Arc Starter 10' / 3.05 m	9-4916
E	Control Cable (only), Power Supply to Arc Starter 15' / 4.5 m	9-4917
E	Control Cable (only), Power Supply to Arc Starter 25' / 7.6 m	9-4918
E	Control Cable (only), Power Supply to Arc Starter 50' / 15.2 m	9-4942
E	Control Cable (only), Power Supply to Arc Starter 75' / 22.9 m	9-4943
E	Control Cable (only), Power Supply to Arc Starter 100' / 30.5 m	9-4944
E	Control Cable (only), Power Supply to Arc Starter 125' / 38.1 m	9-4922
F	Ground Wire, Power Supply to Gas Control Module, 3' / 1 m	9-4923
F	Ground Wire, Power Supply to Gas Control Module, 15' / 4.5 m	9-4924
F	Ground Wire, Power Supply to Gas Control Module, 25' / 7.6 m	9-4925
F	Ground Wire, Power Supply to Gas Control Module, 50' / 15.2 m	9-4926
F	Ground Wire, Power Supply to Gas Control Module, 75' / 22.9 m	9-4927
F	Ground Wire, Power Supply to Gas Control Module, 100' / 30.5 m	9-4928
F	Ground Wire, Power Supply to Gas Control Module, 125' / 38.1 m	9-4929
F	Ground Wire, Power Supply to Gas Control Module, 150' / 45.7 m	9-4930
F1	Ground Wire, Remote Arc Starter to Earth, 3' / 1 m	9-4931
F1	Ground Wire, Remote Arc Starter to Earth, 15' / 4.5 m	9-4932
F1	Ground Wire, Remote Arc Starter to Earth, 25' / 7.6 m	9-4933
F1	Ground Wire, Remote Arc Starter to Earth, 50' / 15.2 m	9-4934
F1	Ground Wire, Remote Arc Starter to Earth, 75' / 22.9 m	9-4935
F1	Ground Wire, Remote Arc Starter to Earth, 100' / 30.5 m	9-4936
F1	Ground Wire, Remote Arc Starter to Earth, 125' / 38.1 m	9-4937
F1	Ground Wire, Remote Arc Starter to Earth, 150' / 45.7 m	9-4938

ULTRA-CUT 100

Key	Description	Catalog #
G	Assembly, Torch Lead, 10' / 3.05 m	4-3053
G	Assembly, Torch Lead, 15' / 4.6 m	4-3009
G	Assembly, Torch Lead, 20' / 6 m	4-3010
G	Assembly, Torch Lead, 25' / 7.6 m	4-3011
G	Assembly, Torch Lead, 30' / 9.1 m	4-3012
G	Assembly, Torch Lead, 35' / 10.6 m	4-3013
G	Assembly, Torch Lead, 40' / 12 m	4-3014
G	Assembly, Torch Lead, 50' / 15.2 m	4-3015
G	Assembly, Torch Lead, 75' / 23 m	4-3016
G	Assembly, Torch Lead, 100' / 30.5 m	4-3017
H,L,Q,R,S,T,U	Auto Gas Lead Set, 10' / 3.05 m (See Note 2)	4-3086
H,L,Q,R,S,T,U	Auto Gas Lead Set, 15' / 4.5 m (See Note 2)	4-3087
H,L,Q,R,S,T,U	Auto Gas Lead Set, 25' / 7.6 m (See Note 2)	4-3088
H,L,Q,R,S,T,U	Auto Gas Lead Set, 35' / 10.6 m (See Note 2)	4-3089
H,L,Q,R,S,T,U	Auto Gas Lead Set, 50' / 15.2 m (See Note 2)	4-3090
H,L,Q,R,S,T,U	Auto Gas Lead Set, 75' / 22.9 m (See Note 2)	4-3091
H,L,Q,R,S,T,U	Auto Gas Lead Set, 100' / 30.5 m (See Note 2)	4-3092
H,L,Q,R,S,T,U	Auto Gas Lead Set, 125' / 38.1 m (See Note 2)	4-3093
H,L,Q,R,S,T,U	Auto Gas Lead Set, 150' / 45.7 m (See Note 2)	4-3094
H,L,Q,R,S,T,U	Auto Gas Lead Set, 175' / 53.3 m (See Note 2)	4-3095
H	Plasma Gas Hose 10' / 3.05 m	9-7103
H	Plasma Gas Hose 15' / 4.5 m	9-7104
H	Plasma Gas Hose 25' / 7.6 m	8-0149
H	Plasma Gas Hose 35' / 10.6 m	9-7270
H	Plasma Gas Hose 50' / 15.2 m	9-7271
H	Plasma Gas Hose 75' / 22.9 m	9-7272
H	Plasma Gas Hose 100' / 30.5 m	9-7273
H	Plasma Gas Hose 125' / 38.1 m	9-7274
H	Plasma Gas Hose 150' / 45.7 m	9-7275
H	Plasma Gas Hose 175' / 53.3 m	9-7276
I	Plasma Lead 4' / 1.22 m (Torch Valve Assembly to Torch)	9-3333
J	Shield Lead 4' / 1.22 m (Torch Valve Assembly to Torch)	9-3334

NOTE 2:

Auto Gas lead sets include Plasma Gas Hose, Shield Gas Hose, Preflow Gas Hose, Water Shield Hose, Fiber Optic Cable, Marking Hose, and Control Cable.

Key	Description	Catalog #
L	Control Cable, Fiber Optic, 3' / 1 m	9-4898
L	Control Cable, Fiber Optic, 15' / 4.5 m	9-4899
L	Control Cable, Fiber Optic, 25' / 7.6 m	9-4900
L	Control Cable, Fiber Optic, 35' / 10.6 m	9-9335
L	Control Cable, Fiber Optic, 50' / 15.2 m	9-4901
L	Control Cable, Fiber Optic, 75' / 22.9 m	9-4902
L	Control Cable, Fiber Optic, 100' / 30.5 m	9-4903
L	Control Cable, Fiber Optic, 125' / 38.1 m	9-4904
L	Control Cable, Fiber Optic, 150' / 45.7 m	9-4905
L	Control Cable, Fiber Optic, 175' / 53.3 m	9-4906
K	Control Cable, Power Supply to Gas Control Module, 3' / 1 m	9-4907
K	Control Cable, Power Supply to Gas Control Module, 15' / 4.5 m	9-4908
K	Control Cable, Power Supply to Gas Control Module, 25' / 7.6 m	9-4909
K	Control Cable, Power Supply to Gas Control Module, 35' / 10.6 m	9-9332
K	Control Cable, Power Supply to Gas Control Module, 50' / 15.2 m	9-4910
K	Control Cable, Power Supply to Gas Control Module, 75' / 22.9 m	9-4911
K	Control Cable, Power Supply to Gas Control Module, 100' / 30.5 m	9-4912
K	Control Cable, Power Supply to Gas Control Module, 125' / 38.1 m	9-4913
K	Control Cable, Power Supply to Gas Control Module, 150' / 45.7 m	9-4914
O	Work Cable, 10' / 3.05 m	9-9300
O	Work Cable, 15' / 4.5 m	9-9301
O	Work Cable, 25' / 7.6 m	9-9302
O	Work Cable, 35' / 10.6 m	9-9303
O	Work Cable, 50' / 15.2 m	9-9304
O	Work Cable, 75' / 22.9 m	9-9305
O	Work Cable, 100' / 30.5 m	9-9306
O	Work Cable, 125' / 38.1 m	9-9307
P	Control Cable, CNC to Power Supply, 25' / 7.6 m	9-8312
P	Control Cable, CNC to Power Supply, 50' / 15.2 m	9-8313
P	Control Cable, CNC to Power Supply, 75' / 22.9 m	9-8315
P	Control Cable, CNC to Power Supply, 100' / 30.5 m	9-8316
P	Control Cable, CNC to Power Supply, 125' / 38.1 m	9-8317

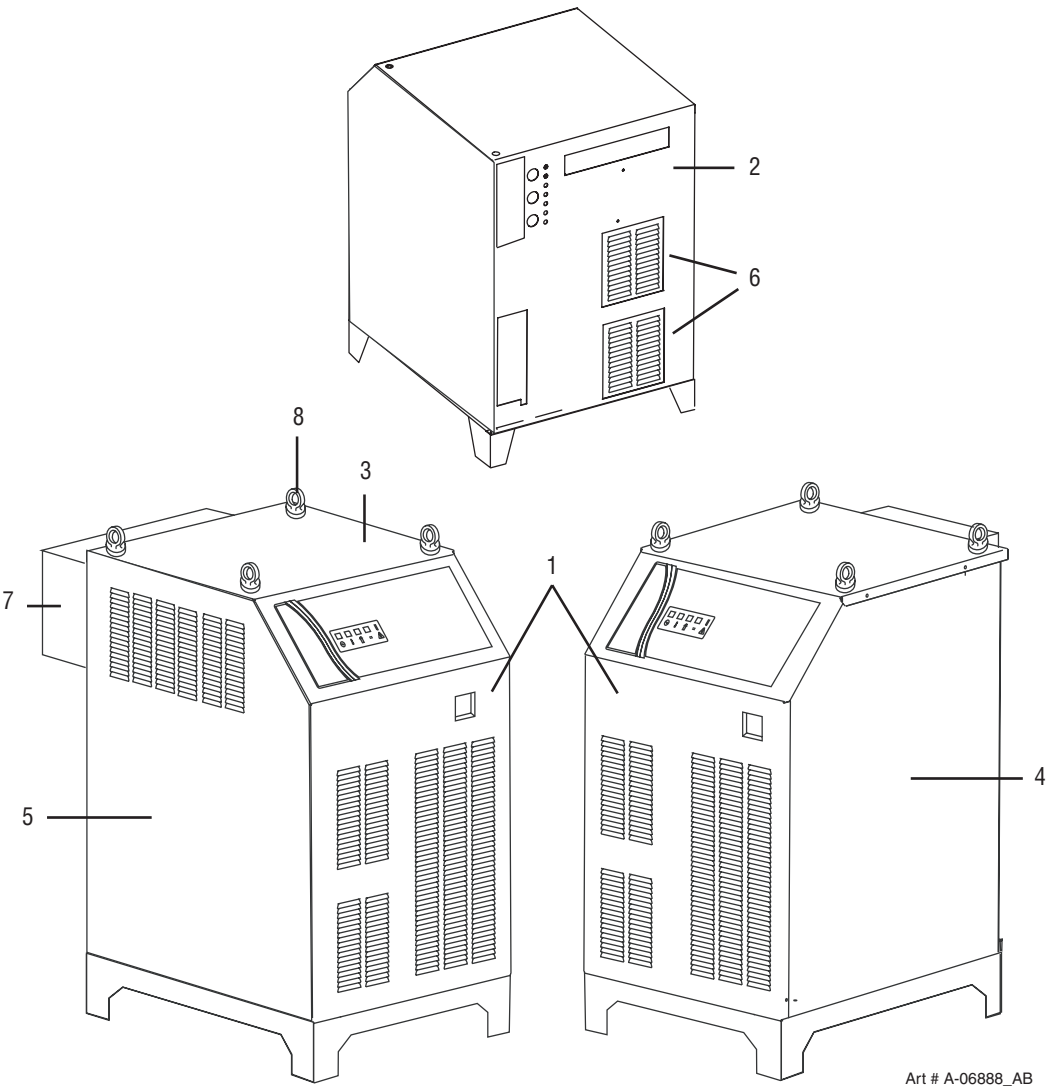
ULTRA-CUT 100

Key	Description	Catalog #
Q	Shield Gas Hose 10' / 3.05 m	9-7277
Q	Shield Gas Hose 15' / 4.5 m	9-7278
Q	Shield Gas Hose 25' / 7.6 m	9-7279
Q	Shield Gas Hose 35' / 10.6 m	9-7280
Q	Shield Gas Hose 50' / 15.2 m	9-7281
Q	Shield Gas Hose 75' / 22.9 m	9-7282
Q	Shield Gas Hose 100' / 30.5 m	9-7283
Q	Shield Gas Hose 125' / 38.1 m	9-7284
Q	Shield Gas Hose 150' / 45.7 m	9-7285
Q	Shield Gas Hose 175' / 53.3 m	9-7286
R	Pilot Gas/Pre-Flow Hose 10' / 3.05 m	9-6956
R	Pilot Gas/Pre-FlowHose 15' / 4.5 m	9-6957
R	Pilot Gas/Pre-Flow Hose 25' / 7.6 m	9-6959
R	Pilot Gas/Pre-FlowHose 35' / 10.6 m	9-6961
R	Pilot Gas/Pre-Flow Hose 50' / 15.2 m	9-6963
R	Pilot Gas/Pre-Flow Hose 75' / 22.9 m	9-6964
R	Pilot Gas/Pre-Flow Hose 100' / 30.5 m	9-7080
R	Pilot Gas/Pre-Flow Hose 125' / 38.1 m	9-7081
R	Pilot Gas/Pre-Flow Hose 150' / 45.7 m	9-4876
R	Pilot Gas/Pre-Flow Hose 175' / 53.3 m	9-4877
S	Control Cable 10' / 3.05 m	9-7093
S	Control Cable 15' / 4.5 m	9-7094
S	Control Cable 25' / 7.6 m	9-7095
S	Control Cable 35' / 10.6 m	9-7096
S	Control Cable 50' / 15.2 m	9-7097
S	Control Cable 75' / 22.9 m	9-7098
S	Control Cable 100' / 30.5 m	9-7099
S	Control Cable 125' / 38.1 m	9-7100
S	Control Cable 150' / 45.7 m	9-7101
S	Control Cable 175' / 53.3 m	9-7102

Key	Description	Catalog #
T	Water Shield 10' / 3.05 m	9-6985
T	Water Shield 15' / 4.5 m	9-6986
T	Water Shield 25' / 7.6 m	9-6988
T	Water Shield 35' / 10.6 m	9-6996
T	Water Shield 50' / 15.2 m	9-6992
T	Water Shield 75' / 22.9 m	9-6993
T	Water Shield 100' / 30.5 m	9-7073
T	Water Shield 125' / 38.1 m	9-7074
T	Water Shield 150' / 45.7 m	9-7075
T	Water Shield 175' / 53.3 m	9-7076
U	Marking Gas Hose 10' / 3.05 m	9-4725
U	Marking Gas Hose 15' / 4.5 m	9-4726
U	Marking Gas Hose 25' / 7.6 m	9-4728
U	Marking Gas Hose 35' / 10.6 m	9-4730
U	Marking Gas Hose 50' / 15.2 m	9-4730
U	Marking Gas Hose 75' / 22.9 m	9-4732
U	Marking Gas Hose 100' / 30.5 m	9-4733
U	Marking Gas Hose 125' / 38.1 m	9-7077
U	Marking Gas Hose 150' / 45.7 m	9-7078
U	Marking Gas Hose 175' / 53.3 m	9-7079
V	Cable, Remote TSC to CCM 10' / 3.05 m	9-9230
V	Cable, Remote TSC to CCM 15' / 4.5 m	9-9231
V	Cable, Remote TSC to CCM 25' / 7.6 m	9-9232
V	Cable, Remote TSC to CCM 35' / 10.6 m	9-9233
V	Cable, Remote TSC to CCM 50' / 15.2 m	9-9234
V	Cable, Remote TSC to CCM 75' / 22.9 m	9-9235
V	Cable, Remote TSC to CCM 100' / 30.5 m	9-9236
V	Cable, Remote TSC to CCM 125' / 38.1 m	9-9237
V	Cable, Remote TSC to CCM 150' / 45.7 m	9-9238
V	Cable, Remote TSC to CCM 175' / 53.3 m	9-9239
W	Communication Control Cable, CNC 25' / 7.6 m	9-8674
W	Communication Control Cable, CNC 50' / 15.2 m	9-8675
W	Communication Control Cable, CNC 75' / 22.9 m	9-8676
W	Communication Control Cable, CNC 100' / 30.5 m	9-8677
W	Communication Control Cable, CNC 125' / 38.1 m	9-8678
W	Communication Control Cable, CNC 150' / 45.7 m	9-8679
W	Communication Control Cable, CNC 175' / 53.3 m	9-8680

6.05 Power Supply External Replacement Parts

Item #	Qty	Description	Catalog #
1	1	Power Supply Front Panel	N/A
2	1	Power Supply Rear Panel	N/A
3	1	Power Supply Top Panel	9-0445B/9-0445G
4	1	Power Supply Right Side	9-0442B/9-0442G
5	1	Power Supply Left Side	9-0443B/9-0443G
6	1	Louver Panel	N/A
7	1	Power Supply Connections Cover	N/A
8	1	Lifting Eye	9-9373

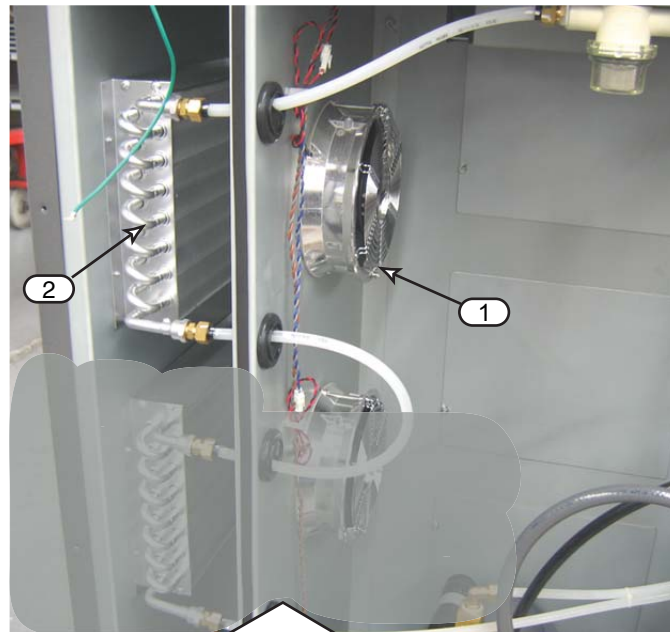


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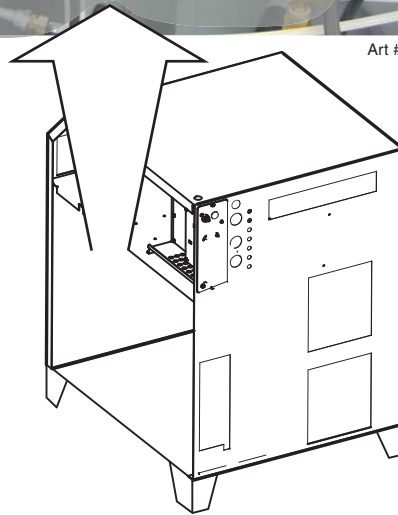
6.06 Power Supply Replacement Parts - Right Side

Item #	Qty	Description	Ref. Des.	Catalog #
1	1	Heat Exchanger Fan	FAN1	9-9338
2	1	Radiator		9-9339

NOTE: Heat Exchanger Fan is the same fan used with the Inverter

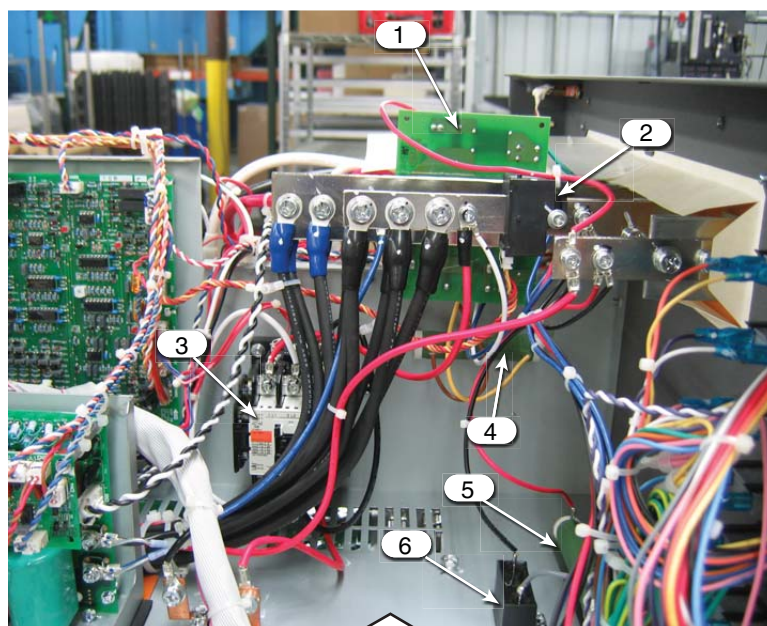


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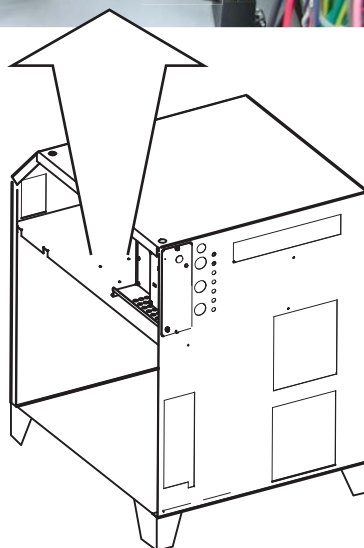


6.07 Power Supply Replacement Parts - Right Side

Item #	Qty	Description	Ref. Des.	Catalog #
1	1	Output Filter PC	PCB8	9-9341
2	1	Sensor	HCT2	9-9368
3	1	Magnetic Contactor, pilot	MC2	9-9343
4	1	Resistor	R6	9-9340
5	1	Resistor	R9	
6	1	Capacitor	C2	

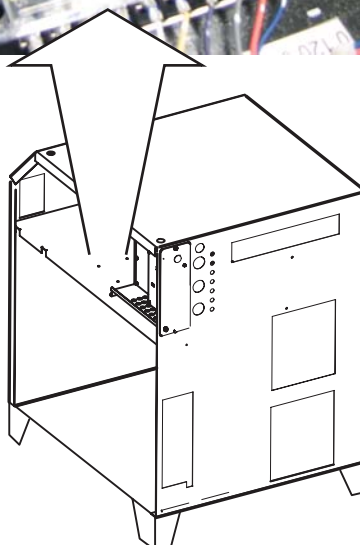
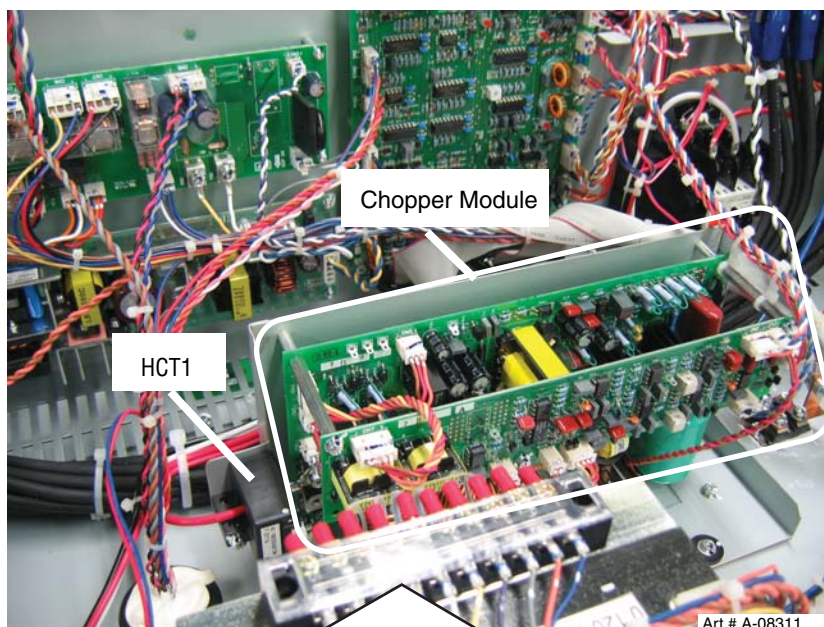


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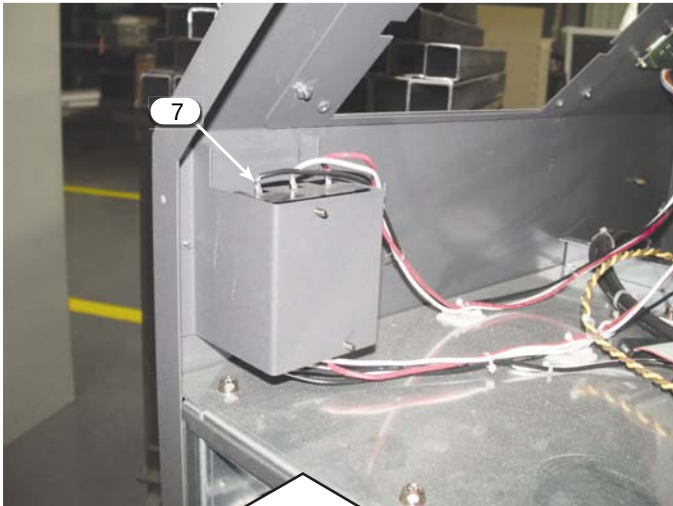
6.08 Power Supply Replacement Parts - Right Side

Item #	Qty	Description	Ref. Des.	Catalog #
	1	Chopper Module		9-9389
	1	Sensor	HCT1	9-9342

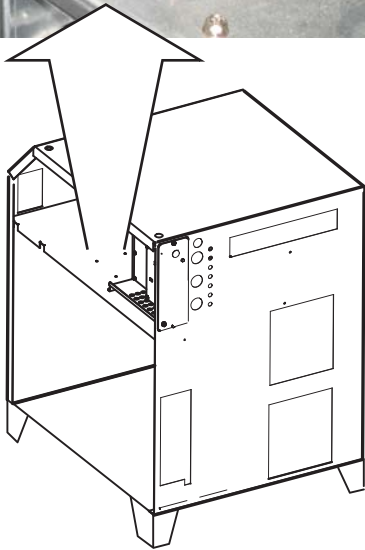


6.09 Power Supply Replacement Parts - Front Panel

Item #	Qty	Description	Ref. Des.	Catalog #
7	1	Main Power Switch ('Circuit Protector') for 230-400-460 VAC units	CP1	9-9344

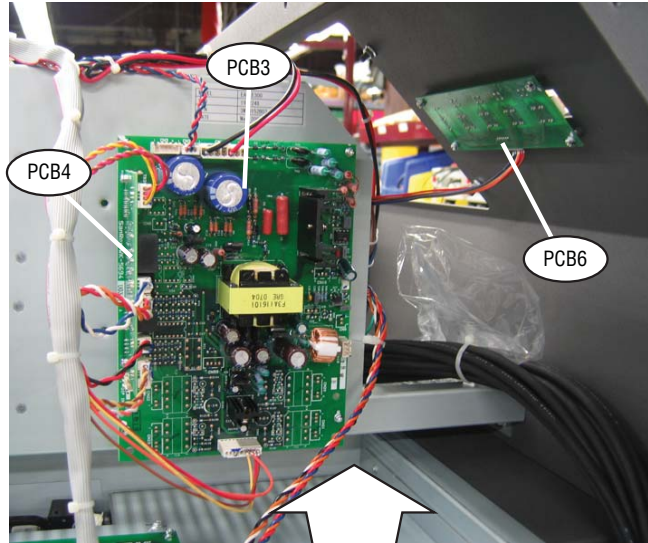


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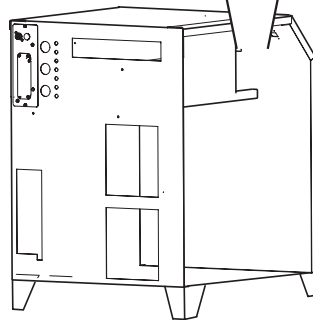


6.10 Power Supply Replacement Parts - Front Panel

Item #	Qty	Description	Ref. Des.	Catalog #
	1	PC Board	PCB3	9-9365
	1	PC Board 230/460VAC	PCB4	9-9370
	1	PC Board 600VAC	PCB4	9-9492
	1	Display PC Board	PCB6	9-9347

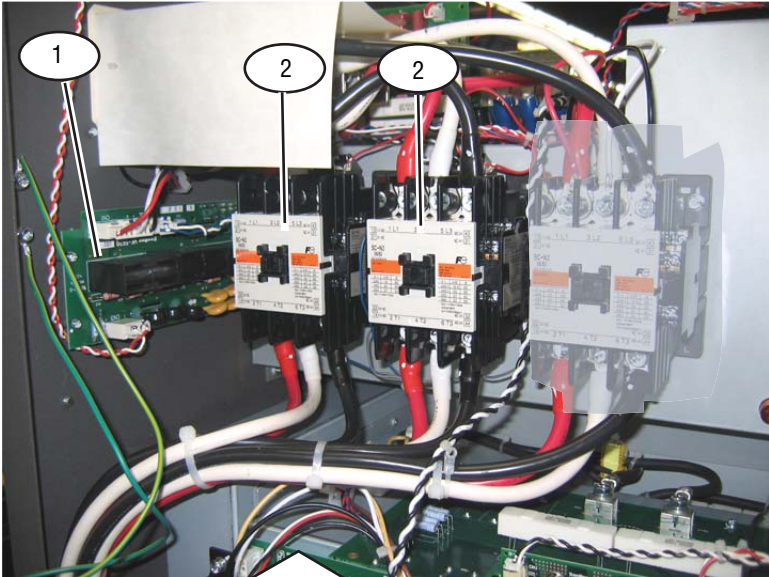


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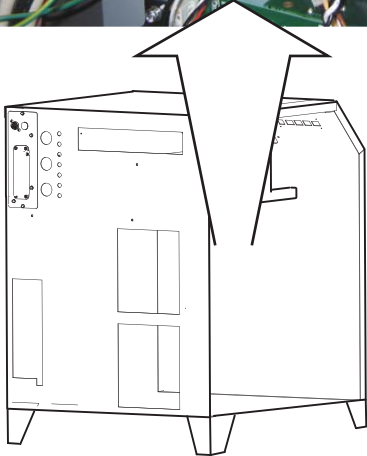


6.11 Power Supply Replacement Parts - Right Side

Item #	Qty	Description	Ref. Des.	Catalog #
1	1	PCBoard 230-400-460VAC	PCB1	9-9362
1	1	PCBoard 600VAC	PCB1	9-9491
2	1	Magnetic Contactor	MC1	9-9364
		ONLY 1 for this unit		

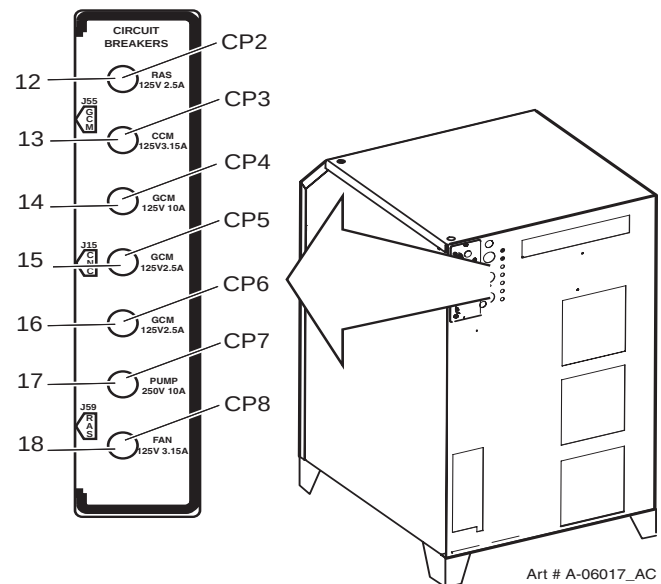
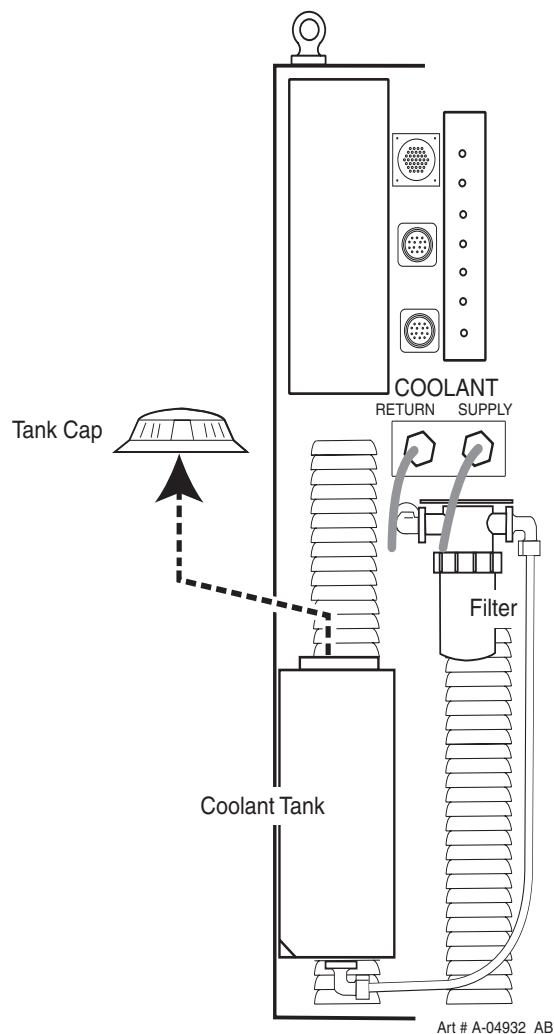


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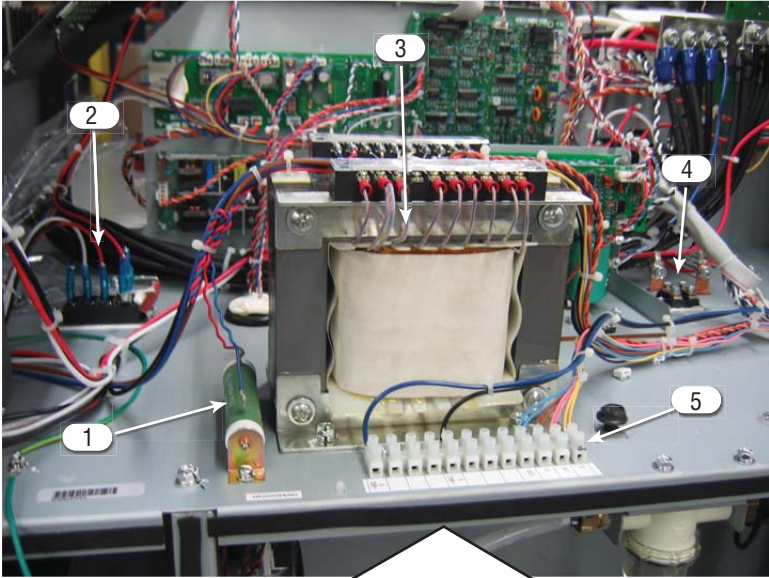
6.12 Power Supply Replacement Parts - Rear Panel

Item #	Qty	Description	Breaker Rating	Circuit Rating	Ref. Des. Catalog #
12	1	Circuit Breaker (RAS)	125V, 2.5 A	120VAC @ 1A	CP2 9-9348
13	1	Circuit Breaker ('TDC')	125V, 3.15A	24VAC @ 2A	CP3 9-9349
14	1	Circuit Breaker (AGC)	125V, 10A	24VAC @ 5A	CP4 9-9350
15	1	Circuit Breaker (AGC)	125V, 2.5A	120VAC @ 1A	CP5 9-9348
16	1	Circuit Breaker (AGC)	125V, 2.5A	15VDC @ 1A	CP6 9-9348
17	1	Circuit Breaker (Pump)	250V, 10A	200VAC @ 3A	CP7 9-9351
18	1	Circuit Breaker (Fan)	125V, 3.15A	24VDC @ 1A	CP8 9-9349
	1	Coolant Tank			9-5948
	1	Tank Cap			8-5142
	1	Filter, Coolant			8-4276

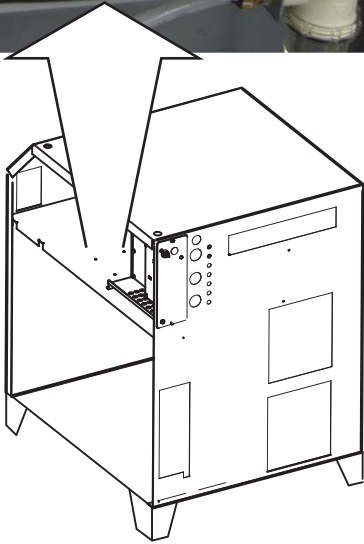


6.13 Power Supply Replacement Parts - Right Side

Item #	Qty	Description	Ref. Des.	Catalog #
1	1	Resistor	R7	9-9363
2	1	Rectifier (Diode)	D1	9-9345
3	1	Transformer for 230/460 VAC units	T1	9-9355
4	1	Diode	D2	9-9486
5	1	Terminal Strip	TB3	N/A



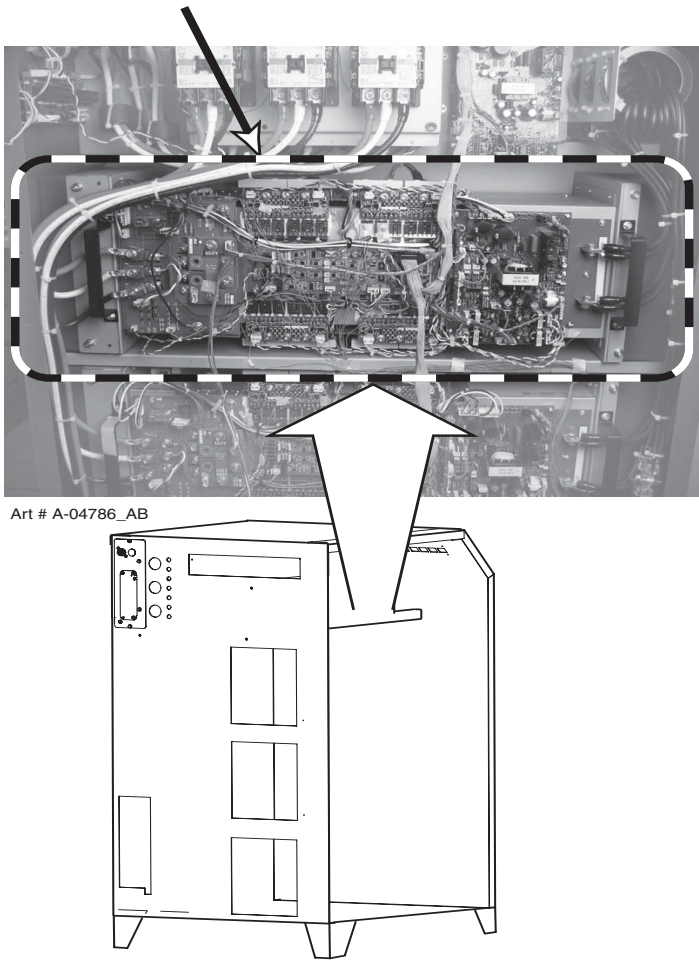
Art # A-08310



6.14 Power Supply Replacement Parts

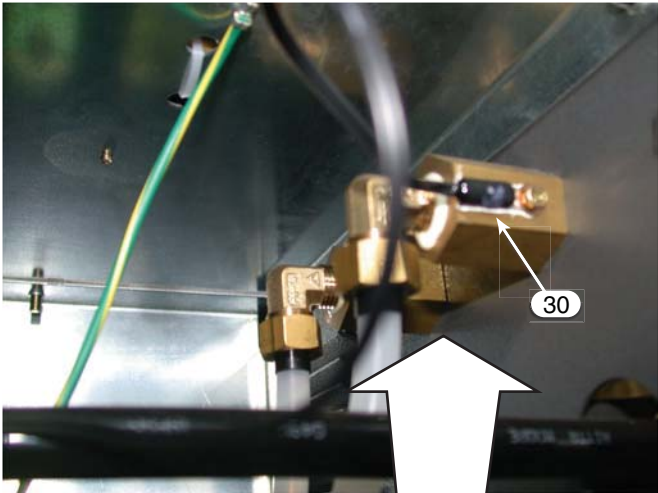
Item #	Qty	Description	Catalog #
29	2	Inverter Module 208/230V	9-9360D
	2	Inverter Module 400V CE and 400V Non CE	9-9482D
	2	Inverter Module 600V	9-9485D
Not Shown:			
	2	Inverter Cooling Fan	9-9338

NOTE: The Inverter Cooling Fan is the same fan used for Heat Exchanger cooling.

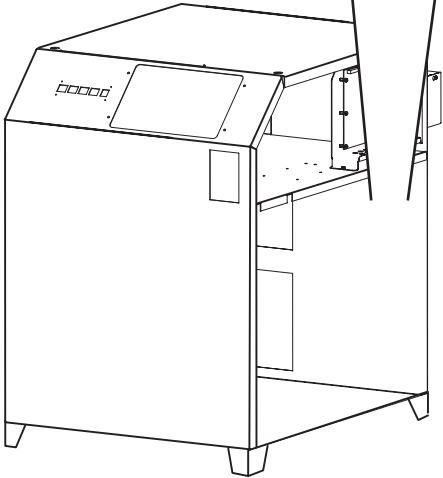


6.15 Power Supply Replacement Parts - Right Side

Item #	Qty	Description	Ref. Des.	Catalog #
30	1	Sensor (Thermistor)	TH1	9-9361

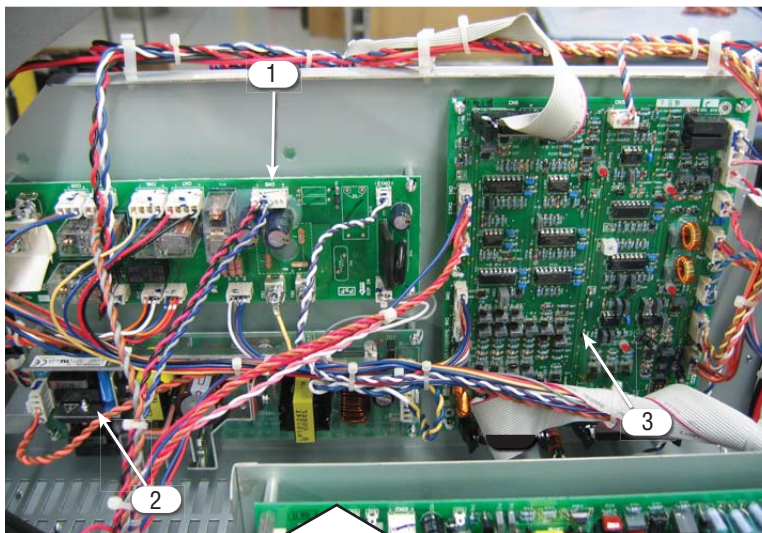


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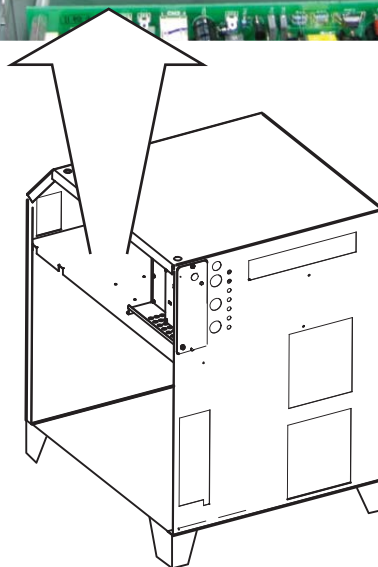


6.16 Power Supply Replacement Parts

Item #	Qty	Description	Ref. Des.	Catalog #
1	1	Relay PC Board	PCB7	9-9366
2	1	24 VDC Power Supply	N/A	N/A
3	1	Internal Control PC Board	PCB5	9-9346



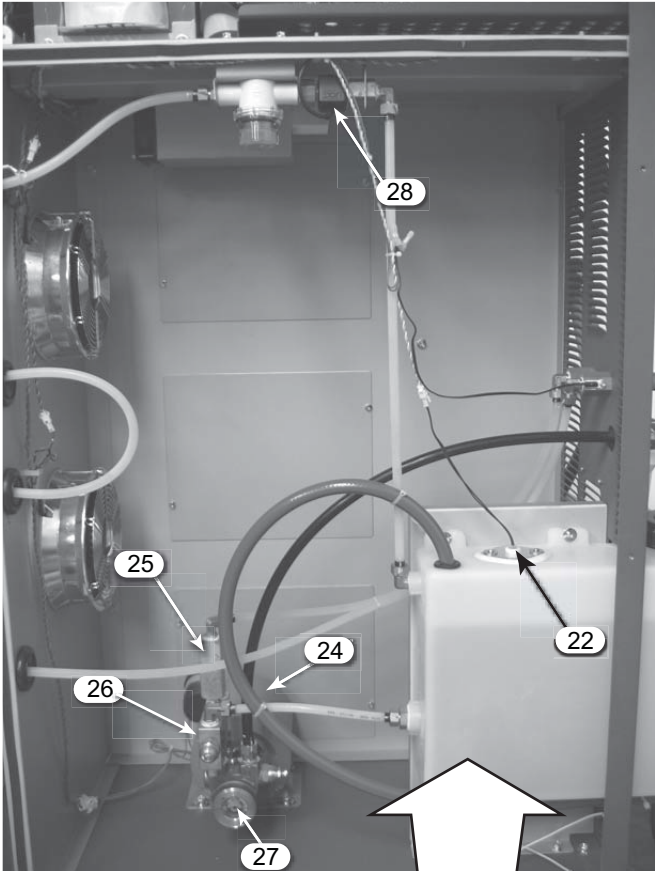
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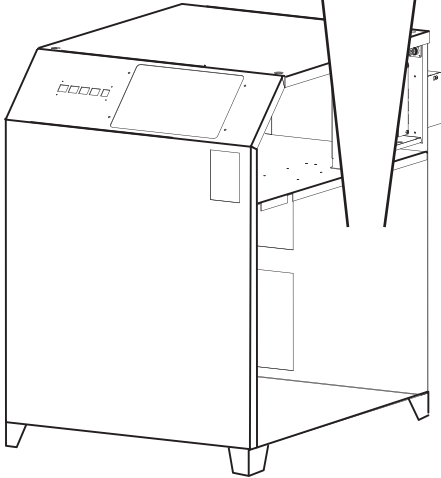
6.17 Power Supply Replacement Parts - Right Side

Item #	Qty	Description	Ref. Des.	Catalog #
22	1	Sensor (Level Switch)	LSW1	9-9354
24	1	Motor	MOT1	9-9381
25	1	Pump/Check Valve/Bypass Valve Assembly		9-9380
26	1	Bypass Valve (part of assembly)		
27	1	Pump (part of assembly)		
28	1	Sensor (Flow Indicator)	F1	9-9359

Illustration is for Reference ONLY



Art # A-08236



6.18 DFC-3000 Automated Gas Control System Replacement Components

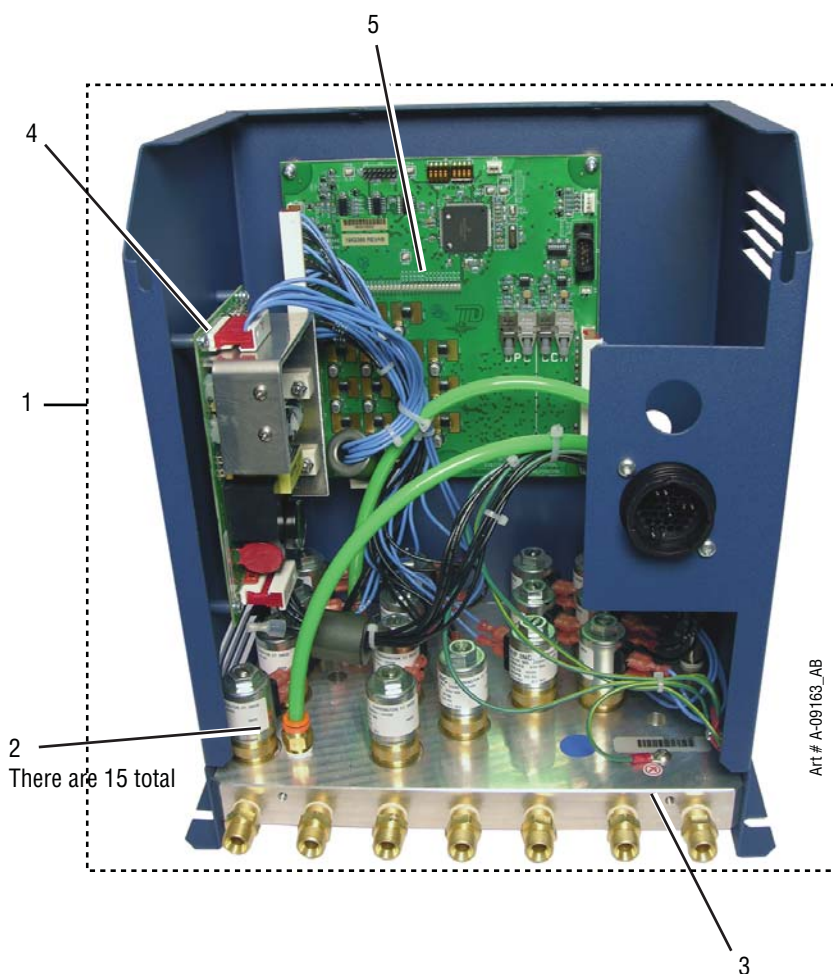
Item #	Qty	Description	Catalog #
1	1	Assy, DMC-3000, Demo	9-9491D
2	1	Assy, DPC-3000, Demo	9-9443D
3	1	Assy, TSC-3000, Fixed Mounting, Demo (Not Shown)	9-9444D
4	1	Assy, TSC-3000, Remote, Demo	9-9490D



Art # A-09135_AB

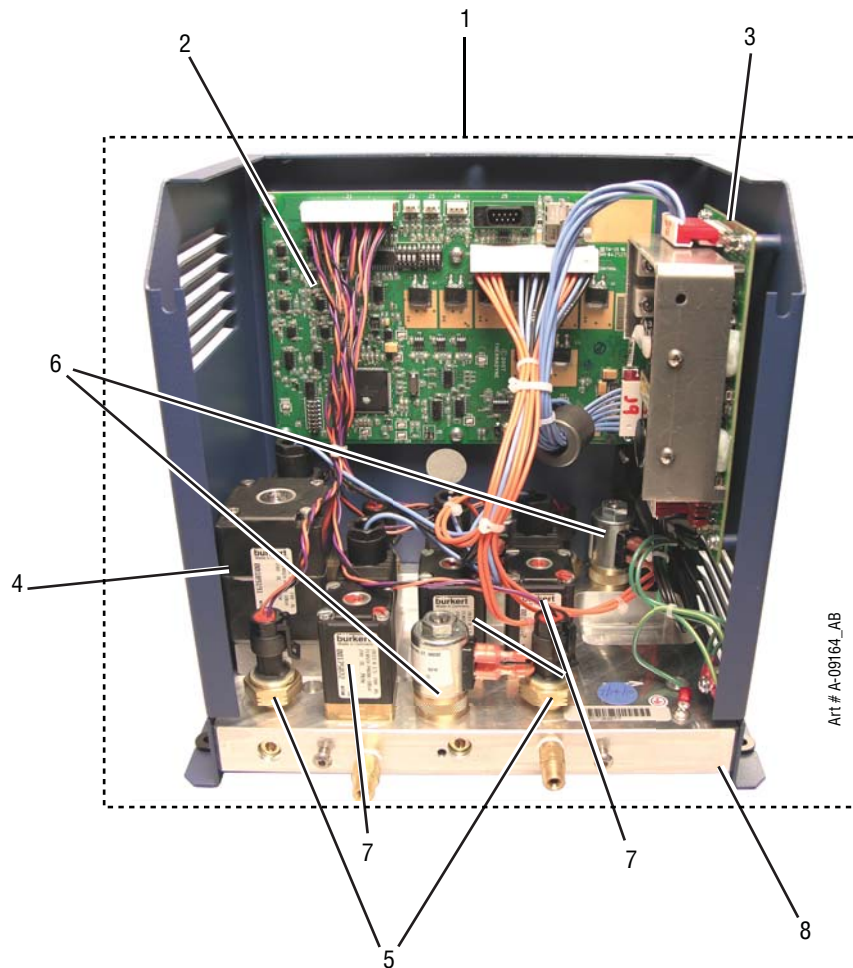
6.19 DMC-3000 Gas Control Module Replacement Parts

Item #	Qty	Description	Catalog #
1	1	DMC-3000 assembly	9-9491D
2	1	Solenoid	9-8264
3	1	Manifold subassembly (includes 15 or item 2)	9-7546
4	1	SMPS PCB	9-8263
5	1	Control PCB	9-7291



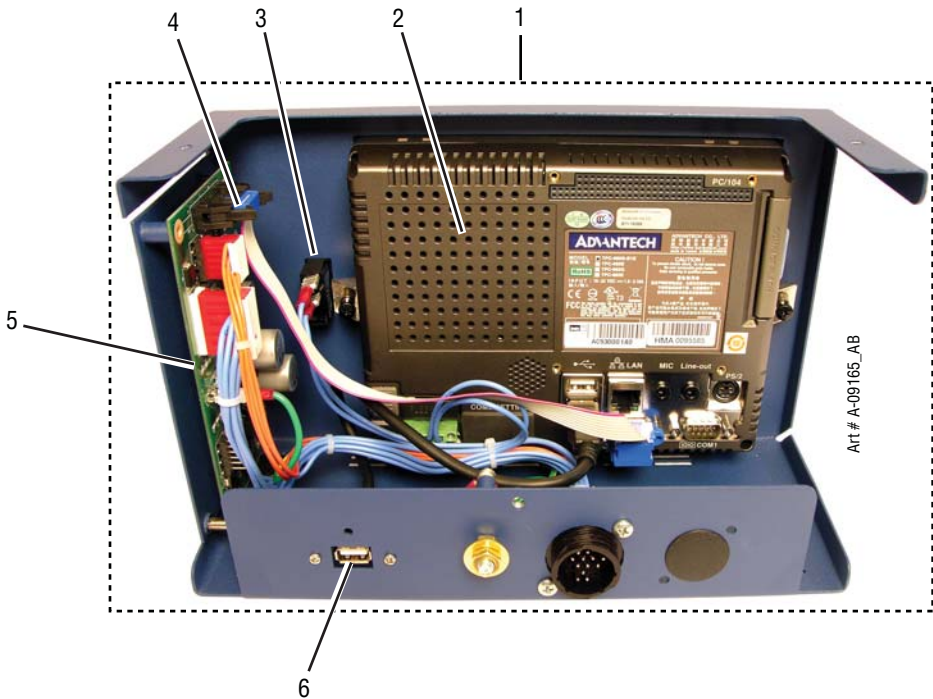
6.20 DPC-3000 Gas Control Module Replacement Parts

Item #	Qty	Description	Catalog #
1	1	Assy, DPC-3000, Demo	9-9443D
2	1	DPC-PCB	9-8262
3	1	SMPS PCB	9-8263
4	1	Shield Gas P Valve	9-8267
5	1	Sensor, Pressure, 1/8mnpt (six total)	9-8269
6	1	Solenoid	9-8264
7	1	Plasma Lo and Pre-Flow P Valve	9-8268
8	1	DPC Manifold Fully Assembled	9-7658



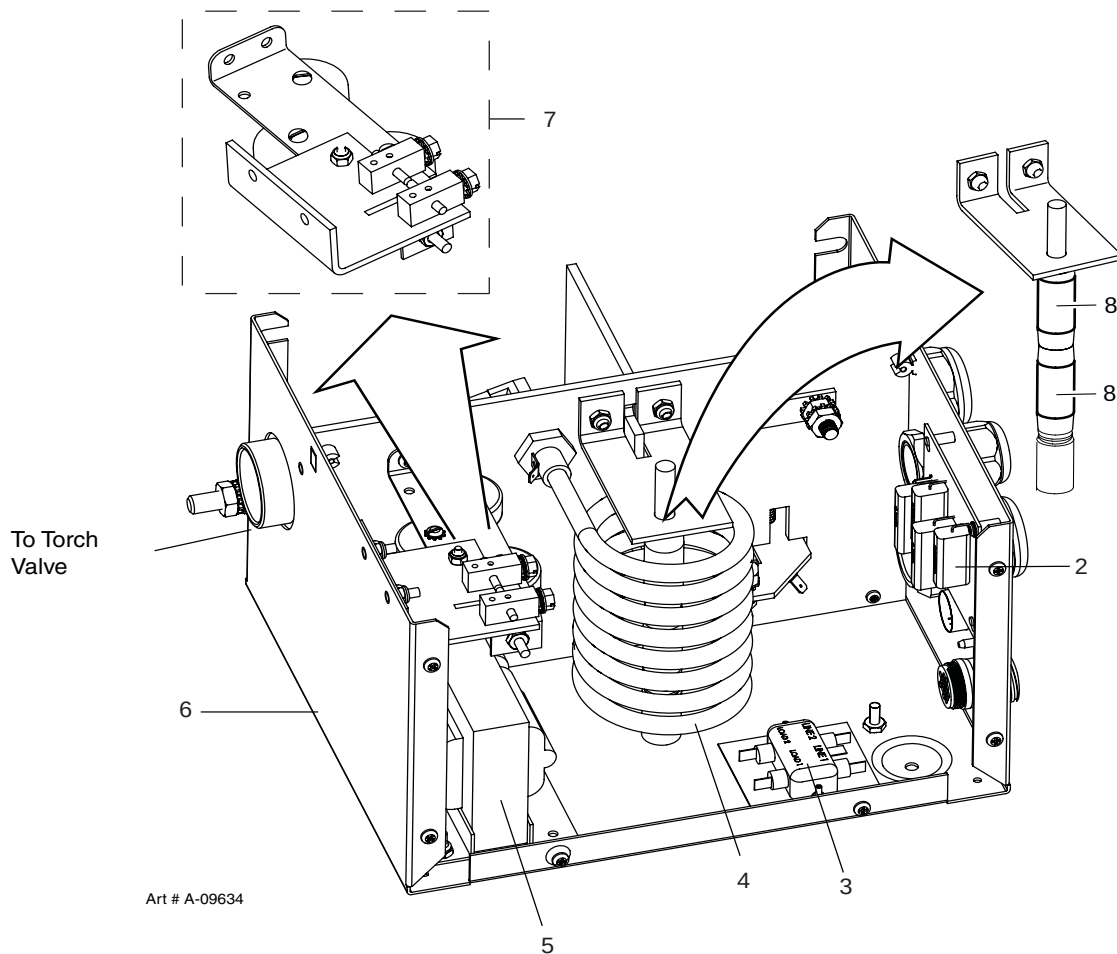
6.21 TSC-3000 Touch Screen Control Remote and Internal Replacement Parts

Item #	Qty	Description	Catalog #
1	1	Assy, TSC-3000, Remote , Demo	9-9490D
1	1	Assy, TSC-3000, Internal , Demo	9-944
2	1	Touch Panel Computer	9-7543
3	1	Switch, Rocker SPST	9-1042
4	1	Ribbon Cable HMI to Interface	N/A
5	1	PCB	9-7547
6	1	USB Panel Mount Extension	9-7545



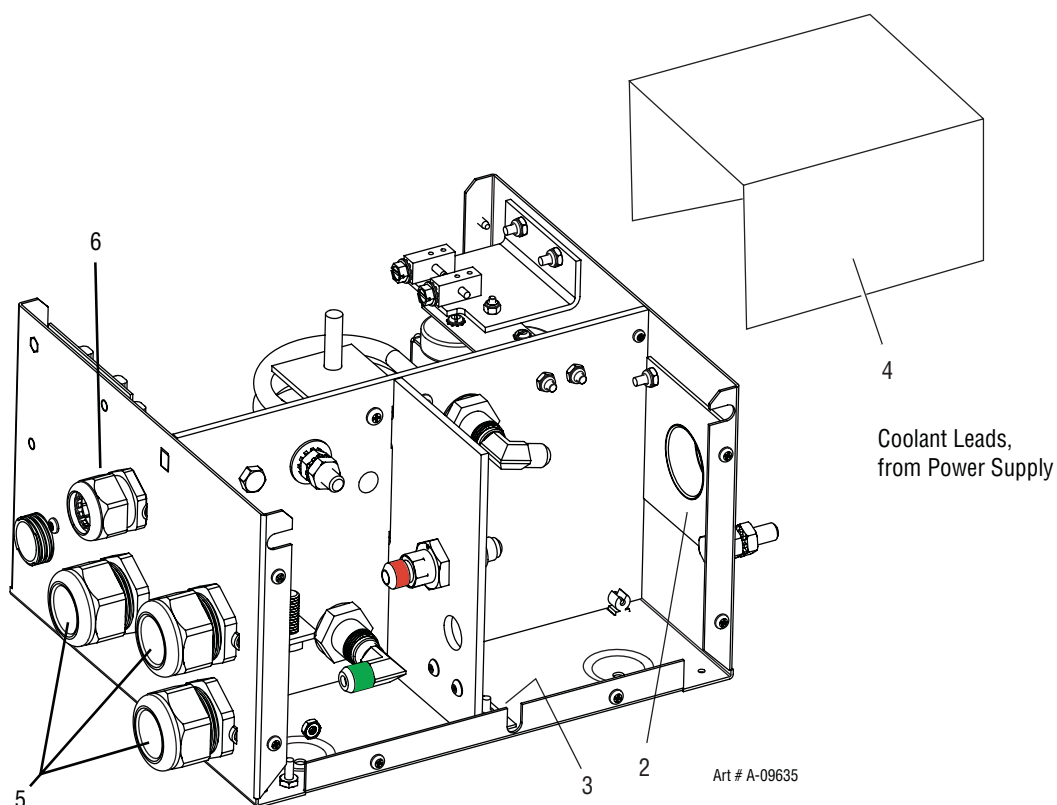
6.22 Remote Arc Starter (RAS-1000) Replacement Parts

Item #	Qty	Description	Catalog #
1	1	Harness, Internal (Not Shown)	9-4956
2	1	Pilot Cap Assembly	9-9423
3	1	EMI Filter	9-1023
4	1	HF Water Cooled Coil Assembly	9-4958
5	1	HF Transformer Assembly, Arc Starter	9-4959
6	1	Enclosure, Arc Starter Base	9-4961
7	1	Spark Gap/Cap Assembly HF	9-4957
8	1	Ferrite Bead 2" Long	9-4965



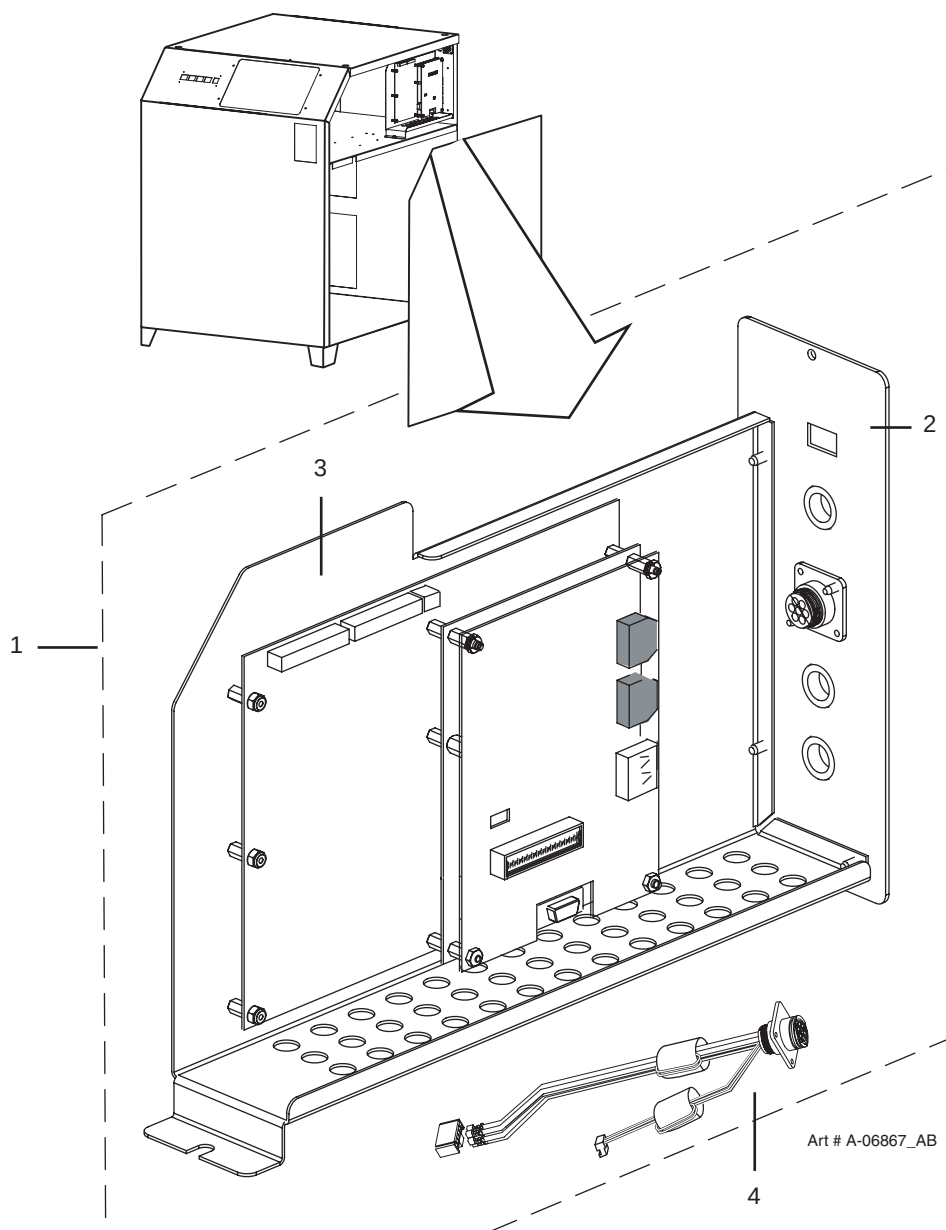
6.23 Remote Arc Starter (RAS-1000) Replacement Parts

Item #	Qty	Description	Catalog #
1	1	Retainer, ABS Wall	9-4964
2	1	(Outer) Torch Lead Ground Assembly	9-4955
3	1	Bracket, Bulkhead Holder	9-4963
4	1	Cover, Arc Starter	9-4962
5	1	Strain Relief, large	9-7571
6	1	Strain Relief, small	8-7072



6.24 Command & Control Module For Auto Gas Control - Replacement Parts

Item #	Qty	Description	Catalog #
1	1	Assembly, CCM, Auto Gas, Demo	9-9417D
2	1	Plate, CCM Face	9-9419
3	1	Panel, CCM Mount	Not Available
4	1	Harness, CCM I/O Board	9-9421

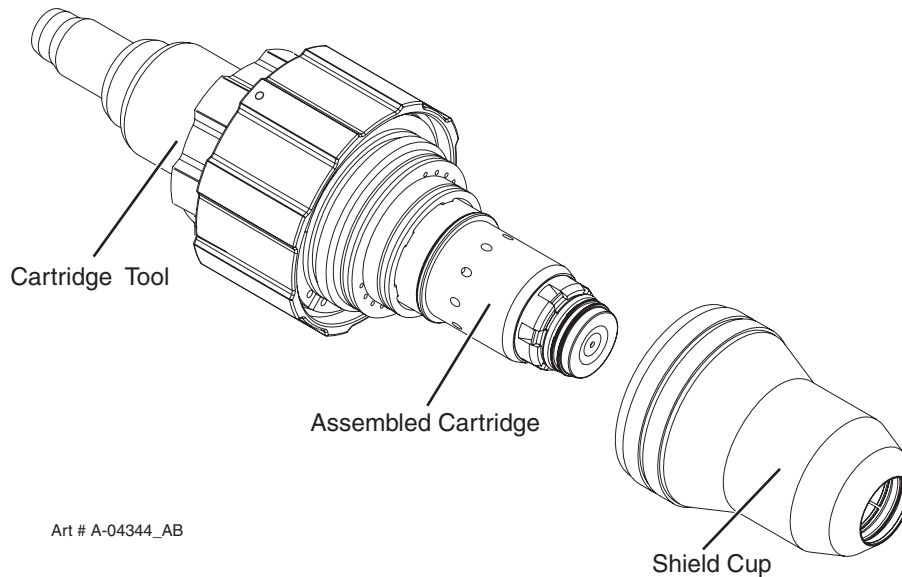


Intentionally Blank

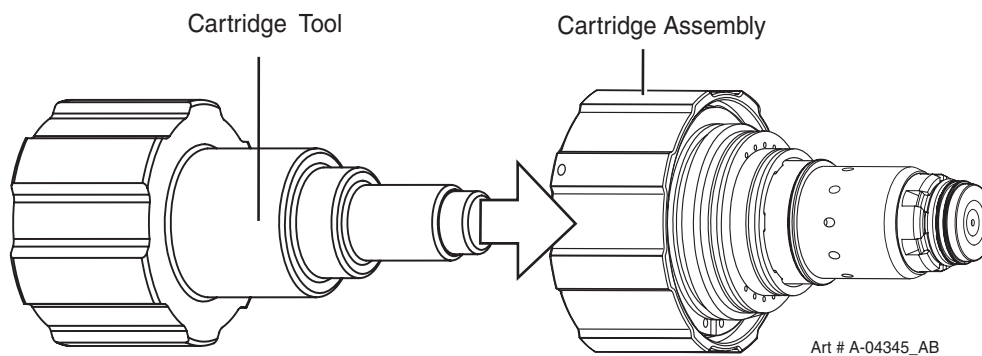
SECTION 7. TORCH MAINTENANCE

7.01 Consumable Removal

1. Use the Cartridge Tool to hold the Shield Cup & Cartridge Assembly stationary. Turn the Shield Cup to remove it from the Cartridge Assembly.

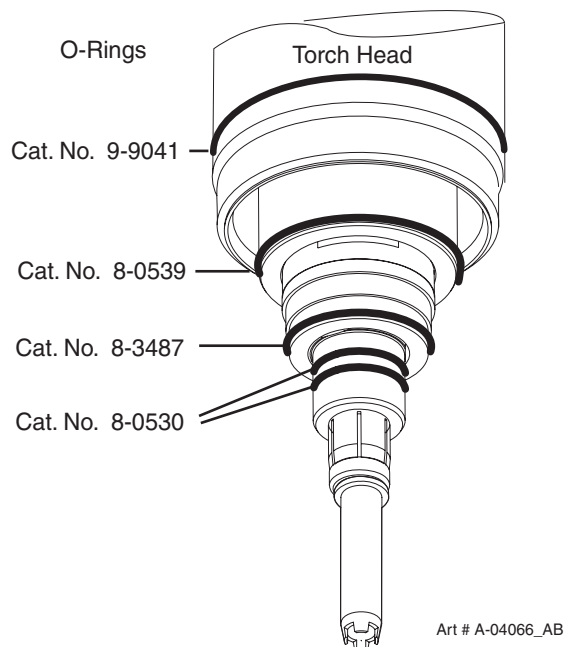
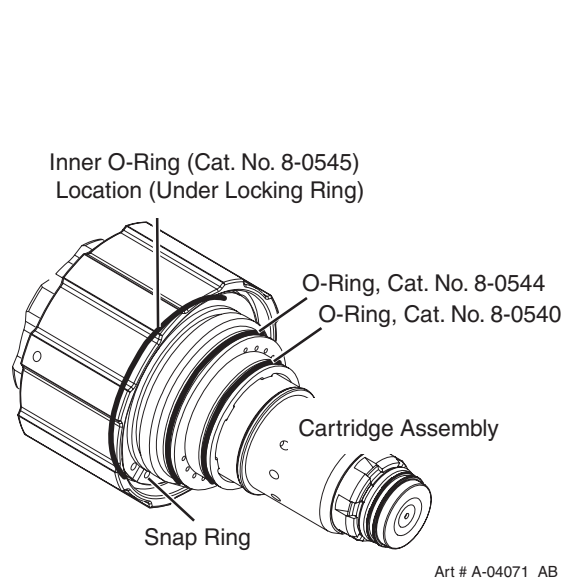


2. Take the Cartridge Tool off the back of the Cartridge Assembly. Use the Cartridge Tool to push the consumable parts out of the Cartridge.



7.02 O-Ring Lubrication

Lubricate all three O-Rings on the Cartridge Assembly and all three O-Rings on the Torch Head periodically with O-Ring Lubricant supplied. Remove the snap ring on the cartridge assembly and slide the locking ring downward for access to the O-Ring under the locking ring.



CAUTION

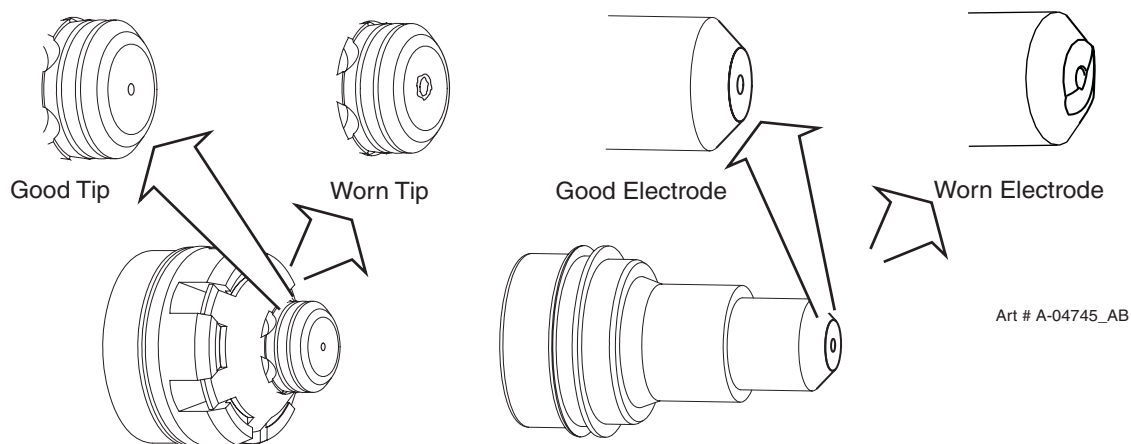
Use only Thermal Dynamics No. 9-4893 O-Ring Lubricant (Christo Lube MCG-129) with this torch part. Use of other lubricants may cause irreparable damage to the torch.

7.03 Parts Wear

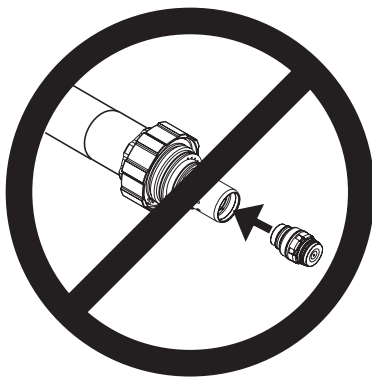
Replace the Gas Distributor if it is charred or cracked.

Replace the Gas Distributor if the flange is damaged in any way.

Replace the tip and/or electrode if they are worn.



7.04 Torch Consumables Installation

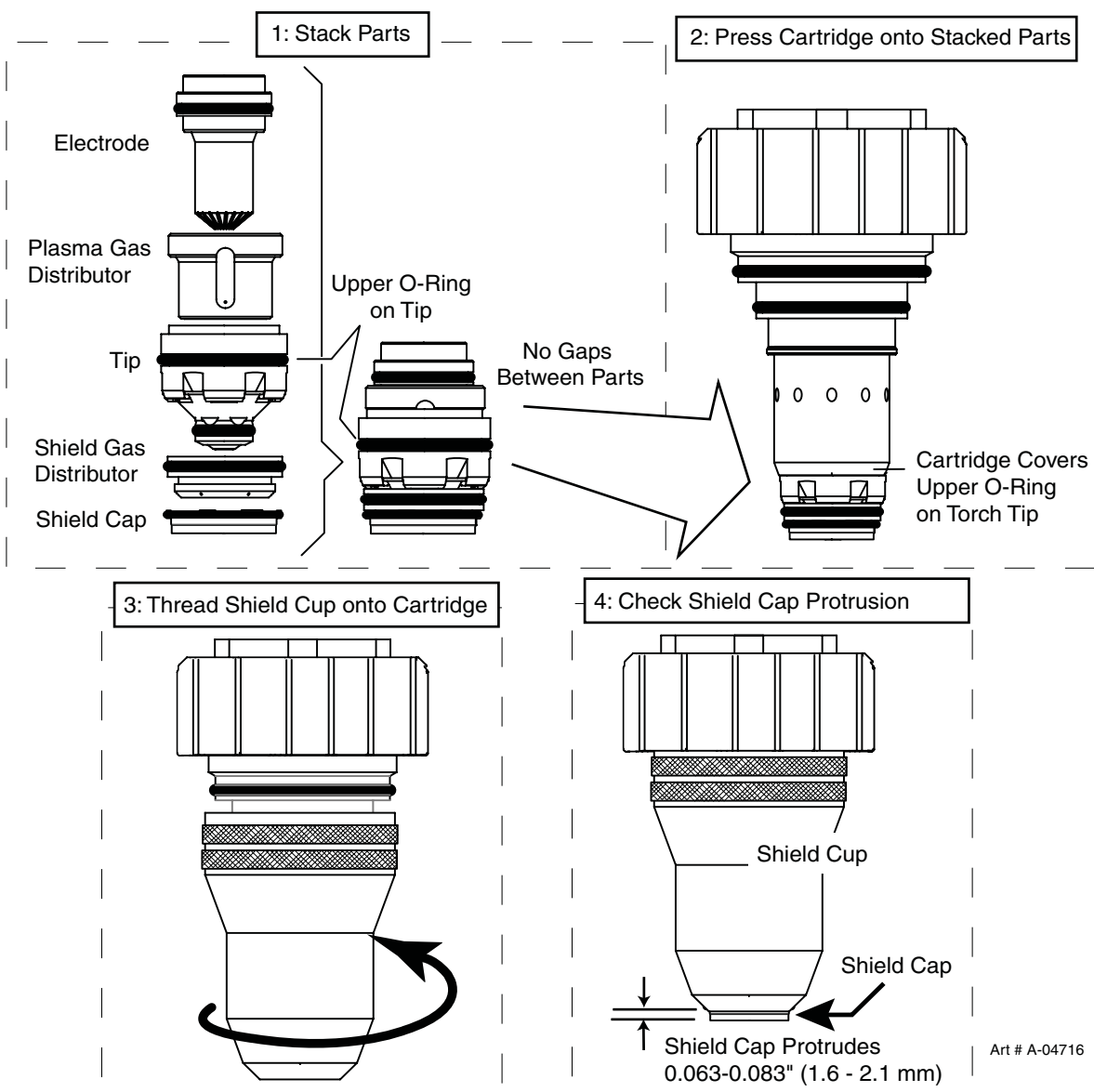


WARNINGS

Do not install consumables into the Cartridge while the Cartridge is attached to the Torch Head. Keep foreign materials out of the consumables and Cartridge. Handle all parts carefully to avoid damage, which may affect torch performance.

Art # A-03887

1. Install the consumables as follows:



Art # A-04716

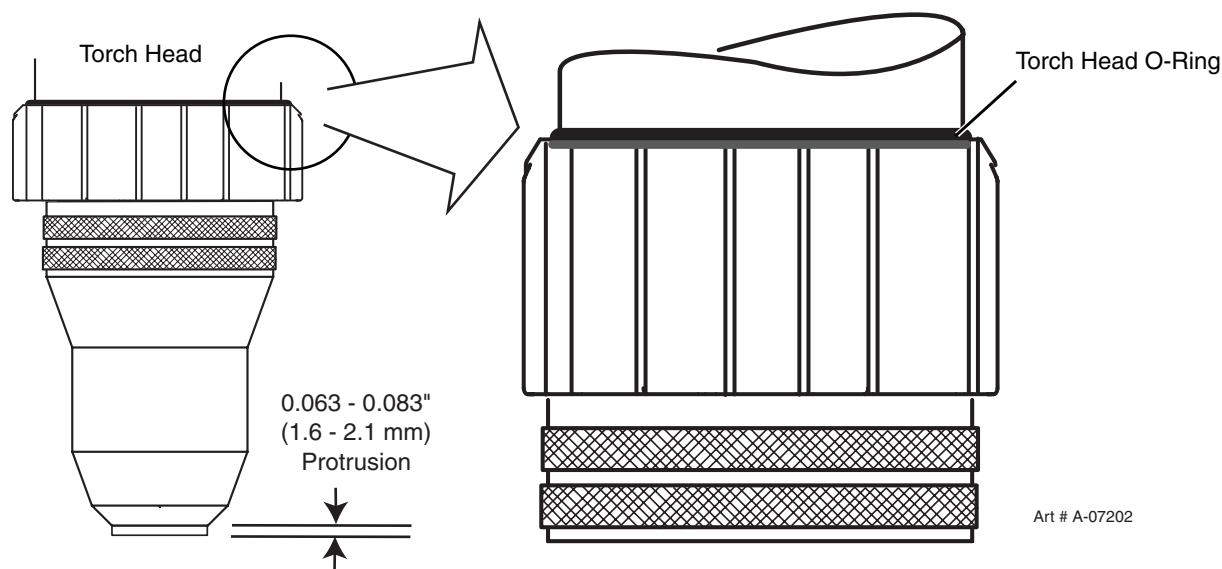
2. Remove the Cartridge Tool from the Cartridge and install the assembled Cartridge onto the Torch Head.



CAUTION

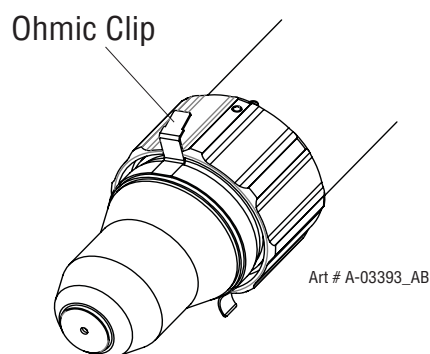
The cartridge assembly must cover the O-Ring on the torch head.

Do not force the cartridge if it will not tighten fully. Remove the cartridge assembly and gently clean the threads on the torch head with a wire brush. Apply oxygen-compatible lubricant (supplied with the torch) to the threads.



Installing Assembled Cartridge Onto Torch Head

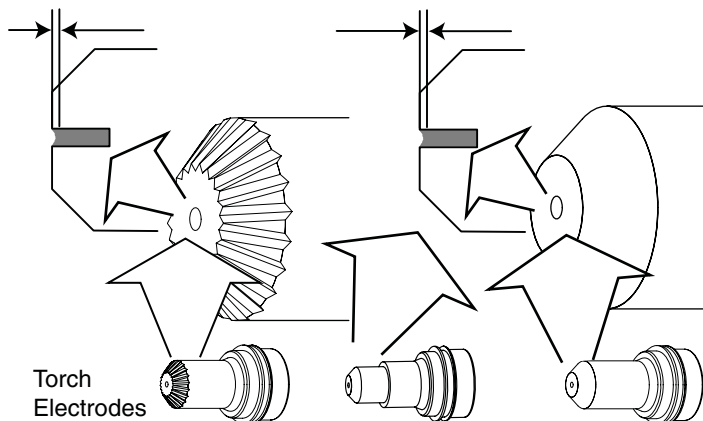
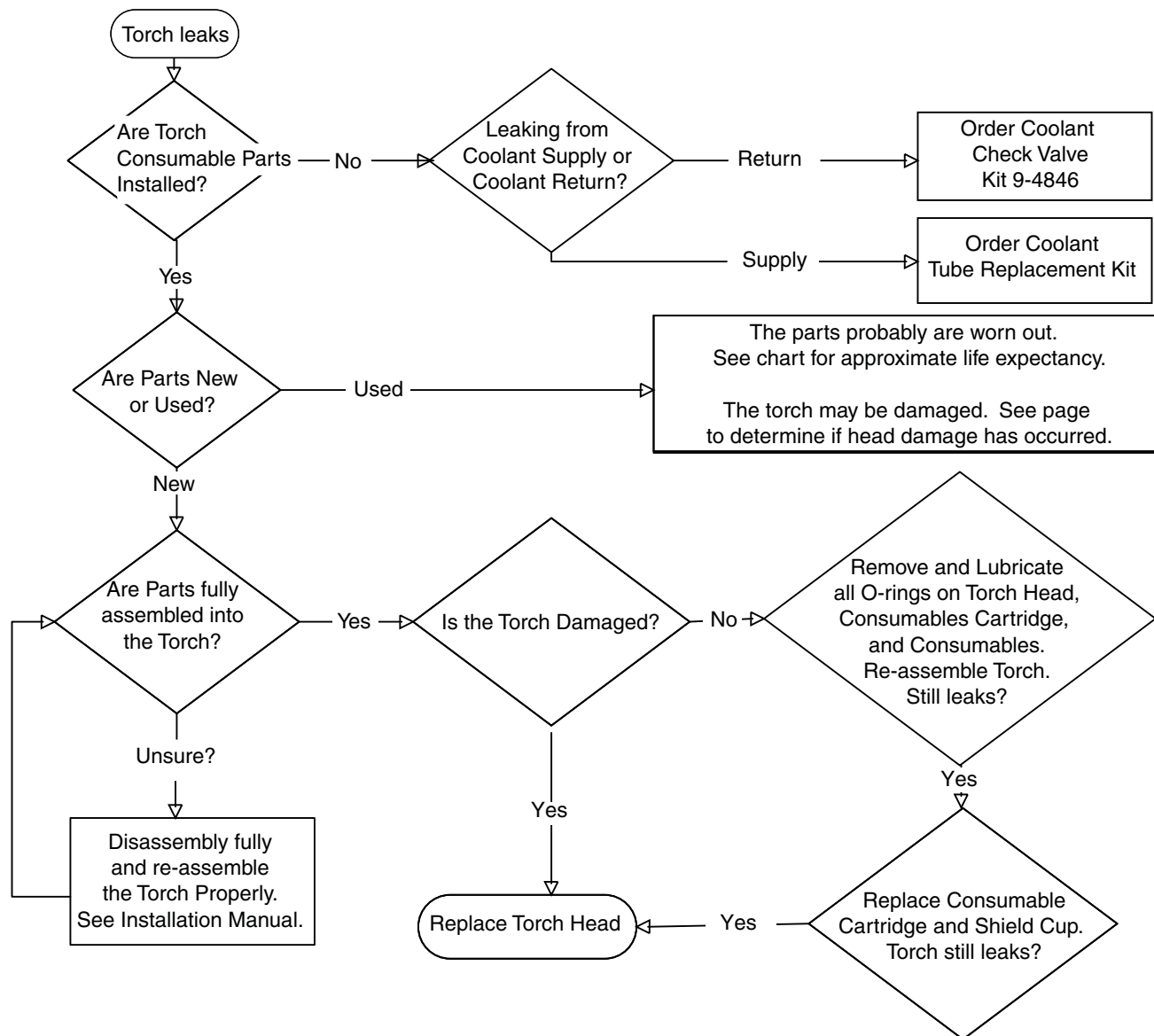
3. Slide the ohmic clip over the shield cup if using ohmic torch height control sensing.



4. Connect the wire lead from the height finder to the ohmic clip.

7.05 Coolant Leak Trouble-Shooting

Never operate the system if coolant leaks from the torch. A steady drip indicates that torch parts are damaged or installed improperly. Operating the system in this condition can damage the torch head. Refer to the following chart for guidance on coolant leakage from the torch head.

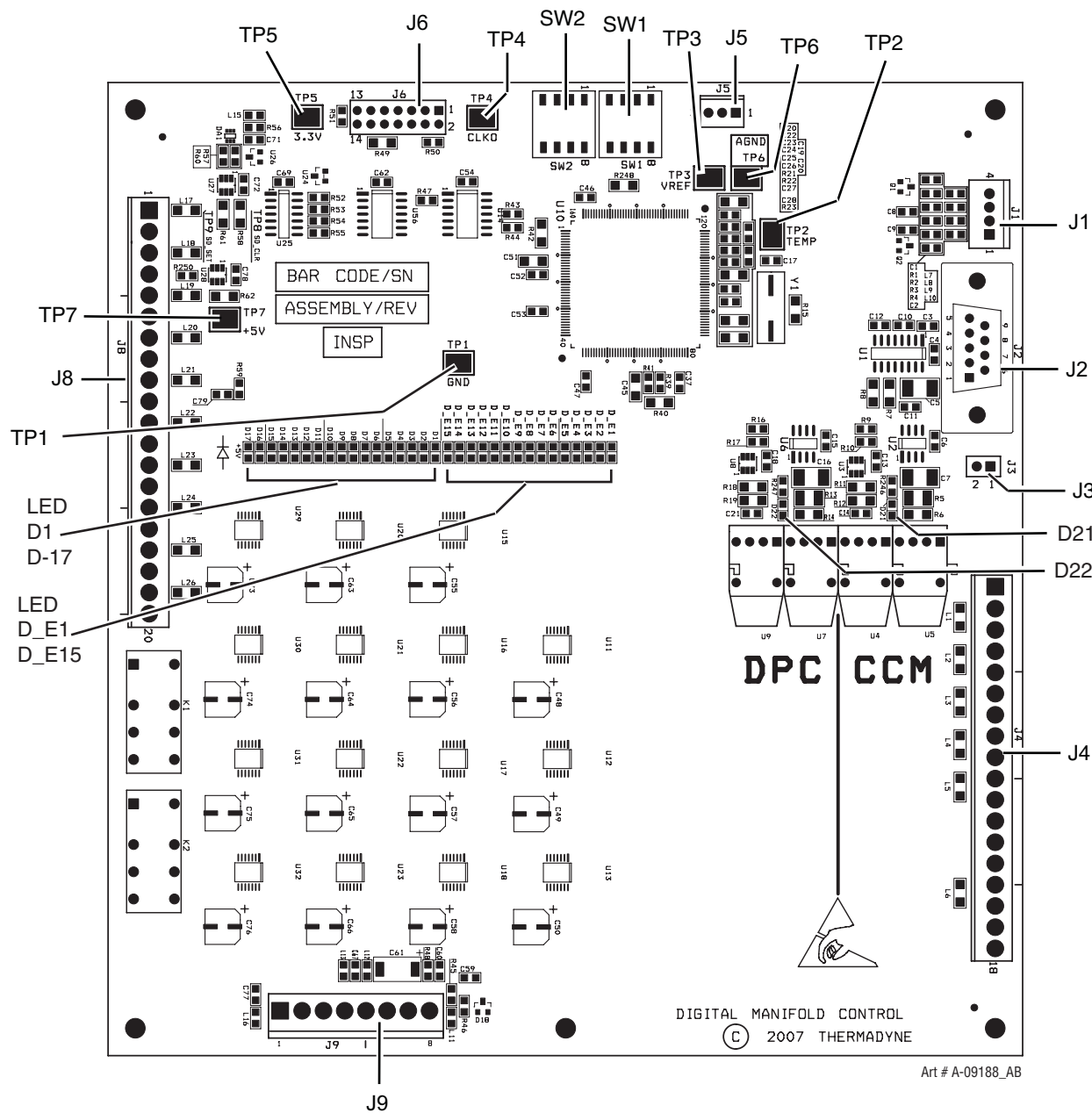


Amperage	Plasma Gas	Recommended Wear Depth for Replacement	
		Inch	mm
30	O2	0.04	1
	Air	0.04	2
50	O2	0.04	1
	Air	0.08	2
70	O2	0.04	1
	Air	0.08	2
100	O2	0.04	1
	H35	0.08	2

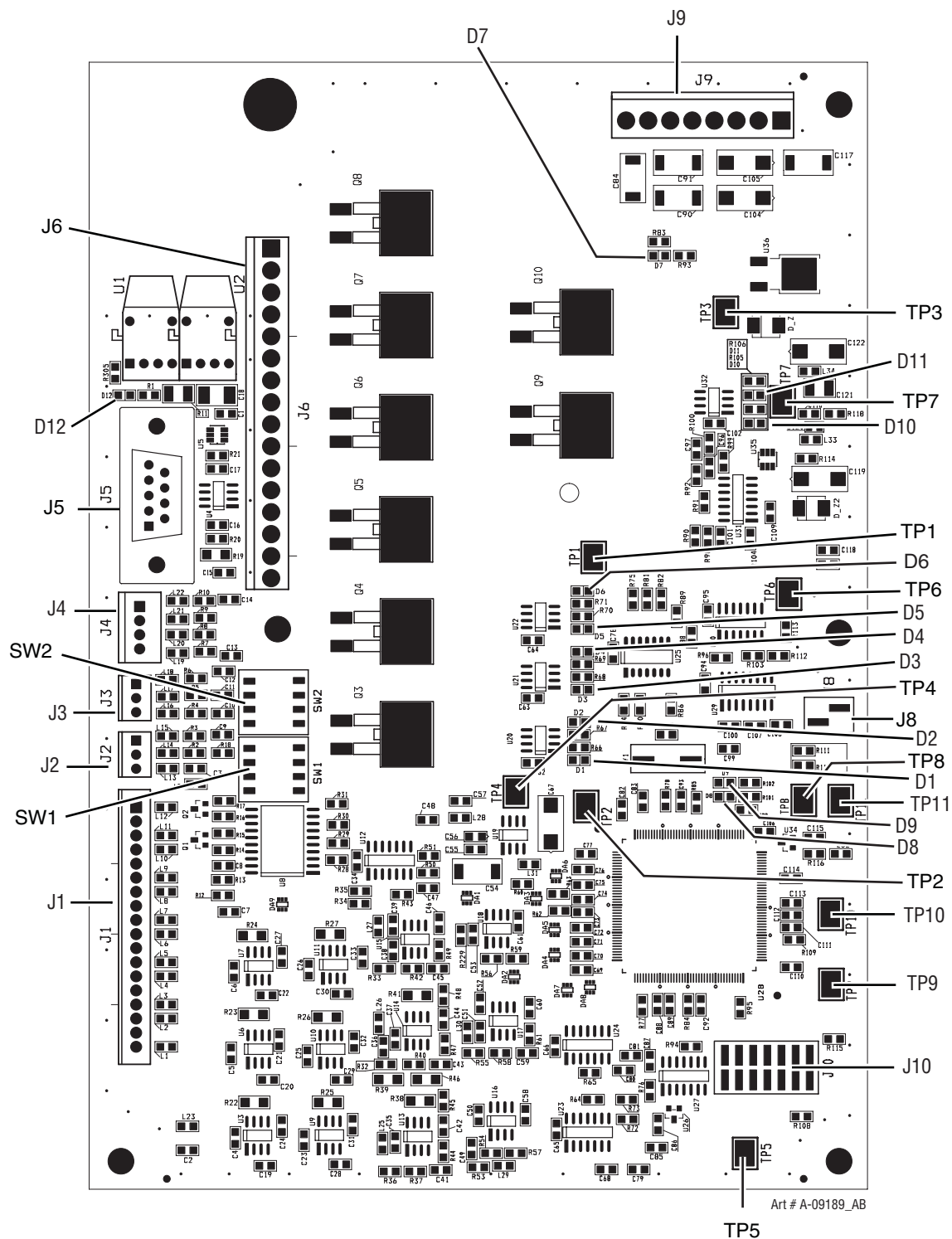
Art # A-04704_AB



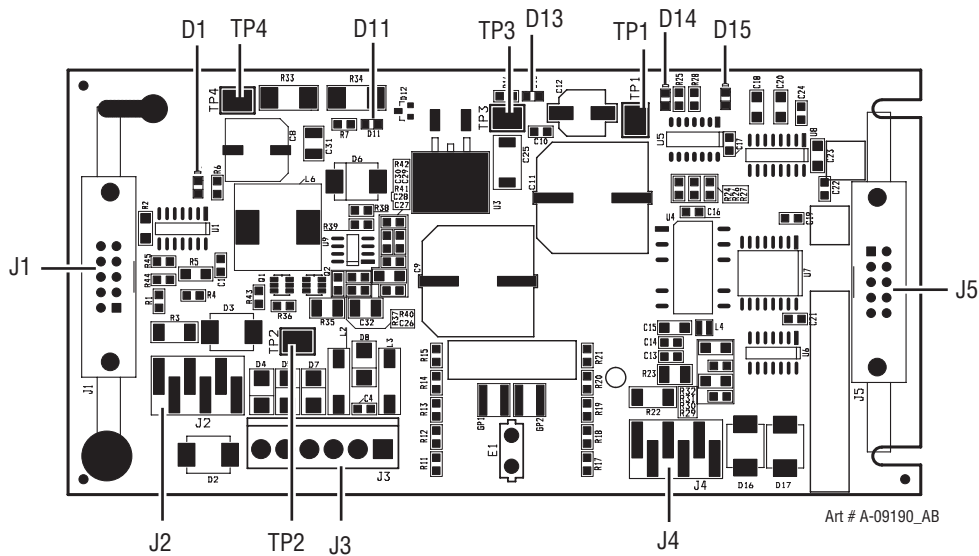
APPENDIX 2: DMC-3000 PCB Layout

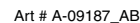


APPENDIX 3: DPC-3000 PCB Layout

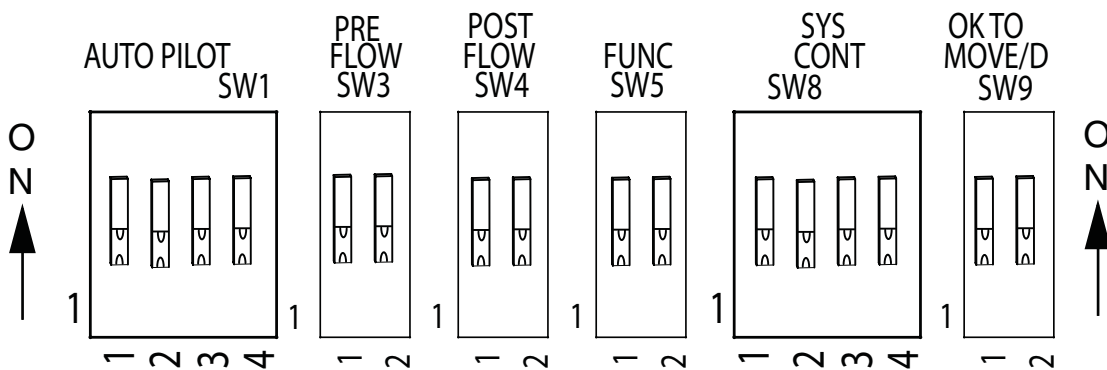
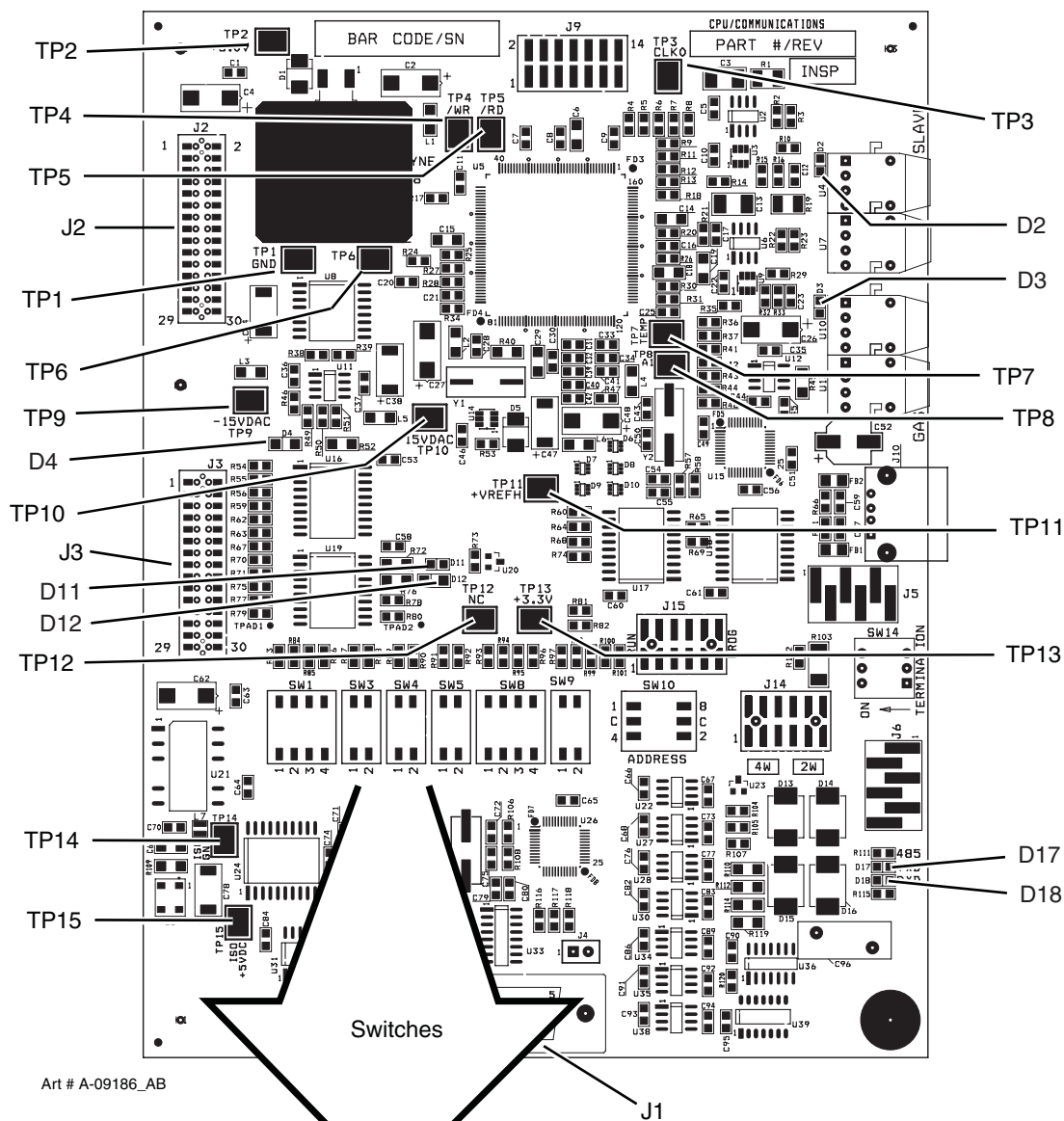


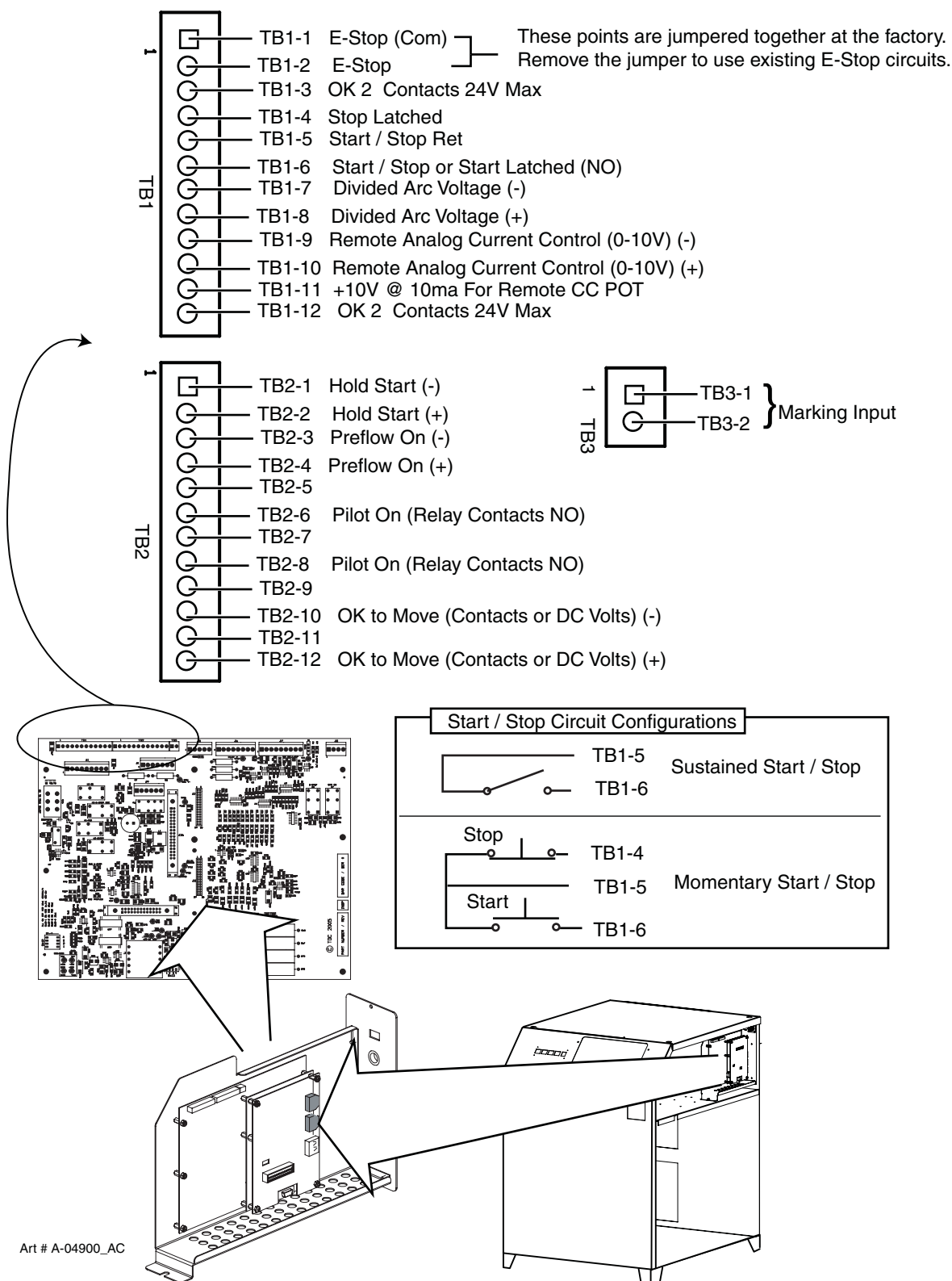
APPENDIX 4: TSC-3000 PCB Layout





APPENDIX 6: CCM CPU PCB Layout





APPENDIX 7: CNC

A7.01 CNC functions

CNC I/O circuits provide at least 1000V galvanic isolation from the plasma power supply.

While the CNC circuits are isolated from the power supply, many of the signal returns on J15 and TB1 & 2 are common to each other. J15 pins 1, 4, 5 & 10 and TB1-1, 5, 7, 9, and TB2-1 & 3 are all common. J15 pin 12 and TB2-10 are also connected to the others when SW6 (OK to Move select) is set for voltage.

Rear Panel CNC Connector J15:

14 Circuit (Amp CPC) Remote Standard:

Chassis gnd (for SC-11 cable shield)	1
¹ Start/Stop	3 (+); 4 (-)
¹ Ok to Move (contacts or voltage ²)	12 (-); 14 (+)
¹ Divided Arc volts (selectable ratio 50:1; 40:1; 30:1; 16.6:1)	5 (-); 6 (+)
Arc Volts (w /100K series res)	7 (-); 9 (+)
¹ Analog Current Control (0-10V) ³	10 (-); 11 (+)
Isolated Circuit Comm (for SC-11)	8
Chassis Gnd	13

¹ These are also duplicated on TB1 & TB2, use one or the other not both. Additional functions are only available on TB1 & 2.

² SW6 on CCMI/O PCB selects OK to Move for isolated contact closure or DC Volts (15-18V) at <100ma. When set for contacts, OK to Move circuit is rated for 120 VAC / 28 VDC @ 1A.

Internal CNC connections. TB1 & TB2 on CCM module.

Connections are provided on the CCM module TB1 & TB2 terminal blocks including most of the rear panel functions plus some additional features. All these signals are isolated from the plasma power supply but signals marked (comm.) and (-) are common to each other.

Users are expected to install their own CNC cable to these connections. Knockout hole is provided in rear panel of CCM module. User shall provide strain relief / cord grip for user installed cable.

Function	Connection
E-Stop	TB1-1 (comm.) & TB1-2
OK To Move 2 (Contact) ⁵	TB1-3; TB1-12
Stop Latched (NC) ⁴	TB1-4
Start/Stop Ret ⁴	TB1-5 (comm.)
Start /Stop or Start Latched (NO) ⁴	TB1-6
Divided Arc Voltage	TB1-7(-); TB1- 8(+)
Remote Analog Current Control (0-10V)	TB1-9(-); TB1-10(+)
+10V @ 10ma For Remote CC POT ⁵	TB1-11
Hold Start	TB2-1(-); TB2-2(+)
Preflow On	TB2-3(-); TB2-4(+)
Pilot On (Relay Contacts NO)	TB2-6; TB2-8

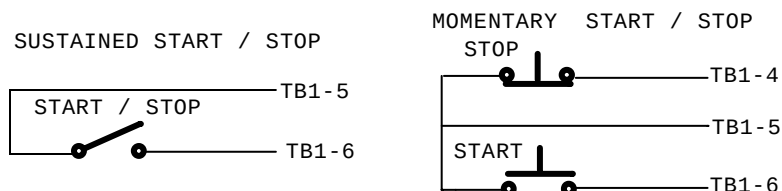
Function	Connection
OK to Move (Contacts or DC Volts)	TB2-10(-); TB2-12(+)
Plasma Marking Remote Select	TB3-1: TB3-2

A7.02 CNC Input / Output Descriptions

E-Stop input— Requires closed connection rated for 35ma. @ 20VDC for unit to operate. Factory installed jumper between TB1-1&2 must be removing when connecting user supplied E-Stop circuit.

⁴**Start/Stop input**—Switch (momentary or sustained) rating 35ma. @ 20 VDC

Start / Stop circuit configurations. Momentary Start / Stop (Latched) is only available at TB1.



Divided Arc Voltage output — Arc Voltage signal is isolated from plasma supply, however (-) is common with other isolated CNC signals. Max Divided Arc Voltage signal level depends on actual arc voltage times divide ratio however can not exceed approximately 12 V.

³**Analog Current Control input**— Analog Current Control includes analog isolation module, separate isolation module not usually required however it's low input is common with the other isolated CNC inputs. Scaling of Analog Current Control input is 0V = 0A, 10V. = MAX output and is linear in between. However MIN output is 10A. User is responsible for setting correct analog voltage to maintain at least 10A output. To use Analog Current Control on the I/O PCB set SW 11 to down position and on the CPU PCB set SW8-2 ON (up).

Hold Start input—Normally open, close to hold start. Circuit rating 10 ma. @ 20VDC. Delays pilot ignition, gas preflow continues. Used for synchronizing starts when multiple plasma supplies are used on same cutting table. User supplies circuit to keep Hold Start inputs active until all torches have found height.

Preflow On input— Normally open, close to start preflow prior to normal START signal. Circuit rating 10 ma. @ 20VDC. Torch Height Controls (THC) normally issue START signal to plasma supply after torch height has been found. Then the plasma takes 1-2 seconds (or more) to perform preflow before igniting pilot. Some THCs have an output that can start preflow early during height finding saving 1-2 seconds on each cut.

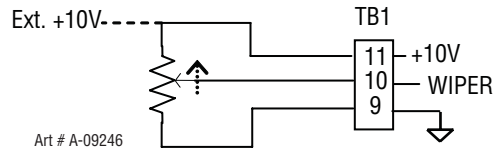
Pilot On output – Relay contacts rated 1A @ 120 VAC / 28 VDC. Contacts close when pilot on. Can be wired parallel with Ok to Move contacts to start machine motion when pilot established. Used when starting over holes. Starting over holes requires setting SW8-1 ON (up) on the CPU PCB for extended pilot time. Using extended pilot time to start over holes or for cutting over holes will reduce parts life.

OK to Move output — Active when cutting arc is established, arc is transferred. Used to signal cutting table to start X-Y motion. Relay contacts rated 1A @ 120 VAC or 28 VDC when SW6 set for contacts. When SW6 is set for DCV, output supplies 15-18 VDC @ 100 ma. May be wired parallel with Pilot On to start cutting machine motion as soon as pilot established.

⁵ **The following features are available in Ultracut CCM revision level BA and higher or S/N 05517201 and higher.**

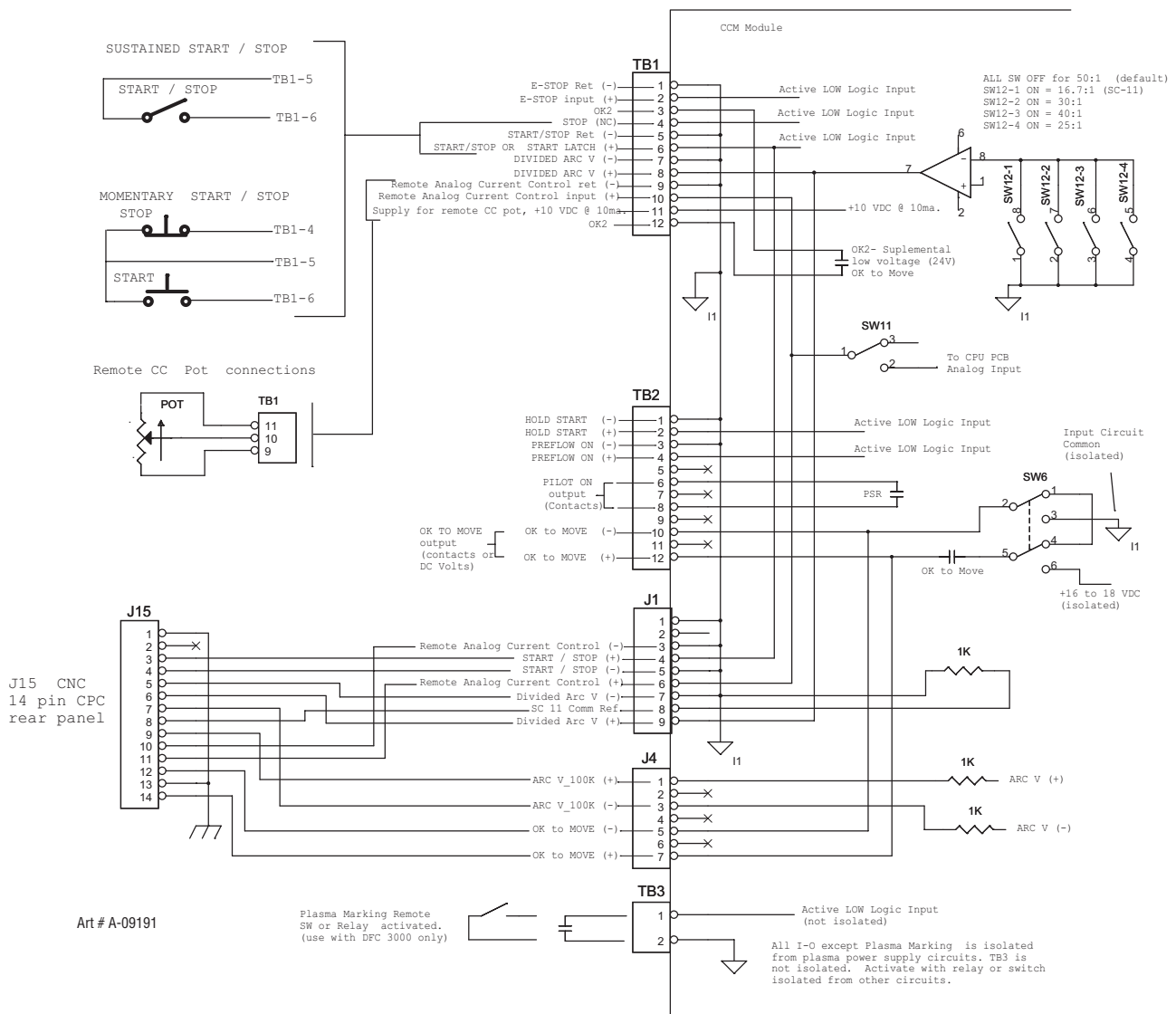
⁵ **OK to Move2** – Provides a second set of N.O. contacts that close when arc transfer is detected. Contacts are rated for maximum of 24 VAC/DC @ 1A.

⁵ **+10V @ 10ma. For Remote CC Pot** – Previously CCM versions if one wanted to use a potentiometer for the Remote Analog Current Control (CC) input an external 10 V supply was required for Pot High.. Now an isolated (from main plasma circuits) 10V supply is provided. Recommended value of the pot is 5K or 10K.

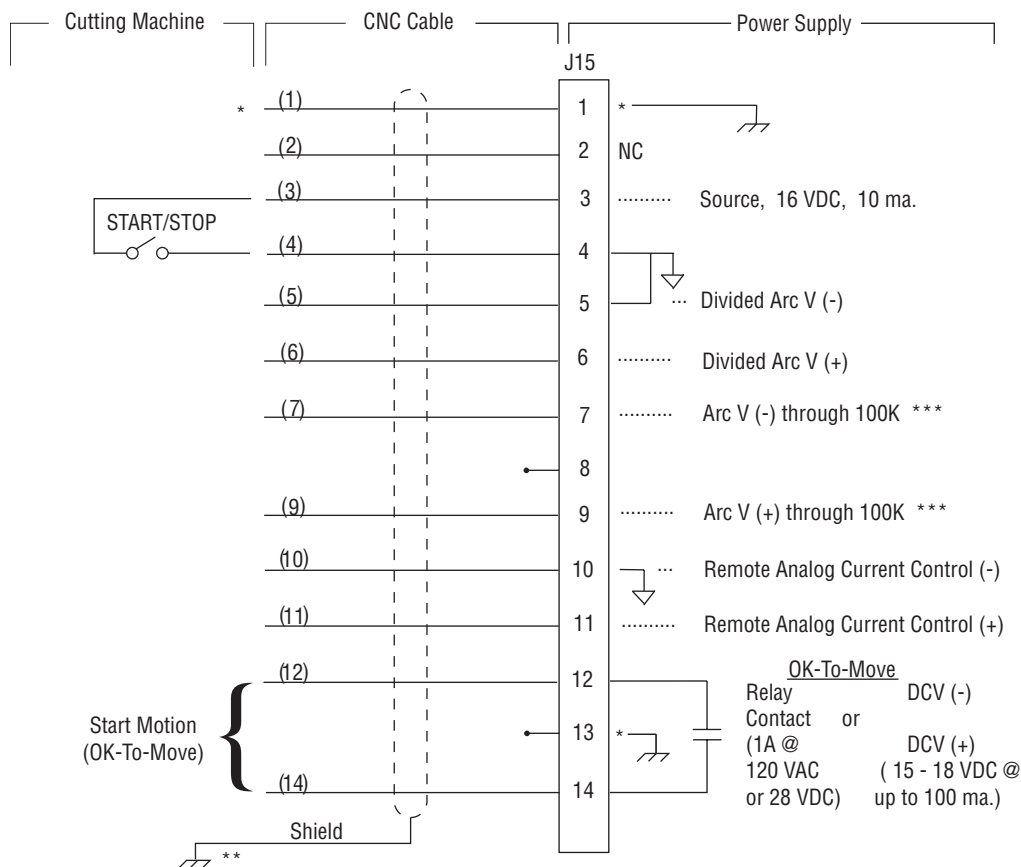


⁵ **Plasma Marking Select (Remote)** – Plasma Marking, available only with DFC 3000, may be activated with a contact closure between TB3-1 & TB3-2 if SW8-4, DIP switch on the CPU board (smaller of the 2 CCM boards), is also on. Opening the connection between TB3-1 & TB3-2 switched back to normal cutting mode. For Ultracut power supplies It is OK to leave SW8-4 on whether you are marking or not.

A7.03 Simplified CNC Circuit.



A7.04 CNC Connections.



A-07632

* Power Supply Gnd not used for CNC cable
Do not connect wire #1 to anything.

** Cable Shield drain wire must be
connected to ground at cutting machine.

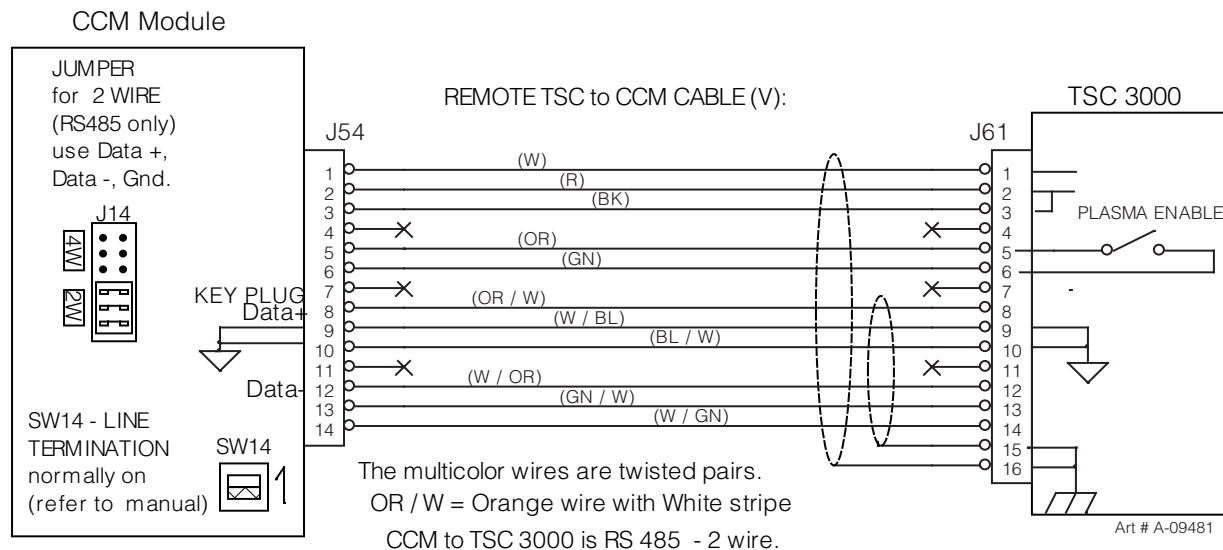
*** Used only for special Torch Height Control

APPENDIX 8: SERIAL COMMUNICATIONS

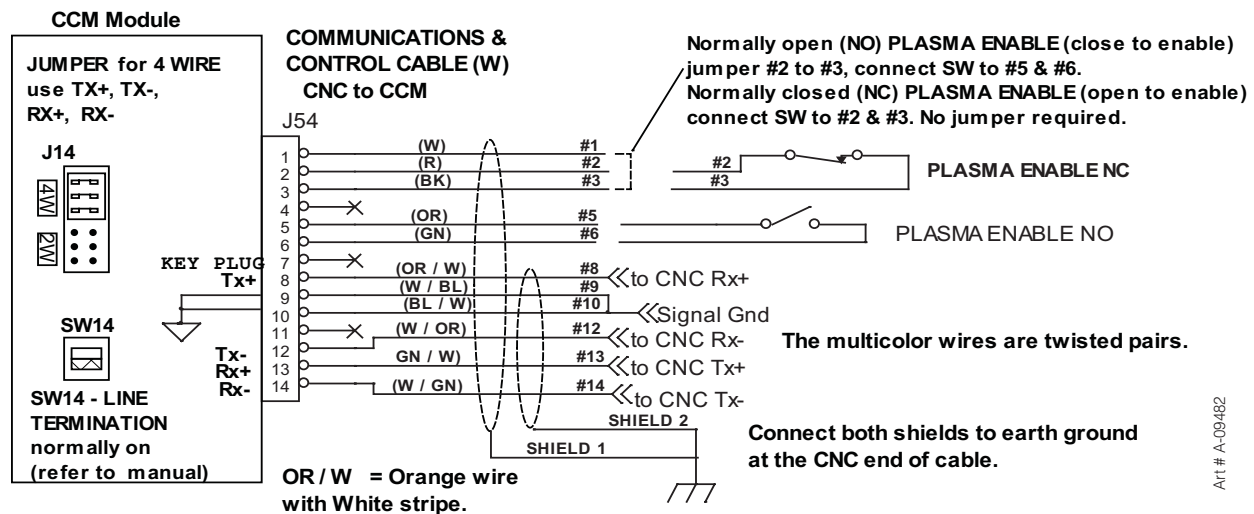
A8.01 Cables and Switch Settings for 2 and 4 Wire

The CCM communicates with the TSC 3000 (touch screen HMI panel) using RS 485 (2 wire, half duplex). When a CNC controller is used in place of the TSC the CCM can be reconfigured for RS 422 (4 wire, half duplex) if supported by the CNC. RS 422 is normally full duplex but the CCM only supports half duplex. RS 232 is not directly supported. Converters are available from various sources to convert RS 232 to RS 485/422. When possible, the 4 wire configuration is recommended to use the 4 wire configuration when possible for easier troubleshooting. The serial communication port of the CCM is provided with 3KV isolation from the rest of the plasma supply circuitry.

RS 485 wiring (2 wire, half duplex)



RS 422 wiring (4 wire, half duplex) (also called RS 485 4 wire)



Recommend using the 4 wire when possible as it is easier to troubleshoot if there is a problem. Note that CCM Tx+ connects to CNC Rx+ and CCM TX- connects to CNC RX- for 4 wire half duplex. If using 2 wire RS 485 connect CCM Tx+ (also called D+ or Data +) to CNC Tx+ and CCM Tx- (also called D- or Data-) to CNC Tx-. The Rx wires are not used for 2 wire.

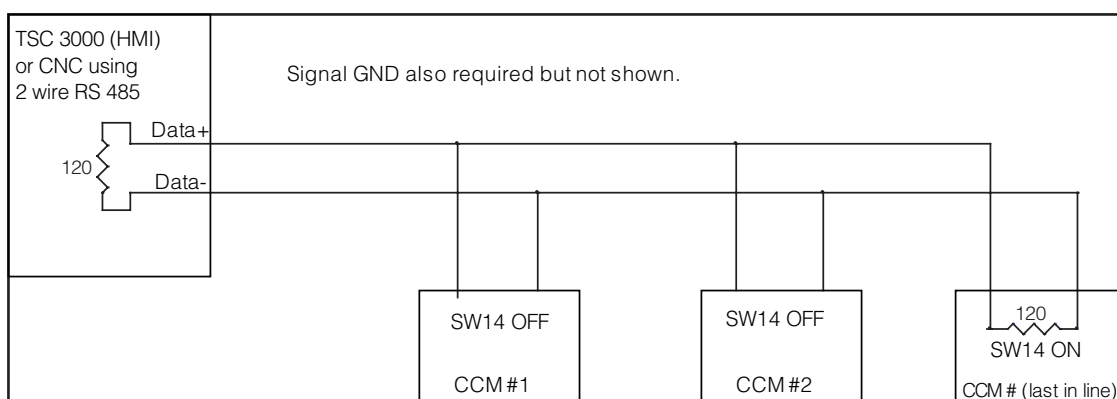
Line termination:

RS 485 and RS 422 are both “multi-drop protocols, that is there can be multiple devices on the same line. We do not currently support more than one CCM. Until we do line termination should always be ON.

For RS 485 it is recommended that the communication lines be terminated at each end with the line’s characteristic impedance. For RS 422 it is recommended the line be terminated at the receiver end.

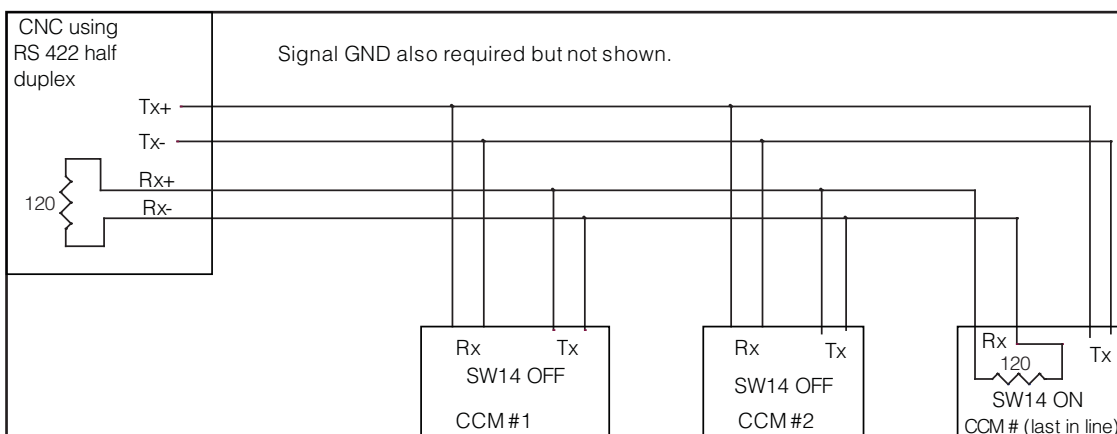
The CCM has a line termination switch SW14 whose default position is ON. For CCMs that are not at the end of the line (like CCM #1 & #2 below) switch off SW14.

RS 485 with multiple CCMs:



Art # A-09483

RS 422 4 wire half duplex:

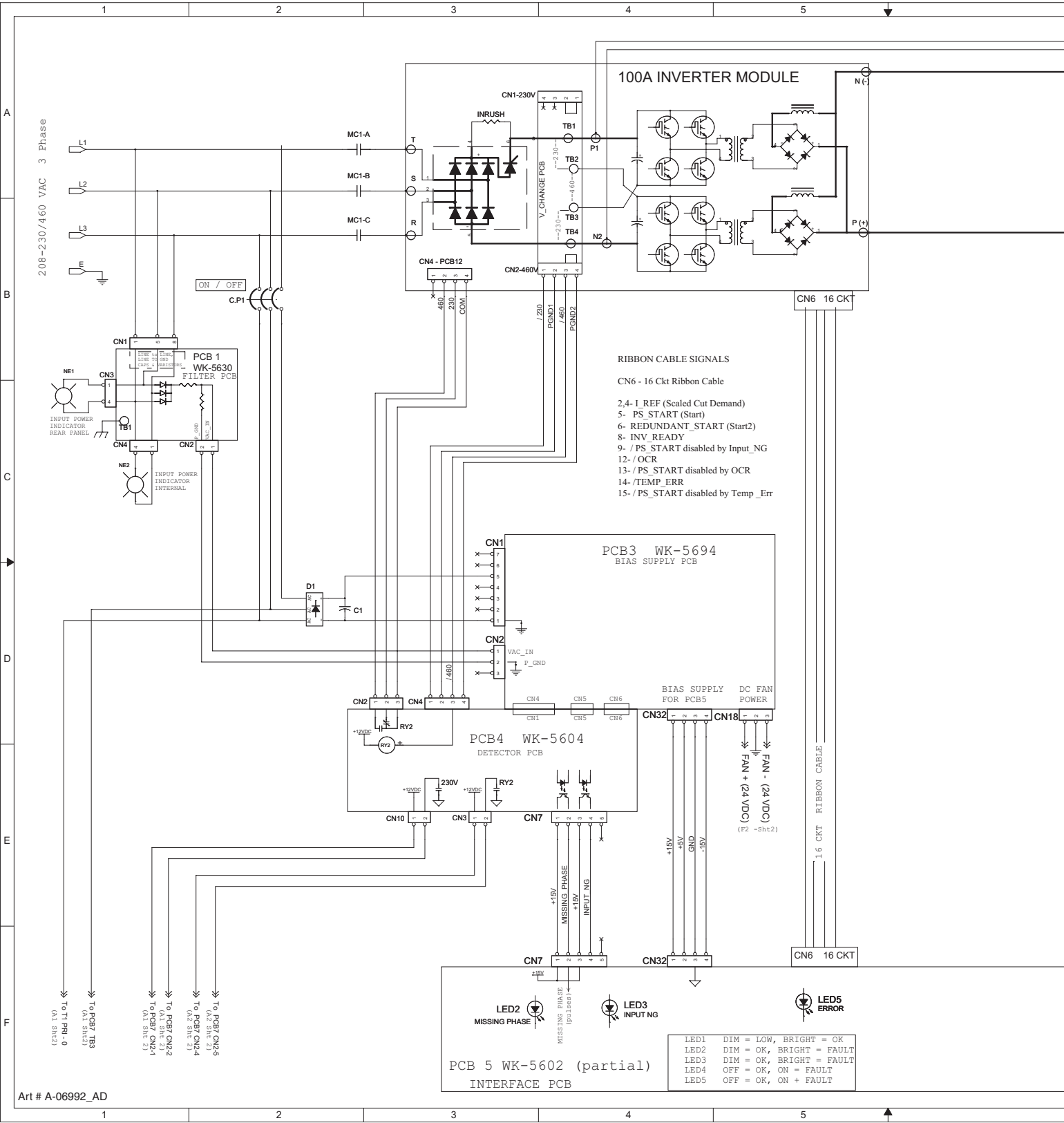


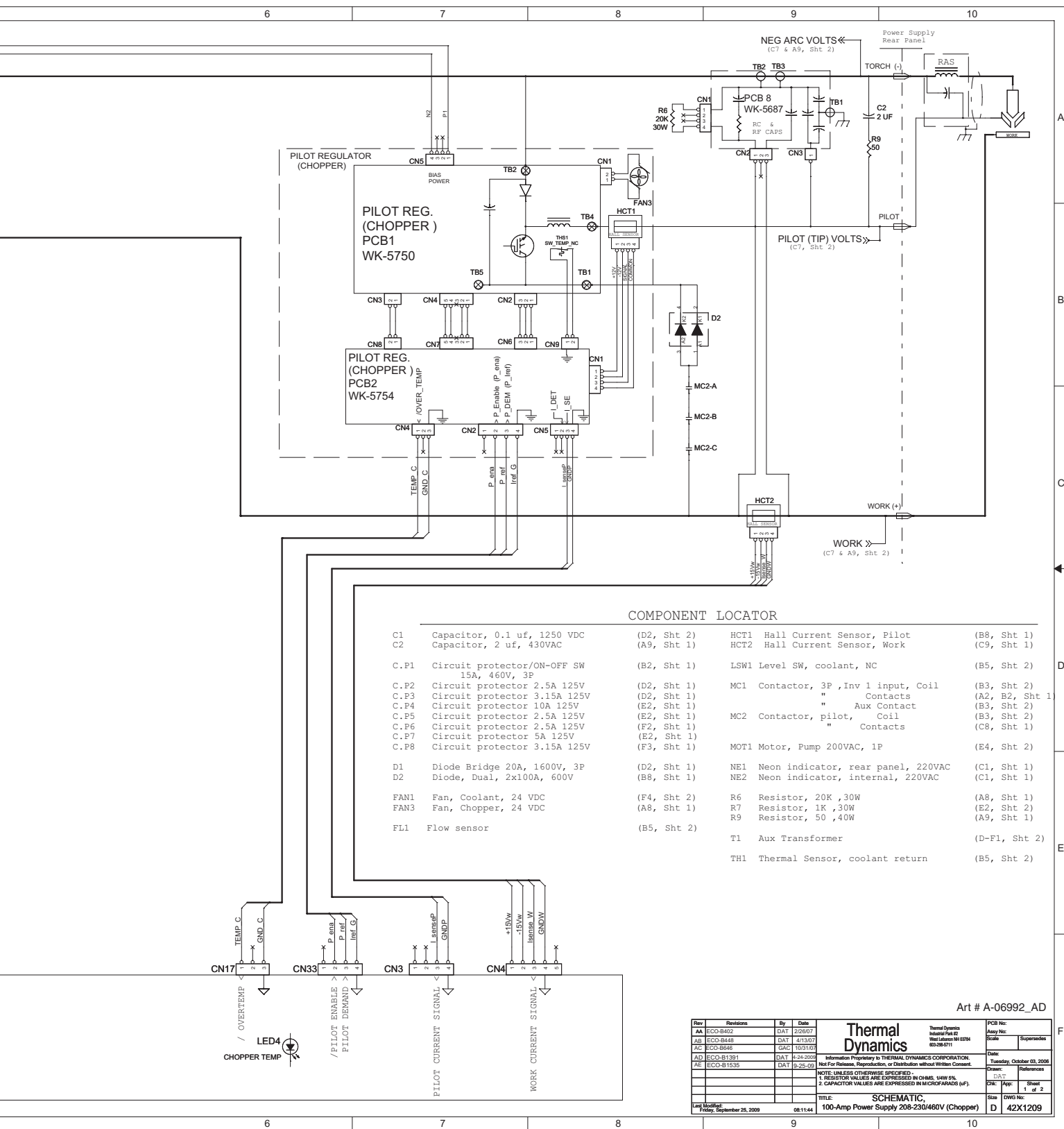
Art # A-09484

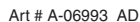
CCM Address:

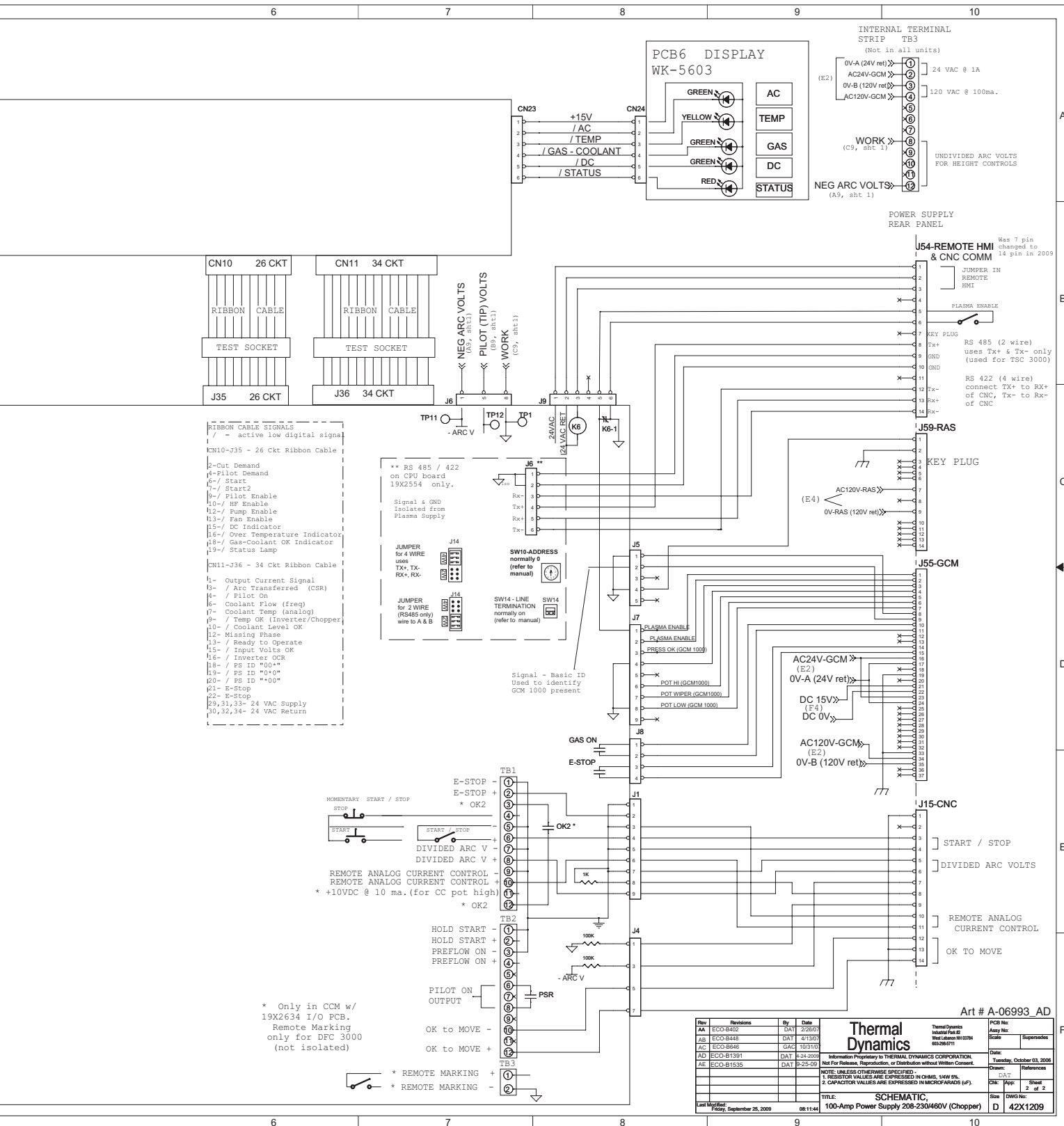
When more than one device is used in parallel on a serial communication line each has to have a unique address. The CCM has a switch SW10 to set the address of each CCM. The “0” factory setting is correct for system with one plasma (one CCM). We do not currently support CCMs in parallel. When that becomes available details for setting other addresses will be included in an updated manual explaining installation and set up of parallel systems.

APPENDIX 9: Schematic, 230-460V, Chopper Pilot

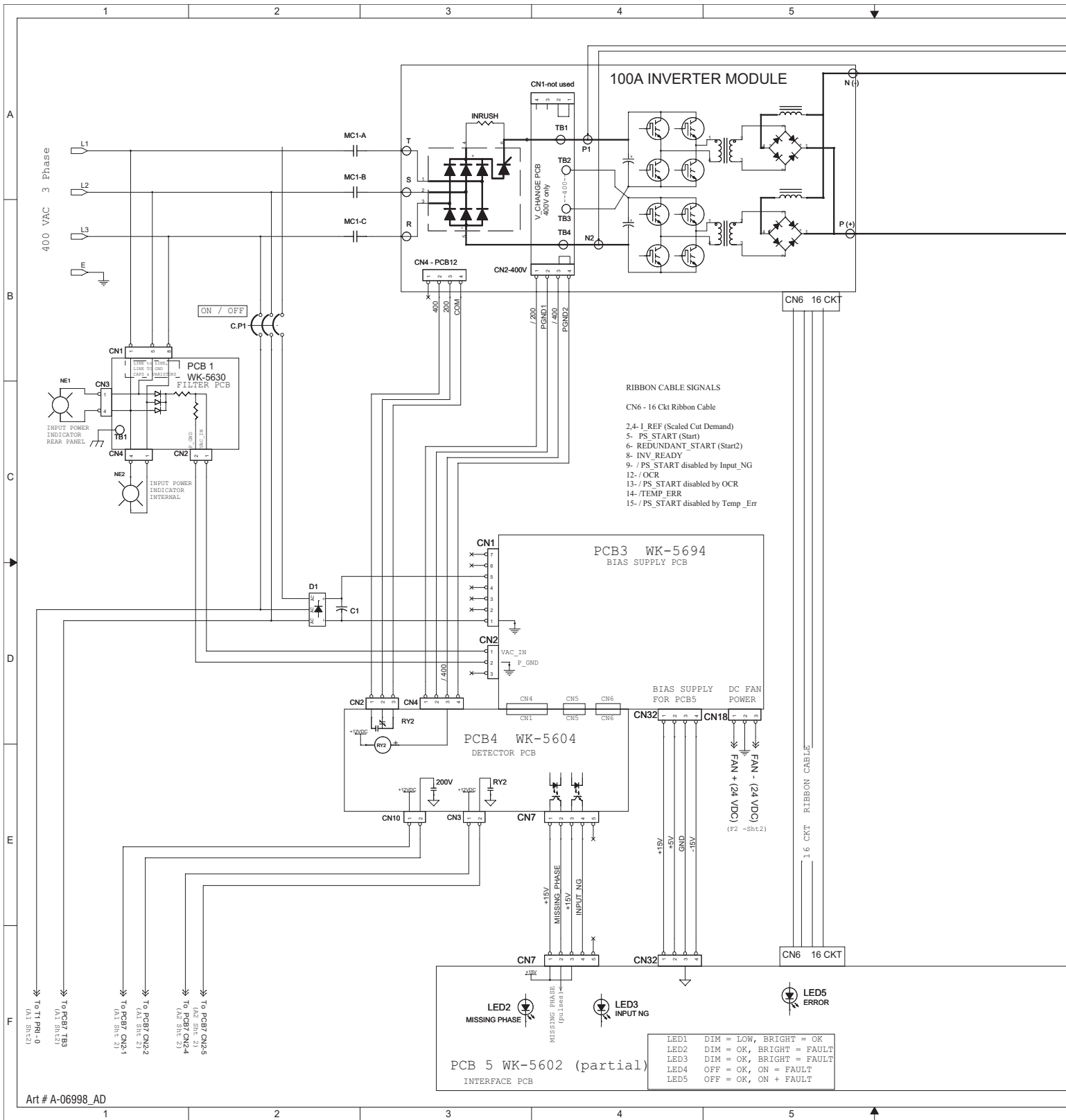


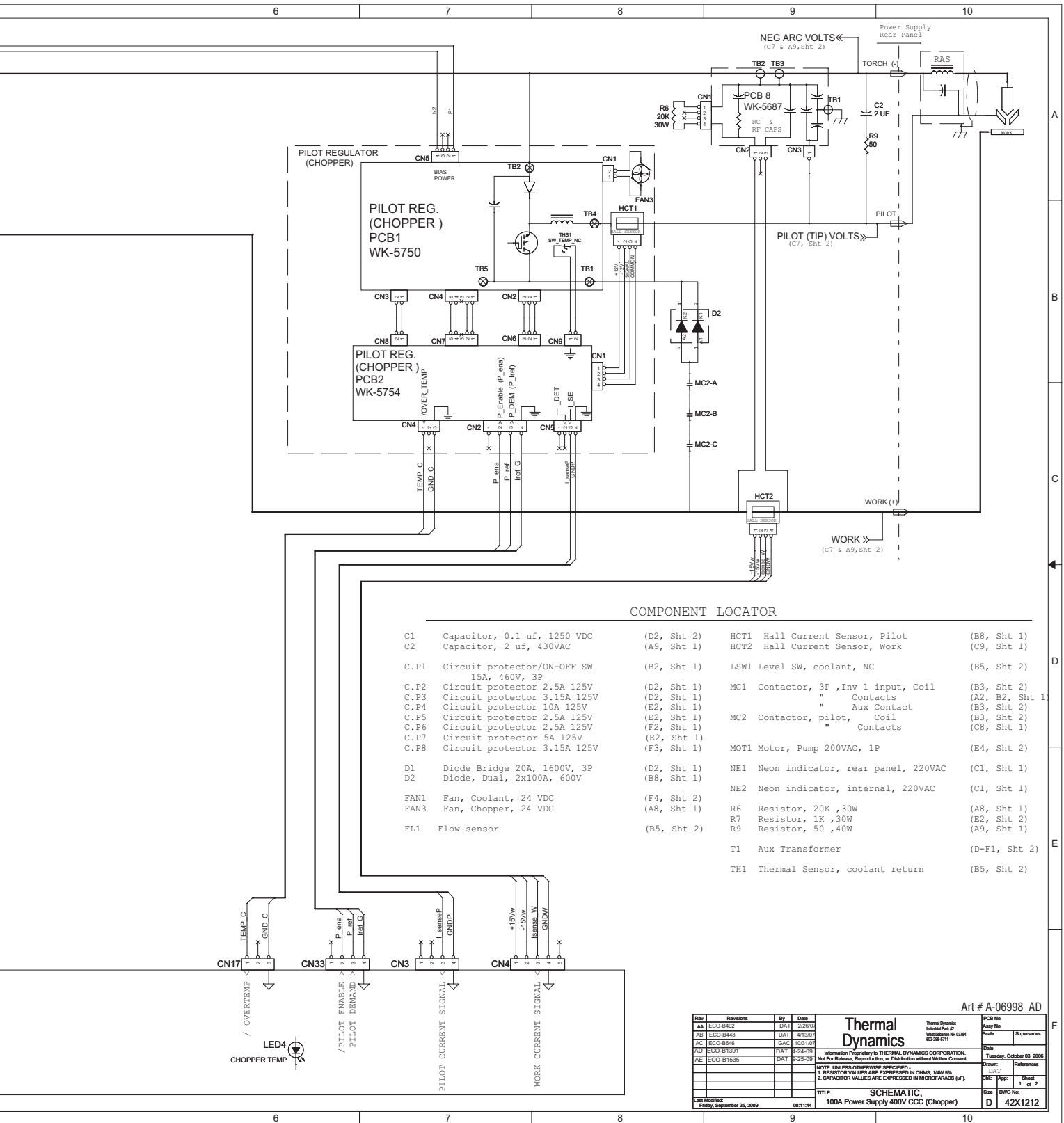




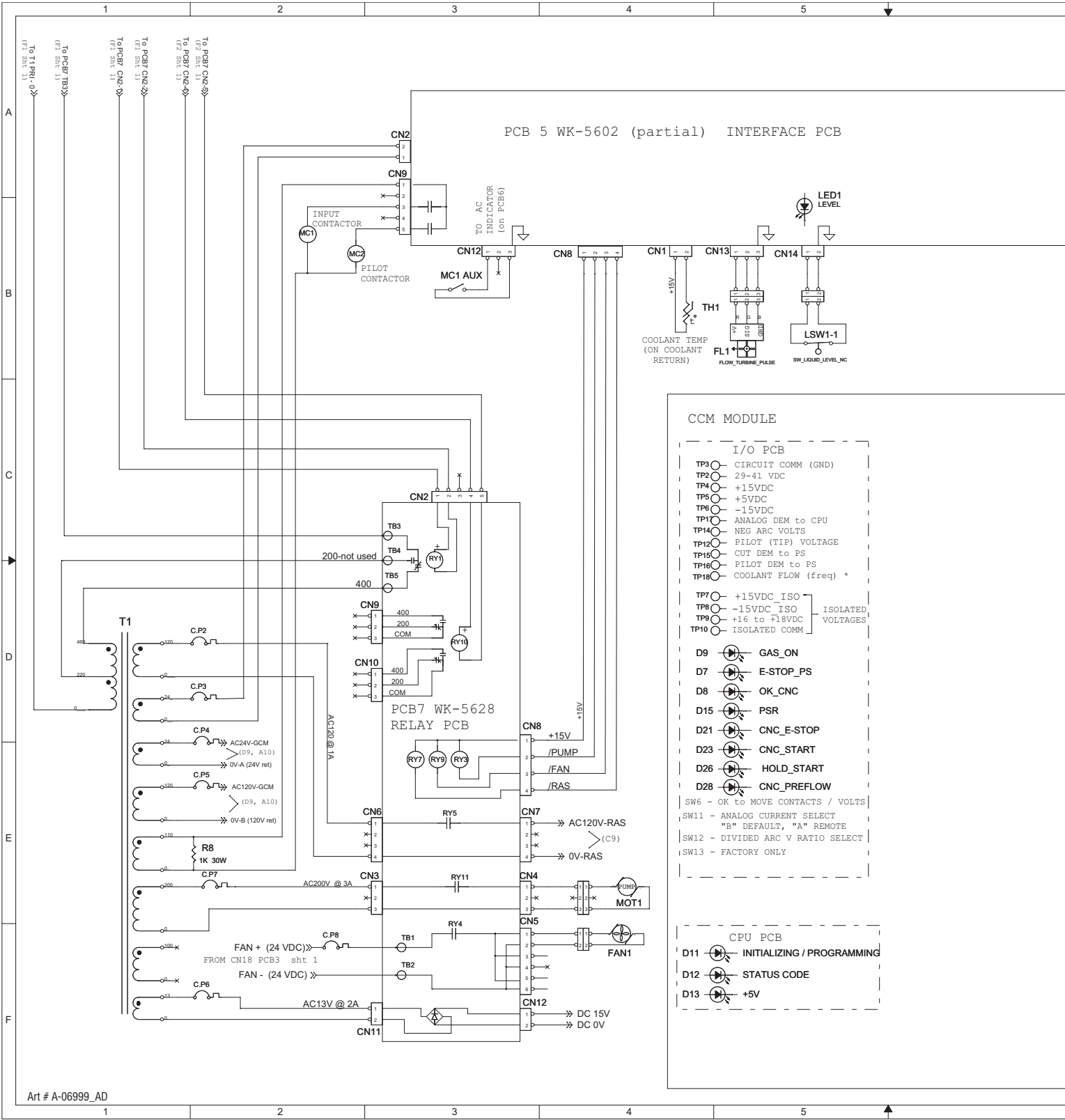


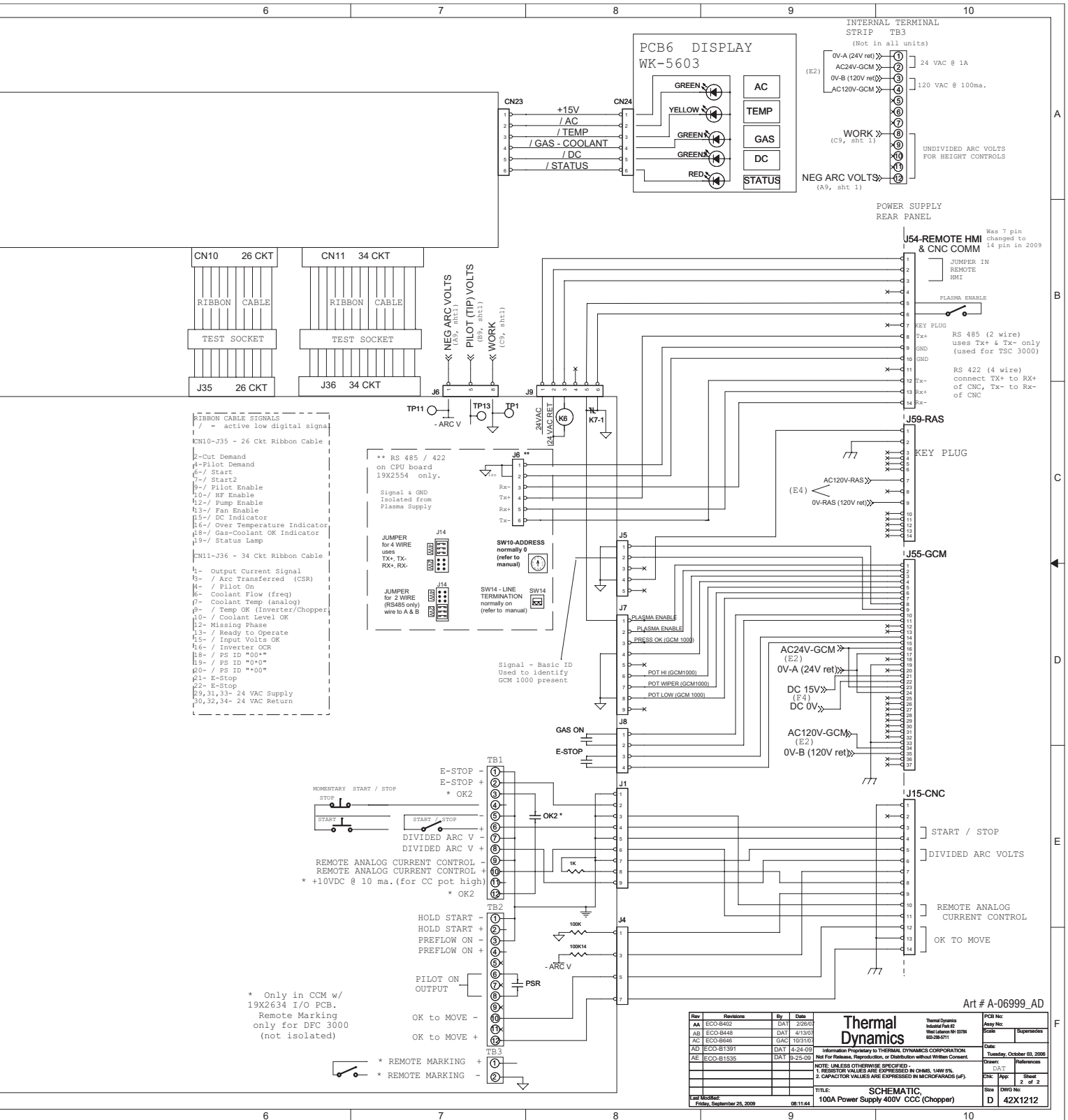
APPENDIX 10: Schematic, CCC 400V, w/ Chopper Pilot



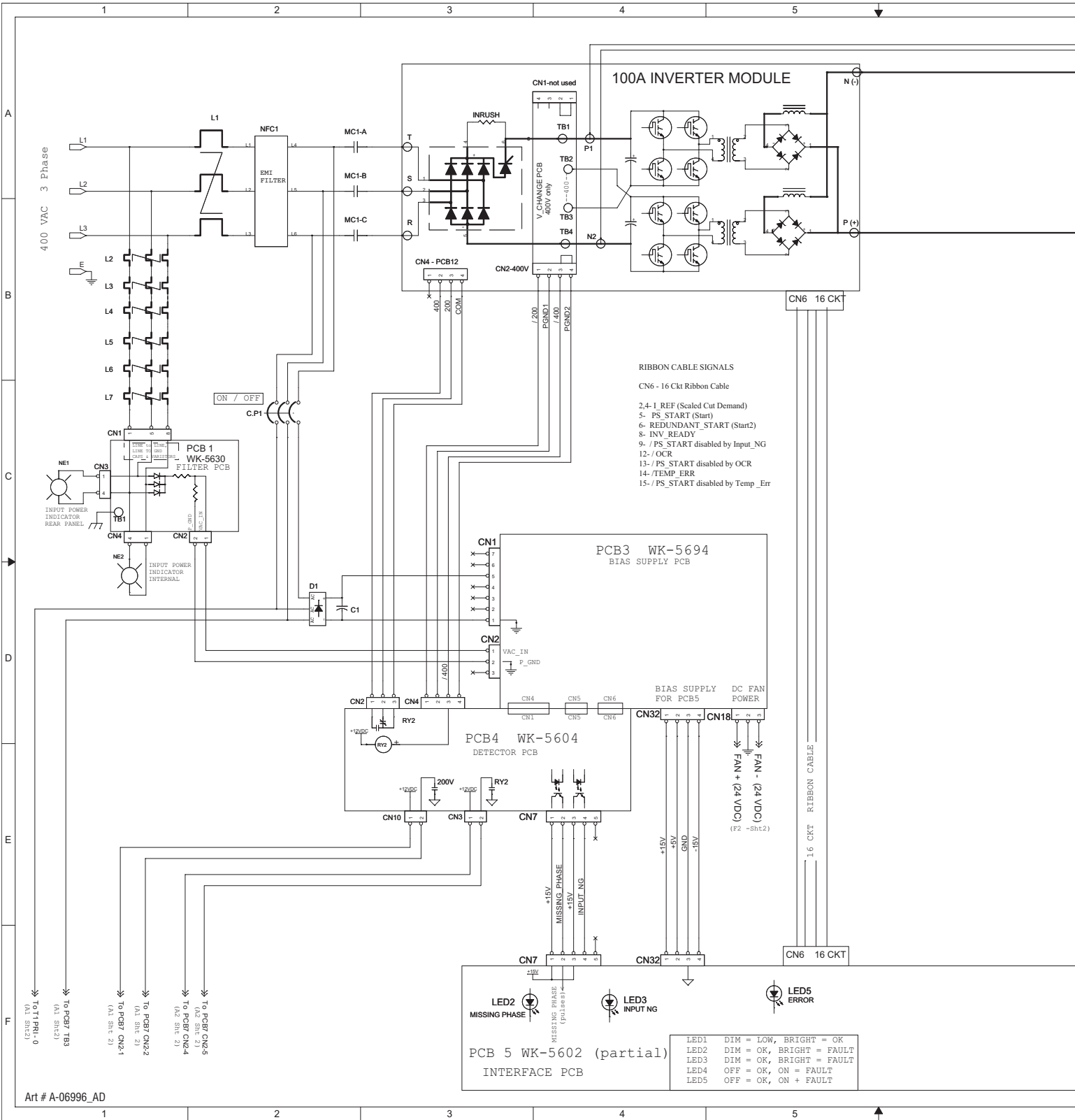


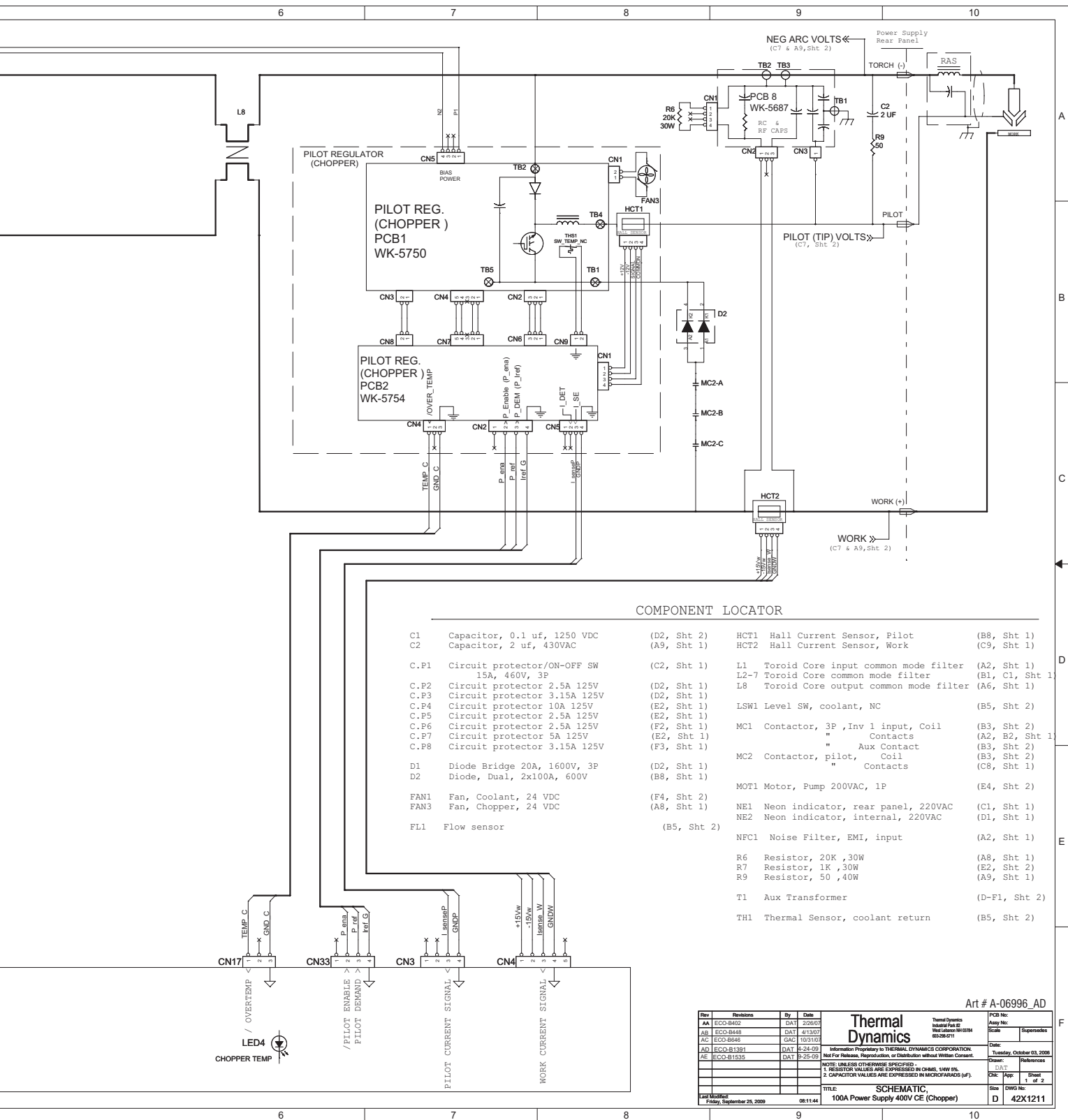
APPENDIX 10: Schematic, CCC 400V, w/ Chopper Pilot Cont.



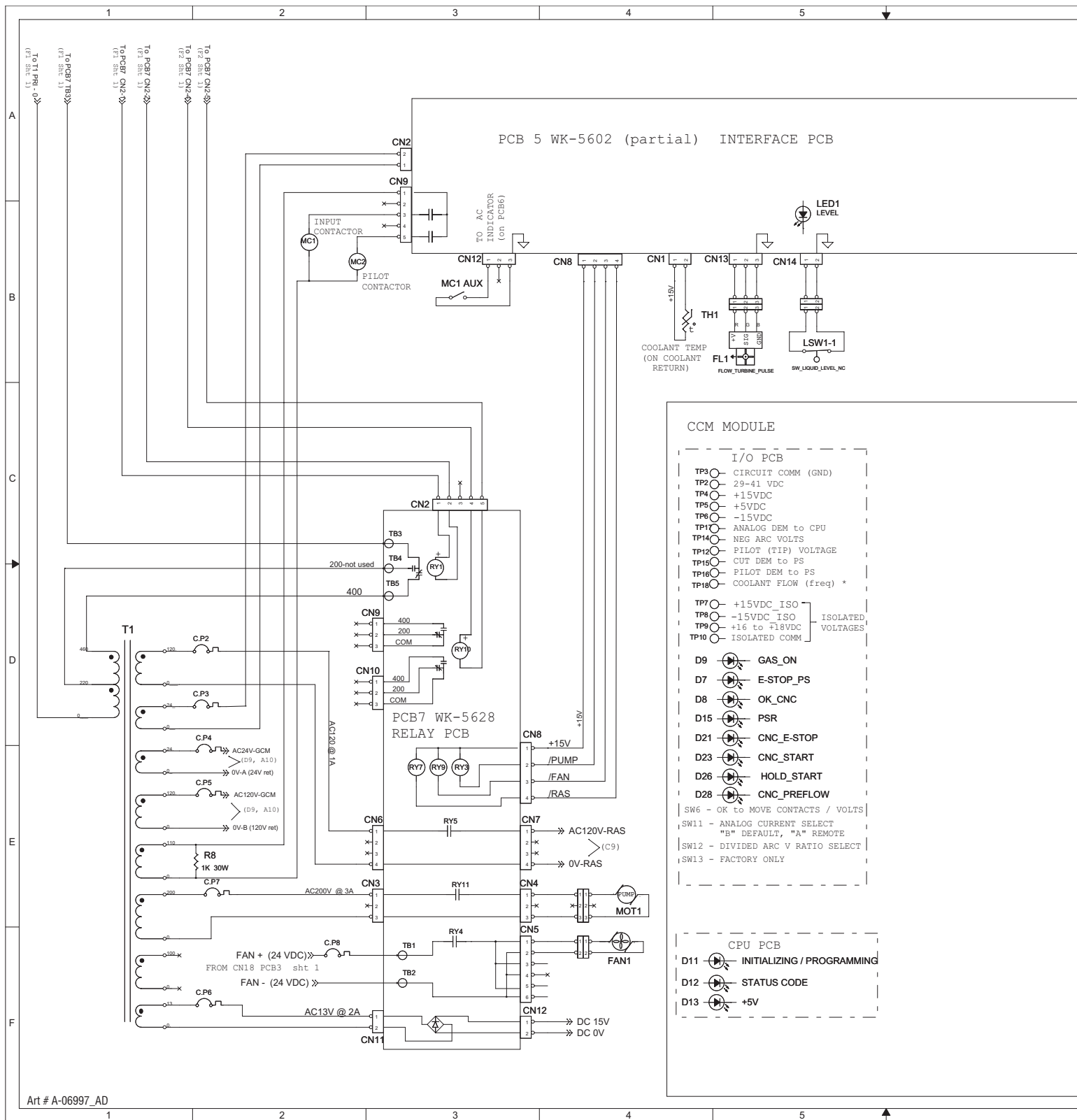


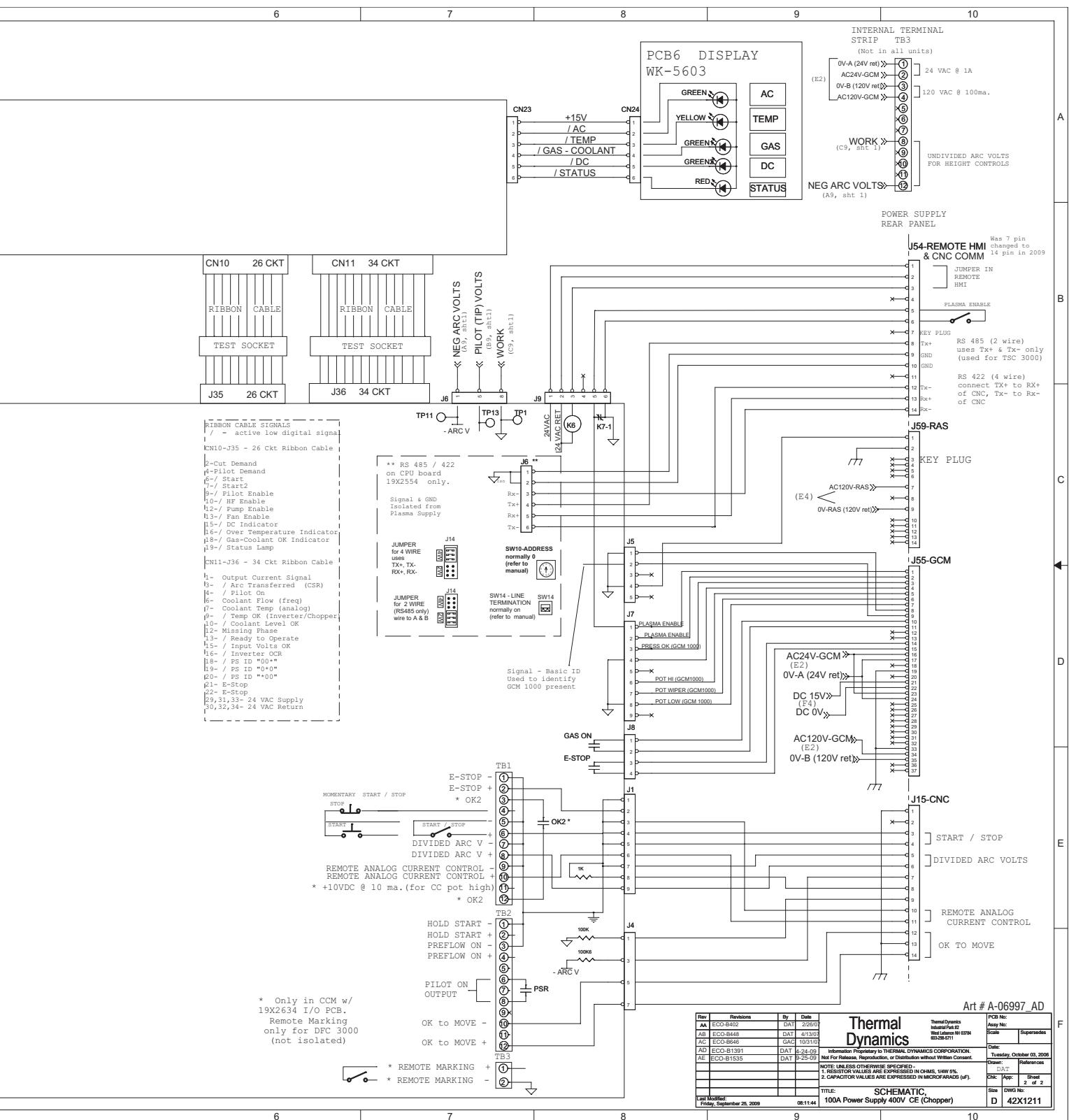
APPENDIX 11: Schematic, CE 400V, Chopper Pilot



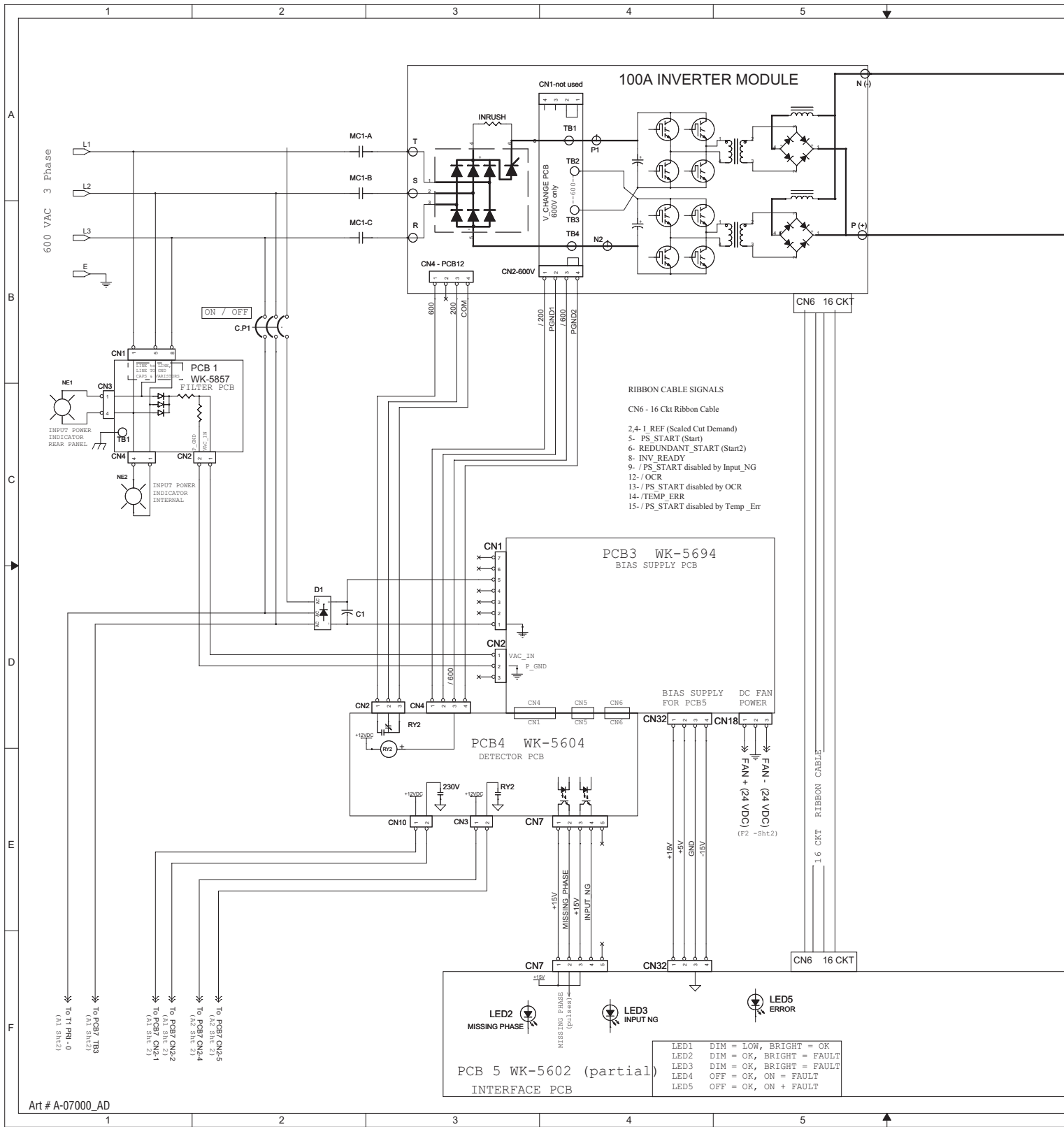


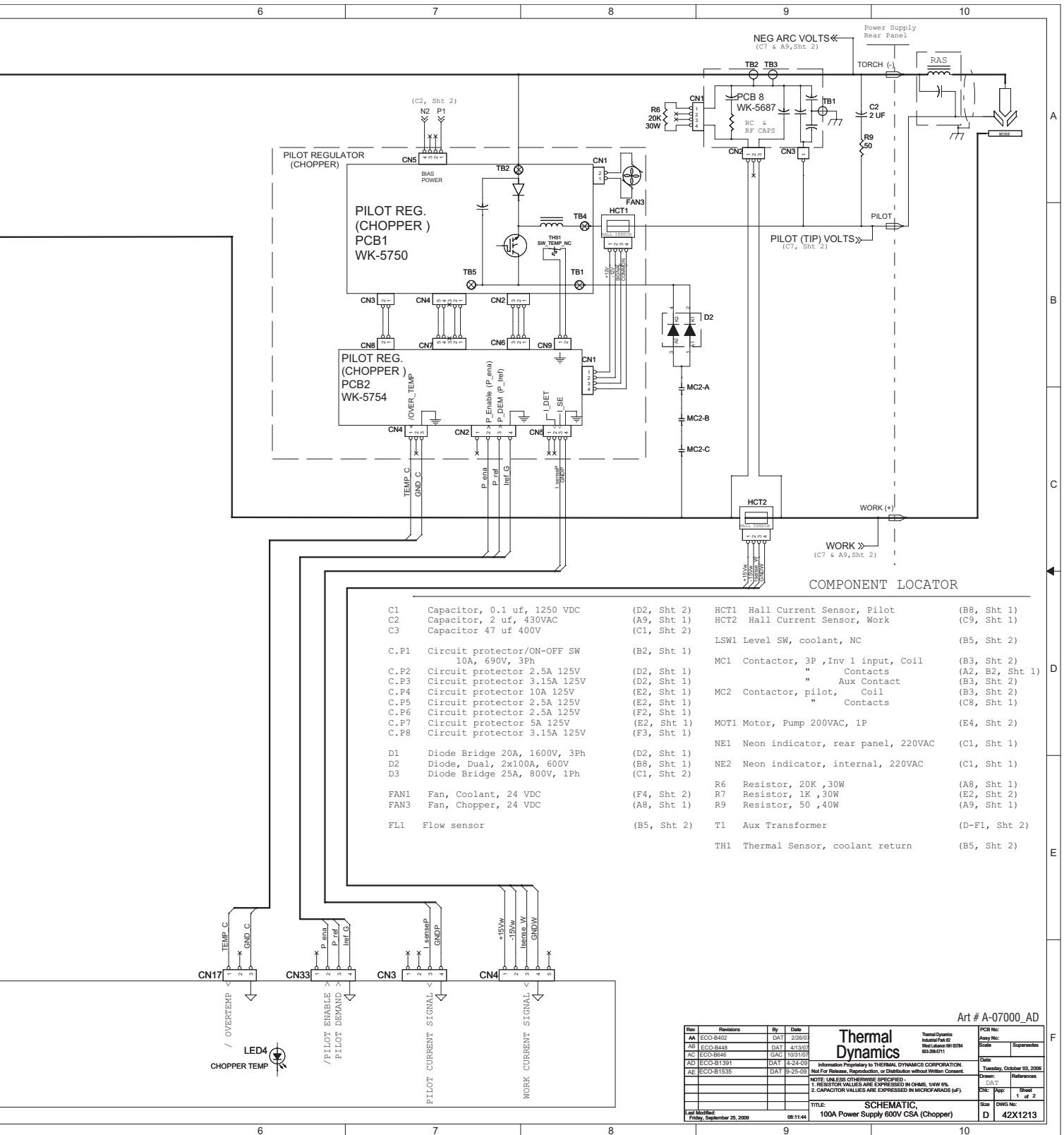
APPENDIX 11: Schematic, CE 400V, Chopper Pilot Cont.



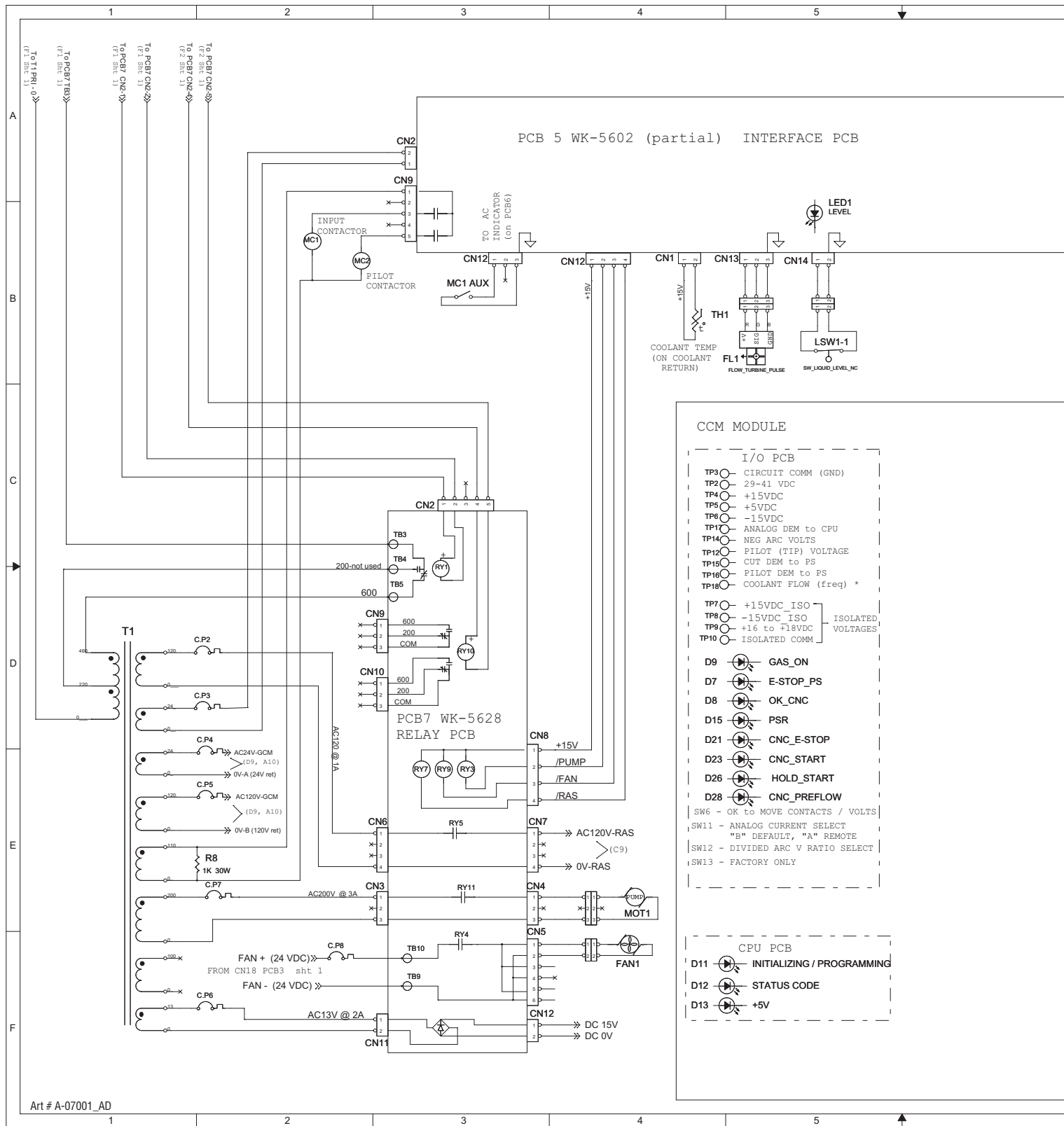


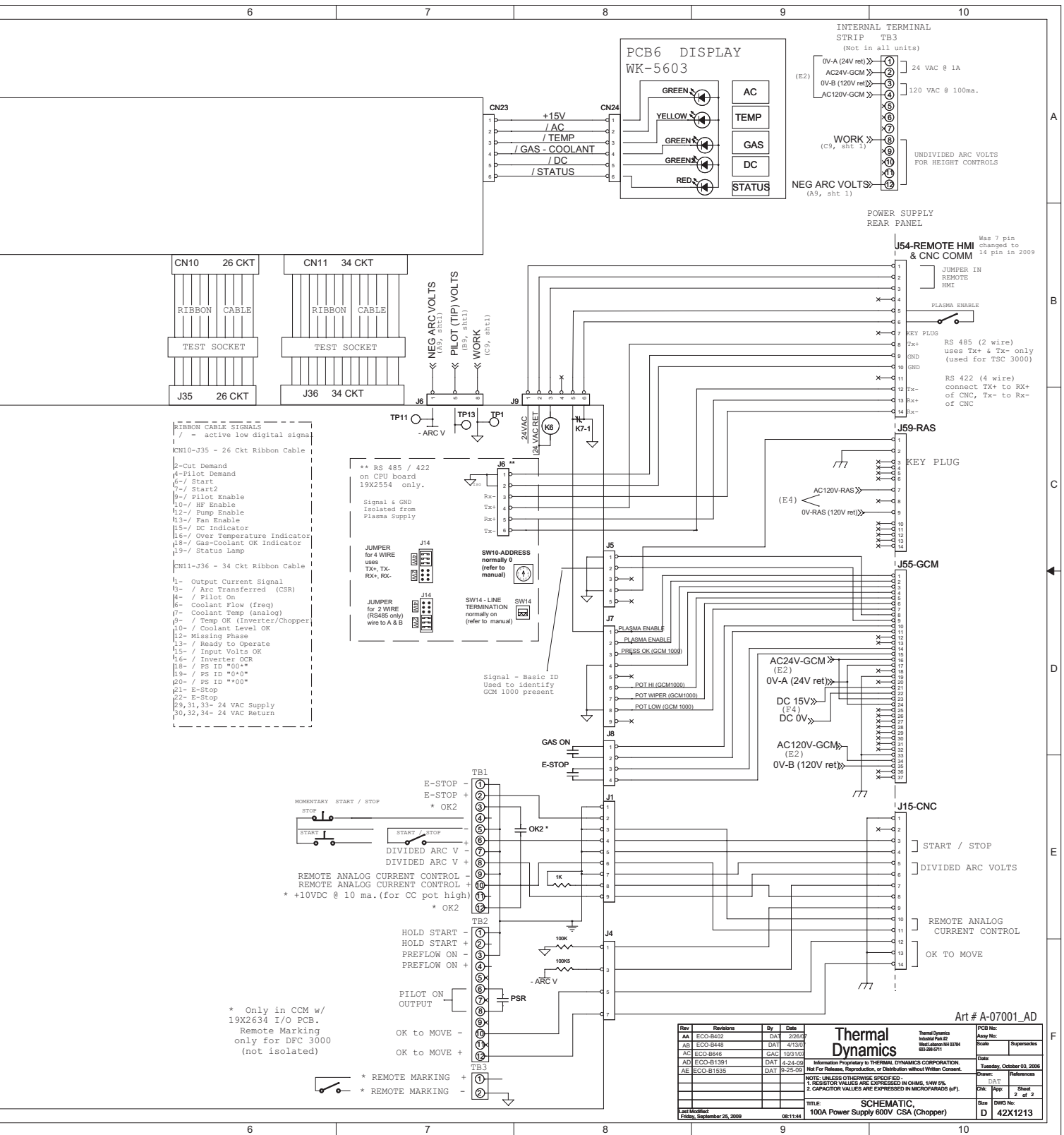
APPENDIX 12: Schematic, CSA 600V, Chopper Pilot



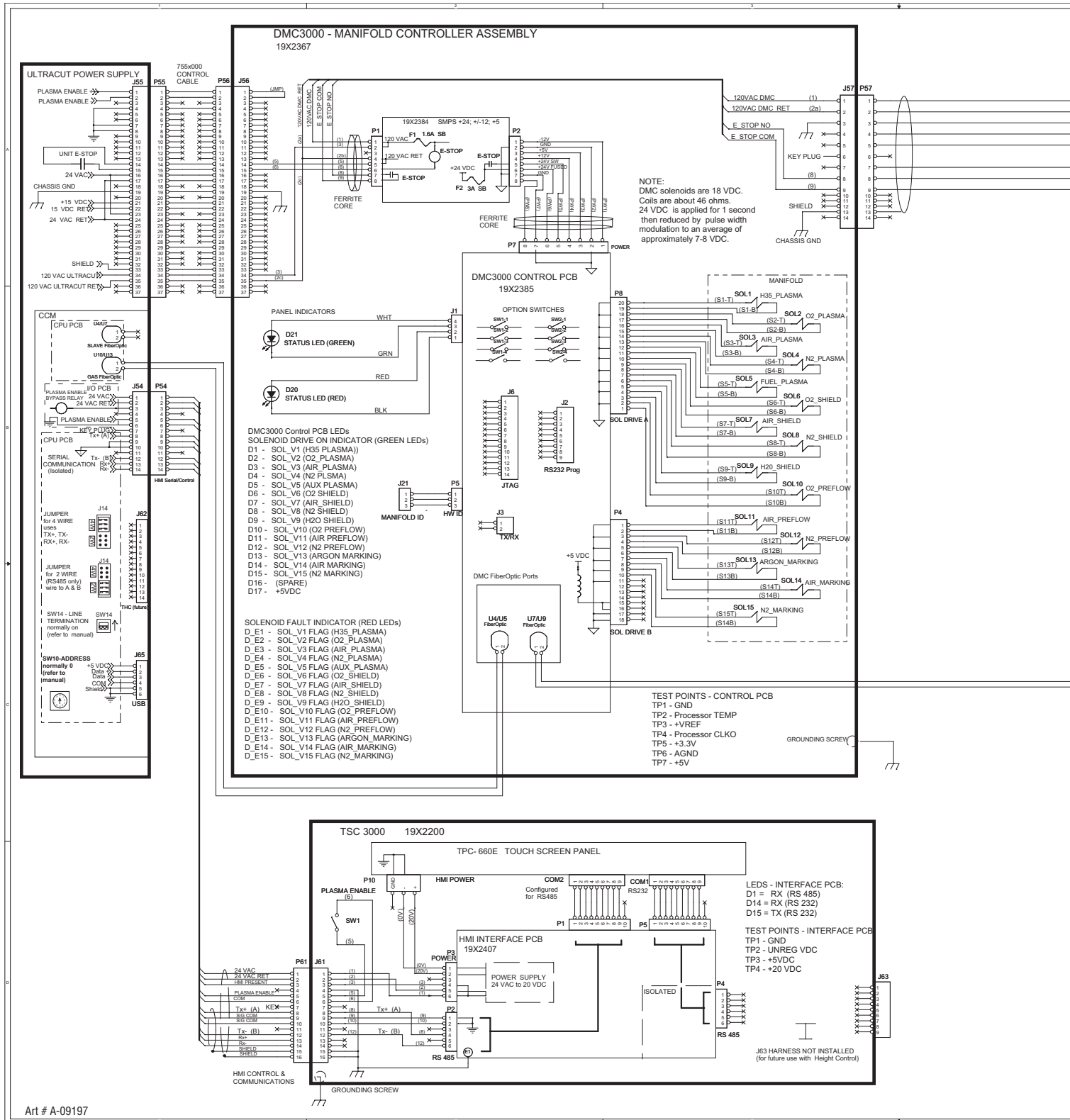


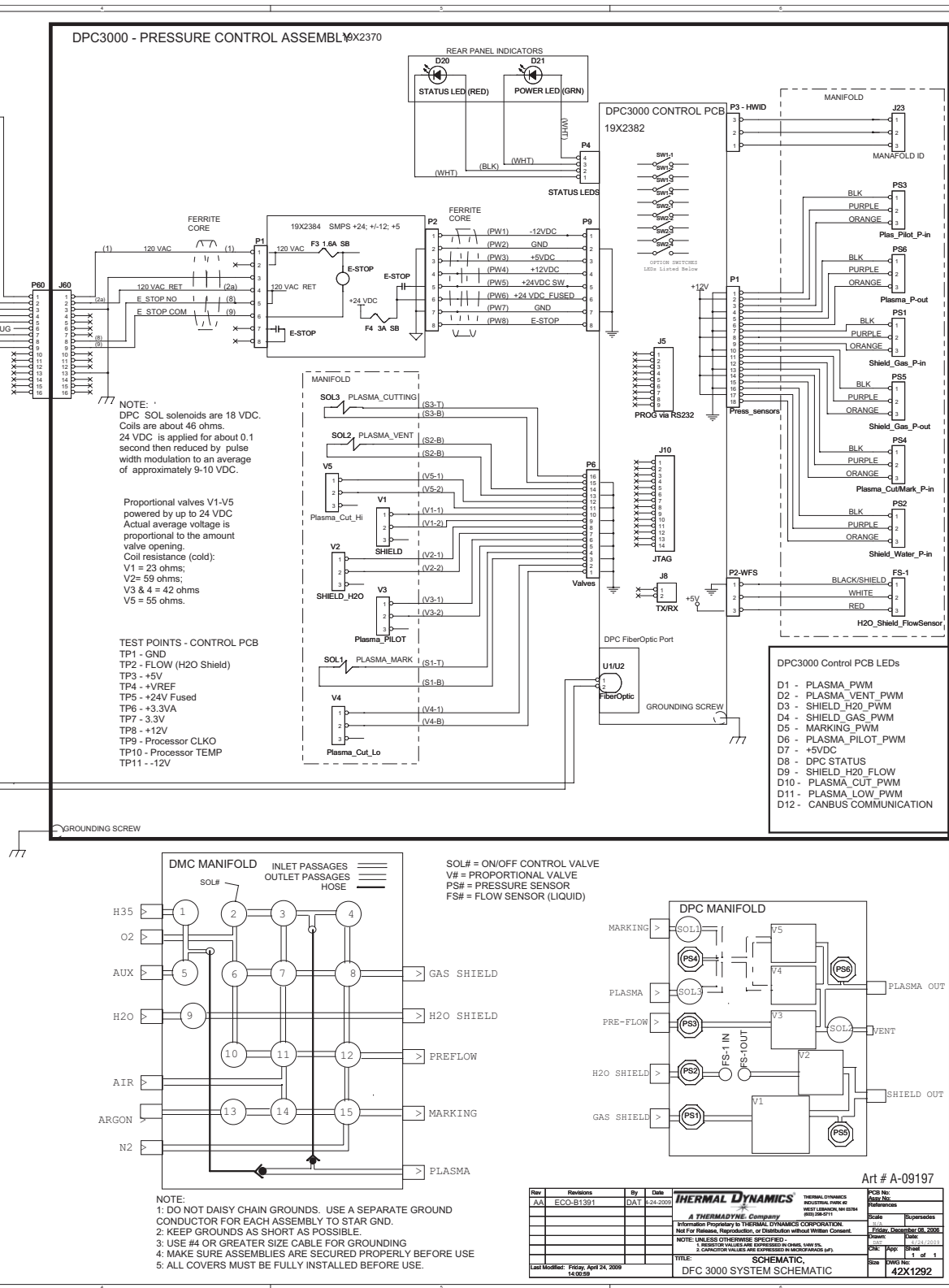
APPENDIX 12: Schematic, CSA 600V, Chopper Pilot Cont.





APPENDIX 13: Schematic, DFC-3000 Auto Gas Box System





Art # A-09197

Rev	Revisions	By	Date
AA	ECO-B1391	DAT	8-24-2009

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TITLE: SCHEMATIC DFC 3000 SYSTEM SCHEMATIC		Friday, December 18, 2008 1 of 1 42X1292

APPENDIX 14: PUBLICATION HISTORY

Cover Date	Rev	Change(s)
May 14, 2009	AA	First issue
Sept. 29, 2009	AB	Updated Appendix with new CNC Functions information and updated input power cable charts sections 2 and 3 and system schematics with serial communication connector and updated TB1 and TB3 information in Appendix 7 per ECOB1535.
Oct, 28, 2009	AC	Updated art in section 3 and 6 to show “W” in system layout being grounded and CNC communication cable having 11 not 8 wires per ECOB1566.
Nov. 25, 2009	AD	Corrected chart on pg. 4-21 to show all information in group 6, per ECOB1598.
Apr. 16, 2010	AE	Updated the appendix with serial communication information Appdx 8, per ECOB1706.
Aug. 26, 2010	AF	Updated sections 3 and 6 with new RAS-1000 part numbers per ECOB1704 and sections 6 with external sheet metal part numbers per ECOB1737. Updated section 6 DPC-3000 parts list per ECOB1713.
Oct. 28, 2010	AG	Updated section 6 by removing ref. to CCM type 2 module and corrected part numbers for TSC-3000 per ECOB1844.

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Canada Customer Care
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